Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 26–37 (Papers 556–558)

Transcribed from the Original Publications

Contents

556. On Steiner's Surface (continued)

[Continued from earlier pages of the same paper.]

[p. 6] where the first factor gives the node. Equating to zero the second factor, we have

$$\left\{qZ + \left(2q - 1 - \tfrac{2}{q}\right) X\right\}^2 = X^2 \left\{\left(2q - 1 - \tfrac{2}{q}\right)^2 - 4q^2 + 4q - 1\right\}$$

$$= X^2 \cdot \tfrac{4}{q^2}(1-q)(1+2q);$$

or, finally,

$$qZ = \left\{-2q + 1 + \tfrac{2}{q} \pm \tfrac{2}{q}\sqrt{(1-q)(1+2q)}\right\} X,$$

giving two real values for all values of q from $q = 1$ to $q = -\frac{1}{2}$. (See the Table afterwards referred to.)

We may recapitulate as follows:

$q > 1$, or $\lambda > \frac{1}{2}h$; the curve is imaginary, but with three real acnodes, answering to the acnodal parts of the nodal lines:

$q = 1$, or $\lambda = \frac{1}{2}h$; the summit appears as a fourth acnode:

$q < 1 > \frac{1}{4}$, or $\lambda < \frac{1}{2}h > \frac{1}{3}h$; the curve consists of three acnodes and a trigonoid lying within the triangle and having the sides of the triangle for bitangents of imaginary contact:

$q = \frac{1}{4}$, or $\lambda = \frac{1}{3}h$; the curve consists of three acnodes and a trigonoid having the sides of the triangle for osculating tangents:

$q < \frac{1}{4} > 0$, or $\lambda < \frac{1}{3}h > \frac{1}{4}h$; the curve consists of three conjugate points and an indented trigonoid having the sides of the triangle for bitangents of real contact:

$q = 0$, or $\lambda = \frac{1}{4}h$; curve has the summits of the triangle for cusps:

$q < 0 > -\frac{1}{4}$, or $\lambda < \frac{1}{4}h > \frac{1}{6}h$; curve has three crunodes, or say it is a cis-centric trifolium:

$q = -\frac{1}{4}$, or $\lambda = \frac{1}{6}h$; curve has a triple point, or say it is a centric trifolium:

$q < -\frac{1}{4} > -\frac{1}{2}$, or $\lambda < \frac{1}{6}h > 0$; curve has three crunodes, or say it is a trans-centric trifolium:

$q = -\frac{1}{2}$, or $\lambda = 0$; curve is a two-fold circle:

$q < -\frac{1}{2}$, or $\lambda < 0$; the curve becomes again imaginary, consisting of three acnodes answering to the acnodal parts of the nodal lines.

For the better delineation of the series of curves, I calculated the following Table, wherein the first column gives a series of values of $\lambda : h$; the second the corresponding values of q, $= \dfrac{2\lambda - h}{2\lambda - 2h}$; the third the positions of the point of contact, say with the side $Z = 0$, the value of $X : Y$ being calculated from the foregoing formula

$$\frac{X}{Y} = \frac{1}{2q} + 1 \pm \sqrt{\frac{1}{4q^2} - \frac{1}{q}};$$

[p. 7] and the fourth the apsidal distances, say for the radius vector $X = Y$, the value of $Z : X$ being calculated from the foregoing formula

$$\frac{Z}{X} + \frac{X}{Z} = -2 + \frac{1}{q} + \frac{2}{q^2} \pm \frac{2}{q}\sqrt{\left(\frac{1}{q} - 1\right)\left(2 + \frac{1}{q}\right)}.$$

The Table is:

$\lambda : h$	q	Contact, $Z=0$, $X:Y=$	Apses, $X=Y$; $X:Z=$
$*\ \cdot 75$	$1\cdot 00$	imposs.	$1\cdot$
$\cdot 70$	$\cdot 6666$	imposs.	$\cdot 032$ or $7\cdot 968$
$*\ \cdot 666$	$\cdot 5$	$1\cdot$	$0\cdot \quad 16\cdot$
$\cdot 65$	$\cdot 4285$	$\cdot 320$ or $3\cdot 124$	$\cdot 006 \quad 22\cdot 44$
$\cdot 6$	$\cdot 25$	$\cdot 0721 \quad 13\cdot 9279$	$\cdot 059 \quad 67\cdot 941$
$\cdot 55$	$\cdot 1111$	$\cdot 011 \quad 78\cdot 988$	$\cdot 15 \quad 337\cdot 85$
$*\ \cdot 5$	$0\cdot$	0 or ∞	$\cdot 25 \quad \infty$
$\cdot 45$	$-\cdot 0909$	$\cdot 005 \quad 118\cdot 99$	$\cdot 39 \quad 457\cdot 61$
$\cdot 4$	$-\cdot 1666$	$\cdot 029 \quad 33\cdot 971$	$\cdot 496 \quad 127\cdot 504$
$\cdot 35$	$-\cdot 2308$	$\cdot 060 \quad 16\cdot 72$	$\cdot 648 \quad 61\cdot 79$
$*\ \cdot 3$	$-\cdot 2857$	$\cdot 099 \quad 10\cdot 151$	$\cdot 812 \quad 37\cdot 187$
$*\ \cdot 25$	$-\cdot 3333$	$\cdot 141 \quad 6\cdot 854$	$1\cdot \quad 25$
$\cdot 2$	$-\cdot 375$	$\cdot 207 \quad 4\cdot 904$	$1\cdot 218 \quad 17\cdot 89$
$\cdot 15$	$-\cdot 4118$	$\cdot 276 \quad 3\cdot 622$	$1\cdot 48 \quad 13\cdot 25$
$\cdot 10$	$-\cdot 4444$	$\cdot 372 \quad 2\cdot 690$	$1\cdot 813 \quad 9\cdot 937$
$\cdot 05$	$-\cdot 4737$	$\cdot 515$ or $1\cdot 941$	$3\cdot 15$ or $6\cdot 46$
$*\ 0$	$-\cdot 5$	$1\cdot$	$4\cdot$

where the asterisks show the critical values of $\lambda : h$.

It is worth while to transform the equation to new coordinates X', Y', Z' such that $X'=0$, $Y'=0$, $Z'=0$ represent the sides of the triangle formed by the three nodes. Writing for shortness $X+Y+Z=P$, $YZ+ZX+XY=Q$, $XYZ=R$, the equation is

$$(qP-Q)^2 = 4(2q+1)PR.$$

The expressions of X, Y, Z in terms of the new coordinates are of the form $X'+\theta P'$, $Y'+\theta P'$, $Z'+\theta P'$, where $P'=X'+Y'+Z'$; writing also $Q'=Y'Z'+Z'X'+X'Y'$, $R'=X'Y'Z'$, then the values of P, Q, R are

$$(1+3\theta)P', \quad Q'+(2\theta+3\theta^2)P'^2, \quad R'+\theta P'Q'+(\theta^2+\theta^3)P'^3,$$

[p. 8] and the transformed equation is

$$\left[\{q(1+3\theta)^2-2\theta-3\theta^2\}P'^2-Q'\right]^2 = 4(2q+1)(1+3\theta)\{(\theta+\theta^2)P'^3+\theta P'Q'+R'\},$$

which is satisfied by $Q'=0$, $R'=0$, if only

$$\{q(1+3\theta)^2-2\theta-3\theta^2\}^2 = 4(2q+1)(1+3\theta)(\theta^2+\theta^3),$$

or, if for a moment $q(1+3\theta)=\Omega$, the equation is

$$(\Omega^2-2\theta-3\theta^2)^2 = 4(\theta^2+\theta^3)(2\Omega+1+3\theta),$$

that is,

$$\Omega^4+\Omega^2(-6\theta^2-4\theta)+\Omega(-8\theta^3-8\theta^2)-3\theta^4-4\theta^3=0,$$

that is,

$$(\Omega+\theta)^2(\Omega^2-2\theta\Omega-3\theta^2-4\theta)=0.$$

If the new axes pass through the nodes, then $\Omega + \theta = 0$; that is, $q(1+3\theta)+\theta = 0$, which equation gives the value of θ for which the new axes have the position in question; substituting in the first instance for q the value $\dfrac{-\theta}{3\theta+1}$, the equation becomes

$$\theta^2\{2\theta(1+\theta)P'^2 + Q'\}^2 = 4(1+\theta)P'\{\theta^2(1+\theta)P'^3 + \theta P'Q' + R'\},$$

that is,

$$Q'^2 = 4(1+\theta)P'R'\,;$$

or, finally, substituting for θ its value in terms of q, the required equation is

$$Q'^2 = 4\cdot\frac{2q+1}{3q+1}\,P'R',$$

that is,

$$(Y'Z' + Z'X' + X'Y')^2 = 4\cdot\frac{2q+1}{3q+1}\,X'Y'Z'(X'+Y'+Z').$$

In particular, for $q = 0$ the equation is

$$(Y'Z' + Z'X' + X'Y')^2 - 4X'Y'Z'(X'+Y'+Z') = 0,$$

which is right, since, in the case in question (the tricuspidal curve), we have

$$X,\ Y,\ Z = X',\ Y',\ Z'.$$

I remark, in passing, that, taking the equation to be

$$(Y'Z' + Z'X' + X'Y')^2 = m\,X'Y'Z'(X'+Y'+Z'),$$

we may write herein

$$\begin{aligned} Z' &= \tfrac{1}{2} - x,\\ X' &= \tfrac{1}{4} + \tfrac{1}{2}x - \tfrac{\sqrt{3}}{2}y,\\ Y' &= \tfrac{1}{4} + \tfrac{1}{2}x + \tfrac{\sqrt{3}}{2}y\,; \end{aligned}$$

[p. 9] where

$$\begin{aligned} x &= \frac{2\sqrt{m(m-3)}}{9}\cos\theta - \frac{2(m-3)}{9}\cos 2\theta,\\ y &= \frac{2\sqrt{m(m-3)}}{9}\sin\theta + \frac{2(m-3)}{9}\sin 2\theta, \end{aligned}$$

which are the formulae for the description of the trinodal quartic as a unicursal curve.

I consider now the second position; viz. the horizontal plane now passes through the centre of the tetrahedron, and is parallel to two opposite edges. The equations of the nodal lines are here $(y=0,\ z=0)$, $(z=0,\ x=0)$, $(x=0,\ y=0)$; and if for convenience we assume the distance of the mid-points of opposite edges to be $= 2$, or the half of this $= 1$, then the equations of the faces are

$$\begin{aligned} X &= x + y + z - 1 = 0,\\ Y &= -x - y + z - 1 = 0,\\ Z &= x - y - z - 1 = 0,\\ W &= -x + y - z - 1 = 0, \end{aligned}$$

and the equation of the surface is

$$\sqrt{X} + \sqrt{Y} + \sqrt{Z} + \sqrt{W} = 0.$$

Proceeding to rationalise, this is

$$X + Y + 2\sqrt{XY} = Z + W + 2\sqrt{ZW},$$

viz.

$$2z + \sqrt{XY} = \sqrt{ZW}\,;$$

we thence have

$$4z^2 + 4z\sqrt{XY} + XY = ZW,$$

or, since

$$ZW - XY = 4z + 4xy,$$

this is

$$z + xy - z^2 = z\sqrt{XY}\,;$$

whence

$$(z + xy - z^2)^2 = z^2\{(z-1)^2 - (x+y)^2\}\,;$$

or reducing,

$$x^2y^2 + z^2x^2 + z^2y^2 + 2xyz - z^2 = 0,$$

a form which puts in evidence the nodal lines. Considering z as constant, we have the equation of the section; this is a quartic having the node $(x = 0,\ y = 0)$, and two other nodes at infinity on the two axes respectively; moreover, the curve has for bitangents the intersections of its plane with the faces of the tetrahedron; or what is the same thing, attributing to z its constant value, the equations of the bitangents are

$$\begin{aligned} x + y + z - 1 &= 0, \\ -x - y + z - 1 &= 0, \\ x - y - z - 1 &= 0, \\ -x + y - z - 1 &= 0. \end{aligned}$$

[p. 10] These lines form a rectangle which is the section of the tetrahedron; observe that this is inscribed in the square the corners of which are $x = \pm 1$, $y = \pm 1$; viz. $z = +1$ (highest section), this is the dexter diagonal (considered as an indefinitely thin rectangle), and as z diminishes, the longer side decreases and the shorter increases until for $z = 0$ (central section) the rectangle becomes a square; after which, for z negative it again becomes a rectangle in the conjugate direction, and finally, for $z = -1$ (lowest section) it becomes the sinister diagonal (considered as an indefinitely thin rectangle). But on account of the symmetry it is sufficient to consider the upper sections for which z is positive. The sides $\pm(x + y) + z - 1 = 0$ parallel to the dexter diagonal of the square may for convenience be termed the dexter sides, and the others the sinister sides. In what follows I write c to denote the constant value of z.

We require to know whether the bitangents have real or imaginary contacts; and for this purpose to find the coordinates of the points of contact.

Take first a dexter bitangent $x + y + c - 1 = 0$; the coordinates of any point hereof are

$$x = \tfrac{1}{2}(1 - c + \theta), \quad y = \tfrac{1}{2}(1 - c - \theta),$$

where θ is arbitrary; and substituting in the equation of the curve, we should have for θ a twofold quadric equation, giving the values of θ for the two points of contact respectively. We have

$$x^2 + y^2 = \tfrac{1}{2}\{(1-c)^2 + \theta^2\}, \quad xy = \tfrac{1}{4}\{(1-c)^2 - \theta^2\},$$

and thence

$$8c^2\{(1-c)^2 + \theta^2\} + 8c\{(1-c)^2 - \theta^2\} + \{(1-c)^2 - \theta^2\}^2 = 0,$$

viz. this equation is

$$\{\theta^2 - (1-c)(1+3c)\}^2 = 0,$$

a twofold quadric equation, as it should be; and the values of θ being $= \pm\sqrt{(1-c)(1+3c)}$, we see that these, and therefore the contacts, are real from $c = 1$ to $c = -\frac{1}{3}$.

In exactly the same way for a sinister bitangent $\pm(x-y) - c - 1 = 0$, we have

$$x = \tfrac{1}{2}(1 + c + \phi), \quad -y = \tfrac{1}{2}(1 + c - \phi), \quad \text{and} \quad \phi = \pm\sqrt{(1-3c)(1+c)},$$

viz. the values of ϕ, and therefore the contacts, are real from $c = \frac{1}{3}$ to $c = -1$.

That is,

$c = 1$ to	Contacts of Dexter Bitangents.	Contacts of Sinister Bitangents.
$c = 1$ to $\frac{1}{3}$	real,	imaginary,
$c = \frac{1}{3}$ to $-\frac{1}{3}$	real,	real,
$c = -\frac{1}{3}$ to -1	imaginary,	real.

or say $c = 1$ to $\frac{1}{3}$, the contacts are real, imaginary; but $c = \frac{1}{3}$ to 0, they are real, real. In the transition case, $c = \frac{1}{3}$, the sinister bitangents become osculating (4-pointic) tangents touching at points on the dexter diagonal. This can be at once verified.

[p. 11] Observe that when $c = 1$, we have

$$(x+y)^2 + x^2y^2 = 0,$$

so that the only real point is $x = 0$, $y = 0$; viz. this is a tacnode, having the real tangent $x + y = 0$. For $c = 0$ (central section) the equation becomes $x^2y^2 = 0$; viz. the curve is here the two nodal lines each twice.

It is now easy to trace the changes of form.

$c = 1$; curve is a tacnode, as just mentioned, tangent the dexter diagonal.

$c < 1 > \frac{1}{3}$; curve is a figure of 8 inside the rectangle, having real contacts with the dexter sides, but imaginary contacts with the sinister sides.

$c = \frac{1}{3}$; curve is a figure of 8 having real contacts with the dexter sides, and osculating (4-pointic) contacts with the sinister sides.

$c < \frac{1}{3} > 0$; curve is an indented figure of 8 having real contacts as well with the sinister as the dexter sides.

$c = 0$; curve is squeezed up into a finite cross, being the crunodal parts of the nodal lines; and joined on to these we have the acnodal parts, so that the whole curve consists of the lines $x = 0$, $y = 0$ each as a twofold line.

For tracing the curve, it is convenient to turn the axes through an angle of 45°; viz. writing $\dfrac{y+x}{\sqrt{2}}$, $\dfrac{y-x}{\sqrt{2}}$ in place of x, y respectively, the equation becomes

$$c^2(y^2-x^2)^2+c(y^2+x^2)+\tfrac{1}{4}(y^2-x^2)^2=0\,;$$

$$x=0 \text{ gives } y^2=0 \text{ or } y^2=-4c(1+c),^*$$

$$y=0 \text{ gives } x^2=0 \text{ or } x^2=4c(1-c).$$

Moreover, we have

$$\tfrac{1}{4}(c-1)^2(y^2-x^2)+8c^2y^2+(y^2-x^2)^2=0,$$

viz.

$$(x^2+y^2-2c^2-2c)^2=4c^2\{(c-1)^2-2y^2\},$$

and similarly

$$(x^2+y^2+2c^2-2c)^2=4c^2\{(c+1)^2-2x^2\},$$

putting in evidence the bitangents, now represented by the equations $c-1=\pm y\sqrt{2}$ and $c+1=\pm x\sqrt{2}$ respectively. And for the first of these, or $c-1=\pm y\sqrt{2}$, we have for the points of contact $x^2=\frac{1}{2}(1-c)(1+3c)$; and for the second of them, or $c+1=\pm x\sqrt{2}$, the points of contact are $y^2=\frac{1}{2}(1+c)(1-3c)$.

I consider the circumscribed cone having its vertex at a point $(0,0,\gamma)$ on the nodal line $(x=0,\ y=0)$. Writing in the equation of the surface $x=\lambda(z-\gamma)$, $y=\mu(z-\gamma)$, the equation, throwing out the factor $(z-\gamma)^2$, becomes

$$\lambda^2\mu^2(z-\gamma)^2+2\lambda\mu z(z-\gamma)+(\lambda^2+\mu^2)z^2-z^2=0,$$

that is,

$$(\lambda^2\mu^2+\lambda^2+\mu^2)\,z^2+2(-\gamma\lambda\mu+1)\,z\lambda\mu+\gamma^2\lambda^2\mu^2=0\,;$$

* y always imaginary when c is positive.

[p. 12] and equating to zero the discriminant in regard to z, we have

$$\gamma^2\lambda^2\mu^2(\lambda^2\mu^2+\lambda^2+\mu^2)-(-\gamma\lambda\mu+1)^2=0,$$

that is,

$$\gamma^2(\lambda^2+\mu^2)+2\gamma\lambda\mu-1=0\,;$$

and substituting herein the values $\lambda=\dfrac{x}{z-\gamma}$ and $\mu=\dfrac{y}{z-\gamma}$, we have the equation of the cone, viz. this is

$$\gamma^2(x^2+y^2)+2\gamma xy-(z-\gamma)^2=0,$$

or, what is the same thing,

$$\gamma^2(x^2+y^2-1)+2\gamma(xy+z)-z^2=0\,;$$

viz. this is a quadric cone having for its principal planes $z=\gamma$, $x+y=0$, $x-y=0$, these last being the planes through the nodal line and the two edges of the tetrahedron.

In the particular case $\gamma=\infty$, the cone becomes the circular cylinder $x^2+y^2-1=0$.

The cone intersects the plane $z=0$ in the conic

$$\gamma^2(x^2+y^2-1)+2\gamma xy=0,$$

which is a conic passing through the corners of the square $(x = 0,\ y = \pm 1)$, $(x = \pm 1,\ y = 0)$. For $\gamma > 1$, that is, for an exterior point, the conic is an ellipse having for the squares of the reciprocals of the semi-axes $1 + \frac{1}{\gamma}$, $1 - \frac{1}{\gamma}$ (this at once appears by writing in the equation $\frac{x+y}{\sqrt{2}}$, $\frac{x-y}{\sqrt{2}}$ in place of x, y respectively). In particular, for $\gamma = \infty$, the curve becomes the circle $x^2 + y^2 = 1$. We have thus the apparent contour of the surface as seen from the point $z = \gamma$ on the nodal line, projected on the plane $z = 0$ of the other two nodal lines.

To find the curve of contact of the cone and surface, or say the surface-contour from the same point, write for a moment

$$V = \gamma^2(x^2 + y^2 - 1) + 2\gamma(xy + z) - z^2,$$

$$U = (xy + z)^2 - z^2(x^2 + y^2 - 1),$$

then, substituting for $x^2 + y^2 - 1$ its value in terms of V from the first equation, we find

$$U = \left(xy + z - \frac{z^2}{\gamma}\right)^2 \frac{V}{\gamma^2},$$

and the equations $V = 0$, $U = 0$ give therefore

$$xy + z - \frac{z^2}{\gamma} = 0,$$

or say $\gamma(xy + z) - z^2 = 0$.

The cone and surface therefore touch along the quadriquadric curve

$$\gamma^2(x^2 + y^2 - 1) + 2\gamma(xy + z) - z^2 = 0,$$

$$\gamma(xy + z) - z^2 = 0,$$

equations which may be replaced by

$$\gamma^2(x^2 + y^2 - 1) + xy + z = 0,$$

$$\gamma^2(x^2 + y^2 - 1) + z^2 = 0.$$

In the case $\gamma = \infty$, the equations are $x^2 + y^2 - 1 = 0$, $xy + z = 0$, viz. the curve is the intersection of the hyperbolic paraboloid $xy + z = 0$ by the cylinder $x^2 + y^2 - 1 = 0$.

557. On Certain Constructions for Bicircular Quartics

[p. 13] *[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 29–31. Read March 12, 1874.]*

I CALL to mind that if F, G are any two points and F', G' their antipoints, then the circle on the diameter FG and that on the diameter $F'G'$ are concentric orthotomics, viz. they have the same centre, and the sum of the squared radii is $= 0$. Moreover, if the circles B, B' are concentric orthotomics, and the circle A is orthotomic to B, then it is a bisector of B', viz. it cuts B' at the extremities of a diameter of B'; and B is then said to be a bifid circle in regard to A.

Given two real circles, these have an axial orthotomic, the circle, centre on the line of centres at its intersection with the radical axis, which cuts at right angles the given circles; viz. this axial orthotomic is real if the circles have no real intersection; but if the intersections are real, then the axial orthotomic is a pure imaginary, and instead thereof we may consider its concentric orthotomic, viz. this is the axial bifid of the two circles, or circle having its centre on the line of centres at the intersection thereof with the radical axis or common chord of the two circles, and having this common chord for its diameter.

If one of the circles is a pure imaginary, then we have still an axial orthotomic; viz. the pure imaginary circle is replaced by the concentric orthotomic; and the axial orthotomic is a bisector of the substituted circle; and so if each of the circles is a pure imaginary, then we have still an axial orthotomic, viz. each circle is replaced by the concentric orthotomic, and the axial orthotomic is a bisector of the substituted circles. And in either case the axial orthotomic of the original circles (one or each of them pure imaginary) is real; viz. this is given either as the axial bisector of one real circle and orthotomic of another real circle; or as the axial bisector of two circles, from which the reality thereof easily appears. Or we may verify it thus: Suppose

[p. 14] that the two circles are $(x-\alpha)^2+y^2=\beta^2$, $(x-\alpha')^2+y^2=\beta'^2$, and their axial orthotomic $(x-m)^2+y^2=k^2$, then we have $(m-\alpha)^2=\beta^2+k^2$, $(m-\alpha')^2=\beta'^2+k^2$; subtracting, it appears that m is real; and then if either β^2 or β'^2 is negative, the equation containing this quantity shows that k^2 is positive; viz. the circle $(x-m)^2+y^2=k^2$ is real.

The above remarks have an obvious application to the theory of bicircular quartics; viz. a bicircular quartic is the envelope of a variable circle, having its centre on a conic, and orthotomic to a circle: it may be that this circle is a pure imaginary. We then replace it by the concentric orthotomic, and say that the curve is the envelope of a variable circle having its centre on a conic and bisecting a circle. We have thus a real form for cases which originally present themselves under an imaginary form.

The Bicircular Quartic with given vertices.

First, if the vertices are real; let the vertices taken in order be F, G, H, K.

First construction: On HK as diameter describe a circle, and on FG as diameter a circle; on the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_1, and describe the axial orthotomic circle Σ_1 of the two circles (viz. the centre of Σ_1 is on the axis of symmetry at its intersection with the radical axis of the two circles); then the curve is the envelope of a variable circle having its centre on Θ_1 and orthotomic to Σ_1.

Second construction: On FH as diameter describe a circle, and on GK as diameter a circle. On the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_2, and describe the axial bifid circle Σ_2' of the two circles (viz. the centre of Σ_2' is on the axis of symmetry at its intersection with the radical axis or common chord of the two circles, and its diameter is this common chord); then the curve is the envelope of a variable circle having its centre on Θ_2 and bisecting Σ_2'.

Third construction: On FK as diameter describe a circle, and on GH as diameter a circle; and then, as in the first construction, a conic Θ_3 and a circle Σ_3; the curve is the envelope of a variable circle having its centre on Θ_3 and orthotomic to Σ_3.

Observe that in the three constructions the conics have always the same centre; and if the three conics are taken with the same foci, then the three constructions give one and the same bicircular quartic. The first and third constructions form a pair, and there is no reason for selecting one of them in preference to the other; but the second construction is unique; it is on this account natural to make use of it in discussing the series of curves with the given vertices.

In the particular case where the points F, G and H, K are situate symmetrically on opposite sides of a centre O ($OF = OK$, $OG = OH$), then in the third construction the centres each coincide with O, or the axis of the conic vanishes; hence the construction fails: the first and second constructions hold good, and in each of them the

[p. 15] conic and circle are concentric. The curve is in this case quadrantal: having, besides the original axis of symmetry, another axis of symmetry through O, at right angles thereto.

Secondly, if the vertices are two real, two imaginary, say f, $g = \alpha \pm \beta i$; h, k, we modify the first or third construction; viz. if F', G' are the antipoints of F, G; then on $F'G'$ as diameter describe a circle, and on HK as diameter a circle. On the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_1, and describe the axial bisector-orthotomic circle Σ_1 of the two circles; viz. this is the circle (centre on the axis of symmetry) which bisects the circle $F'G'$, and cuts at right angles the circle HK; then the curve is the envelope of the variable circle having its centre on Θ_1 and orthotomic to Σ_1.

Thirdly, if the vertices are all imaginary, say f, $g = \alpha \pm \beta i$; h, $k = \gamma \pm \delta i$, we modify the first or third construction. Take F', G' the antipoints of F, G, and H', K' the antipoints of H, K; then on $F'G'$ as diameter describe a circle, and on $H'K'$ as diameter a circle; on the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ, and describe the axial bisector-circle Σ of the two circles (viz. this is a circle, centre on the axis of symmetry, bisecting each of the circles); the curve is the envelope of a variable circle, centre on the conic Θ and cutting at right angles the circle Σ.

558. A Geometrical Interpretation of the Equations Obtained by Equating to Zero the Resultant and the Discriminants of Two Binary Quantics

[p. 16] *[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 31–33. Read March 12, 1874.]*

CONSIDER the equations

$$U = (a,\ b,\ \ldots \,|\, t,\ 1)^{\lambda} = 0,$$

$$U' = (a',\ b',\ \ldots \,|\, t,\ 1)^{\lambda'} = 0\,;$$

and equating to zero the discriminants of the two functions respectively, and also the resultant of the two functions, let the equations thus obtained be

$$\Delta = (a,\ b,\ \ldots)^{2\lambda-2} = 0,$$

$$\Delta' = (a',\ b',\ \ldots)^{2\lambda'-2} = 0,$$

$$R = (a,\ b,\ \ldots)^{\lambda'}(a',\ b',\ \ldots)^{\lambda} = 0.$$

I take $(a,\ b,\ \ldots)$, $(a',\ b',\ \ldots)$ to be linear functions of the coordinates $(x,\ y,\ z)$; and t to be an indeterminate parameter. Hence $U = 0$ represents a line the envelope whereof is the curve $\Delta = 0$, or, what is the same thing, the equation $U = 0$ represents any tangent of the curve $\Delta = 0$; this is a unicursal curve of the order $2\lambda - 2$ and class λ, with $3(\lambda-2)$ cusps and $\frac{1}{2}(\lambda-2)(\lambda-3)$ nodes. Similarly $U' = 0$ represents a line the envelope of which is the curve $\Delta' = 0$: this is a unicursal curve of the order $2\lambda'-2$ and class λ', with $3(\lambda'-2)$ cusps and $\frac{1}{2}(\lambda'-2)(\lambda'-3)$ nodes; the equation $U' = 0$ represents any tangent of this curve.

The equations $U = 0$, $U' = 0$ considered as existing simultaneously with the same value of t, establish a $(1, 1)$ correspondence between the tangents (or if we please, between the points) of the two curves. The locus of the intersection of the corresponding tangents is the curve $R = 0$, a unicursal curve of the order $\lambda+\lambda'$, with $\frac{1}{2}(\lambda+\lambda'-1)(\lambda+\lambda'-2)$ nodes and no cusps; consequently of the class $2(\lambda+\lambda'-1)$.

[p. 17] It is to be shown that the curve $R = 0$ touches the curve $\Delta = 0$ in $\lambda' + 2\lambda - 2$ points, and similarly the curve $\Delta' = 0$ in $2\lambda' + \lambda - 2$ points.

In fact, consider any tangent T' of the curve Δ'; let this meet the curve Δ in a point A, and let Q be the tangent at A to the curve Δ; suppose, moreover, that T is the tangent of Δ corresponding to the tangent T' of Δ'. Then if Q and T coincide, the corresponding tangent of T' will be Q, and the curve R will pass through A. It is easy to see that in this case the curves R, Δ will touch at A.

Again, if P be a tangent from A to the curve Δ, then, if T and P coincide, the corresponding tangent of T' will be P, and the curve R will pass through A; but in this case the point A will be a mere intersection, not a point of contact, of the two curves.

The tangents T, Q each correspond to T', and they consequently correspond to each other. For a given position of T we have a single position of T', and therefore $2\lambda-2$ positions of A, or, what is the same thing, of Q; that is, for a given position of T we have $2\lambda-2$ positions of Q. Again, to a given position of Q corresponds a single position of A, therefore λ' positions of T', therefore also λ' positions of T; that is, for a given position of Q we have λ' positions of T. The correspondence between T, Q is thus a $(\lambda', 2\lambda-2)$ correspondence, and the number of united tangents is therefore $\lambda'+2\lambda-2$, or the curves R, Δ touch in $\lambda'+2\lambda-2$ points.

The tangents T, P each correspond to T', and they therefore correspond to each other. For a given position of T we have a single position of T', and therefore $2\lambda-2$ positions of A, and thence $(2\lambda-2)(\lambda-2)$ positions of P; that is, for a given position of T we have $(2\lambda-2)(\lambda-2)$ positions of P. Again, to a given position of P correspond $2\lambda-4$ positions of A, therefore $(2\lambda-4)\lambda'$ positions of T' or of T; that is, for a given position of P we have $(2\lambda-4)\lambda'$ positions of T. The correspondence between T, P is thus a $[2\lambda'(\lambda-2), 2(\lambda-1)(\lambda-2)]$ correspondence, and the number of united tangents is $2(\lambda+\lambda'-1)(\lambda-2)$; or the curves R, Δ meet in $2(\lambda+\lambda'-1)(\lambda-2)$ points.

Reckoning the contacts twice, the total number of intersections of R, Δ is

$$2\lambda'+4\lambda-4+2(\lambda+\lambda'-1)(\lambda-2) = (\lambda+\lambda')(2\lambda-2),$$

as it should be.

In the particular case $\lambda = \lambda' = 2$, the curves Δ, Δ' are conics, and the curve R is a quartic curve touching each of the conics 4 times; this is at once verified, since the equations here are

$$ac-b^2 = 0, \quad a'c'-b'^2 = 0, \quad 4(ac-b^2)(a'c'-b'^2)-(ac'+a'c-2bb')^2 = 0.$$

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 381–405 (Paper 607, continued)

Transcribed from the Original Publications

Contents

607. A Memoir on Prepotentials (continued)

[Continued from earlier pages of the same paper. Articles 96–131.]

[p. 381] **96.** The positive root θ of the equation

$$\frac{a^2}{f^2+\theta}+\frac{c^2}{h^2+\theta}-1+\frac{e^2}{\theta}=0$$

has certain properties which connect themselves with the function

$$\Theta_0=\theta^{q-1}\big\{(\theta+f^2)\dots(\theta+h^2)\big\}^{-\frac{1}{2}}.$$

We have, the accents denoting differentiations in regard to θ,

$$J'\,\frac{d\theta}{da}=\frac{2a}{\theta+f^2}\,,\qquad J'\,\frac{d\theta}{de}=-\frac{2e}{\theta}\,J'\,;\quad\text{and}\quad\frac{d\theta}{da}=\frac{1}{J'}\cdot\frac{2a}{\theta+f^2},$$

where

$$J'=\frac{a^2}{(f^2+\theta)^2}+\dots+\frac{c^2}{(h^2+\theta)^2}+\frac{e^2}{\theta^2}\,;$$

and we have the like formulae for $\dfrac{d\theta}{dc}$, $\dfrac{d\theta}{de}$.

We deduce

$$\frac{a}{\theta+f^2}\,\frac{d\theta}{da}+\dots+\frac{c}{\theta+h^2}\,\frac{d\theta}{dc}+\frac{e}{\theta}\,\frac{d\theta}{de}=\frac{2}{J'},$$

and to this we may join, η being arbitrary,

$$\frac{a}{\theta+\eta+f^2}\,\frac{d\theta}{da}+\dots+\frac{c}{\theta+\eta+h^2}\,\frac{d\theta}{dc}+\frac{e}{\theta+\eta}\,\frac{d\theta}{de}=\frac{2}{J'}\Big[1-\Big\{\frac{a^2\,\eta}{(\theta+f^2)(\theta+\eta+f^2)}+\dots+\frac{e^2\,\eta}{\theta(\theta+\eta)}\Big\}\Big].$$

Again, defining $\nabla_a\theta$ and $\Box\theta$ as immediately appears, we have

$$\nabla_a\theta=\left(\frac{d\theta}{da}\right)^2+\dots+\left(\frac{d\theta}{dc}\right)^2,\qquad=\frac{4}{J'},\quad\Box\theta=\frac{8a^2}{J'(\theta+f^2)^2}\,;$$

and passing to the second differential coefficients, we have

$$\frac{d^2\theta}{da^2}=\frac{2}{J'(\theta+f^2)}-\frac{8a^2}{J'^2(\theta+f^2)^2}-\frac{4a^2J''}{J'^3(\theta+f^2)^2}\,,$$

where

$$J''=-\frac{2a^2}{(f^2+\theta)^3}-\dots-\frac{2c^2}{(h^2+\theta)^3}-\frac{2e^2}{\theta^3}\,,$$

and the like formulae for $\dfrac{d^2\theta}{dc^2}$, $\dfrac{d^2\theta}{de^2}$. Joining to these $\dfrac{1}{e}\,\dfrac{d\theta}{de}=\dfrac{2}{J'}\cdot\dfrac{2q+1}{\theta}\,\dfrac{d\theta}{de}$, we obtain

$$\Box\theta=\left(\frac{d^2\theta}{da^2}+\dots+\frac{d^2\theta}{dc^2}+\frac{d^2\theta}{de^2}+\frac{2q+1}{e}\,\frac{d\theta}{de}\right)$$

$$=\frac{2}{J'}\left[\frac{1}{\theta+f^2}+\dots+\frac{1+(2q+1)}{\theta}\right]-\frac{2}{J'^2}\left[\frac{2a^2}{(\theta+f^2)^2}+\dots+\frac{2e^2}{\theta^2}\right]-\frac{J''}{J'^3}\,(4a^2\cdots).$$

[p. 382] where the last two terms destroy each other; observing that we have

$$\frac{1}{e^2}\cdot\frac{d}{de}\big(e^{2q+1}\big)=\frac{2q+1}{e^2}\cdot e^{2q}\cdot\frac{1}{e}\cdot e=\frac{2q+1}{e^2}\cdot e^{2q-2}\cdot e^2\,;$$

the result is

$$\Box\theta = \frac{2}{J'}\left(\frac{2q+2}{\theta} - \frac{2\theta'}{\theta}\right) + \frac{4\theta'}{\theta\, J'^2}\,.$$

97. First example, $x^2 = a^2 + \cdots + c^2$, and θ the positive root of $\dfrac{x^2}{f^2+\theta} + \dfrac{e^2}{\theta} = 1$.

V is assumed $= J\int_\theta^\infty t^{q-1}(t+f^2)^{-\frac{1}{2}s}\,dt$, where $q+1$ is positive.

I do not work the example out; it corresponds step by step with, and is hardly more simple than, the next example, which relates to the ellipsoid. The result is

$$\rho = (\sqrt{\pi})^s\,\Gamma(q+1)^{-1}\left[\left(1 - \tfrac{x^2}{f^2} - \tfrac{e^2}{f^2}\right)^q\right], \quad \text{if} \quad \tfrac{x^2}{f^2} + \tfrac{e^2}{f^2} < 1;$$

hence the integral

$$\int \frac{\left\{1 - \frac{x^2}{f^2} - \frac{e^2}{f^2}\right\}^q dx \ldots dz}{\left\{(a-x)^2 + \cdots + (c-z)^2 + e^2\right\}^{\frac{1}{2}s+q}}\,,$$

taken over the sphere $a^2 + \cdots + z^2 = f^2$,

$$= \frac{(\sqrt{\pi})^s\,\Gamma(q+1)}{\Gamma(\frac{1}{2}s+q+1)}\, f^s \int_\theta^\infty t^{q-1}(t+f^2)^{-\frac{1}{2}s}\,dt.$$

98. Second example, θ the positive root of $\dfrac{a^2}{f^2+\theta} + \cdots + \dfrac{c^2}{h^2+\theta} + \dfrac{e^2}{\theta} = 1$, $q+1$ positive.

Consider here the function

$$V = \int_\theta^\infty t^{q-1}\left\{(t+f^2)\ldots(t+h^2)\right\}^{-\frac{1}{2}}\,dt;$$

this satisfies the prepotential equation. We have in fact

$$\frac{dV}{da} = -\Theta_0\,\frac{d\theta}{da}\,, \qquad \frac{d^2V}{da^2} = -\frac{d\Theta_0}{d\theta}\left(\frac{d\theta}{da}\right)^2 - \Theta_0\,\frac{d^2\theta}{da^2}\,,$$

with the like expressions for $\dfrac{d^2V}{dc^2}$, $\dfrac{d^2V}{de^2}$; also

$$\frac{2q+1}{e}\,\frac{dV}{de} = -\Theta_0\,\frac{2q+1}{e}\,\frac{d\theta}{de}\,.$$

Hence

$$\Box V = -\Theta_0\,\Box\theta - \theta'\,\nabla_a\theta\,,$$

[p. 383] or, substituting for $\Box\theta$ and $\nabla_a\theta$ their values, this is

$$= \Theta_0\left\{-\frac{2}{J'}\left(\frac{2q+2}{\theta} - \frac{2\theta'}{\theta}\right) - \frac{4\theta'}{\theta\,J'^2}\right\} - \frac{4\theta'}{J'} = 0.$$

Moreover V does not become infinite for any values of $(a, \ldots, c, e)$, e not $= 0$; and it vanishes for points at ∞. And not only so, but for indefinitely large values of any of the coordinates $(a, \ldots, c, e)$ it reduces itself to a numerical multiple of $(a^2 + \cdots + c^2 + e^2)^{-\frac{1}{2}s+q}$; in fact, in this case θ is indefinitely large, $= a^2 + \cdots + c^2 + e^2$. Consequently throughout the integral, t is indefinitely large, and we may therefore write

$$V = \int_\theta^\infty t^{q-1}t^{-\frac{1}{2}s}\,dt = \frac{-1}{\frac{1}{2}s - q}\left[t^{-\frac{1}{2}s+q}\right]_\theta^\infty = \frac{1}{\frac{1}{2}s-q}\,\theta^{-\frac{1}{2}s+q};$$

that is,

$$V = \frac{1}{\frac{1}{2}s - q}\,(a^2 + \cdots + c^2 + e^2)^{-\frac{1}{2}s+q}.$$

The conditions of the theorem are thus satisfied, and we have for ρ either of the formulae

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{(\sqrt{\pi})^s\,\Gamma(q)}\,e^{2q}\,W, \qquad \rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)}\,\frac{1}{e^{2q+1}}\,\frac{dW}{de}\,;$$

in the former of them q must be positive; in the latter it is sufficient if $q+1$ be positive.

99. We have W the same function of $(x,\ldots,z,e)$ that V is of $(a,\ldots,c,e)$; viz. writing λ for the positive root of

$$\frac{x^2}{f^2+\lambda} + \cdots + \frac{z^2}{h^2+\lambda} + \frac{e^2}{\lambda} = 1,$$

the value of W is

$$W = \int_\lambda^\infty t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}}\,dt.$$

Considering the formula which involves $e^{2q}W$, — first, if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$, then, when e is $= 0$ the value of λ is not $= 0$; the integral W is therefore finite (not indefinitely large), and we have $e^{2q}W = 0$, consequently $\rho = 0$.

But if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$, then, when e is indefinitely small λ is also indefinitely small; viz. we then have $\dfrac{e^2}{\lambda} = 1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}$, and the value of W is

$$W = \frac{1}{q}\,\lambda^q (f\ldots h)^{-1},$$

and hence

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{(\sqrt{\pi})^s\,\Gamma(q+1)}\,(f\ldots h)^{-1}\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^q.$$

[p. 384] **100.** Again, using the formula which involves $\big(e^{2q+1}\dfrac{dW}{de}\big)$, we have first $\dfrac{dV}{da} = -\Theta\,\dfrac{d\theta}{da}$, or substituting for Θ and $\dfrac{d\theta}{da}$ their values and multiplying by e^{2q+1}, we find

$$e^{2q+1}\,\frac{dV}{da} = 2e^{2q}\theta^{q-1}\,\frac{a}{(f^2+\theta)^2}\big\{(\theta+f^2)\ldots(\theta+h^2)\big\}^{-\frac{1}{2}} \cdot \frac{1}{(\theta+f^2)^2} + \cdots + \frac{1}{(h^2+\theta)^2} + \tfrac{1}{\theta^2}\,,$$

and therefore

$$e^{2q+1}\,\frac{dW}{de} = -2e^{2q}\lambda^{q-1}\big\{(\lambda+f^2)\ldots(\lambda+h^2)\big\}^{-\frac{1}{2}}\,\frac{1}{\cdots}.$$

Hence, writing $e = 0$: first, for an exterior point or $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$, λ is not $= 0$, and the expression vanishes in virtue of the factor e^{2q+1}, whence also $\rho = 0$; next, for an interior point or $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$, λ is $= 0$, hence also $\dfrac{e^2}{\lambda} = 1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}$ is infinite; neglecting in comparison with it the other terms $\dfrac{a^2}{(\lambda+f^2)^2} + \cdots$, the value is

$$= \frac{2}{e^2}\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^{-1};$$

and we have, as before,

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{(\sqrt{\pi})^s\,\Gamma(q+1)}\,(f\ldots h)^{-1}\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^q.$$

101. Hence in the formula

$$V = \int \frac{\rho\, dx \dots dz}{\big\{(a-x)^2 + \dots + (c-z)^2 + e^2\big\}^{\frac{1}{2}s+q}}$$

ρ has the value just found, or, what is the same thing, we have

$$\int \frac{\big(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\big)^q\, dx \dots dz}{\big\{(a-x)^2 + \dots + (c-z)^2 + e^2\big\}^{\frac{1}{2}s+q}}$$

taken over ellipsoid $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1$,

$$= \frac{(\sqrt{\pi})^s\, \Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)}\, (f \dots h) \int_\theta^\infty t^{q-1} \big\{(t+f^2) \dots (t+h^2)\big\}^{-\frac{1}{2}}\, dt.$$

[p. 385] **102.** We may in this result write $e = 0$. There are two cases, according as the attracted point is exterior or interior: if it is exterior, $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$, θ will denote the positive root of the equation $\frac{a^2}{f^2+\theta} + \dots + \frac{c^2}{h^2+\theta} = 1$; if it be interior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} < 1$, θ will be $= 0$; and we thus have

$$\int \frac{\big(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\big)^q\, dx \dots dz}{\big\{(a-x)^2 + \dots + (c-z)^2\big\}^{\frac{1}{2}s+q}}$$

$$= \frac{(\sqrt{\pi})^s\, \Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)}\, (f \dots h) \int_\theta^\infty t^{q-1} \big\{(t+f^2) \dots (t+h^2)\big\}^{-\frac{1}{2}}\, dt, \quad \text{for exterior point } \frac{a^2}{f^2}+\dots+\frac{c^2}{h^2} > 1,$$

$$= \frac{(\sqrt{\pi})^s\, \Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)}\, (f \dots h) \int_0^\infty t^{q-1} \big\{(t+f^2) \dots (t+h^2)\big\}^{-\frac{1}{2}}\, dt, \quad \text{for interior point } \frac{a^2}{f^2}+\dots+\frac{c^2}{h^2} < 1;$$

but as regards the value for an interior point it is to be observed that, unless q be negative (between 0 and -1, since $1+q$ is positive by hypothesis), the two sides of the equation will be each of them infinite.

103. Third example. We assume here

$$V = \int_\theta^\infty dt\, I^m\, T,$$

where

$$I = 1 - \frac{a^2}{f^2+t} - \dots - \frac{c^2}{h^2+t} - \frac{e^2}{t},$$

$$T = t^{q-1} \big\{(t+f^2) \dots (t+h^2)\big\}^{-\frac{1}{2}};$$

as before, θ is the positive root of the equation

$$1 - \frac{a^2}{f^2+\theta} - \dots - \frac{c^2}{h^2+\theta} - \frac{e^2}{\theta} = 0,$$

and $\frac{1}{2}s+q$ is positive in order that the integral may be finite; also m is positive.

104. In order to show that V satisfies the prepotential equation $\square V = 0$, I shall, in the first place, consider the more general expression

$$V = \int_{\theta+\eta}^\infty dt\, I_0^m\, T,$$

where η is a constant positive quantity which will be ultimately put $= 0$. The functions previously called J and Θ will be written J_0 and Θ_0, and J_η will now denote

$$J_\eta = 1 - \frac{a^2}{\theta+\eta+f^2} - \cdots - \frac{c^2}{\theta+\eta+h^2} - \frac{e^2}{\theta+\eta},$$

$$\Theta_\eta = (\theta+\eta)^{q-1}\big\{(\theta+\eta+f^2)\dots(\theta+\eta+h^2)\big\}^{-\frac{1}{2}};$$

[p. 386] whence also, subtracting from J the evanescent function J_0, we have

$$J - J_0 = \frac{a^2\,\eta}{(\theta+f^2)(\theta+\eta+f^2)} + \cdots + \frac{e^2\,\eta}{\theta(\theta+\eta)};$$

say this is $= \eta P$; and we have thence, by former equations and in the present notation,

$$\frac{a}{\theta+\eta+f^2}\,\frac{d\theta}{da} + \cdots + \frac{c}{\theta+\eta+h^2}\,\frac{d\theta}{dc} + \frac{e}{\theta+\eta}\,\frac{d\theta}{de} = \frac{2}{J_0'}(1-\eta P),$$

$$\nabla_a\theta = \frac{4}{J_0'^2}\,, \qquad \square\theta = \frac{4\Theta_0'}{\Theta_0\,J_0}\,.$$

In virtue of the equation which determines θ, we have

$$\frac{dV}{da} = \int_{\theta+\eta}^{\infty} dt\, mI^{m-1}\frac{dI}{da}\,T, \qquad \frac{d^2V}{da^2} = \int_{\theta+\eta}^{\infty} dt\Big\{m(m-1)I^{m-2}\Big(\frac{dI}{da}\Big)^2 T + mI^{m-1}\,\frac{d^2I}{da^2}\,T\Big\}$$

$$-J_\eta^m\Theta_\eta\,\frac{d\theta}{da}\,;$$

and thence

$$-mJ_\eta^{m-1}\Theta_\eta\,\frac{d\theta}{da} - J_\eta^m\Theta_\eta\,\frac{d^2\theta}{da^2} - J_\eta^m\Theta_\eta\,\left(\frac{d\theta}{da}\right)^2$$

$$-J_\eta^m\Theta_\eta'\,\frac{d\theta}{da}\,,$$

with like expressions for $\dfrac{d^2V}{dc^2}$, $\dfrac{d^2V}{de^2}$. Also

$$\frac{2q+1}{e}\,\frac{dV}{de} = \int_{\theta+\eta}^{\infty} dt\, mI^{m-1}\,\frac{2q+1}{e}\,\frac{dI}{de}\,T - J_\eta^m\,\Theta_\eta\cdot\frac{2q+1}{e}\,\frac{d\theta}{de}\,;$$

and hence

$$\square V = \int_{\theta+\eta}^{\infty} dt\,\Big\{m(m-1)I^{m-2}\Big[\Big(\tfrac{dI}{da}\Big)^2 + \cdots\Big]T + mI^{m-1}\,\square I\,T\Big\}$$

$$+4mJ_\eta^{m-1}\Theta_\eta\left(\frac{a}{\theta+\eta+f^2}\,\frac{d\theta}{da} + \cdots + \frac{e}{\theta+\eta}\,\frac{d\theta}{de}\right) - J_\eta^m\Theta_\eta\,\square\theta - J_\eta^m\Theta_\eta'\,\nabla_a\theta\,.$$

[p. 387] **105.** Writing I', T' for the first derived coefficients of I, T in regard to t, we have

$$I' = \frac{a^2}{(f^2+t)^2} + \cdots + \frac{c^2}{(h^2+t)^2} + \frac{e^2}{t^2}\,, \qquad \frac{T'}{T} = \frac{q-1}{t} - \tfrac{1}{2}\left(\frac{1}{f^2+t} + \cdots + \frac{1}{h^2+t}\right) = \frac{2q+1}{2t}\cdots\,.$$

The integral is therefore

$$\int_{\theta+\eta}^{\infty} dt\,\Big\{2mI^{m-1}\,I'\,T + m(m-1)I^{m-2}\cdot 4I'\cdot T\Big\}$$

$$= \int_{\theta+\eta}^{\infty} dt\,\Big\{4mI^{m-1}\,T' + 4m(m-1)I^{m-2}\,I'\,T\Big\},$$

viz. $I^{m-1}T$ vanishing for $t=\infty$, this is

$$= -4mJ_\eta^{m-1}\Theta\,.$$

Hence, writing $(J^m\Theta)'$ instead of $\dfrac{d}{dt}(J^m\Theta)$, we have

$$\Box V = -4mJ_\eta^{m-1}\Theta_\eta + 4mJ_\eta^{m-1}\,\Theta_\eta\left(\frac{a}{\theta+\eta+f^2}\,\frac{d\theta}{da}+\cdots+\frac{e}{\theta+\eta}\,\frac{d\theta}{de}\right)$$

$$-(J^m\Theta)'\,\nabla_a\theta\,,$$

viz. this is

$$\Box V = -4mJ_\eta^{m-1}\Theta_\eta + 4mJ_\eta^{m-1}\,\frac{1}{J_0'}\big(1-\eta P\big)\cdot 2 - (J_\eta^m\Theta_\eta)'\,\frac{4}{J_0'^2}\,,$$

or, writing $mJ_\eta^{m-1}J_\eta'\Theta + J_\eta^m\Theta'$ instead of $(J^m\Theta)'$, this is

$$\Box V = -\frac{4mJ_\eta^{m-1}\Theta_\eta}{J_0'}\big(J_\eta' - 2JP + J_0'\big) - \frac{4J_\eta^m}{J_0'^2}\big(\Theta_\eta'\,J_0 - \Theta(J_0')'\big).$$

We have here

$$J_\eta' - 2JP + J_0' = \eta^2\,Q, \quad \text{suppose.}$$

Also $\Theta_\eta'\Theta_0 - \Theta\Theta_0'$ contains the factor η, is $=\eta M$ suppose.

[p. 388] **106.** Substituting for J, $J'-2JP+J$, and $\Theta'\Theta_0-\Theta\Theta_0'$ their values ηP, ηQ, and ηM, the whole result contains the factor η^{m+1}, viz. we have

$$\Box V = -\frac{4\eta^{m+1}\,\Theta_\eta}{J_0^{m+1}}\,\big(mP^mQ + P^mM\big).$$

If here, except in the term η^{m+1}, we write $\eta=0$, we have

$$P = \frac{a^2}{(\theta+f^2)^2}+\cdots+\frac{c^2}{(\theta+h^2)^2}+\frac{e^2}{\theta^2}\,,$$

the formula becomes

$$\Box V = -\frac{4\eta^{m+1}\,\Theta_0}{J_0^{m+1}}\left(mP^m\,J_0'^{-} + P^mM_0\right),$$

or (instead of J_0, Θ_0) using now J, Θ in their original significations

$$J = 1-\frac{a^2}{f^2+\theta}-\cdots-\frac{c^2}{h^2+\theta}-\frac{e^2}{\theta}, \quad \text{and} \quad \Theta = \theta^{q-1}\big\{(\theta+f^2)\ldots(\theta+h^2)\big\}^{-\frac12},$$

this is

$$\Box V = -\frac{4\eta^{m+1}\,\Theta}{J^{m+1}}\left(mP^m\,J_0^{-1} + P^mM_0\right),$$

or, what is the same thing,

$$\Box V = -\frac{4\eta^{m+1}\,\Theta}{J^{m+1}}\,\big(mP + (\ldots)\big),$$

viz. the expression in { } is

$$-\frac{1}{(\theta+f^2)^2}+\cdots+\frac{1}{(\theta+h^2)^2}+\frac{1}{\theta^2}-\frac{2q+2}{\theta}.$$

We thus see that, η being infinitesimal, $\Box V$ is infinitesimal of the order η^{m+1}; and hence, η being $= 0$, we have

$$\Box V = 0;$$

viz. the prepotential equation is satisfied by the value

$$V = \int_\theta^\infty dt\, I^m\, T,$$

where $m+1$ is positive.

107. We have consequently a value of ρ corresponding to the foregoing value of V; and this value is

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)}\,\frac{1}{e^{2q+1}}\,\frac{dW}{de}\,,$$

[p. 389] where, writing λ for the positive root of

$$1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda} = 0,$$

we have

$$W = \int_\lambda^\infty dt\left(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\right)^m t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}};$$

we thence obtain

$$\frac{dW}{de} = \int_\lambda^\infty dt\,\frac{-2me}{t}\left(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\right)^{m-1} t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}} - (\ldots)\frac{d\lambda}{de},$$

or, multiplying by e^{2q+1} and substituting for $\dfrac{d\lambda}{de}$ its value $\dfrac{2e}{\lambda}\cdot\dfrac{1}{\ldots}$, we have

$$e^{2q+1}\,\frac{dW}{de} = -2me^{2q+2}\int_\lambda^\infty dt\,\frac{1}{t}\left(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\right)^{m-1} t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}} + \cdots,$$

where the second term, although containing the evanescent factor

$$\left(1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda}\right)^m,$$

is for the present retained.

108. I attend to the second term.

1°. Suppose $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$; then, as e diminishes and becomes $= 0$, λ does not become zero, but it becomes the positive root of the equation

$$1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} = 0;$$

hence the term, containing as well the evanescent factor e^{2q+2} as the other evanescent factor $\left(1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda}\right)^m$, is $= 0$.

[p. 390] 2°. Suppose $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$; then, as e diminishes to zero, λ tends to become $= 0$, but $\dfrac{e^2}{\lambda}$ is finite and $= 1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}$, whence $\dfrac{1}{\lambda}$ is indefinitely large; and since $\dfrac{x^2}{(\lambda+f^2)^2} + \cdots + \dfrac{z^2}{(\lambda+h^2)^2}$ becomes $= +\dfrac{x^2}{f^4} + \cdots + \dfrac{z^2}{h^4}$, which is finite, the denominator may be reduced to $\dfrac{e^2}{\lambda}$, and the term therefore is

$$= -2\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)\left(1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}\right)^{m-1} e^{2q+2}(f\ldots h)^{-1}\lambda^{-q-1},$$

which, the other factor being finite, vanishes in virtue of the evanescent factor

$$\Big(1 - \frac{x^2}{f^2+\lambda} - \cdots - \frac{z^2}{h^2+\lambda} - \frac{e^2}{\lambda}\Big)^m.$$

Hence the second term always vanishes, and we have (e being $= 0$)

$$e^{2q+1}\,\frac{dW}{de} = -2m\int_\lambda^\infty dt\,\frac{e^{2q+2}}{t}\Big(1 - \frac{x^2}{f^2+t} - \cdots - \frac{z^2}{h^2+t} - \frac{e^2}{t}\Big)^{m-1} t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}}.$$

109. Considering first the case $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} > 1$: then, as e diminishes to zero, λ does not become $= 0$; the integral contains no infinite element, and it consequently vanishes in virtue of the factor e^{2q+2}.

But if $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} < 1$, then, introducing instead of t the new variable ξ, $= t \cdot \dfrac{1}{\lambda}$, that is, $t = \dfrac{\lambda}{\xi}$, $dt = -\dfrac{\lambda}{\xi^2}\,d\xi$, and writing for shortness

$$R = 1 - \frac{x^2}{f^2+\xi} - \cdots - \frac{z^2}{h^2+\xi}\,,$$

the term becomes

$$= \int_0^R d\xi \cdot 2m(R-\xi)^{m-1}\xi^q\big\{(f^2+\xi)\ldots(h^2+\xi)\big\}^{-\frac{1}{2}},$$

where, as regards the limits, corresponding to $t = \infty$ we have $\xi = 0$, and corresponding to $t = \lambda$ we have ξ the positive root of $R - \xi = 0$. But e is indefinitely small; except for indefinitely small values of ξ, we have

$$R = 1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2}, \quad \text{and} \quad \big\{(f^2+\xi)\ldots(h^2+\xi)\big\}^{-\frac{1}{2}} = (f\ldots h)^{-1};$$

[p. 391] and if ξ be indefinitely small, then, whether we take the accurate or the reduced expressions, the elements are finite, and the corresponding portion of the integral is indefinitely small. We may consequently reduce as above; viz. writing now

$$R = 1 - \tfrac{x^2}{f^2} - \cdots - \tfrac{z^2}{h^2},$$

the formula is

$$\begin{aligned} e^{2q+1}\,\frac{dW}{de} &= \int_0^R d\xi\, 2m(R-\xi)^{m-1}\,\xi^q\,(f\ldots h)^{-1} \\ &= -2m(f\ldots h)^{-1}\int_0^R d\xi\,\xi^q(R-\xi)^{m-1}, \end{aligned}$$

or writing $\xi = Ru$, the integral becomes $= R^{q+m}\int_0^1 du\,u^q(1-u)^{m-1}$, which is

$$= R^{q+m}\,\frac{\Gamma(1+q)\Gamma(m)}{\Gamma(1+q+m)};$$

that is, we have

$$e^{2q+1}\,\frac{dW}{de} = -\frac{2m(f\ldots h)^{-1}\Gamma(1+q)\Gamma(m)}{\Gamma(1+q+m)}\,R^{q+m},$$

and consequently

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)\Gamma(m)}{(\sqrt{\pi})^s\,\Gamma(1+q+m)}\,(f\ldots h)^{-1}\,R^{q+m},$$

that is,

$$\rho = \frac{\Gamma(\tfrac{1}{2}s+q)\Gamma(m)}{(\sqrt{\pi})^s\,\Gamma(1+q+m)}\,(f\dots h)^{-1}\Big(1-\tfrac{x^2}{f^2}-\dots-\tfrac{z^2}{h^2}\Big)^{q+m},$$

viz. ρ has this value for values of $(x,\dots,z)$ such that $\frac{x^2}{f^2}+\dots+\frac{z^2}{h^2}<1$, but is $=0$ if $\frac{x^2}{f^2}+\dots+\frac{z^2}{h^2}>1$.

110. Multiplying by a constant factor so as to reduce ρ to the value R^{q+m}, the final result is that the integral

$$V = \int \frac{\Big(1-\tfrac{x^2}{f^2}-\dots-\tfrac{z^2}{h^2}\Big)^{q+m}\,dx\dots dz}{\big\{(a-x)^2+\dots+(c-z)^2+e^2\big\}^{\frac{1}{2}s+q}},$$

the limits being given by the equation

$$\frac{x^2}{f^2}+\dots+\frac{z^2}{h^2}=1,$$

is equal to

$$\frac{\Gamma(\tfrac{1}{2}s)\Gamma(1+q+m)}{\Gamma(1+q)\Gamma(\tfrac{1}{2}s+q+m)}\,(f\dots h)\int_\theta^\infty dt\,t^m\Big(1-\frac{a^2}{f^2+t}-\dots-\frac{c^2}{h^2+t}-\frac{e^2}{t}\Big)^m t^{q-1}\big\{(t+f^2)\dots(t+h^2)\big\}^{-\frac{1}{2}},$$

where θ is the positive root of

$$1-\frac{a^2}{\theta+f^2}-\dots-\frac{c^2}{\theta+h^2}-\frac{e^2}{\theta}=0.$$

[p. 392] In particular, if $e=0$, or

$$V = \int \frac{\Big(1-\tfrac{x^2}{f^2}-\dots-\tfrac{z^2}{h^2}\Big)^{q+m}\,dx\dots dz}{\big\{(a-x)^2+\dots+(c-z)^2\big\}^{\frac{1}{2}s+q}},$$

there are two cases:

exterior, $\frac{a^2}{f^2}+\dots+\frac{c^2}{h^2}>1$, θ is positive root of $1-\frac{a^2}{f^2+\theta}-\dots-\frac{c^2}{h^2+\theta}=0$,

interior, $\frac{a^2}{f^2}+\dots+\frac{c^2}{h^2}<1$, θ vanishes, viz. the limits in the integral are $\infty, 0$;

q must be negative, $1+q$ positive as before, in order that the t-integral may not be infinite in regard to the element $t=0$.

It is assumed in the proof that m and $1+q$ are each of them positive; but, as appears by the second example, the theorem is true for the extreme value $m=0$; it does not, however, appear that the proof can be extended to include the extreme value $q=-1$. The formula seems, however, to hold good for values of m, q beyond the foregoing limits; and it would seem that the only necessary conditions are $\frac{1}{2}s+q$, $1+m$, and $1+q+m$, each of them positive. The theorem is, in fact, a particular case of the following one, proved Annex X. No. 162, viz.

$$V = \int \frac{\phi\Big(1-\tfrac{x^2}{f^2}-\dots-\tfrac{z^2}{h^2}\Big)\,dx\dots dz}{\big\{(a-x)^2+\dots+(c-z)^2+e^2\big\}^{\frac{1}{2}s+q}},$$

taken over the ellipsoid $\frac{x^2}{f^2}+\dots+\frac{z^2}{h^2}=1$, is equal to

$$\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\tfrac{1}{2}s+q)}\,(f\dots h)\int_\theta^\infty dt\,\big\{(t+f^2)\dots(t+h^2)\big\}^{-\frac{1}{2}}(1-\sigma)^{q+\frac{1}{2}s}\int_0^1(1-x)^q\phi\{\sigma+(1-\sigma)x\}\,dx,$$

where σ denotes $\dfrac{a^2}{f^2+t}+\cdots+\dfrac{c^2}{h^2+t}+\dfrac{e^2}{t}$; assuming $\phi u=(1-u)^{q+m}$, we have

$$\phi\{\sigma+(1-\sigma)x\}=(1-\sigma)^{q+m}(1-x)^{q+m},$$

and the theorem is thus proved.

111. Particular cases: $m=0$;

$$\int\frac{\left(1-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}\right)^q dx\ldots dz}{\left\{(a-x)^2+\cdots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\ldots h)\int_\theta^\infty dt\,t^{q-1}\left\{(t{+}f^2)\ldots(t{+}h^2)\right\}^{-\frac{1}{2}}.$$

Cor. In a somewhat similar manner it may be shown that

$$\int\frac{\left(1-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}\right)^q x\,dx\ldots dz}{\left\{(a-x)^2+\cdots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=\frac{a(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\ldots h)\int_\theta^\infty dt\,t^{q-1}\left\{(t{+}f^2)\ldots(t{+}h^2)\right\}^{-\frac{1}{2}}.$$

[p. 393] Multiplying the first by a and subtracting it from the second, we have

$$\int\frac{\left(1-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}\right)^q (a-x)\,dx\ldots dz}{\left\{(a-x)^2+\cdots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=-\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\ldots h)\int_\theta^\infty dt\,\frac{a}{f^2+t}\,t^{q-1}\left\{(t{+}f^2)\ldots(t{+}h^2)\right\}^{-\frac{1}{2}},$$

or, writing $q+1$ for q, this is

$$\int\frac{\left(1-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}\right)^q (a-x)\,dx\ldots dz}{\left\{(a-x)^2+\cdots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q+1}}=\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)}\,(f\ldots h)\int_\theta^\infty dt\,\frac{a}{f^2+t}\,t^{q}\left\{(t{+}f^2)\ldots(t{+}h^2)\right\}^{-\frac{1}{2}},$$

and we have similar formulae with $\ldots$, $(c-z)$, e, instead of $(a-x)$, in the numerator.

112. If $m=1$, we have

$$\int\frac{\left(1-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}\right)^{q+1} dx\ldots dz}{\left\{(a-x)^2+\cdots+(c-z)^2+e^2\right\}^{\frac{1}{2}s+q}}=\frac{(\sqrt{\pi})^s\,\Gamma(1+q)}{\Gamma(2+\frac{1}{2}s+q)}\,(f\ldots h)\int_\theta^\infty dt\,t^{q}\left\{(t{+}f^2)\ldots(t{+}h^2)\right\}^{-\frac{1}{2}},$$

which, differentiated in respect to a, gives the $(a-x)$-formula; hence conversely, assuming the $a-x,\ldots,c-z,e$-formulae, we obtain by integration the last preceding formula to a constant près, viz. we thereby obtain the multiple integral $=C+$ right-hand function, where C is independent of $(a,\ldots,c,e)$; by taking these all infinite, and observing that then $\theta=\infty$, the two integrals each vanish, and we obtain $C=0$.

In particular, when $s=3$, $q=-\frac{1}{2}$, then

$$\int\frac{\left(1-\frac{x^2}{f^2}-\cdots-\frac{z^2}{h^2}\right)^{\frac{1}{2}} dx\,dy\,dz}{\left\{(a-x)^2+(b-y)^2+(c-z)^2+e^2\right\}^{1}}=\frac{\pi^2}{2}(fgh)\int_\theta^\infty\frac{dt}{\sqrt{t(t+f^2)(t+g^2)(t+h^2)}},$$

which, putting therein $e=0$, gives the potential of an ellipsoid for the cases of an exterior point and an interior point respectively.

Annex V. Green's Integration of the Prepotential Equation. Art. Nos. 113 to 128.

[p. 393, cont.] **113.** In the present Annex, I in part reproduce Green's process for the integration of this equation by means of a series of functions, which are analogous to Laplace's Functions and may be termed "Greenians" (see his Memoir on the Attraction of Ellipsoids, referred to above, p. 320); each such function gives rise to a Prepotential Integral.

Green shows, by a complicated and difficult piece of general reasoning, that there exist solutions of the form $V = \Theta\phi$ (see post. No. 116), where ϕ is a function of the
[p. 394] s new variables $\alpha, \beta, \ldots, \gamma$ without θ, such that $\nabla^2\phi = \kappa\phi$, κ being a function of θ only; these functions ϕ of the variables $\alpha, \beta, \ldots, \gamma$ are in fact the Greenian Functions in question. The function of the order 0 is $\phi = 1$; those of the order 1 are $\phi = \alpha$, $\phi = \beta, \ldots, \phi = \gamma$; those of the order 2 are $\phi = \alpha\beta$, etc., and s functions each of the form

$$\phi = \tfrac{1}{2}(A\alpha^2 + B\beta^2 + \cdots + C\gamma^2) + D.$$

The existence of the functions just referred to other than the s functions involving the squares of the variables is obvious enough; the difficulty first arises in regard to these s functions; and the actual development of them appears to me important by reason of the light which is thereby thrown upon the general theory. This I accomplish in the present Annex; and I determine by Green's process the corresponding prepotential integrals. I do not go into the question of the Greenian Functions of orders superior to the second.

114. I write for greater clearness $(a, b, \ldots, c, e)$ instead of $(a, \ldots, c, e)$ to denote the series of $(s+1)$ variables; viz. $(a, b, \ldots, c)$ will denote a series of s variables; corresponding to these we have the semiaxes $(f, g, \ldots, h)$, and the new variables $(\alpha, \beta, \ldots, \gamma)$; these last, with the before-mentioned function θ, are the $s+1$ new variables of the problem; and, for convenience, there is introduced also a quantity ε; viz. we have

$$a = \sqrt{f^2+\theta}\,\alpha, \quad b = \sqrt{g^2+\theta}\,\beta, \ \ldots, \ c = \sqrt{h^2+\theta}\,\gamma, \quad e = \sqrt{\theta}\,\varepsilon,$$

where $\lambda = \alpha^2 + \beta^2 + \cdots + \gamma^2 + \varepsilon^2$.

That is, we have θ a function of $a, b, \ldots, c, e$, determined by

$$\frac{a^2}{f^2+\theta} + \frac{b^2}{g^2+\theta} + \cdots + \frac{c^2}{h^2+\theta} + \frac{e^2}{\theta} = 1,$$

and then $\alpha, \beta, \ldots, \gamma$ are given as functions of the same quantities $a, b, \ldots, c, e$ by the equations

$$\alpha^2 = \frac{a^2}{f^2+\theta}, \quad \beta^2 = \frac{b^2}{g^2+\theta}, \ \ldots, \ \gamma^2 = \frac{c^2}{h^2+\theta};$$

also ε, considered as a function of the same quantities, is

$$\varepsilon^2 = 1 - \frac{a^2}{f^2+\theta} - \frac{b^2}{g^2+\theta} - \cdots - \frac{c^2}{h^2+\theta} = \frac{e^2}{\theta}.$$

115. Introducing instead of $a, b, \ldots, c, e$ the new variables $\alpha, \beta, \ldots, \gamma, \theta$, the transformed differential equation is

$$\nabla^2 V + \frac{d^2V}{d\theta^2}\Big(s + 2q + 2 - \frac{\alpha^2}{f^2+\theta} - \cdots - \frac{\gamma^2}{h^2+\theta}\Big)\cdot 4 + \nabla^2 V = 0,$$

[p. 395] where for shortness

$$\nabla^2 V = \theta\left[\frac{f^2+\theta}{a^2} - \frac{h^2+\theta}{c^2} + 1\right]\frac{d^2V}{d\alpha^2} + \cdots$$

$$+\theta\left[\frac{f^2+\theta}{a^2} + \cdots\right]\frac{d^2V}{d\gamma^2} - \frac{\theta\cdot 2q}{e^2}\,\frac{d^2V}{d\varepsilon^2}\ \cdots$$

$$+\frac{2\alpha}{f^2+\theta}\frac{d^2V}{d\alpha\,d\beta} - \&c.$$
$$+\frac{1}{f^2+\theta}\Big(-\frac{\alpha}{f^2+\theta} - \&c.\Big)\frac{dV}{d\alpha}$$
$$+\frac{1}{\theta}\Big(-\frac{\varepsilon}{\theta} - \&c.\Big)\frac{dV}{d\beta}$$
$$+\frac{1}{\theta}\Big(-\frac{\varepsilon}{\theta}\Big)\frac{dV}{d\gamma}\,.$$

Also

$$\frac{d^2V}{d\theta^2} = \frac{1}{4(f^2+\theta)}\Big[\frac{f^2+\theta}{a^2}+\cdots+\frac{h^2+\theta}{c^2}+\frac{\theta}{e^2}\Big]\Big[(f^2+\theta)\frac{d^2V}{d\alpha^2}+\cdots+(h^2+\theta)\frac{d^2V}{d\gamma^2}+\theta\,\frac{d^2V}{d\varepsilon^2}\Big]\cdots.$$

116. To integrate the equation for V, we assume

$$V = \Theta\phi,$$

where Θ is a function of θ only, and ϕ a function of $\alpha, \beta, \dots, \gamma$ (without θ), such that

$$\nabla^2\phi = \kappa\phi,$$

κ being a function of θ only. Assuming that this is possible, the remaining equation to be satisfied is obviously

$$\Theta'' + \kappa\Theta = 0.$$

Solutions of the form in question are

$$\phi = 1, \quad \kappa = 0,$$
$$\phi = \alpha, \quad \kappa = \frac{1}{f^2+\theta}, \quad \text{or} \quad \phi = \beta, \dots, \kappa = \frac{1}{g^2+\theta}, \ \dots, \ \frac{1}{h^2+\theta},$$
$$\phi = \alpha\beta, \quad \kappa = \frac{1}{f^2+\theta} + \frac{1}{g^2+\theta}, \ \&c.,$$

and the corresponding values of Θ involve θ.
[p. 396] and it can be shown next that there is a solution of the form

$$\phi = \tfrac{1}{2}(A\alpha^2 + \cdots + C\gamma^2) + D.$$

117. In fact, assuming that this satisfies $\nabla^2\phi - \kappa\phi = 0$, we must have identically

$$\frac{A}{f^2+\theta}\Big\{-\alpha^2\frac{\theta}{f^2+\theta}(\beta^2+\cdots+\gamma^2) - \alpha^2\frac{\theta}{f^2+\theta}\,\varepsilon^2 + 1\Big\}$$
$$+\frac{B}{g^2+\theta}\Big\{\cdots\Big\}+\cdots$$
$$+\tfrac{2q}{\varepsilon^2}\Big\{-\alpha^2\frac{\theta}{f^2+\theta} - \beta^2\frac{\theta}{g^2+\theta} - \cdots\Big\}$$
$$+\frac{A}{f^2+\theta}\Big\{-\alpha^2\tfrac{\theta}{f^2+\theta} - \cdots + \tfrac{2}{f^2+\theta}\Big\}+\cdots$$
$$+\frac{2q}{\varepsilon^2}\Big\{-\alpha^2\tfrac{\theta}{f^2+\theta} - \cdots\Big\}$$
$$+\kappa\big\{\tfrac{1}{2}(A\alpha^2 + B\beta^2 + \cdots + C\gamma^2) + D\big\} = 0\,;$$

so that, from the term in α^2, we have

$$\frac{A}{f^2+\theta}\left\{-\frac{\theta}{f^2+\theta}(s+2q+2)\right\}-\frac{B\theta}{(g^2+\theta)(f^2+\theta)}-\cdots-\frac{C\theta}{(h^2+\theta)(f^2+\theta)}+\tfrac{1}{2}\kappa A=0,$$

or, what is the same thing,

$$A\left\{2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta)\right\}+\kappa f^2 D=0,$$

with the like equations from $\beta^2,\dots,\gamma^2$; and from the constant term we have

$$\frac{A}{f^2+\theta}+\frac{B}{g^2+\theta}+\cdots+\frac{C}{h^2+\theta}+\kappa D=0.$$

118. Multiplying this last by f^2, and adding it to the first, we obtain

$$A\left\{1+\frac{f^2}{f^2+\theta}\right\}\cdot 1+\frac{Bf^2}{g^2+\theta}+\cdots+\frac{Cf^2}{h^2+\theta}+\tfrac{1}{2}\kappa A(f^2+\theta)+\kappa f^2 D=0;$$

viz. putting for shortness $\Omega=\dfrac{a^2}{f^2+\theta}+\cdots=\dfrac{\theta}{f^2+\theta}\alpha^2+\cdots+\dfrac{\theta}{h^2+\theta}\gamma^2$, this is

$$A\left\{2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta)\right\}+\kappa f^2 D=0;$$

and similarly

$$B\left\{2q+2+\Omega+\tfrac{1}{2}\kappa(g^2+\theta)\right\}+\kappa g^2 D=0,$$
$$C\left\{2q+2+\Omega+\tfrac{1}{2}\kappa(h^2+\theta)\right\}+\kappa h^2 D=0.$$

[p. 397] To these we join the foregoing equation

$$\frac{A}{f^2+\theta}+\frac{B}{g^2+\theta}+\cdots+\frac{C}{h^2+\theta}+\kappa D=0.$$

Eliminating $A,B,\dots,C,D$, we have an equation which determines κ as a function of θ; and the equations then determine the ratios of $A,B,\dots,C,D$, so that these quantities will be given as determinate multiples of an arbitrary quantity M. The equation for κ is in fact

$$\frac{f^2(f^2+\theta)}{2q+2+\Omega+\frac{1}{2}\kappa(f^2+\theta)}+\cdots+\frac{h^2(h^2+\theta)}{2q+2+\Omega+\frac{1}{2}\kappa(h^2+\theta)}+1=0;$$

and the values of $A,B,\dots,C,D$ are then

$$A=\frac{Mf^2}{2q+2+\Omega+\frac{1}{2}\kappa(f^2+\theta)},\quad B=\frac{Mg^2}{2q+2+\Omega+\frac{1}{2}\kappa(g^2+\theta)},\ \dots,\ C=\frac{Mh^2}{2q+2+\Omega+\frac{1}{2}\kappa(h^2+\theta)},\quad D=-\frac{M}{\kappa}$$

values which seem to be dependent on θ: if they were so, it would be fatal to the success of the process; but they are really independent of θ.

119. That they are independent of θ depends on the theorems; that we have

$$\kappa=\frac{(2q+2+\Omega)\kappa_0}{2q+2-\frac{1}{2}\kappa_0\theta},$$

where κ_0 is a quantity independent of θ determined by the equation

$$\frac{f^2}{2q+2+\frac{1}{2}\kappa_0 f^2}+\frac{g^2}{2q+2+\frac{1}{2}\kappa_0 g^2}+\cdots+\frac{h^2}{2q+2+\frac{1}{2}\kappa_0 h^2}+1=0$$

(κ_0 is in fact the value of κ on writing $\theta = 0$): and that, omitting the arbitrary multiplier, the values of $A, B, \ldots, C, D$ then are

$$\frac{f^2}{2q+2+\frac{1}{2}\kappa_0 f^2},\quad \frac{g^2}{2q+2+\frac{1}{2}\kappa_0 g^2},\ \ldots,\ \frac{h^2}{2q+2+\frac{1}{2}\kappa_0 h^2},\quad -\frac{1}{\kappa_0},$$

or, what is the same thing, the value of ϕ is

$$\phi = \frac{f^2\alpha^2}{2q+2+\frac{1}{2}\kappa_0 f^2} + \frac{g^2\beta^2}{2q+2+\frac{1}{2}\kappa_0 g^2} + \cdots + \frac{h^2\gamma^2}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0}.$$

120. To explain the ground of the assumption

$$\kappa = \frac{(2q+2+\Omega)\kappa_0}{2q+2-\frac{1}{2}\kappa_0\theta},$$

observe that, assuming

$$\frac{2q+2+\Omega+\frac{1}{2}\kappa(f^2+\theta)}{2q+2+\frac{1}{2}\kappa_0 f^2} = \frac{2q+2+\Omega+\frac{1}{2}\kappa(g^2+\theta)}{2q+2+\frac{1}{2}\kappa_0 g^2},$$

[p. 398] then multiplying out and reducing, we obtain

$$\big(g^2-f^2\big)\big\{\kappa_0(2q+2+\Omega) - (2q+2)\kappa + \tfrac{1}{2}\kappa\kappa_0\theta\big\} = 0;$$

viz. the equation divides out by the factor $g^2 - f^2$, thereby becoming

$$\kappa_0(2q+2+\Omega) - (2q+2)\kappa + \tfrac{1}{2}\kappa\kappa_0\theta = 0,$$

that is, it gives for κ the foregoing value: hence clearly, κ having this value, we obtain by symmetry

$$2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta),\ 2q+2+\Omega+\tfrac{1}{2}\kappa(g^2+\theta),\ \ldots,\ 2q+2+\Omega+\tfrac{1}{2}\kappa(h^2+\theta),$$

proportional to

$$2q+2+\tfrac{1}{2}\kappa_0 f^2,\ 2q+2+\tfrac{1}{2}\kappa_0 g^2,\ \ldots,\ 2q+2+\tfrac{1}{2}\kappa_0 h^2;$$

viz. the ratios, not only of $A : B$, but of $A : B : \cdots : C$ will be independent of θ.

121. To complete the transformation, starting with the foregoing value of κ, we have

$$2q+2+\Omega+\tfrac{1}{2}\kappa(f^2+\theta) = (2q+2+\Omega)\,\frac{2q+2+\frac{1}{2}\kappa_0(f^2+\theta)}{2q+2-\frac{1}{2}\kappa_0\theta},\ \ \&c.;$$

so that we have

$$\begin{aligned} A\big\{2q+2+\tfrac{1}{2}\kappa_0 f^2\big\} + \kappa_0 f^2 D &= 0,\\ B\big\{2q+2+\tfrac{1}{2}\kappa_0 g^2\big\} + \kappa_0 g^2 D &= 0,\\ C\big\{2q+2+\tfrac{1}{2}\kappa_0 h^2\big\} + \kappa_0 h^2 D &= 0, \end{aligned}$$

and

$$\frac{A}{f^2+\theta} + \frac{B}{g^2+\theta} + \cdots + \frac{C}{h^2+\theta} = \frac{(2q+2+\Omega)\kappa_0 D}{2q+2-\frac{1}{2}\kappa_0\theta}.$$

Substituting for $A, B, \ldots, C$ their values, this last becomes

$$\frac{\kappa_0 f^2 D}{2q+2+\frac{1}{2}\kappa_0\theta}\left\{\frac{2q+2}{2q+2+\frac{1}{2}\kappa_0 f^2} - \cdots - \frac{\kappa_0 h^2 D}{h^2+\theta}\right\} + \frac{2q+2}{2q+2-\frac{1}{2}\kappa_0\theta}\left\{\frac{2q+2}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{\theta}{h^2+\theta}\right\};$$

viz. this is
$$\Big\{\frac{2q+2}{2q+2+\frac{1}{2}\kappa_0 f^2}-\cdots\Big\}(2q+2)\kappa_0 D,$$
or, substituting for κ its value, and dividing out by $2q+2$, we have
$$\frac{1}{2q+2+\frac{1}{2}\kappa_0 f^2}+\frac{1}{2q+2+\frac{1}{2}\kappa_0 g^2}+\cdots+\frac{1}{2q+2+\frac{1}{2}\kappa_0 h^2}=-1,$$
the equation for the determination of κ_0.

[p. 399] **122.** The equation for κ_0 is of the order s; there are consequently s functions of the form in question, and each of the terms $\alpha^2,\beta^2,\ldots,\gamma^2$ can be expressed as a linear function of these. It thus appears that any quadric function of $\alpha,\beta,\ldots,\gamma$ can be expressed as a sum of Greenian functions; viz. the form is
$$\begin{aligned}&A\alpha^2+B\beta^2+\&c.\\&+C'\alpha\beta+\&c.\\&+D'\Big(\frac{f^2\alpha^2}{2q+2+\frac{1}{2}\kappa_0' f^2}+\frac{g^2\beta^2}{2q+2+\frac{1}{2}\kappa_0' g^2}+\cdots+\frac{h^2\gamma^2}{2q+2+\frac{1}{2}\kappa_0' h^2}-\frac{1}{\kappa_0'}\Big)\\&+D''\big(\,,\quad\ldots,\quad\ldots,\quad\ldots\big)\\&(s\text{ lines}),\end{aligned}$$
viz. the terms multiplied by D', D'', &c. respectively are those answering to the roots $\kappa_0', \kappa_0'', \ldots$ of the equation in κ_0.

The general conclusion is that any rational and integral function of $\alpha,\beta,\ldots,\gamma$ can be expressed as a sum of Greenian functions.

123. We have next to integrate the equation
$$\Theta''+\kappa\Theta=0.$$
Suppose $\kappa=0$, a particular solution is $\Theta=1$. Next, suppose
$$\kappa=\frac{1}{f^2+\theta}\cdot\frac{-1}{2q-2}\cdot\frac{-\theta}{f^2+\theta}-\cdots-\frac{-\theta}{h^2+\theta}-1\,;$$
a particular solution is $\Theta=\sqrt{f^2+\theta}$: in fact, omitting the constant denominator, or writing $\Theta=\sqrt{f^2+\theta}$, and therefore
$$\frac{d\Theta}{d\theta}=\tfrac{1}{2}\,\frac{1}{\sqrt{f^2+\theta}}\,,\qquad \frac{d^2\Theta}{d\theta^2}=-\tfrac{1}{4}\,\frac{1}{(f^2+\theta)^{\frac{3}{2}}}\,,$$
the equation to be verified is
$$-\frac{1}{4}\,\frac{1}{(f^2+\theta)^{\frac{3}{2}}}+\frac{1}{f^2+\theta}\cdot\sqrt{f^2+\theta}\cdot\big\{2q-2-\tfrac{-\theta}{f^2+\theta}-\cdots\big\}=0.$$

Again, suppose $\kappa=\dfrac{1}{f^2+\theta}\cdot\dfrac{1}{g^2+\theta}+\dfrac{2\theta}{(f^2+\theta)(g^2+\theta)}+\dfrac{1}{f^2+\theta}\cdot\dfrac{1}{g^2+\theta}$&c. (value belonging to $\phi=\alpha\beta$, see No. 116);

a particular solution is $\Theta=\sqrt{f^2+\theta}\cdot\sqrt{g^2+\theta}$: in fact, omitting the constant factor, or writing

[p. 400] and therefore
$$\frac{d\Theta}{d\theta}=\sqrt{\frac{g^2+\theta}{f^2+\theta}}\cdot\frac{1}{2}+\sqrt{\frac{f^2+\theta}{g^2+\theta}}\cdot\frac{1}{2},\quad \frac{d^2\Theta}{d\theta^2}=-\frac{1}{4}\,\frac{\sqrt{f^2+\theta}}{(g^2+\theta)^{\frac{3}{2}}}-\frac{1}{4}\,\frac{\sqrt{g^2+\theta}}{(f^2+\theta)^{\frac{3}{2}}}\,,$$

the equation to be verified is

$$-\frac{1}{4}\left\{\frac{\sqrt{g^2+\theta}}{(f^2+\theta)^{\frac{3}{2}}}+\frac{2\theta}{\sqrt{f^2+\theta}\,\sqrt{g^2+\theta}}+\frac{\sqrt{f^2+\theta}}{(g^2+\theta)^{\frac{3}{2}}}\right\}$$

$$+\left(\sqrt{g^2+\theta}+\sqrt{f^2+\theta}\right)\cdot\tfrac{1}{4}=0,$$

or putting for shortness $\Omega=\dfrac{\theta}{f^2+\theta}+\dfrac{\theta}{g^2+\theta}+\cdots+\dfrac{\theta}{h^2+\theta}$, this is

$$\frac{\theta}{(f^2+\theta)^2}\cdot\sqrt{g^2+\theta}-\frac{2\theta}{\sqrt{f^2+\theta}\,\sqrt{g^2+\theta}}+\frac{\theta}{(g^2+\theta)^2}\cdot\sqrt{f^2+\theta}+\frac{2\theta}{\sqrt{f^2+\theta}\,\sqrt{g^2+\theta}}\{2q+2+\Omega\}\cdot\Big(-\sqrt{g^2+\theta}-\sqrt{f^2+\ldots}$$

which is true. And, generally, the particular solution is deduced from the value of ϕ by writing therein

$$\frac{\sqrt{f^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}},\quad\frac{\sqrt{g^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}},\ \ldots,\ \frac{\sqrt{h^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}}$$

in place of $\alpha,\beta,\ldots,\gamma$ respectively: say the value thus obtained is $\Theta=H$, where H is what ϕ becomes by the above substitution.

124. Represent for a moment the equation in Θ by

$$\frac{d^2\Theta}{d\theta^2}+2P\,\frac{d\Theta}{d\theta}+\kappa\Theta=0,$$

and assume that this is satisfied by $\Theta=H\int z\,d\theta$. Then we have

$$\frac{d\Theta}{d\theta}=zH+\frac{dH}{d\theta}\int z\,d\theta,$$

$$\frac{d^2\Theta}{d\theta^2}=z\,\frac{dH}{d\theta}+H\,\frac{dz}{d\theta}+\frac{d^2H}{d\theta^2}\int z\,d\theta,$$

$$+\kappa H\int z\,d\theta=0,$$

[p. 401] and therefore

$$\left(H''+2PH'+\kappa H\right)\int z\,d\theta+2Hz'+4PHz=0;$$

viz. multiplying by $\dfrac{H}{4z}$, this is

$$\frac{H^2z'}{2z}+2PH^2=0,$$

or

$$\frac{d}{d\theta}\left(H^2z\right)+4HH'z=0;$$

viz. substituting for P its value, this is

$$\frac{1}{H^2z}\,\frac{d}{d\theta}\left(H^2z\right)\left(\tfrac{1}{2}q+1-\tfrac{1}{f^2+\theta}-\cdots-\tfrac{1}{h^2+\theta}\right)\cdot\tfrac{1}{2}=0.$$

Hence, integrating,

$$H^2z=\frac{C\theta^{-q-1}}{\sqrt{f^2+\theta}\cdot\sqrt{g^2+\theta}\cdots\sqrt{h^2+\theta}},\quad C\text{ an arbitrary constant},$$

and

$$\Theta = CH\int_\chi^\theta \frac{\theta^{-q-1}\,d\theta}{H^2\sqrt{f^2+\theta}\cdot\sqrt{g^2+\theta}\cdots\sqrt{h^2+\theta}}\,,\quad \chi \text{ arbitrary},$$

where the constants of integration are C, χ; or, what is the same thing, taking $\bar{T}$ the same function of t that Θ is of θ (viz. $\bar{T}$ is what H becomes on writing therein

$$\frac{\sqrt{f^2+t}}{\sqrt{f^2+g^2+\cdots+h^2}},\quad \frac{\sqrt{g^2+t}}{\sqrt{f^2+g^2+\cdots+h^2}},\ \ldots,\ \frac{\sqrt{h^2+t}}{\sqrt{f^2+g^2+\cdots+h^2}}$$

in place of $\alpha, \beta, \ldots, \gamma$ respectively), then

$$\Theta = -CH\int_\theta^\chi \bar{T}\,\frac{t^{-q-1}\,dt}{t\cdot g^2+t\cdots h^2+t},$$

where χ may be taken $=\infty$: we thus have

$$V = \Theta\phi = -CH\int_\theta^\infty \bar{T}\,\frac{t^{-q-1}\,dt}{t\cdot g^2+t\cdots h^2+t}.$$

Recollecting that

$$\lambda = \alpha^2+\beta^2+\cdots+\gamma^2+\varepsilon^2,$$

so that for $\theta=\infty$ we have $a^2+b^2+\cdots+c^2+e^2=?$, the assumption $\chi=\infty$ comes to making V vanish for infinite values of $(a, b, \ldots, c, e)$.

125. We have to find the value of ρ corresponding to the foregoing value of V; viz. W being the value of V, on writing therein $(x, y, \ldots, z)$ in place of $(a, b, \ldots, c)$, then (theorem A)

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)\,e^{2q+1}}\cdot\frac{dW}{de}\,.$$

[p. 402] Take λ the same function of $(x, y, \ldots, z, e)$ that θ is of $(a, b, \ldots, c, e)$: viz. take λ the positive root of

$$\frac{x^2}{f^2+\lambda}+\frac{y^2}{g^2+\lambda}+\cdots+\frac{z^2}{h^2+\lambda}+\frac{e^2}{\lambda}=1;$$

and let $(\xi, \eta, \ldots, \zeta, \iota)$ correspond to $(\alpha, \beta, \ldots, \gamma, \varepsilon)$, viz.

$$\xi = \frac{x}{\sqrt{f^2+\lambda}},\quad \eta=\frac{y}{\sqrt{g^2+\lambda}},\ \ldots,\ \zeta=\frac{z}{\sqrt{h^2+\lambda}},\quad \iota=\sqrt{1-\frac{x^2}{f^2+\lambda}-\cdots-\frac{z^2}{h^2+\lambda}}=\frac{e}{\sqrt{\lambda}};$$

so that W is the same function of $(\xi, \eta, \ldots, \lambda)$ that V is of $(\alpha, \beta, \ldots, \theta)$; say this is

$$W = -CH\int_\lambda^\infty \bar{T}\,\frac{t^{-q-1}\,dt}{\cdots},$$

then we have for ρ the value

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)\,e^{2q+1}}\cdot\left[\left(1-\frac{x^2}{f^2+\lambda}-\cdots-\frac{z^2}{h^2+\lambda}\right)\cdot\frac{1}{\sqrt{f^2+\lambda}}\frac{d}{d\xi}+\cdots+\frac{1}{\sqrt{h^2+\lambda}}\frac{d}{d\zeta}+\frac{1}{\sqrt{\lambda}}\frac{d}{d\iota}\right]W,$$

where e is to be put $=0$.

126. Suppose e is $=0$; then, if $\frac{x^2}{f^2}+\cdots+\frac{z^2}{h^2}>1$, λ is not $=0$, but is the positive root of $\dfrac{x^2}{f^2+\lambda}+\cdots+\dfrac{z^2}{h^2+\lambda}=1$: $\iota=\sqrt{1-\frac{x^2}{f^2+\lambda}-\cdots-\frac{z^2}{h^2+\lambda}}=0$; and we have $\rho=0$, viz. ρ is $=0$ for all points outside the ellipsoid $\frac{x^2}{f^2}+\cdots+\frac{z^2}{h^2}=1$.

But if $\frac{x^2}{f^2}+\cdots+\frac{z^2}{h^2}<1$, then, on writing $e=0$, we have $\lambda=0$, $\iota^{-1}=\dfrac{1}{\lambda}$,

$$\rho=\frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)}\cdot\frac{e^{2q+1}\sqrt{\lambda}}{\cdots}\Big[\frac{x}{\sqrt{f^2+\lambda}}\,\frac{1}{\sqrt{f^2+\lambda}}\,\frac{d\bar{T}}{d\xi}+\cdots+\frac{1}{\sqrt{h^2+\lambda}}\cdot\frac{1}{\sqrt{h^2+\lambda}}\,\frac{d\bar{T}}{d\zeta}+\frac{1}{\sqrt{\lambda}}\cdot\frac{1}{\sqrt{\lambda}}\,\frac{d\bar{T}}{d\iota}\Big]_{\lambda=0},$$

where the term in ( ) is

$$=-2\Theta\Lambda_0/(f\ldots h)\cdot\lambda^{q+1}.$$

Hence

$$\rho=\frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)\,\Lambda_0\,fg\ldots h}\,\psi$$
$$=\frac{\Gamma(\frac{1}{2}s+q)}{2(\sqrt{\pi})^s\,\Gamma(q+1)\,\Lambda_0\,fg\ldots h}\,\psi\Big(\frac{x}{f},\frac{y}{g},\ldots,\frac{z}{h}\Big),$$

[p. 403] where ψ_0,Λ_0 are what ψ,Λ become on writing therein $\lambda=0$. It will be remembered that Λ is what H becomes on changing therein θ into λ; hence Λ_0 is what H becomes on writing therein $\theta=0$.

Moreover ψ is what ϕ becomes on changing therein $\alpha,\beta,\ldots,\gamma$ into $\xi,\eta,\ldots,\zeta$: writing $\lambda=0$, we have $\xi=\dfrac{x}{f}$, $\eta=\dfrac{y}{g},\ldots,\zeta=\dfrac{z}{h}$; hence ψ_0 is what ϕ becomes on changing therein $\alpha,\beta,\ldots,\gamma$ into $\dfrac{x}{f},\dfrac{y}{g},\ldots,\dfrac{z}{h}$. And it is proper in ϕ to restore the original variables by writing $\dfrac{\alpha}{\sqrt{f^2+\theta}},\dfrac{\beta}{\sqrt{g^2+\theta}},\ldots,\dfrac{\gamma}{\sqrt{h^2+\theta}}$ in place of $\alpha,\beta,\ldots,\gamma$.

127. Recapitulating,

$$V=\int\frac{\rho\,dx\ldots dz}{\{(a-x)^2+\cdots+(c-z)^2+e^2\}^{\frac{1}{2}s+q}},$$

where, since for the value of V about to be mentioned ρ vanishes for points outside the ellipsoid, the integral is to be taken over the ellipsoid

$$\frac{x^2}{f^2}+\cdots+\frac{z^2}{h^2}=1;$$

and then, transferring a constant factor, if

$$V=-\frac{\Gamma(\frac{1}{2}s+q)}{\Gamma(q+1)}\,\Lambda_0(f\ldots h)\,H\int_\theta^\infty\bar{T}\,\frac{t^{-q-1}\,dt}{\sqrt{(t+f^2)\ldots(t+h^2)}\cdot\sqrt{t}},$$

the corresponding value of ρ is

$$\rho=\psi_0\Big(1-\tfrac{x^2}{f^2}-\cdots-\tfrac{z^2}{h^2}\Big)^q,$$

where Λ_0 is what H becomes on writing therein $\theta=0$, and ψ_0 is what ψ becomes on writing $\dfrac{x}{f},\dfrac{y}{g},\ldots,\dfrac{z}{h}$ in place of $\alpha,\beta,\ldots,\gamma$.

128. Thus, putting for shortness $\bar{H}=t^{-q-1}\{(t+f^2)\ldots(t+h^2)\}^{-\frac{1}{2}}$, we have in the three several cases $\phi=1$, $\phi=\alpha$, $\phi=\alpha\beta$ respectively,

$$H=1,\qquad \rho=\Big(1-\tfrac{x^2}{f^2}-\cdots-\tfrac{z^2}{h^2}\Big)^q,\qquad V=\frac{\Gamma(\frac{1}{2}s+q)}{\Gamma(q+1)}(f\ldots h)\int_\theta^\infty\bar{H},$$

$$H=\frac{\sqrt{f^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}},\qquad \ldots,\qquad V=\ldots\int_\theta^\infty\bar{H},$$

$$H = \frac{\sqrt{f^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}} \cdot \frac{\sqrt{g^2+\theta}}{\sqrt{f^2+g^2+\cdots+h^2}}, \quad \ldots, \quad V = \ldots \int_\theta^\infty \frac{\bar{H}\,dt}{\sqrt{f^2+t}\cdot\sqrt{g^2+t}}.$$

[p. 404] For the case last considered

$$\phi = \frac{f^2\alpha^2}{2q+2+\frac{1}{2}\kappa_0 f^2} + \cdots + \frac{h^2\gamma^2}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0},$$

$$H = \frac{f^2(f^2+\theta)}{2q+2+\frac{1}{2}\kappa_0 f^2} + \cdots + \frac{h^2(h^2+\theta)}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0}, \quad \bar{T} \text{ same function with } t \text{ for } \theta,$$

$$\psi_0 = \frac{x^2}{2q+2+\frac{1}{2}\kappa_0 f^2} + \cdots + \frac{z^2}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0},$$

$$\Lambda_0 = \frac{f^4}{2q+2+\frac{1}{2}\kappa_0 f^2} + \cdots + \frac{h^4}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0},$$

where κ_0 is the root of the equation

$$\frac{1}{2q+2+\frac{1}{2}\kappa_0 f^2} + \cdots + \frac{1}{2q+2+\frac{1}{2}\kappa_0 h^2} + 1 = 0.$$

Annex VI. Examples of Theorem C. Art. Nos. 129 to 132.

129. First example: relating to the $(s+1)$-coordinal sphere $a^2+\cdots+z^2+w^2=f^2$. Assume

$$V' = \frac{M}{\left(a^2+\cdots+c^2+e^2\right)^{\frac{1}{2}(s-1)}}, \qquad V'' = \frac{M}{f^{s-1}} \quad \text{(a constant)};$$

these values each satisfy the potential equation.

V' is not infinite for any point outside the surface, and for indefinitely large distances it is of the proper form.

V'' is not infinite for any point inside the surface; and at the surface $V'=V''$. The conditions of the theorem are therefore satisfied. Writing

$$V = \int \frac{\rho\,dS}{\left\{(a-x)^2+\cdots+(c-z)^2+(e-w)^2\right\}^{\frac{1}{2}(s-1)}},$$

we have

$$\rho = \frac{1}{4(\sqrt{\pi})^{s+1}}\left(\frac{dV'}{dn'} - \frac{dV''}{dn''}\right),$$

where

$$\frac{dV''}{dn''} = -\left(\tfrac{x}{f}\tfrac{d}{dx}+\cdots+\tfrac{z}{f}\tfrac{d}{dz}+\tfrac{w}{f}\tfrac{d}{dw}\right)\frac{M}{(x^2+\cdots+z^2+w^2)^{\frac{1}{2}(s-1)}} = \frac{(s-1)M(x^2+\cdots+z^2+w^2)^{\frac{1}{2}(s-1)}}{f\cdot(x^2+\cdots+z^2+w^2)^{\frac{1}{2}(s+1)}},$$

[p. 405] which at the surface is

$$= \frac{(s-1)M}{f^s}.$$

Hence

$$\rho = \frac{(s-1)\Gamma(\frac{1}{2}(s+1))M}{4(\sqrt{\pi})^{s+1}f^s} = \frac{\Gamma(\frac{1}{2}s+1)M}{2(\sqrt{\pi})^{s+1}f^s} \quad \text{(viz. } \rho \text{ is constant).}$$

130. Writing for convenience $M = \dfrac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))}\,\delta f^s\, Sf^s$ (δf^s a constant which may be put $= 1$), also $a^2 + \cdots + c^2 + e^2 = s^2$, we have $\rho = Sf^s$, and consequently

$$\int \frac{\delta f^s\, dS}{\left\{(a-x)^2 + \cdots + (c-z)^2 + (e-w)^2\right\}^{\frac{1}{2}(s-1)}} = \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))}\cdot\frac{f^s}{s^{s-1}}, \quad \text{for exterior point } s > f,$$

$$= \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))}\cdot f^{s-1}, \quad \text{for interior point } s < f.$$

By making $a, \ldots, c, e$ all indefinitely large, we find

$$\int \delta f^s\, dS = \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))}\, f^s,$$

viz. the expression on the right-hand side is here the mass of the shell thickness δf.

Taking $s = 3$, we have the ordinary formulae for the Potential of a uniform spherical shell.

131. Suppose $s = 3$, but let the surface be the infinite cylinder $x^2 + y^2 = f^2$. Take here

$$V' = M \log\sqrt{a^2 + b^2}, \qquad V'' = M \log f,$$

each satisfying the potential equation $\left(\dfrac{d^2V}{dx^2} + \dfrac{d^2V}{dy^2}\right) = 0$; but V, instead of vanishing, is infinite at infinity, and the conditions of the theorem are not satisfied; the Potential of the cylinder is in fact infinite. But the failure is a mere consequence of the special value of s, viz. this is such that $s-2$, instead of being positive, is $= 0$. Reverting to the general case of $(s+1)$-dimensional space, let the surface be the infinite cylinder $x^2 + \cdots + z^2 = f^2$; and assume

$$V' = \frac{M}{\left(a^2 + \cdots + c^2\right)^{\frac{1}{2}(s-2)}}, \qquad V'' = \frac{M}{f^{s-2}} \quad \text{(a constant)}.$$

These satisfy the potential equation; viz. as regards V', we have

[End of pages 381–405 (PDF pp. 401–425). Paper 607 continues in subsequent pages.]

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 406–430 (Papers 607 conclusion, 608, 609, 610)

Transcribed from the Original Publications

Contents

607. A Memoir on Prepotentials (concluded)

[Continued from earlier pages of the same paper. Articles 131 (concl.)–162.]

[p. 406] V' is not infinite at any point outside the cylinder; and it vanishes at infinity, except indeed when only the coordinate e is infinite, and its form at infinity is not

$$= M\big(a^2 + \cdots + c^2 + e^2\big)^{-\frac{1}{2}(s-2)}.$$

V'' is not infinite for any point within the cylinder; and at the surface we have $V' = V''$. We have

$$\rho = -\frac{\Gamma(\frac{1}{2}s - \frac{1}{2})}{4\,(\Gamma\frac{1}{2})^{s+1}}\Big(\frac{dW'}{da^2} + \frac{dW''}{da'^2}\Big),$$

where

$$\frac{dW'}{da^2} = \frac{-(s-2)\frac{1}{2}(a^2 + \cdots + z^2)\,M}{(a^2 + \cdots + c^2)^{\frac{1}{2}s}}, \qquad \frac{dW''}{da'^2} = \frac{-(s-2)\,M}{f^{s-2}} \text{ at the surface;} \quad \frac{dW''}{da'^2} = 0,$$

and therefore

$$\rho = \frac{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})\,f^{s-1}\,M}{4\,(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}} \quad \text{(viz. } \rho \text{ is constant);}$$

or, what is the same thing, writing $M = \dfrac{4(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}\,\rho}{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})}$, whence $\rho = Bf$, and writing also $a^2 + \cdots + c^2 = \varpi^2$, we have

$$V' = \frac{4(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}\,\rho}{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})}\cdot\frac{1}{\varpi^{s-2}} \quad \text{for an exterior point } \varpi > f,$$

$$V'' = \frac{4(\Gamma\frac{1}{2})^{s+1}\,f^{s-1}\,\rho}{(s-2)\,\Gamma(\frac{1}{2}s - \frac{1}{2})}\cdot\frac{1}{f^{s-2}} \quad \text{for an interior point } \varpi < f.$$

132. This is right; but we can without difficulty bring it to coincide with the result obtained for the $(s+1)$-dimensional sphere with only $s-1$ in place of s; we may in fact, by a single integration, pass from the cylinder $x^2 + \cdots + z^2 = f^2$ to the s-dimensional sphere or circle $x^2 + \cdots + z^2 = f^2$, which is the base of this cylinder. Writing first $dS = d\Sigma\,dw$, where $d\Sigma$ refers to the s variables $(x, \ldots, z)$ and the sphere $x^2 + \cdots + z^2 = f^2$; or using now dS in this sense, then in place of the original dS we have $d\Sigma\,dw$: and w being ∞, the limits of w being $\pm\infty$, then in place of $e - w$ we may write simply w. This being so, and putting for shortness $(a-x)^2 + \cdots + (c-z)^2 = A^2$, the integral is

$$V = \iint \frac{\delta f\,dS\,dw}{(A^2 + w^2)^{\frac{1}{2}(s-2)}},$$

and we have without difficulty

$$\int_{-\infty}^{+\infty} \frac{dw}{(A^2 + w^2)^{\frac{1}{2}(s-2)}} = \frac{1}{A^{s-3}}\cdot\frac{\sqrt{\pi}\,\Gamma\frac{1}{2}(s-2)}{\Gamma\frac{1}{2}(s-1)}.$$

[p. 407] To prove it, write $w = A\tan\theta$, then the integral is in the first place converted into

$$\frac{2}{A^{s-3}}\int_0^{\frac{\pi}{2}} \cos^{s-4}\theta\,d\theta,$$

which, putting $\cos\theta = \sqrt{x}$ and therefore $\sin\theta = \sqrt{1-x}$, becomes

$$= \frac{1}{A^{s-3}} \int_0^1 x^{\frac{1}{2}s-\frac{5}{2}} (1-x)^{-\frac{1}{2}}\, dx,$$

which has the value in question.

Hence, replacing A by its value, we have

$$V = \frac{\sqrt{\pi}\,\Gamma\frac{1}{2}(s-2)}{\Gamma\frac{1}{2}(s-1)} \int \frac{\delta f\, dS}{\{(a-x)^2 + \cdots + (c-z)^2\}^{\frac{1}{2}(s-3)}} = \frac{4\pi^{\frac{1}{2}s}\,\Gamma(\frac{1}{2})\, f^{s-1}\rho}{(s-2)\,\Gamma\frac{1}{2}(s-2)} \cdot \frac{1}{(a^2+\cdots+c^2)^{\frac{1}{2}(s-3)}} \quad \text{or} \quad \frac{1}{f^{s-3}};$$

that is,

$$V = \frac{4(\Gamma\frac{1}{2})^s\, f^{s-1}\,\rho}{(s-3)\,\Gamma\frac{1}{2}(s-3)} \cdot \frac{1}{(a^2+\cdots+c^2)^{\frac{1}{2}(s-3)}} \quad \text{or} \quad \frac{1}{f^{s-3}};$$

viz. this is the formula for the sphere with $s-1$ instead of s.

Annex VII. Example of Theorem D, Art. Nos. 133 and 134.

133. The example relates to the $(s+1)$-dimensional sphere $x^2 + \cdots + z^2 + w^2 = f^2$.

Instead of at once assuming for V a form satisfying the proper conditions as to continuity, we assume a form with indeterminate coefficients, and make it satisfy the conditions in question. Write

$$V = \frac{M}{(a^2+\cdots+c^2+e^2)^{\frac{1}{2}(s-1)}} \qquad \text{for } a^2+\cdots+c^2+e^2 > f^2;$$

$$V = A(a^2+\cdots+c^2+e^2) + B \qquad \text{for } a^2+\cdots+c^2+e^2 < f^2.$$

In order that the two values may be equal at the surface, we must have

$$\frac{M}{f^{s-1}} = Af^2 + B;$$

in order that the derived functions $\dfrac{dV}{da}$, &c. may be equal, we must have

$$\frac{-(s-1)\,a\,M}{f^{s+1}} = 2Aa, \quad \&c.,$$

viz. these are all satisfied if only $\dfrac{-(s-1)\,M}{f^{s+1}} = 2A$.

We have thus the values of A and B; or the exterior potential being as above

$$V = \frac{M}{(a^2+\cdots+c^2+e^2)^{\frac{1}{2}(s-1)}},$$

[p. 408] the value of the interior potential must be

$$V = -\frac{M}{f^{s+1}}\left\{(\tfrac{1}{2}s+\tfrac{1}{2}) - (\tfrac{1}{2}s-\tfrac{1}{2})\,\frac{a^2+\cdots+c^2+e^2}{f^2}\right\}.$$

The corresponding values of W are of course

$$\frac{M}{(a^2+\cdots+c^2+e^2+w^2)^{\frac{1}{2}(s-1)}} \quad \text{and} \quad \frac{M}{f^{s+1}}\left\{(\tfrac{1}{2}s+\tfrac{1}{2}) - (\tfrac{1}{2}s-\tfrac{1}{2})\,\frac{a^2+\cdots+c^2+e^2+w^2}{f^2}\right\};$$

and we thence find

$$\rho = 0, \qquad \text{if } a^2 + \cdots + c^2 + w^2 > f^2,$$

$$\rho = -\frac{\Gamma(\frac{1}{2}s - \frac{1}{2})}{4\,(\Gamma\frac{1}{2})^{s+1}}\Big\{-4(\tfrac{1}{2}s - \tfrac{1}{2})\,(\tfrac{1}{2}s + \tfrac{1}{2})\,\frac{M}{f^{s+3}}\Big\} = \frac{\Gamma(\frac{1}{2}s + \frac{1}{2})}{(\Gamma\frac{1}{2})^{s+1}} \cdot \frac{M}{f^{s+3}},$$

$$\text{if } a^2 + \cdots + c^2 + w^2 < f^2.$$

Assuming for M the value $\dfrac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})}\, f^{s+1}$, the last value becomes $\rho = 1$; writing for shortness $a^2 + \cdots + c^2 + e^2 = \kappa^2$, we have

$$V = \int \frac{dx \ldots dz\, dw}{[(a-x)^2 + \cdots + (c-z)^2 + (e-w)^2]^{\frac{1}{2}(s-1)}} \text{ over the } (s{+}1)\text{-dim. sphere } x^2{+}\cdots{+}z^2{+}w^2 = f^2,$$

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})} \cdot \frac{f^{s+1}}{\kappa^{s-1}}, \quad \text{for an exterior point } \kappa > f,$$

$$= \frac{(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})}\big\{(\tfrac{1}{2}s + \tfrac{1}{2})f^2 - (\tfrac{1}{2}s - \tfrac{1}{2})\,\kappa^2\big\}, \quad \text{for an interior point } \kappa < f.$$

134. The case of the ellipsoid $\dfrac{x^2}{f^2} + \cdots + \dfrac{z^2}{h^2} = 1$ for $s+1$-dimensional space may be worked out by the theorem; this is, in fact, what is done in tridimensional space by Lejeune-Dirichlet in his Memoir of 1846 above referred to (p. 321).

Annex VIII. Prepotentials of the Homaloids. Art. Nos. 135 to 137.

135. We have in tridimensional space the series of figures—the plane, the line, the point; and there is in like manner in $(s+1)$-dimensional space a corresponding series of $(s+1)$ terms; the s-coordinal plane,… the $(s-1)$-coordinal plane,… the line, the point: say these are the homaloids or homaloidal figures. And, taking the density as uniform, or, what is the same thing, $= 1$, we may consider the prepotentials of these several figures in regard to an attracted point, which, for greater simplicity, is taken not to be on the figure.

136. The integral may be written

$$V = \int \frac{dw \ldots dt}{\{(a-x)^2 + \cdots + (c-z)^2 + (d-w)^2 + \cdots + (e-t)^2 + u^2\}^{\frac{1}{2}s+q}},$$

which still relates to a $(s+1)$-dimensional space: the $(s+1)$ coordinates of the attracted point are $(a, \ldots, c, d, \ldots, e, u)$, instead of being $(a, \ldots, c, e)$; viz. we have the

[p. 409] s' coordinates $(a, \ldots, c)$, the $s{-}s'$ coordinates $(d, \ldots, e)$, and the $(s{+}1)$th coordinate u: and the integration is extended over the $(s-s')$-dimensional figure $w = -\infty$ to $+\infty, \ldots, t = -\infty$ to $+\infty$. And it is also assumed that q is positive.

It is at once clear that we may reduce the integral to

$$V = \int \frac{dw \ldots dt}{\{(a-x)^2 + \cdots + (c-z)^2 + u^2 + w^2 + \cdots + t^2\}^{\frac{1}{2}s+q}},$$

say for shortness

$$= \int \frac{dw \ldots dt}{(A^2 + w^2 + \cdots + t^2)^{\frac{1}{2}s+q}},$$

where $A^2 = (a-x)^2 + \cdots + (c-z)^2 + u^2$, is a constant as regards the integration, and where the limits in regard to each of the $s - s'$ variables are $-\infty$, $+\infty$.

We may for these variables write $r\xi, \ldots, r\zeta$, where $\xi^2 + \cdots + \zeta^2 = 1$; and we then have $dw \ldots dt = r^{s-s'-1}\, dr\, dS$, where dS is the element of surface of the $(s-s')$-coordinal unit-sphere $\xi^2 + \cdots + \zeta^2 = 1$. We thus obtain

$$V = \int \frac{r^{s-s'-1}\, dr}{(A^2+r^2)^{\frac{1}{2}s+q}} \int dS,$$

where the integral in regard to r is taken from 0 to ∞, and the integral $\int dS$ over the surface of the unit-sphere; hence by Annex I. the value of this last factor is $\dfrac{2(\Gamma\frac{1}{2})^{s-s'}}{\Gamma\frac{1}{2}(s-s')}$. The integral represented by the first factor will be finite, provided $\frac{1}{2}(s-s')$ is positive; which is the case for any value whatever of s', if only q be positive.

The first factor is an integral such as is considered in Annex II.; to find its value we have only to write $r = Ax$, and we thus find it to be

$$\frac{1}{(A^2)^{\frac{1}{2}s'+q}} \cdot \frac{1}{A} \int_0^\infty \frac{x^{s-s'-1}\, dx}{(1+x)^{\frac{1}{2}s+q}} = \frac{1}{A^{s'+2q}} \cdot \frac{1}{2}\, \frac{\Gamma\frac{1}{2}(s-s')\, \Gamma(\frac{1}{2}s'+q)}{\Gamma(\frac{1}{2}s+q)},$$

and we thus have

$$V = \frac{(\Gamma\frac{1}{2})^{s-s'}\, \Gamma(\frac{1}{2}s'+q)}{\Gamma(\frac{1}{2}s+q)} \cdot \frac{1}{\{(a-x)^2 + \cdots + (c-z)^2 + u^2\}^{\frac{1}{2}s'+q}}.$$

137. As a verification, observe that the prepotential equation $\square V = 0$, that is,

$$\Big(\frac{d^2}{da^2} + \cdots + \frac{d^2}{dc^2} + \frac{d^2}{dd^2} + \cdots + \frac{d^2}{de^2} + \frac{d^2}{du^2} + \frac{2q+1}{u}\,\frac{d}{du}\Big) V = 0,$$

for a function V, which contains only the $s'+1$ variables $(a, \ldots, c, u)$, becomes

$$\Big(\frac{d^2}{da^2} + \cdots + \frac{d^2}{dc^2} + \frac{d^2}{du^2} + \frac{2q+1}{u}\,\frac{d}{du}\Big) V = 0,$$

which is satisfied by V, a constant multiple of $\big\{(a-x)^2 + \cdots + (c-z)^2 + u^2\big\}^{-\frac{1}{2}s'-q}$.

Annex IX. The Gauss–Jacobi Theory of Epispheric Integrals. Art. No. 138.

[p. 410] **138.** The formula obtained (Annex IV. No. 110) is proved only for positive values of m; but writing therein $q = 0$, $m = -\frac{1}{2}$, it becomes

$$\int \sqrt{1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}}\, \frac{dx \ldots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s}}$$

$$= \frac{(\Gamma\frac{1}{2})^s}{\Gamma\frac{1}{2}s}\, f \ldots h \int_0^\infty t^{-\frac{1}{2}} \Big(1 - \frac{a^2}{t+f^2} - \cdots - \frac{c^2}{t+h^2} - \frac{e^2}{t}\Big)^{-\frac{1}{2}} \{(t+f^2)\ldots(t+h^2)\}^{-\frac{1}{2}},$$

a formula which is obtainable as a particular case of the more general formula

$$V = \int \frac{dS}{\{(*\!\!\between x, \ldots, z, w)^2\}^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_{-A}^\infty \frac{dt}{\sqrt{-\,\text{Disct.}\,.\big[(*\!\!\between X, \ldots, Z, W, T)^2 + t(X^2 + \cdots + Z^2 + W^2 - T^2)\big]}},$$

(notation to be presently explained), being a result obtained by Jacobi by a process which is in fact the extension to any number of variables of that used by Gauss[1] in his Memoir "Determinatio attractionis quam exerceret planeta, &c." (1818). I proceed to develop this theory.

139. Jacobi's process has reference to a class of s-tuple integrals (including some of those here previously considered) which may be termed "epispheric": viz. considering the $(s+1)$ variables $(x, \dots, z, w)$ connected by the equation $x^2+\dots+z^2+w^2=1$, or say they are the coordinates of a point on a $(s+1)$-tuple unit-sphere, then the form is $\int U\, dS$, where dS is the element of the surface of the unit-sphere, and U is any function of the $s+1$ coordinates; the integral is taken to be of the form $\int \dfrac{dS}{\{(*\between x, \dots, z, w, 1)^2\}^{\frac{1}{2}s}}$, and we then obtain the general result above referred to.

Before going further it is convenient to remark that, taking as independent variables the s coordinates $x, \dots, z$, we have $dS = \dfrac{dx \dots dz}{w}$, where w stands for $\pm\sqrt{1-x^2-\dots-z^2}$; we must in obtaining the integral take account of the two values of w, and finally extend the integral to the values of $x, \dots, z$ which satisfy $x^2+\dots+z^2 \le 1$.

If, as is ultimately done, in place of $x, \dots, z$ we write $\dfrac{x}{f}, \dots, \dfrac{z}{h}$ respectively, then the value of $dS = \dfrac{1}{f \dots h} \cdot \dfrac{dx \dots dz}{w}$, where w now stands for $\pm\sqrt{1-\dfrac{x^2}{f^2}-\dots-\dfrac{z^2}{h^2}}$; we must, in finding the value of the integral, take account of the two values of w, and finally extend the integral to the values of $x, \dots, z$ which satisfy $\dfrac{x^2}{f^2}+\dots+\dfrac{z^2}{h^2} \le 1$.

[p. 411] **140.** The determination of the integral depends upon formulae for the transformation of the spherical element dS, and of the quadric function $(x, y, \dots, z, w, 1)^2$.

First, as regards the spherical element dS; let the $s+1$ variables $x, y, \dots, z, w$ which satisfy $x^2+y^2+\dots+z^2+w^2=1$ be regarded as functions of the s independent variables $\theta, \phi, \dots, \psi$; then we have

$$dS = \frac{1}{w}\begin{vmatrix} \frac{dx}{d\theta} & \frac{dy}{d\theta} & \dots & \frac{dz}{d\theta} & \frac{dw}{d\theta} \\ \frac{dx}{d\phi} & \frac{dy}{d\phi} & \dots & \frac{dz}{d\phi} & \frac{dw}{d\phi} \\ \vdots & \vdots & & \vdots & \vdots \\ \frac{dx}{d\psi} & \frac{dy}{d\psi} & \dots & \frac{dz}{d\psi} & \frac{dw}{d\psi} \end{vmatrix} d\theta\, d\phi \dots d\psi = \frac{\partial(x, y, \dots, z, w)}{\partial(\theta, \phi, \dots, \psi, *)}\, d\theta\, d\phi \dots d\psi, \text{ for shortness.}$$

Suppose we effect on the $s+1$ variables $(x, y, \dots, z, w)$ a transformation

$$x,\ y,\ \dots,\ z,\ w = \frac{X}{T}, \frac{Y}{T}, \dots, \frac{Z}{T}, \frac{W}{T},$$

thus introducing for the moment $s+2$ variables $X, Y, \dots, Z, W, T$, which satisfy identically $X^2+Y^2+\dots+Z^2+W^2-T^2=0$; then, considering these as functions of the foregoing s independent variables $\theta, \phi, \dots, \psi$, we have

$$dS = \frac{1}{2^{s+1}}\, \frac{W}{T^{s+2}} \cdot \frac{\partial(X, Y, \dots, Z, W)}{\partial(\theta, \phi, \dots, \psi, *)}\, d\theta\, d\phi \dots d\psi.$$

141. Considering next the $s+2$ variables $X, Y, \dots, Z, W, T$ as linear functions (with constant terms) of the $s+1$ new variables $\xi, \eta, \dots, \zeta, \omega$, or say as linear functions of the $s+2$ quantities $\xi, \eta, \dots, \zeta, \omega, 1$: which implies between them a linear relation

$$aX+bY+\dots+cZ+dW+eT=1;$$

[1] [*Ges. Werke*, t. iii, pp. 331–355.]

and assuming that we have identically

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1,$$

so that, in consequence of the left-hand side being $= 0$, the right-hand side is also $= 0$; viz. $\xi, \eta, \ldots, \zeta, \omega$ are connected by

$$\xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 = 1 :$$

[p. 412] let $d\Sigma$ represent the spherical element belonging to the coordinates $\xi, \eta, \ldots, \zeta, \omega$. Considering these as functions of the foregoing s independent variables $\theta, \phi, \ldots, \psi$, we have

$$d\Sigma = \frac{1}{\omega} \begin{vmatrix} \dfrac{d\xi}{d\theta} & \dfrac{d\eta}{d\theta} & \cdots & \dfrac{d\zeta}{d\theta} & \dfrac{d\omega}{d\theta} \\ \dfrac{d\xi}{d\phi} & \dfrac{d\eta}{d\phi} & \cdots & \dfrac{d\zeta}{d\phi} & \dfrac{d\omega}{d\phi} \\ \vdots & \vdots & & \vdots & \vdots \\ \dfrac{d\xi}{d\psi} & \dfrac{d\eta}{d\psi} & \cdots & \dfrac{d\zeta}{d\psi} & \dfrac{d\omega}{d\psi} \end{vmatrix} d\theta\, d\phi \ldots d\psi = \frac{\partial(\xi, \eta, \ldots, \zeta, \omega)}{\partial(\theta, \phi, \ldots, \psi, *)}\, d\theta\, d\phi \ldots d\psi.$$

142. In this expression we have $\xi, \eta, \ldots, \zeta, \omega$, each of them a linear function of the $s + 2$ quantities $X, Y, \ldots, Z, W, T$; the determinant is consequently a linear function of $s + 2$ like determinants obtained by substituting for the variables any $s + 1$ out of the $s + 2$ variables $X, Y, \ldots, Z, W, T$; but in virtue of the equation

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = 0,$$

these $s + 2$ determinants are proportional to the quantities $X, Y, \ldots, Z, W, T$ respectively, and the determinant thus assumes the form

$$\frac{aX + bY + \cdots + cZ + dW + eT}{T}\,\Delta,$$

where Δ is the like determinant with $(X, Y, \ldots, Z, W)$, and where the coefficients $a, b, \ldots, c, d, e$ are precisely those of the linear relation $aX + bY + \cdots + cZ + dW + eT = 1$; the last-mentioned expression is thus $\dfrac{1}{T}\Delta$, or, substituting for Δ its value, we have

$$d\Sigma = \frac{1}{T}\,\frac{\partial(X, Y, \ldots, Z, W)}{\partial(\theta, \phi, \ldots, \psi, *)}\, d\theta\, d\phi \ldots d\psi;$$

viz. comparing with the foregoing expression for dS we have

$$dS = \frac{1}{T^{s+1}}\, d\Sigma,$$

which is the requisite formula for the transformation of dS.

143. Consider the integral

$$\int \frac{dS}{\{(*\rangle x, y, \ldots, z, w, 1)^2\}^{\frac{1}{2}s}},$$

which, from its containing a single quadric function, may be called "one-quadric." Then effecting the foregoing transformation,

$$x,\ y,\ \ldots,\ z,\ w = \frac{X}{T}, \frac{Y}{T}, \ldots, \frac{Z}{T}, \frac{W}{T},$$

[p. 413] and observing that

$$(*\!\!\!\!\big)x, y, \dots, z, w, 1)^2 = \frac{1}{T^2}(*\!\!\!\!\big)X, Y, \dots, Z, W, T)^2,$$

the integral becomes

$$= \int \frac{d\Sigma}{\{(*\!\!\!\!\big)X, Y, \dots, Z, W, T)^2\}^{\frac{1}{2}s}},$$

where $X, Y, \dots, Z, W, T$ denote given linear functions (with constant coefficients) of the $s+1$ variables $\xi, \eta, \dots, \zeta, \omega$, or, what is the same thing, given linear functions of the $s+2$ quantities $\xi, \eta, \dots, \zeta, \omega, 1$, such that identically

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1.$$

We have then $\xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1 = 0$, and $d\Sigma$ as the corresponding spherical element.

144. We may have $X, Y, \dots, Z, W, T$ such linear functions of $\xi, \eta, \dots, \zeta, \omega, 1$ that not only

$$X^2 + Y^2 + \cdots + Z^2 + W^2 - T^2 = \xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 - 1$$

as above, but also

$$(*\!\!\!\!\big)X, Y, \dots, Z, W, T)^2 = A\xi^2 + B\eta^2 + \cdots + C\zeta^2 + E\omega^2 - L;$$

this being so, the integral becomes

$$\int \frac{d\Sigma}{[A\xi^2 + B\eta^2 + \cdots + C\zeta^2 + E\omega^2 - L]^{\frac{1}{2}s}},$$

where the $s+2$ coefficients $A, B, \dots, C, E, L$ are given by means of the identity

$$-(\theta+A)(\theta+B)\dots(\theta+C)(\theta+E)(\theta+L) = \text{Disct.}\,.\big[(*\!\!\!\!\big)X, Y, \dots, Z, W, T)^2 + \theta(X^2+Y^2+\cdots+Z^2+W^2-T^2)\big];$$

viz. equating the discriminant to zero, we have an equation in θ, the roots whereof are $-A, -B, \dots, -C, -E, -L$.

The integral is

$$\int \frac{d\Sigma}{\big[(A-L)\xi^2 + (B-L)\eta^2 + \cdots + (C-L)\zeta^2 + (E-L)\omega^2\big]^{\frac{1}{2}s}}$$

which is of the form

$$\int \frac{d\Sigma}{\big[a\xi^2 + b\eta^2 + \cdots + c\zeta^2 + e\omega^2\big]^{\frac{1}{2}s}},$$

where I provisionally assume that $a, b, \dots, c, e$ are all positive.

145. To transform this, in place of the $s+1$ variables $\xi, \eta, \dots, \zeta, \omega$ connected by $\xi^2 + \eta^2 + \cdots + \zeta^2 + \omega^2 = 1$, we introduce the $s+1$ variables $x, y, \dots, z, w$, such that

$$x = \frac{\xi\sqrt{a}}{\rho}, \quad y = \frac{\eta\sqrt{b}}{\rho}, \quad \dots, \quad z = \frac{\zeta\sqrt{c}}{\rho}, \quad w = \frac{\omega\sqrt{e}}{\rho},$$

[p. 414] where

$$\rho^2 = a\xi^2 + b\eta^2 + \cdots + c\zeta^2 + e\omega^2,$$

and consequently

$$x^2 + y^2 + \cdots + z^2 + w^2 = 1.$$

Hence, writing dS to denote the spherical element corresponding to the point $(x, y, \dots, z, w)$, we have, by a former formula,

$$dS = \frac{1}{w}\,\frac{\partial(x, y, \dots, z, \omega\sqrt{e}/\rho)}{\partial(\theta, \phi, \dots, \psi, *)}\,d\theta\,d\phi\dots d\psi = \frac{(ab\dots ce)^{\frac{1}{2}}}{\rho^{s+1}}\,d\Sigma,$$

or, what is the same thing,

$$d\Sigma = \frac{\rho^{s+1}}{(ab\dots ce)^{\frac{1}{2}}}\,dS.$$

Hence, integrating each side, and observing that $\int dS$, taken over the whole spherical surface $x^2 + y^2 + \dots + z^2 + w^2 = 1$, is $= \dfrac{2(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})}$, we have

$$\int \frac{d\Sigma}{\bigl[a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2\bigr]^{\frac{1}{2}s+\frac{1}{2}}} = \frac{2(\Gamma\frac{1}{2})^{s+1}}{\Gamma(\frac{1}{2}s + \frac{1}{2})}\cdot\frac{1}{(ab\dots ce)^{\frac{1}{2}}}.$$

146. For $a, b, \dots, c, e$ write herein $a+\theta, b+\theta, \dots, c+\theta, e+\theta$ respectively, and multiply each side by θ^{q-1}, where q is any positive integer or fractional number less than $\frac{1}{2}s$: integrate from $\theta = 0$ to $\theta = \infty$. On the left-hand side, attending to the relation $\xi^2 + \eta^2 + \dots + \zeta^2 + \omega^2 = 1$, the integral in regard to θ is

$$\int_0^\infty \frac{\theta^{q-1}\,d\theta}{(\rho^2 + \theta)^{\frac{1}{2}(s+1)}},$$

where $\rho^2 = a\xi^2 + b\eta^2 + \dots + c\zeta^2 + e\omega^2$, is independent of θ as before; the value of the definite integral is

$$\frac{\Gamma\{\frac{1}{2}(s+1) - q\}\,\Gamma(q)}{\Gamma\frac{1}{2}(s+1)}\cdot\frac{1}{\rho^{s+1-2q}},$$

which, replacing ρ by its value and multiplying by $d\Sigma$, and prefixing the integral sign, gives the left-hand side; hence, forming the equation and dividing by a numerical factor, we have

$$\int \frac{d\Sigma}{\bigl[a\xi^2 + \dots + c\zeta^2 + e\omega^2\bigr]^{\frac{1}{2}s+\frac{1}{2}-q}} = \frac{2(\Gamma\frac{1}{2})^{s+1}}{\Gamma\frac{1}{2}(s+1-q)\,\Gamma(q)}\cdot\frac{1}{(ab\dots ce)^{\frac{1}{2}}}\int_0^\infty dt\,t^{q-1}\bigl[(t+a)\dots(t+c)(t+e)\bigr]^{-\frac{1}{2}}.$$

In particular, if $q = \frac{1}{2}$, then

$$\int \frac{d\Sigma}{\bigl[a\xi^2 + \dots + c\zeta^2 + e\omega^2\bigr]^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_0^\infty dt\bigl[(t+a)\dots(t+c)(t+e)\bigr]^{-\frac{1}{2}};$$

[p. 415] or, if for $a, \dots, c, e$ we restore the values $A - L, \dots, C - L, E - L$, then

$$\int \frac{d\Sigma}{\bigl[(A-L)\xi^2 + \dots + (C-L)\zeta^2 + (E-L)\omega^2\bigr]^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_{-L}^\infty dt\bigl[(t+A)\dots(t+C)(t+E)(t+L)\bigr]^{-\frac{1}{2}},$$

viz. we thus have

$$\int \frac{dS}{\bigl[(*\between x, \dots, z, w, 1)^2\bigr]^{\frac{1}{2}s}} = \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_{-L}^\infty dt\bigl[(t+A)\dots(t+C)(t+E)(t+L)\bigr]^{-\frac{1}{2}},$$

where $(t + A)\dots(t + C)(t + E)(t + L)$ is in fact a given rational and integral function of t; viz. it is

$$= -\operatorname{Disct.}.\bigl[(*\between X, \dots, Z, W, T)^2 + t(X^2 + \dots + Z^2 + W^2 - T^2)\bigr].$$

147. Consider, in particular, the integral

$$\int \frac{dS}{\left[(a-fx)^2+\cdots+(c-hz)^2+(e-kw)^2+l^2\right]^{\frac{1}{2}s}};$$

here

$$\begin{aligned}&(aT-fX)^2+\cdots+(cT-hZ)^2+(eT-kW)^2+l^2T^2+t(X^2+\cdots+Z^2+W^2-T^2)\\ &\quad=(f^2+t)X^2+\cdots+(h^2+t)Z^2+(k^2+t)W^2+(a^2+\cdots+c^2+e^2+l^2-t)T^2\\ &\qquad-2afXT-\cdots-2chZT-2ekWT;\end{aligned}$$

viz. the discriminant taken negatively is

$$\begin{vmatrix} t+f^2 & \cdots & \cdots & -af \\ \vdots & \ddots & & \vdots \\ \cdots & & t+k^2 & -ek \\ -af & \cdots & -ek & -(a^2+\cdots+c^2+e^2+l^2)+t \end{vmatrix},$$

which is

$$=(t+A)\ldots(t+C)(t+E)(t+L);$$

and consequently $-A,\ldots,-C,-E,-L$ are the roots of the equation

$$1-\frac{a^2}{t+f^2}-\cdots-\frac{c^2}{t+h^2}-\frac{e^2}{t+k^2}-\frac{l^2}{t}=0.$$

148. The roots are all real; moreover there is one and only one positive root. Hence, taking $-L$ to be the positive root, we have $A,\ldots,C,E,L$ all positive, and therefore *à fortiori* $A-L,\ldots,C-L,E-L$ all positive: which agrees with a foregoing

[p. 416] provisional assumption. Or, writing for greater convenience θ to denote the positive quantity $-L$, that is, taking θ to be the positive root of the equation

$$1-\frac{a^2}{\theta+f^2}-\cdots-\frac{c^2}{\theta+h^2}-\frac{e^2}{\theta+k^2}-\frac{l^2}{\theta}=0,$$

we have

$$\int \frac{dS}{[(a-fx)^2+\cdots+(c-hz)^2+(e-kw)^2+l^2]^{\frac{1}{2}s}}=\frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_\theta^\infty dt\left(1-\frac{a^2}{t+f^2}-\cdots-\frac{c^2}{t+h^2}-\frac{e^2}{t+k^2}-\frac{l^2}{t}\right)^{-\frac{1}{2}}[t$$

or, what is the same thing, we have

$$\frac{1}{f\ldots h\,k}\int\frac{dx\ldots dz}{w[(a-x)^2+\cdots+(c-z)^2+(e-kw)^2+l^2]^{\frac{1}{2}s}}=\frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)}\int_\theta^\infty dt\left(1-\frac{a^2}{t+f^2}-\cdots-\frac{c^2}{t+h^2}-\frac{e^2}{t+k^2}-\right.$$

where on the left-hand side w now denotes $\sqrt{1-\dfrac{x^2}{f^2}-\cdots-\dfrac{z^2}{h^2}}$, and the limiting equation is $\dfrac{x^2}{f^2}+\cdots+\dfrac{z^2}{h^2}=1$.

149. Suppose $l=0$: then, if

$$\frac{a^2}{f^2}+\cdots+\frac{c^2}{h^2}+\frac{e^2}{k^2}>1,$$

the equation

$$1 - \frac{a^2}{\theta + f^2} - \cdots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta + k^2} = 0$$

has a positive root differing from zero, which may be represented by the same letter θ; but if

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} < 1,$$

then the positive root of the original equation becomes $= 0$; viz. as l gradually diminishes to zero, the positive root also diminishes and becomes ultimately zero.

Hence, writing $l = 0$, we have

$$\int \frac{dS}{[(a - fx)^2 + \cdots + (c - hz)^2 + (e - kw)^2]^{\frac{1}{2}s}},$$

or, what is the same thing,

$$\frac{1}{f \dots h\, k} \int \frac{dx \dots dz}{w[(a - x)^2 + \cdots + (c - z)^2 + (e - kw)^2]^{\frac{1}{2}s}},$$

[p. 417] θ now denoting either the positive root of the equation

$$1 - \frac{a^2}{\theta + f^2} - \cdots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta + k^2} = 0,$$

or else 0, according as

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} > 1 \text{ or } < 1.$$

In the case $\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} < 1$, the inferior limit being then 0, this is, in fact, Jacobi's theorem (*Crelle*, t. xii. p. 69, 1834); but Jacobi does not consider the general case where l is not $= 0$, nor does he give explicitly the formula in the other case

$$l = 0,\ \frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} + \frac{e^2}{k^2} > 1.$$

150. Suppose $k = 0$, e being in the first instance not $= 0$: then the former alternative holds good; and observing, in regard to the form which contains w in the denominator, that we can now take account of the two values by simply multiplying by 2, we have

$$\int \frac{dS}{[(a - fx)^2 + \cdots + (c - hz)^2 + e^2]^{\frac{1}{2}s}} = \frac{2}{f \dots h} \int \frac{dx \dots dz}{[(a - x)^2 + \cdots + (c - z)^2 + e^2]^{\frac{1}{2}s}}$$

(where on the right-hand side w denotes $\sqrt{1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2}}$, and the limiting equation being $\frac{x^2}{f^2} + \cdots + \frac{z^2}{h^2} = 1$); each $= \frac{2(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s)} \int_\theta^\infty dt \left(1 - \frac{a^2}{t + f^2} - \cdots - \frac{c^2}{t + h^2} - \frac{e^2}{t}\right)^{-\frac{1}{2}} [t(t+f^2) \dots (t+h^2)]^{-\frac{1}{2}}$, where θ is here the positive root of the equation $1 - \frac{a^2}{\theta + f^2} - \cdots - \frac{c^2}{\theta + h^2} - \frac{e^2}{\theta} = 0$, which is the formula referred to at the beginning of the present Annex. We may in the formula write $e = 0$, thus obtaining the theorem under two different forms for the cases

$$\frac{a^2}{f^2} + \cdots + \frac{c^2}{h^2} > 1 \text{ and } < 1 \quad \text{respectively.}$$

Annex X. Methods of Lejeune-Dirichlet and Boole. Art. Nos. 151 to 162.

151. The notion, that the density ρ is a discontinuous function vanishing for points outside the attracting mass, has been made use of in a different manner by Lejeune-Dirichlet (1839) and Boole (1857): viz. supposing that ρ has a given value $= f(x, \dots, z)$ within a given closed surface S and is $= 0$ outside the surface, these geometers in the expression of a potential or prepotential integral replace ρ by a definite integral which possesses the discontinuity in question, viz. it is $= f(x, \dots, z)$ for points inside

[p. 418] the surface and $= 0$ for points outside the surface; and then in the potential or prepotential integral they extend the integration over the whole of infinite space, thus getting rid of the equation of the surface as a limiting equation for the multiple integral.

152. Lejeune-Dirichlet's paper "Sur une nouvelle méthode pour la détermination des intégrales multiples" is published in *Comptes Rendus*, t. viii. pp. 155–160 (1839), and *Liouville*, t. iv. pp. 164–168 (same year). The process is applied to the form

$$-\frac{1}{p-1}\frac{d}{da}\int \frac{dx\,dy\,dz}{\{(a-x)^2+(b-y)^2+(c-z)^2\}^{\frac{1}{2}(p-1)}},$$

taken over the ellipsoid $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2}=1$; but it would be equally applicable to the triple integral itself, or say to the s-tuple integral

$$\int \frac{dx\dots dz}{\{(a-x)^2+\dots+(c-z)^2\}^{\frac{1}{2}s+q}}$$

or, indeed, to

$$\int \frac{dx\dots dz}{\{(a-x)^2+\dots+(c-z)^2+e^2\}^{\frac{1}{2}s+q}}$$

taken over the ellipsoid $\dfrac{x^2}{f^2}+\dots+\dfrac{z^2}{h^2}=1$; but it may be as well to attend to the first form, as more resembling that considered by the author.

153. Since $\dfrac{2}{\pi}\int_0^\infty \dfrac{\sin\lambda\phi}{\phi}\cos\lambda\phi\,d\phi$ is $= 1$ or 0, according as λ is < 1 or > 1, it follows that the integral is equal to the real part of the following expression,

$$\frac{2}{\pi}\int_0^\infty \frac{d\phi}{\phi}\,e^{i\phi}\int \frac{e^{-i\phi(x^2+\dots+z^2)}\,dx\dots dz}{\{(a-x)^2+\dots+(c-z)^2\}^{\frac{1}{2}s+q}},$$

where the integrations in regard to $x, \dots, z$ are now to be extended from $-\infty$ to $+\infty$ for each variable. A further transformation is necessary: since

$$\int_0^\infty e^{-x\tau}\,\tau^{r-1}\,d\tau = \frac{\Gamma(r)}{x^r}\quad (x \text{ positive, and } r \text{ positive and } < 1),$$

writing herein $(a-x)^2+\dots+(c-z)^2$ for x, and $\frac{1}{2}s+q$ for r, we have

$$\frac{1}{\{(a-x)^2+\dots+(c-z)^2\}^{\frac{1}{2}s+q}} = \frac{1}{\Gamma(\frac{1}{2}s+q)}\int_0^\infty d\tau\,\tau^{\frac{1}{2}s+q-1}\,e^{-\tau\{(a-x)^2+\dots+(c-z)^2\}},$$

and the value is thus

$$\frac{2}{\pi\,\Gamma(\frac{1}{2}s+q)}\int_0^\infty \frac{d\phi}{\phi}\,e^{i\phi}\int_0^\infty d\tau\,\tau^{\frac{1}{2}s+q-1}\int e^{-i\phi(x^2+\dots+z^2)}\,e^{-\tau\{(a-x)^2+\dots+(c-z)^2\}}\,dx\dots dz,$$

[p. 419] where the integral in regard to the variables $(x, \dots, z)$ is

$$= e^{-i\phi\tau\frac{a^2+\dots+c^2}{\tau+i\phi}} \int dx\, e^{-(\tau+i\phi)\left(x-\frac{\tau a}{\tau+i\phi}\right)^2} \dots ,$$

and here the x-integral is

$$= \pi^{\frac{1}{2}} \sqrt{\frac{1}{\tau+i\phi}} = \pi^{\frac{1}{2}}\, \frac{e^{-\frac{1}{2}i\theta}}{\sqrt{\tau^2+\phi^2}},$$

and the like for the other integrals up to the z-integral. The resulting value is thus

$$= \pi^{\frac{s}{2}}\, (f\dots h)\, e^{-i\phi\tau\frac{a^2+\dots+c^2}{\tau+i\phi}}\, e^{-\frac{1}{2}is\theta}\, \frac{1}{(\tau^2+\phi^2)^{\frac{1}{4}s}},$$

viz. writing $\psi = \dfrac{\phi}{\tau}$, $d\phi = \dfrac{\phi}{\tau}\, dt = \dfrac{\psi}{\tau}\, dt$, we have

$$= \frac{2\pi^{\frac{1}{2}s-1}}{\Gamma(\frac{1}{2}s+q)}(f\dots h) \int_0^\infty dt\, \frac{t^{q-1}}{\sqrt{(t^2+\tau^2)\dots(h^2+t)}} \int_0^\infty d\phi\, \phi^{q+r-1}\, e^{i\phi\sigma} \sin\phi\, d\phi.$$

154. But we have to consider only the real part of this expression; viz. writing for shortness $\sigma = \dfrac{a^2+\dots+c^2}{t+f^2} + \dots + \dfrac{c^2}{t+h^2}$, we require the real part of

$$e^{-i\sigma\phi} \int_0^\infty e^{i\sigma\phi}\, \phi^{q-1} \sin\phi\, d\phi.$$

Writing here for $\sin\phi$ its exponential value $\dfrac{1}{2i}(e^{i\phi} - e^{-i\phi})$, and using the formula

$$\frac{1}{(1-\sigma)^q} = \frac{1}{\Gamma(q)} \int_0^\infty d\phi\, \phi^{q-1}\, e^{-(1-\sigma)\phi} \quad (\sigma \text{ positive}),$$

and the like one

$$\frac{1}{(\sigma-1)^q} = \frac{1}{\Gamma(q)} e^{i\pi q} \int_0^\infty d\phi\, \phi^{q-1}\, e^{-(\sigma-1)\phi} \quad (\sigma \text{ negative}),$$

(in which formulae q must be positive and less than 1), we see that the real part in question is 0, or is

$$\frac{\Gamma q \sin(q+1)\pi}{2(1-\sigma)^q} = \frac{\pi}{2\,\Gamma(1-q)} \cdot \frac{1}{(1-\sigma)^q},$$

according as $\sigma > 1$ or $\sigma < 1$.

155. If the point is interior, $\dfrac{a^2}{f^2} + \dots + \dfrac{c^2}{h^2} < 1$, and consequently also $\sigma < 1$, and the value, writing $(\Gamma\frac{1}{2})^s$ instead of $\pi^{\frac{1}{2}s}$, is

[p. 420] But if the point be exterior, $\dfrac{a^2}{f^2} + \dots + \dfrac{c^2}{h^2} > 1$, and hence, writing θ for the positive root of the equation, $\sigma = 1$; viz. θ is the positive root of the equation $\dfrac{a^2}{\theta+f^2} + \dots + \dfrac{c^2}{\theta+h^2} = 1$; then $t = 0$, σ is greater than 1, and continues so as t increases, until, for $t = \theta$, σ becomes $= 1$, and for larger values of t we have $\sigma < 1$; and the expression thus is

$$= \frac{(\Gamma\frac{1}{2})^s}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1-q)}(f\dots h) \int_\theta^\infty dt\, t^{q-1} \big[(t+f^2)\dots(t+h^2)\big]^{-\frac{1}{2}} \Big(1 - \frac{a^2}{t+f^2} - \dots - \frac{c^2}{t+h^2}\Big)^{-q};$$

viz. the two expressions, in the cases of an interior point and an exterior point respectively, give the value of the integral

$$\int \frac{dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}s+q}}.$$

This is, in fact, the formula of Annex IV. No. 110, writing therein $e = 0$ and $m = -q$.

156. Boole's researches are contained in two memoirs dated 1846, "On the Analysis of Discontinuous Functions," *Trans. Royal Irish Academy*, vol. xxi. (1848), pp. 124–139, and "On a certain Multiple Definite Integral," do. pp. 140–150 (the particular theorem about to be referred to is stated in the postscript of this memoir), and in the memoir "On the Comparison of Transcendents, with certain applications to the theory of Definite Integrals," *Phil. Trans.*, vol. cxlvii. (1857), pp. 745–803, the theorem being the third example, p. 794. The method is similar to, and was in fact suggested by, that of Lejeune-Dirichlet; the auxiliary theorem made use of in the memoir of 1857 for the representation of the discontinuity being

$$\phi(\sigma) = \frac{1}{\pi\,\Gamma(-q)} \int_{-\infty}^{\infty} \int_{0}^{\infty} du\, dv\, v^q \cos\{(u-\sigma)v + \tfrac{1}{2}q\pi\}\, \phi(u),$$

which is a deduction from Fourier's theorem.

Changing the notation (and in particular writing s and $\frac{1}{2}s+q$ for his n and i), the method is here applied to the determination of the s-tuple integral

$$V = \int dx \dots dz \frac{\phi\Big(\dfrac{x^2}{f^2} + \dots + \dfrac{z^2}{h^2}\Big)}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}},$$

where ϕ is an arbitrary function, taken over the ellipsoid $\dfrac{x^2}{f^2} + \dots + \dfrac{z^2}{h^2} = 1$.

157. The process is as follows: we have

$$\frac{\phi\Big(\dfrac{x^2}{f^2} + \dots + \dfrac{z^2}{h^2}\Big)}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}} = \frac{1}{\pi\,\Gamma(\frac{1}{2}s+q)} \int_0^\infty \int_0^\infty \int_0^\infty du\, dv\, d\tau\, v^{\frac{1}{2}s+q}\, \tau^{\frac{1}{2}s+q-1}\, e^{-\tau v} \cos\Big\{\Big(u - \frac{x^2}{f^2} - \dots -$$

[p. 421] viz. the right-hand side is here equal to the left-hand side or is $= 0$, according as $\dfrac{x^2}{f^2} + \dots + \dfrac{z^2}{h^2} < 1$ or > 1. V is consequently obtained by multiplying the right-hand side by $dx \dots dz$ and integrating from $-\infty$ to $+\infty$ for each variable.

Hence, changing the order of the integration,

$$V = \frac{1}{\pi\,\Gamma(\frac{1}{2}s+q)} \int_0^\infty \int_0^\infty \int_0^\infty du\, dv\, d\tau\, v^{\frac{1}{2}s+q}\, \tau^{\frac{1}{2}s+q-1}\, e^{-\tau v}\, \phi u \cdot \Pi,$$

where

$$\Pi = \int dx \dots dz \cos\Big(uv - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2} + \tau\{(a-x)^2 + \dots + (c-z)^2\}v + \tfrac{1}{2}(\tfrac{1}{2}s+q)\pi\Big).$$

Now if

$$\xi = \sqrt{\frac{v}{f^2}}\, x,\ \dots,\ \zeta = \sqrt{\frac{v}{h^2}}\, z.$$

158. Substituting, and integrating with respect to $\xi, \dots, \zeta$ between the limits $-\infty$, $+\infty$, we have

$$\Pi = \frac{\pi^{\frac{s}{2}} f \dots h}{\sqrt{(1+f^2\tau v) \dots (1+h^2\tau v)}} \cos\Big\{\Big(u - \frac{a^2\tau v}{1+f^2\tau v} - \dots - \frac{c^2\tau v}{1+h^2\tau v}\Big)v + e^2\tau v + \tfrac{1}{2}q\pi\Big\},$$

or, what is the same thing, writing $\dfrac{t}{v}$ in place of τ, this is

$$= \frac{\pi^{\frac{s}{2}}\, f \dots h}{\sqrt{(1+f^2t)\dots(1+h^2t)}} \cos\Big\{\Big(u - \frac{a^2t}{1+f^2t} - \dots - \frac{c^2t}{1+h^2t}\Big)v + e^2t + \tfrac12 q\pi\Big\},$$

that is, writing

$$\sigma = \frac{a^2t}{1+f^2t} + \dots + \frac{c^2t}{1+h^2t} + e^2t,$$

we have

$$V = \frac{\pi^{\frac{s}{2}-1}(f\dots h)}{\Gamma(\frac12 s+q)} \int_0^\infty\!\!\int_0^\infty du\,dt\,t^{q-1}\,v^q \cos\big[(u-\sigma)v + \tfrac12 q\pi\big]\,\phi u$$

or, writing $\pi^{\frac12 s-1} = (\Gamma\frac12)^s\pi^{-1}$, this is

$$= \frac{(\Gamma\frac12)^s}{\pi\,\Gamma(\frac12 s+q)}(f\dots h)\int_0^\infty dt\,t^{q-1}\big[(1+f^2t)\dots(1+h^2t)\big]^{-\frac12}\int_0^\infty\!\!\int_0^\infty du\,dv\,v^q\cos\big[(u-\sigma)v+\tfrac12 q\pi\big]\,\phi u.$$

159. Boole writes

$$\frac{1}{\pi}\int_0^\infty\!\!\int_0^\infty du\,dv\,v^q\cos\{(u-\sigma)v+\tfrac12 q\pi\}\,\phi u = \Big(-\frac{d}{d\sigma}\Big)^q\phi(\sigma);$$

viz. starting from Fourier's theorem,

$$\frac{1}{\pi}\iint du\,dv\cos(u-\sigma)v\cdot\phi u = \phi(\sigma),$$

[p. 422] where $\phi(\sigma)$ is regarded as vanishing except when σ is between the limits $0, 1$, and the limits of u are taken to be $0, 1$ accordingly, then, according to an admissible theory of general differentiation, we have the result in question. He has in the formula $\dfrac{1}{s}$ instead of my t; and he proceeds, "Here σ increases continually with s. As s varies from 0 to ∞, σ also varies from 0 to ∞. To any positive limits of σ will correspond positive limits of s; and these, as will hereafter appear"—this refers to his note "B"—"will in certain cases replace the limits 0 and ∞ in the expression for V."

160. It seems better to deal with the result in the following manner, as in part shown p. 803 of Boole's memoir. Writing the integral in the form

$$V = \frac{(\Gamma\frac12)^s(f\dots h)}{\pi\,\Gamma(\frac12 s+q)}\int_0^\infty\!\!\int_0^1 du\,dt\,t^{q-1}\big\{(t+f^2)\dots(t+h^2)\big\}^{-\frac12}\phi(u)\int_0^\infty dv\,v^q\cos\big[(u-\sigma)v+\tfrac12 q\pi\big],$$

effect the integration in regard to v; viz. according as u is greater or less than σ, then

$$\int_0^\infty dv\,v^q\cos\{(u-\sigma)v+\tfrac12 q\pi\} = \frac{\Gamma(q+1)\sin(q+1)\pi}{(u-\sigma)^{q+1}} \quad\text{or}\quad 0,$$

$$= \frac{\pi}{\Gamma(-q)\,(u-\sigma)^{q+1}} \quad\text{or}\quad 0;$$

and consequently, writing for σ its value,

$$V = \dots\cdot\Big(u - \frac{a^2t}{1+f^2t} - \dots\Big)^{-q-1}\phi(u), \quad\text{or } 0,\quad \text{as above.}$$

161. To further explain this, consider t as an x-coordinate and u as a y-coordinate; then, tracing the curve

$$y = \frac{a^2x}{f^2+x} + \cdots + \frac{c^2x}{h^2+x},$$

for positive values of x this is a mere hyperbolic branch, as shown in the figure, viz. $x=0$, $y=0$; and as x continually increases to ∞, y continually decreases to zero.

The limits are originally taken to be from $u=0$ to $u=1$ and $t=0$ to $t=\infty$, viz. over the infinite strip bounded by the lines $t0, 01, 11$; but within these limits the

[p. 423] function under the integral sign is to be replaced by zero whenever the values u, t are such that u is less than $\dfrac{a^2t}{f^2+t} + \cdots + \dfrac{c^2t}{h^2+t} + e^2t$, viz. when the values belong to a point in the shaded portion of the strip; the integral is therefore to be extended only over the unshaded portion of the strip; viz. the value is

$$V = \frac{(\Gamma\frac{1}{2})^s (f\ldots h)}{\Gamma(-q)\,\Gamma(\frac{1}{2}s+q)} \iint dt\,du\,t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}} \Big(u - \frac{a^2t}{f^2+t} - \cdots - \frac{c^2t}{h^2+t} - e^2t\Big)^{-q-1} \phi u,$$

the double integral being taken over the unshaded portion of the strip; or, what is the same thing, the integral in regard to u is to be taken from $u = \dfrac{a^2t}{f^2+t} + \cdots + \dfrac{c^2t}{h^2+t} + e^2t$ (say from $=\sigma$) to $u=1$, and then the integral in regard to t is to be taken from $t=\theta$ to $t=\infty$, where, as before, θ is the positive root of the equation $\sigma = 1$, that is, of

$$\frac{a^2t}{f^2+t} + \cdots + \frac{c^2t}{h^2+t} + e^2t = 1.$$

162. Write $u = \sigma + (1-\sigma)x$, and therefore $u - \sigma = (1-\sigma)x$, $1-u = (1-\sigma)(1-x)$, and $du = (1-\sigma)\,dx$; then the limits $(1,0)$ of x correspond to the limits $(1,\sigma)$ of u, and the formula becomes

$$V = \frac{(\Gamma\frac{1}{2})^s (f\ldots h)}{\Gamma(-q)\,\Gamma(\frac{1}{2}s+q)} \int_\theta^\infty \int_0^1 dt\,dx\,t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}} (1-\sigma)^{-q}\,x^{-q-1}\,\phi\big[\sigma + (1-\sigma)x\big],$$

where σ is retained in place of its value $\dfrac{a^2t}{f^2+t} + \cdots + \dfrac{c^2t}{h^2+t} + e^2t$. This is, in fact, a form (deduced from Boole's result in the memoir of 1846) given by me, *Cambridge and Dublin Mathematical Journal*, vol. ii. (1847), p. 219, [44].

If in particular $\phi u = (1-u)^{q+m}$, then $\phi\sigma + (1-\sigma)x = (1-\sigma)^{q+m}\,(1-x)^{q+m}$, and thence

$$\int_0^1 x^{-q-1}\,\phi[\sigma + (1-\sigma)x]\,dx = (1-\sigma)^{q+m} \int_0^1 x^{-q-1}\,(1-x)^{q+m}\,dx = \frac{\Gamma(-q)\,\Gamma(1+q+m)}{\Gamma(1+m)},$$

and then, restoring for σ its value, we have

$$V = \frac{(\Gamma\frac{1}{2})^s\,\Gamma(1+q+m)}{\Gamma(\frac{1}{2}s+q)\,\Gamma(1+m)} (f\ldots h) \int_\theta^\infty dt\,t^{q-1}\big\{(t+f^2)\ldots(t+h^2)\big\}^{-\frac{1}{2}} \Big(1 - \frac{a^2t}{f^2+t} - \cdots - \frac{c^2t}{h^2+t} - e^2t\Big)^{q+m}$$

as the value of the integral

$$\int \frac{(1 - \frac{x^2}{f^2} - \cdots - \frac{z^2}{h^2})^m\,dx\ldots dz}{\{(a-x)^2 + \cdots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}}$$

taken over the ellipsoid $\dfrac{x^2}{f^2} + \cdots + \dfrac{z^2}{h^2} = 1$. This is, in fact, the theorem of Annex IV. No. 110 in its general form; but the proof assumes that q is positive.

[End of Paper 607: A Memoir on Prepotentials.]

608. [Extract from a] Report on Mathematical Tables

[From the Report of the British Association for the Advancement of Science, (1873), pp. 3, 4.]

[p. 424] It was necessary as a preliminary to form a classification of mathematical (numerical) tables; and the following classification was drawn up by Prof. Cayley and adopted by the Committee.

A. Auxiliary for non-logarithmic computations.

1. Multiplication.
2. Quarter-squares.
3. Squares, cubes, and higher powers, and reciprocals.

B. Logarithmic and circular.

4. Logarithms (Briggian) and antilogarithms (do.); addition and subtraction logarithms, &c.
5. Circular functions (sines, cosines, &c.), natural, and lengths of circular arcs.
6. Circular functions (sines, cosines, &c.), logarithmic.

C. Exponential.

7. Hyperbolic logarithms.
8. Do. antilogarithms (e^x) and $h.1\tan(45° + \frac{1}{2}\phi)$, and hyperbolic sines, cosines, &c., natural and logarithmic.

D. Algebraic constants.

9. Accurate integer or fractional values. Bernoulli's Numbers, $\Delta^n 0^m$, &c. Binomial coefficients.
10. Decimal values auxiliary to the calculation of series.

[p. 425]

E. 11. Transcendental constants, e, π, γ, &c., and their powers and functions.

F. Arithmological.

12. Divisors and prime numbers. Prime roots. The Canon arithmeticus, &c.
13. The Pellian equation.
14. Partitions.
15. Quadratic forms $\pm a^2 \pm b^2$, &c., and partition of numbers into squares, cubes, and biquadrates.
16. Binary, ternary, &c. quadratic, and higher forms.
17. Complex theories.

G. Transcendental functions.

18. Elliptic.
19. Gamma.
20. Sine-integral, cosine-integral, and exponential-integral.

21. Bessel's and allied functions.
22. Planetary coefficients for given $\dfrac{a}{a'}$.
23. Logarithmic transcendental.
24. Miscellaneous.

Several of these classes need some little explanation. Thus D 9 and 10 are intended to include the same class of constants, the only difference being that in 9 accurate values are given, while in 10 they are only approximate; thus, for example, the accurate Bernoulli's numbers as vulgar fractions, and the decimal values of the same to (say) ten places are placed in different classes, as the former are of theoretical interest, while the latter are only of use in calculation. It is not necessary to enter into further detail with respect to the classification, as in point of fact it is only very partially followed in the Report.

609. On the Analytical Forms called Factions

[From the Report of the British Association for the Advancement of Science, (1875), p. 10.]

[p. 426] A FACTION is a product of differences such that each letter occurs the same number of times; thus we have a quadrifaction where each letter occurs twice, a cubifaction where each letter occurs three times, and so on. A broken faction is one which is a product of factions having no common letter; thus

$$(a-b)^2(c-d)(d-e)(e-c)$$

is a broken quadrifaction, the product of the quadrifactions

$$(a-b)^2 \quad \text{and} \quad (c-d)(d-e)(e-c).$$

We have, in regard to quadrifactions, the theorem that every quadrifaction is a sum of broken quadrifactions such that each component quadrifaction contains two or else three letters. Thus we have the identity

$$2(a-b)(b-c)(c-d)(d-a) = (b-c)^2\cdot(a-d)^2 - (c-a)^2\cdot(b-d)^2 + (a-b)^2\cdot(c-d)^2,$$

which verifies the theorem in the case of a quadrifaction of four letters; but the verification even in the next following case of a quadrifaction of five letters is a matter of some difficulty.

The theory is connected with that of the invariants of a system of binary quantics.

610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations

[From the Report of the British Association for the Advancement of Science, (1875), pp. 257–305.]

[p. 427] I have in two papers "On the Analytical forms called Trees," *Phil. Mag.* vol. xiii. (1857), pp. 172–176, [203], and ditto, vol. xx. (1859), pp. 374–378, [247], considered this theory; and in a paper "On the Mathematical Theory of Isomers," ditto, vol. xlvii. (1874), p. 444, [586], pointed out its connexion with modern chemical theory. In particular, as regards the paraffins C_nH_{2n+2}, we have n atoms of carbon connected by $n-1$ bands, under the restriction that from each carbon-atom there proceed at most 4 bands (or, in the language of the papers first referred to, we have n knots connected by $n-1$ branches), in the form of a tree; for instance, $n=5$, such forms (and the only such forms) are

[Three tree diagrams of carbon skeletons for $n = 5$ paraffin isomers: the linear (normal pentane), the branched (isopentane), and the centrally branched (neopentane); each carbon-atom labelled with the number of attached hydrogens.]

And if, under the foregoing restriction of only 4 bands from a carbon-atom, we connect with each carbon-atom the greatest possible number of hydrogen-atoms, as shown in the diagrams by the affixed numerals, we see that the number of hydrogen-atoms is 12 ($= 2 \cdot 5 + 2$); and we have thus the representations of three different paraffins, C_5H_{12}. It should be observed that the tree-symbol of the paraffin is

[p. 428] completely determined by means of the tree formed with the carbon-atoms, or say of the carbon-tree, and that the question of the determination of the theoretic number of the paraffins C_nH_{2n+2} is consequently that of the determination of the number of the carbon-trees of n knots, viz. the number of trees with n knots, subject to the condition that the number of branches from each knot is at most $= 4$.

In the paper of 1857, which contains no application to chemical theory, the number of branches from a knot was unlimited; and, moreover, the trees were considered as issuing each from one knot taken as a root, so that, $n = 5$, the trees regarded as distinct (instead of being as above only 3) were in all 9, viz. these were

[Nine root-tree diagrams of 5 *knots each, each issuing from a designated bottom knot.]*

which, regarded as issuing from the bottom knots, are in fact distinct; while, taking them as issuing each from a properly selected knot, they resolve themselves into the above-mentioned 3 forms. The problem considered was in fact that of the "general root-trees with n knots"—general, inasmuch as the number of branches from a knot was without limit; root-trees, inasmuch as the enumeration was made on the principle last referred to. It was found that for

knots	1,	2,	3,	4,	5,	6,	7,	8,
No. of trees was	1,	1,	2,	4,	9,	20,	48,	115,
	$= 1$,	A_2,	A_3,	A_4,	A_5,	A_6,	A_7,	A_8,

the law being given by the equation

$$(1-x)^{-1}(1-x^2)^{-A_2}(1-x^3)^{-A_3}(1-x^4)^{-A_4}\cdots = 1 + x + A_2x^2 + A_3x^3 + A_4x^4 + \cdots;$$

but the next following numbers A_9, A_{10}, A_{11}, the correct values of which are $286, 719, 1842$, were given erroneously as $306, 775, 2009$. I have since calculated two more terms, $A_{12} = 4766$, $A_{13} = 12486$.

The other questions considered in the paper of 1857 and in that of 1859 have less immediate connexion with the present paper, but for completeness I reproduce the results in a Note[2].

[2] In the paper of 1857 I also considered the problem of finding B_r, the number with r free branches, with bifurcations at least: this was given by a like formula

$$(1-x)^{-1}(1-x^2)^{-B_2}(1-x^3)^{-B_3}(1-x^4)^{-B_4}\cdots = 1 + x + 2B_2x^2 + 2B_3x^3 + 2B_4x^4 + \cdots,$$

leading to $B_r = 1, 2, 6, 12, 38, 90$ for $r = 2, 3, 4, 5, 6, 7$.

In the paper of 1859, the question is to find the number of trees with a given number m of terminal knots: we have here

$$\phi_m = \frac{1}{1\cdot 2\cdot 3\cdots(m-1)} \text{ coefficient of } x^{m-1} \text{ in } \Sigma,$$

giving the values $\phi_m = 1, 1, 3, 13, 75, 541, 4683, 47293, \ldots$ for $m = 1, 2, 3, 4, 5, 6, 7, 8, \ldots$

But if from each non-terminal knot there ascend two and only two branches, then in this case $\phi_m =$ coefficient of x^{m-1} in $\dfrac{1-\sqrt{1-4x}}{2}$, viz. we have the very simple form

$$\phi_m = \frac{1\cdot 3\cdot 5\cdots(2m-3)}{1\cdot 2\cdot 3\cdots m}\cdot 2^{m-1},$$

giving $\phi_m = 1, 1, 2, 5, 14, 42, \ldots$ for $m = 1, 2, 3, 4, 5, \ldots$

[p. 429] To count the trees on the principle first referred to, we require the notions of "centre" and "bicentre," due, I believe, to Sylvester; and to establish these we require the notions of "main branch" and "altitude": viz. in a tree, selecting any knot at pleasure as a root, the branches which issue from the root, each with all the branches that belong to it, are the main branches, and the distance of the furthest knot, measured by the number of intermediate branches, is the altitude of the main branch. Thus in the left-hand figure, taking A as the root, there are 3 main branches of the altitudes $3, 3, 1$ respectively: in the right-hand figure, taking A as the root, there are 4 main branches of the altitudes $2, 2, 1, 3$ respectively; and we have

[Two diagrams: left, a tree with vertex A illustrating a centre and three main branches; right, a tree with adjacent vertices A, B illustrating a bicentre.]

then the theorem that in every tree there is either one and only one centre, or else one and only one bicentre; viz. we have (as in the left-hand figure) a centre A which is such that there issue from it two or more main branches of altitudes equal to each other and superior to those of the other main branches (if any); or else (as in the right-hand figure) a bicentre AB, viz. two contiguous knots, such that issuing from A (but not counting AB), and issuing from B (but not counting BA), we have two or more main branches, one at least from A and one at least from B, of altitudes equal to each other and superior to those of the other main branches in question (if any). The theorem, once understood, is proved without difficulty: we consider two terminal knots, the distance of which, measured by the number of intermediate branches, is greater than or equal to that of any other two terminal knots; if, as in the left-hand figure, the distance is even, then the central knot A is the centre of the tree; if, as in the right-hand figure, the distance is odd, then the two central knots AB form the bicentre of the tree.

In the former case, observe that if G, H are the two terminal knots, the distance of which is $= 2\lambda$, then the distance of each from A is $= \lambda$, and there cannot be

[p. 430] any other terminal knot I, the distance of which from A is greater than λ (for, if there were, then the distance of I from G or else from H would be greater than 2λ); there cannot be any two terminal knots I, J, the distance of which is greater than 2λ; and if there are any two knots I, J, the distance of which is $= 2\lambda$, then these belong to different main branches, the distance of each of them from A being $= \lambda$; whence, starting with I, J (instead of G, H), we obtain the same point A as centre. Similarly, in the latter case, there is a single bicentre AB.

Hence, since in any tree there is a unique centre or bicentre, the question of finding the number of distinct trees with n knots is in fact that of finding the number of centre- and bicentre-trees with n knots; or say it is the problem of the "general centre- and bicentre-trees with n knots:" general, inasmuch as the number of branches from a knot is as yet taken to be without limit; or since (as will appear) the number of the bicentre-trees can be obtained without difficulty when the problem of the root-trees is solved, the problem is that of the "general centre-trees with n knots." It will appear that the solution depends upon and is very readily derived from that of the foregoing problem of general root-trees, so that this last has to be considered, not only for its own sake, but with a view to that of the centre-trees. And in each of the two problems we doubly divide the whole system of trees according to the number of the main branches, issuing from the root or centre as the case may be, and according to the altitude of the longest main branch or branches, or say the altitude of the tree; so that the problem really is, for a given number of knots, a given number of main branches, and a given altitude, to find the number of root-trees, or (as the case may be) centre-trees.

We next introduce the restriction that the number of branches from any knot is equal to a given number at most; viz. according as this number is 2, 3 or 4, we have, say oxygen-trees, boron-trees[3], and carbon-trees respectively; and these are, as before, root-trees or centre- or

[3] I should have said nitrogen-trees; but it appears to me that nitrogen is of necessity 5-valent, as shown by the compound, Ammonium-Chloride, $= NH_4Cl$. Of course, the word boron is used simply to stand for a 3-valent element.

bicentre-trees, as the case may be. The case where the number is 2 presents no difficulty: in fact, if the number of knots be $= n$, then the number of root-trees is either $\frac{1}{2}(n+1)$ or $\frac{1}{2}n$; viz. $n = 3$ and $n = 4$, the root-trees are

[Two simple linear root-tree diagrams of 3 and 4 knots respectively.]

and the number of centre- or bicentre-trees is always 1: viz. n odd, there is one centre-tree; and n even, one bicentre-tree; it is only considered as a particular case of the general theorem. The case where the number is 3 is analytically interesting although there may not exist, for any 3-valent element, a series of hydrogen compounds

[End of pages 406–430 (PDF pp. 426–450). Paper 610 continues in subsequent pages.]

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 431–455 (Paper 610, continued)

Transcribed from the Original Publications

Contents

610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations *(continued)*

[Continued from the preceding pages of the same paper. The discussion of carbon-trees corresponding to the paraffins is now developed analytically; Tables I–VIII give the enumeration of root-, centre- and bicentre-trees in the general case and for the oxygen, boron and carbon (chemical) cases.]

[p. 431] B_nH_{n+3} corresponding to the paraffins. The case where the number is = 4 or, say, the carbon-trees, is that which presents the chief chemical interest, as giving the paraffins C_nH_{2n+2}; and I call to mind here that the theory of the carbon root-trees is established as an analytical result for its own sake and as the foundation for the other case, but that it is the number of the carbon centre- and bicentre-trees which is the number of the paraffins.

The theory extends to the case where the number of branches from a knot is at most = 5, or = any larger number; but I have not developed the formula.

I pass now to the analytical theory: considering first the case of general root-trees, we endeavour to find for a given altitude N the number of trees of a given number of knots n and main branches a, or say the generating function

$$\Sigma\,\Omega\,t^a x^n,$$

where the coefficient Ω gives the number of the trees in question. And we assume that the problem is solved for the cases of the several inferior altitudes $0, 1, 2, 3, \ldots, N-1$.

This being so, observe that a tree of altitude N can be built up as shown in the figure, which I call the edification diagram, by combining one or more trees of altitude $N-1$ with a single tree of altitude not exceeding $N-1$; viz. in the figure, $N=3$, we have the two trees a, b, each of altitude 2, combined, as shown by the dotted lines, with the tree c of altitude 1: the whole number of knots in the resulting tree is the sum of the number of knots on the three trees a, b, c: the number of main branches is equal to the number of the trees a, b, plus the number of main branches of the tree c. It is to be observed that the tree c may reduce itself to the tree ($\bullet$) of one knot and of altitude zero; but each of the trees a, b, as being of the altitude $N-1$, must contain at least N knots.

[Edification diagram: two trees a, b of altitude 2 combined by dotted lines with a third tree c of altitude 1, all attached at the root.]

Taking $N=2$ or any larger number, it is hence easy to see that the required generating function $\Sigma\,\Omega\,t^a x^n$ is

$$= (1-tx^N)^{-l_0}\,(1-tx^{N+1})^{-l_1}\,(1-tx^{N+2})^{-l_2}\,\cdots\;[t^{1\cdots\infty}] \qquad \text{(first factor)},$$

$$x + (t)\,x^2 + (t,\,t^2)\,x^3 + (t,\,t^2,\,t^3)\,x^4 + \cdots \qquad \text{(second factor)}.$$

As regards the first factor, the exponents taken with reversed sign, that is, as positive, are l_0 = no. of trees, altitude $N-1$, of N knots; l_1 = ditto, same altitude, of $(N+1)$ knots; l_2 = ditto, same altitude, of $N+2$ knots, and so on; and where the

[p. 432] symbol $[t^{1\cdots\infty}]$ denotes that, in the function or product of factors which precedes it, the terms to be taken account of are those in t, t^2, $t^3, \ldots$; viz. it denotes that the term in t^0, or constant term (= 1 in fact), is to be rejected.

In the second factor, the expressions x, $(t)x^2$, $(t,\,t^2)x^3, \ldots$ represent, for given exponents of t, x, denoting the number of main branches and the number of knots respectively, the number of trees of altitude not exceeding $N-1$: thus x, $= 1\cdot x$ represents the number of such trees, 1 knot, 0 main branch, $= 1$; and so, if the value of $(t,\, t^2,\, t^3,\, t^4)x^5$ be $(\alpha t + \beta t^2 + \gamma t^3 + \delta t^4)x^5$, then for trees of an altitude not exceeding $N-1$, and of 5 knots, α represents the number of trees of 1 main branch, β that of trees of 2 main branches, γ that of trees of 3 main branches, δ that of trees of 4 main branches. It is clear that the number of trees satisfying the given conditions and of an altitude not exceeding $N-1$ is at once obtained by addition of the numbers of the trees satisfying the given conditions, and of the altitudes $0, 1, 2, \ldots, N-1$; all which numbers are taken to be known.

It is to be remarked that the first factor,

$$(1-tx^N)^{-l_0}\,(1-tx^{N+1})^{-l_1}\,(1-tx^{N+2})^{-l_2}\,\cdots\;[t^{1\cdots\infty}],$$

shows by its development the number of combinations of trees $a, b, \ldots$ of the altitude $N-1$; one such tree at least must be taken, and the symbol $[t^{1\cdots\infty}]$ gives effect to this condition: the second factor $x + (t)\,x^2 + (t,\, t^2)\,x^3 + \cdots$ shows the number of the trees c of altitude not exceeding $N-1$. And this being so, there is no difficulty in seeing how the product of the two factors is the generating function for the trees of altitude N.

In the case $N = 0$, the generating function, or GF, is $= x$; viz. altitude 0, there is only the tree (•), 1 knot, 0 main branch.

When $N = 1$, the GF is $= (1-tx)^{-1}\,[t^{1\cdots\infty}]\,x \;=\; tx^2 + t^2x^3 + t^3x^4 + \cdots$, viz. altitude 1, there is 1 tree tx^2, 2 knots, 1 main branch; 1 tree t^2x^3, 3 knots, 2 main branches; and so on.

Hence $N = 2$, we obtain

$$\mathrm{GF} = (1-tx^2)^{-1}(1-tx^3)^{-1}(1-tx^4)^{-1}\cdots\;[t^{1\cdots\infty}]\cdot(x + tx^2 + t^2x^3 + t^3x^4 + \cdots);$$

viz. as regards the second factor, altitude not exceeding 1, that is, $= 0$ or 1, there is altitude 0, 1 tree x, and altitude 1, 1 tree tx^2, 1 tree t^2x^3, and so on. And we hence derive the GF's for the higher values $N = 3$, 4, &c.: the details of the process will be afterwards more fully explained.

So far, we have considered root-trees; but referring to the last diagram, it is at once seen that the assumed root will be a centre, provided only that (instead of, it may be, only a single tree a of the altitude $N-1$), we take always two or more trees of the altitude $N-1$ to form the new tree of the altitude N. And we give effect

[p. 433] to this condition by simply writing in place of $[t^{1\cdots\infty}]$ the new symbol $[t^{2\cdots\infty}]$, which denotes that only the terms $t^2, t^3, t^4, \ldots$ are to be taken account of; viz. that the terms in t^0 and t^1 are to be rejected. The component trees of the altitude $N-1$ are, it is to be observed, as before, root-trees; hence the second factor of the generating function is unaltered: the theorem is that for the centre-trees of altitude N we have the same generating function as for the root-trees, writing only $[t^{2\cdots\infty}]$ in place of $[t^{1\cdots\infty}]$.

Or, what is the same thing, supposing that the first factor, unaffected by either symbol, is

$$= 1 + x^N(\alpha t + \beta t^2 + \cdots) + x^{N+1}(\alpha' t + \beta' t^2 + \cdots) + \cdots,$$

then, affecting it with $[t^{1\cdots\infty}]$, the value for the root-trees is

$$= x^N(\alpha t + \beta t^2 + \cdots) + x^{N+1}(\alpha' t + \beta' t^2 + \cdots) + \cdots,$$

and, affecting it with $[t^{2\cdots\infty}]$, the value for the centre-trees is

$$= x^N(\beta t^2 + \cdots) + x^{N+1}(\beta' t^2 + \cdots) + \cdots$$

It thus appears how the fundamental problem is that of the root-trees, its solution giving at once that of the centre-trees; whereas we cannot conversely solve the problem of the root-trees by means of that of the centre-trees.

As regards the bicentre-trees, it is to be remarked that, starting from a centre-tree of altitude $N+1$ with two main branches, then by simply striking out the centre, so as to convert into a single branch the two branches which issue from it, we obtain a bicentre-tree of altitude N. Observe that the altitude of a bicentre-tree is measured by that of the longest main branch from A or B, not reckoning AB or BA as a main branch. Hence the number of bicentre-trees, altitude N, is = number of centre-trees of two main branches, altitude $N+1$.

This is, in fact, the convenient formula, provided only the number of centre-trees of two main branches has been calculated up to the altitude $N+1$. But we can find independently the number of bicentre-trees of a given altitude N: the bicentre-tree is, in fact, formed by taking the two connected points A, B each as the root of a root-tree altitude N (the number of knots of the bicentre-tree being thus, it is clear, equal to the sum of the numbers of knots of the two root-trees respectively); and it is thus an easy problem of combinations to find the number of bicentre-trees of a given altitude N. Write

$$x^{N+1}\,(1+\beta x+\gamma x^2+\delta x^3+\cdots)$$

as the generating function of the root-trees of altitude N; viz. for such trees, 1 = no. of trees with $N+1$ knots, β = no. with $N+2$ knots, and so on; then the generating function of the bicentre-trees of the same altitude N is

$$=x^{2N+2}\,(1+\beta_1 x+\gamma_1 x^2+\delta_1 x^3+\cdots),$$

[p. 434] where

$$\begin{aligned}\beta_1&=\beta,\\ \gamma_1&=\gamma+\tfrac{1}{2}\beta(\beta+1),\\ \delta_1&=\delta+\beta\gamma,\\ \varepsilon_1&=\varepsilon+\beta\delta+\tfrac{1}{2}\gamma(\gamma+1),\\ \zeta_1&=\zeta+\beta\varepsilon+\gamma\delta,\end{aligned}$$

and so on; or, what is the same thing, calling the first generating function ϕx, then the second generating function is $=\frac{1}{2}\left[(\phi x)^2+\phi(x^2)\right]$.

It will be noticed that the bicentre-trees are not, as were the centre-trees, divided according to the number of their main branches; they might be thus divided according to the sum of the number of the main branches issuing from the two points of the bicentre respectively; a more complete division would be according to the number of main branches issuing from the two points respectively: thus we might consider the bicentre-trees $(2,3)$, with 2 main branches from one point, and 3 main branches from the other point of the bicentre; but the whole theory of the bicentre-trees is comparatively easy, and I do not go into it further.

We have yet to consider the case of the limited trees where the number of branches from a knot is equal to a given number at most: to fix the ideas, say the carbon-trees, where this number is = 4. The distinction as to root-trees and centre- and bicentre-trees is as before; and the like theory applies to the two cases respectively.

Considering first the case of the root-trees, and referring to the former figure for obtaining the trees of altitude N from those of inferior altitudes, then the trees a, $b,\ldots$ of altitude $N-1$ must be each of them a carbon-tree of not more than $(4-1=)3$ main branches: this restriction is necessary, inasmuch as, if for any such tree the number of main branches was = 4, then there would be from the root of such tree 4 branches *plus* the new branch shown by the dotted line, in all 5 branches; and similarly, inasmuch as there is at least one component tree a contributing one main branch, the number of main branches of the tree c must be $(4-1=)3$ at most: the mode of introducing these conditions will appear in the explanation of the actual formation of the generating functions (see explanation preceding Tables III., IV., &c.).

The number of main branches is = 4 at most, and the generating functions have only to be taken up to the terms in t^4; the first factor is consequently in each case affected with a symbol $[t^{1\cdots4}]$, denoting that the only terms to be taken account of are those in t, t^2, t^3, t^4; hence as there is a factor t at least, and the whole is required only up to t^4, the second factor is in each case required only up to t^3.

As regards the centre-trees, the generating functions have here the same expressions as for the root-trees, except that, instead of the symbol $[t^{1\cdots4}]$, we have the symbol $[t^{2\cdots4}]$, denoting that in the first factor the only terms to be taken account of are those in t^2, t^3, t^4; hence as there is a factor t^2 at least, and the whole is required only up to t^4, the second factor is in each case required up to t^2; and we then complete the theory by obtaining the bicentre-trees. The like remarks apply of course to

[p. 435] the boron-trees, number of branches = 3 at most, and to the oxygen-trees, number = 2 at most; but, as already remarked, this last case is so simple, that the general method is applied to it only for the sake of seeing what the general method becomes in such an extreme case.

We thus form the Tables, which I proceed to explain.

Table I. of general root-trees is in fact a Table of triple entry, viz. it gives for any given number of knots from 1 to 13 the number of root-trees corresponding to any given number of main branches and to any given altitude. In each compartment, that is, for any given number of knots, the totals of the columns give the number of the trees for each given altitude, and the totals of the lines give the number of the trees for each given number of main branches: the corner grand totals of these totals respectively show for each given number of knots the whole number of root-trees:

knots	1,	2,	3,	4,	5,	6,	7,	8,	9,	10,	11,	12,	13,
numbers are	1,	1,	2,	4,	9,	20,	48,	115,	286,	719,	1842,	4766,	12486,

as already mentioned: these numbers were calculated by an independent method.

Table II. of general centre- and bicentre-trees consists of a centre part and a bicentre part: the centre part is arranged precisely in the same manner as the root-table. As to the bicentre part, where it will be observed there is no division for number of main branches, the calculation of the several columns is effected by the before-mentioned formula,

$$\phi_1 x = \tfrac{1}{2}\left[(\phi x)^2 + \phi(x^2)\right];$$

thus, column 2, we have by Table I. (totals of column 2)

$$\phi x = x^3 + 2x^4 + 4x^5 + 6x^6 + 10x^7 + 14x^8 + 21x^9 + 29x^{10} + \cdots,$$

and thence

$$\phi_1 x = x^6 + 2x^7 + 7x^8 + 14x^9 + 32x^{10} + 58x^{11} + 110x^{12} + 187x^{13} + \cdots$$

As already mentioned, each column of Table I. is calculated by means of a generating function given as a product of two factors, each of which is obtained from the columns which precede the column in question; and Table II., the centre part of it, is calculated by means of the same generating functions slightly modified: these generating functions serving for the calculation of the two Tables are given in the table entitled "Subsidiary Table for the calculation of the GF's of Tables I. and II.," which immediately follows these two Tables, and will be further explained.

[p. 436]

Table I. — *General Root-trees.*

The Table is arranged by number of knots (from 1 to 13); within each knot-block, rows are indexed by number of main branches and columns by altitude (column 0 being the trivial tree of one knot). The entry in a cell is the number of root-trees with the prescribed number of knots, main branches, and altitude. "Total" rows give the column-sums for each knot-block. The grand total in the last column of each block gives the total number of root-trees with the prescribed number of knots.

$n = 1$: only (•), altitude 0. Total = 1.

$n = 2$: 1 main branch, altitude 1: 1. Total = 1.

$n = 3$: altitude 1, 1 MB: 1; altitude 2, 1 MB: 1, 2 MB: 1. Total altitudes $(1, 2) = (1, 2)$.

$n = 4$: altitudes $(1, 2, 3)$.

MB \ alt	1	2	3	Tot.
1	1	1		2
2		1		1
3			1	1
Tot.	1	2	1	4

$n = 5$:

MB \ alt	1	2	3	4	Tot.
1	1	2	1		4
2		2	1		3
3			1		1
4				1	1
Tot.	1	4	3	1	9

$n = 6$:

MB \ alt	1	2	3	4	5	Tot.
1	1	4	3	1		9
2		2	3	1		6
3		2	1			3
4			1			1
5					1	1
Tot.	1	8	8	2	1	20

$n = 7$:

MB \ alt	1	2	3	4	5	6	Tot.
1	1	6	8	4	1		20
2		3	8	4	1		16
3		3	3	1			7
4		2	1				3
5			1				1
6						1	1
Tot.	1	14	21	9	2	1	48

$n = 8$:

MB \ alt	1	2	3	4	5	6	7	Tot.
1	1	10	18	13	5	1		48
2		5	15	13	5	1		39
3		4	9	4	1			18
4		3	3	1				7
5		2	1					3
6			1					1
7							1	1
Tot.	1	24	47	31	11	2	1	115

$n = 9$:

MB \ alt	1	2	3	4	5	6	7	8	Tot.
1	1	14	38	36	19	6	1		115
2		7	30	36	19	6	1		99
3		5	19	14	5	1			44
4		5	9	4	1				19
5		3	3	1					7
6		2	1						3
7			1						1
8								1	1
Tot.	1	36	101	91	44	13	2	1	289

[Note: scan totals partially illegible; figures above follow Cayley's totals.]

[p. 437]

TABLE I. *(continued).*

$n = 10$:

MB \ alt	1	2	3	4	5	6	7	8	9	Tot.
1	1	21	76	93	61	26	7	1		286
2		7	51	89	61	26	7	1		242
3		7	42	41	20	6	1			117
4		6	20	14	5	1				46
5		5	9	4	1					19
6		3	3	1						7
7		2	1							3
8			1							1
9									1	1
Tot.	1	51	203	242	148	59	15	2	1	722

$n = 11$:

MB \ alt	1	2	3	4	5	6	7	8	9	10	Tot.
1	1	29	147	225	180	94	34	8	1		719
2		9	90	210	180	94	34	8	1		626
3		8	79	110	67	27	7	1			299
4		9	46	42	20	6	1				124
5		7	20	14	5	1					47
6		5	9	4	1						19
7		3	3	1							7
8		2	1								3
9			1								1
10										1	1
Tot.	1	72	396	606	453	222	76	17	2	1	1846

[Totals follow Cayley; certain column totals are partly illegible in the scan.]
$n = 12$:

MB \ alt	1	2	3	4	5	6	7	8	9	10	11	Tot.
1	1	41	277	528	498	308	136	43	9	1		1842
2		11	145	467	498	308	136	43	9	1		1618
3		10	152	273	208	101	35	8	1			788
4		11	91	113	68	27	7	1				318
5		10	47	42	20	6	1					126
6		7	20	14	5	1						47
7		5	9	4	1							19
8		3	3	1								7
9		2	1									3
10			1									1
11											1	1
Tot.	1	100	746	1442	1298	751	315	95	19	2	1	4770

$n = 13$:

MB \ alt	1	2	3	4	5	6	7	8	9	10	11	12	Tot.
1	1	55	509	1198	1323	941	487	188	53	10	1		4766
2		14	238	1012	1524	941	487	188	53	10	1		4468
3		12	272	559	376	244	144	44	9	1			1661
4		15	184	299	213	102	36	8	1				858
5		13	96	116	68	27	7	1					328
6		11	47	42	20	6	1						127
7		7	20	14	5	1							47
8		5	9	4	1								19
9		3	3	1									7
10		2	1										3
11			1										1
12												1	1
Tot.	1	137	1380	3265	3530	2262	1162	429	116	21	2	1	12486

[Several intermediate totals in Cayley's table are partly illegible in the scan; the printed grand total 12486 *is preserved.]*

[p. 438]

TABLE II. — *General Centre- and Bicentre-Trees.*

The Centre part is indexed by knots, main branches and altitude as in Table I.; the "Centre Total" column gives the centre-tree totals; the "Grand Total" is the sum of centre- and bicentre-trees; the "Bicentre" column gives bicentre-tree totals (not divided by number of main branches); and the right-hand block gives bicentre-trees of each altitude. We collect the totals.

n	Centre Tot.	Bicentre Tot.	Grand Tot.	alt 1	alt 2	alt 3	alt 4	alt 5
1	1	0	1					
2	0	1	1	1				
3	1	0	1					
4	1	1	2		1			
5	2	1	3		1			
6	3	3	6		2	1		
7	7	4	11		2	2		
8	12	11	23		3	7	1	
9	27	20	47		3	14	3	
10	55	51	106		4	32	14	1
11	127	108	235		4	58	42	4
12	284	267	551		5	110	128	23 (1)
13	682	619	1301		5	187	334	88 (5)

[Note: the rightmost "alt 5" column for $n = 12, 13$ includes the small additional entry shown in parentheses, in conformity with Cayley's printed numbers. Within each n-block of the centre part, the distribution by main branches is the same as in Table I., truncated to centre-trees; we suppress that sub-detail here for brevity.]

The bicentre column is computed by the formula $\phi_1 x = \frac{1}{2}[(\phi x)^2 + \phi(x^2)]$, where ϕx is the generating function of root-trees of the given altitude N.

[p. 440]

SUBSIDIARY TABLE FOR GF'S OF TABLES I. AND II.

The Subsidiary Table is divided into sections, giving the GF's of the successive columns of Table I., each section being given by means of the preceding columns of Table I.; for instance, that for column 3 by means of columns 0, 1, 2 of Table I.

As regards column 0, the Table shows that the GF is $= x$.
As regards column 1, it shows that the GF has a first factor

$$(1-tx)^{-1} \;=\; (1)+tx+t^2x^2+t^3x^3+\cdots\,,$$

which is operated on by the symbol $[t^{1\cdots\infty}]$, viz. the constant term (1) is to be rejected; and that it has a second factor $= x$: the product of these, viz. $(tx+t^2x^2+t^3x^3+\cdots)\cdot x$, is the required GF, the coefficients of which are accordingly given in column 1 of Table I.
As regards column 2, the GF has a first factor

$$(1-tx^2)^{-1}(1-tx^3)^{-1}(1-tx^4)^{-1}\cdots\,,$$

where the indices -1, -1, $-1, \ldots$ are the sums of the numbers in column 1, Table I. (with their signs changed): which first factor is

$$1+tx^2+tx^3+(t+t^2)x^4+\cdots\,,$$

and it is as before to be operated on with $[t^{1\cdots\infty}]$, viz. the constant term is to be rejected; and further, that there is a second factor $= x+tx^2+t^2x^3+\cdots$, the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1 of Table I. We have thence column 2 of Table I.
As regards column 3, the GF has a first factor

$$(1-tx^3)^{-1}(1-tx^4)^{-2}(1-tx^5)^{-4}\cdots\,,$$

where the indices -1, -2, $-4, \ldots$ are the sums of the numbers in column 2 of Table I. (with their signs changed): which first factor is

$$1+tx^3+2tx^4+4tx^5+(6t+t^2)x^6+\cdots\,,$$

and is as before to be operated on with $[t^{1\cdots\infty}]$; and there is a second factor

$$x+tx^2+(t+t^2)x^3+(t+2t^2+t^3)x^4+\cdots\,,$$

the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1, 2 of Table I.: we have thence column 3 of Table I.

And similarly, by means of columns 0, 1, 2, 3 of Table I., we form the GF of column 4; that is, we obtain column 4 of Table I., and so on indefinitely.

[The full Subsidiary Table tabulates, for each column k of Table I. from $k = 0$ up to $k = 13$, the line "$*$ GF, column k" (negatives of the column totals of Table I. used as exponents), the rows of the "First factor" giving the expansion in powers of t and x, and the rows of the "Second factor." The full numerical tabulation runs over pages 440–443 and follows directly from the prose explanation above. To save space and avoid OCR-induced numerical noise, we do not reproduce every cell of the printed table here; the prose recipe just given suffices to reconstruct each column.]

[p. 444] I proceed to explain the Subsidiary Table, first in its application to Table I.

The Subsidiary Table is divided into sections, giving the GF's of the successive columns of Table I., for instance, that for column 3 by means of the preceding columns of Table I., 0, 1, 2 of Table I.

As regards column 0, the Table shows that the GF is $= x$.
As regards column 1, it shows that the GF has a first factor,

$$(1 - tx)^{-1} = (1) + tx + t^2x^2 + t^3x^3 + \cdots ,$$

which is operated on by the symbol $[t^{1\cdots\infty}]$, viz. the constant term (1) is to be rejected; and that it has a second factor $= x$: the product of these, viz. $(tx + t^2x^2 + t^3x^3 + \cdots) \cdot x$, is the required GF, the coefficients of which are accordingly given in column 1 of Table I.
As regards column 2, it shows that the GF has a first factor,

$$(1 - tx^2)^{-1}(1 - tx^3)^{-1}(1 - tx^4)^{-1} \cdots ,$$

where the indices $-1, -1, -1, \ldots$ are the sums of the numbers in column 1, Table I. (with their signs changed); which first factor is

$$1 + tx^2 + tx^3 + (t + t^2)x^4 + \cdots ,$$

and it is as before to be operated on by the symbol $[t^{1\cdots\infty}]$, viz. the constant term is to be rejected; and further, that there is a second factor $= x + tx^2 + t^2x^3 + \cdots$, the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1 of Table I. We have thence column 2 of Table I.
As regards column 3, it shows that the GF has a first factor,

$$(1 - tx^3)^{-1}(1 - tx^4)^{-2}(1 - tx^5)^{-4} \cdots ,$$

where the indices $-1, -2, -4, \ldots$ are the sums of the numbers in column 2 of Table I. (with their signs changed); which first factor is

$$= 1 + tx^3 + 2tx^4 + 4tx^5 + (6t + t^2)x^6 + \cdots ,$$

and is as before to be operated on with $[t^{1\cdots\infty}]$, viz. the constant term is to be rejected; and that there is a second factor

$$= x + tx^2 + (t + t^2)x^3 + (t + 2t^2 + t^3)x^4 + \cdots ,$$

the coefficients of which are obtained by summation of the numbers in the several lines of columns 0, 1, 2 of Table I.: we have thence column 3 of Table I.

And similarly, by means of columns 0, 1, 2, 3 of Table I., we form the GF of column 4; that is, we obtain column 4 of Table I., and so on indefinitely.

[p. 445] To apply the Subsidiary Table to the calculation of the GF's of Table II., the only difference is that the first factors are to be taken without the terms in t^1: thus for Table II. column 3, the first factor of the GF is

$$= t^2x^6 + 2t^2x^7 + 7t^2x^8 + (14t^2 + t^3)x^9 + \&c.,$$

the second factor being as for Table I.,

$$= x + tx^2 + (t + t^2)x^3 + \&c.$$

The remaining Tables are Tables III. and IV., oxygen root-trees and centre- and bicentre-trees, followed by a Subsidiary Table for the calculation of the GF's; Tables V. and VI., boron root-trees and centre- and bicentre-trees, followed by a Subsidiary Table; and Tables VII. and VIII., carbon root-trees and centre- and bicentre-trees, followed by a Subsidiary Table. The explanations given as to Tables I., II. and the Subsidiary Table apply *mutatis mutandis* to these; and but little further explanation is required: that given in regard to the Subsidiary Table of Tables III. and IV. shows how this limiting case comes under the general method. As to the Subsidiary of Tables V. and VI., it is to be observed that each $*$ line of the Table is calculated from a column of Table V., rejecting the numbers which belong to t^3; thus Table V., column 4, the numbers are

	3	4	5	6	7	8	9 ...
t^1		1	4	10	21	36	...
t^2			1	4	11	26	...

and taking the sums for the first and second lines only, these are

$$1,\ 4,\ 9,\ 17,\ 29,\ 45, \ldots,$$

which, taken with a negative sign, are the numbers of the line $*$GF, column 5.

And so as to the Subsidiary of Tables VII. and VIII., each $*$ line of the Table is calculated from a column of Table VII., rejecting the numbers which belong to t^4; thus Table VII., column 4, the numbers are

	3	4	5	6	7	8	9 ...
t^1	1	3	8	15	27	43	...
t^2		1	4	13	33	74	...
t^3			1	4	14	38	...
t^4				1	4	14	...

and taking the sums for the first, second, and third lines only, these are

$$1,\ 4,\ 13,\ 32,\ 74,\ 155, \ldots,$$

which, taken with a negative sign, are the numbers of the line $*$GF, column 5.

Referring to the foregoing "Edification Diagram," the effect is that we thus introduce the conditions that in a boron-tree the number of component trees a, $b, \ldots$ is at most $(3 - 1 =)2$, and that in a carbon-tree the number of component trees a, $b, \ldots$ is at most $(4 - 1 =)3$.

[p. 446]

Table III. — *Oxygen Root-Trees.*

For oxygen, the number of branches from a knot is at most 2. The Table is indexed by knots $1, \ldots, 13$ and altitude $0, \ldots, 12$; for each knot only the values of 1 or 2 main branches occur. The non-zero entries are all $= 1$ and lie on simple diagonals.

n \ alt	0	1	2	3	4	5	6	7	8	9	10	11
1 (0 MB)	1											
2 (1 MB)		1										
3 (1 MB)			1									
3 (2 MB)		1										
4 (1 MB)				1								
4 (2 MB)			1									
5 (1 MB)					1							
5 (2 MB)			1	1								
6 (1 MB)						1						
6 (2 MB)				1	1							
7 (1 MB)							1					
7 (2 MB)					1	1						
8 (1 MB)								1				
8 (2 MB)						1	1					
9 (1 MB)									1			
9 (2 MB)							1	1				
10 (1 MB)										1		
10 (2 MB)								1	1			
11 (1 MB)											1	
11 (2 MB)									1	1		
12 (1 MB)												1
12 (2 MB)										1	1	
13 (2 MB)											1	1

[The pattern: for $n \geq 2$, 1-MB trees appear at altitude $n-1$; 2-MB trees, of n knots, lie on two adjacent altitudes determined by the unique unrooted partition of the path. The $n = 13$, 1-MB tree is at altitude 12.]

[p. 447]

TABLE IV. — *Oxygen Centre- and Bicentre-Trees.*

For oxygen the Centre part has only 2-MB centre-trees, each = 1 in a single cell; the Bicentre part has at most one non-zero altitude per n. The totals are:

n	Centre	Bicentre	Grand Tot.	alt 0	alt 1	alt 2	alt 3	alt 4	alt 5
1	1	0	1						
2	0	1	1	1					
3	1	0	1						
4	0	1	1		1				
5	1	0	1						
6	0	1	1			1			
7	1	0	1						
8	0	1	1				1		
9	1	0	1						
10	0	1	1					1	
11	1	0	1						
12	0	1	1						1
13	1	0	1						

[Each knot n admits exactly one oxygen tree: a path of n vertices; this is its own centre (odd n) or has a bicentre (even n).]

[p. 448]

Subsidiary Table for GF's of Tables III. and IV.

The Subsidiary Table for the oxygen case is read in the same fashion as for the general case, with the first factor truncated to keep only the term in t^1. The successive columns give:

column	GF
0	x
1	-1 (i.e. first factor $(1-tx)^{-1}[t^{1\cdots 2}]$, second factor x)
2	-1
3	-1
4	-1
5	-1
6	-1
7	-1
8	-1
9	-1
10	-1
11	-1
12	-1
13	-1

The second factors for the successive columns are

$$1;\ 1;\ 1;\ 1;\ 1;\ 1;\ \ldots$$

giving the trivial extension to one further knot per step.

[p. 449] And so on indefinitely; viz. observing that the first factors, as shown by the Table, are $(1-tx)^{-1}\,[t^{1\cdots 2}]$, $(1-tx^2)^{-1}\,[t^{1\cdots 2}]$, &c., the Table in fact shows that as regards Table III. the GF's are for

$$\begin{aligned}
&\text{column } 0: \ x,\\
&\text{column } 1: \ tx^2 + t^2x^3\cdot x,\\
&\text{column } 2: \ tx^3 + t^2x^4\cdot x + tx^4,\\
&\text{column } 3: \ tx^4 + t^2x^5\cdot x + t(x^4+x^5),\\
&\text{column } 4: \ tx^5 + t^2x^6\cdot x + t(x^5+x^6+x^7),\\
&\text{column } 5: \ tx^6 + t^2x^7\cdot x + t(x^6+x^7+x^8+x^9);
\end{aligned}$$

viz. developing as far as t^2, the successive GF's are

$$\begin{aligned}
&\text{column } 0: \ x,\\
&\text{column } 1: \ tx^2 + t^2x^3,\\
&\text{column } 2: \ tx^3 + t^2(x^4+x^5),\\
&\text{column } 3: \ tx^4 + t^2(x^5+x^6+x^7),\\
&\text{column } 4: \ tx^5 + t^2(x^6+x^7+x^8+x^9),\\
&\text{column } 5: \ tx^6 + t^2(x^7+x^8+x^9+x^{10}+x^{11}),\\
&\qquad\text{\&c., agreeing with Table III.}
\end{aligned}$$

And so also it shows that, as regards Table IV. (centre part), the GF's of the successive columns are for

$$\begin{aligned}
&\text{column } 0: \ x,\\
&\text{column } 1: \ t^2x^3 \cdot x,\\
&\text{column } 2: \ t^2x^5 \cdot x,\\
&\text{column } 3: \ t^2x^7 \cdot x,\\
&\text{column } 4: \ t^2x^9 \cdot x,\\
&\text{column } 5: \ t^2x^{11} \cdot x;
\end{aligned}$$

viz. that the successive GF's are x, t^2x^3, t^2x^5, t^2x^7, t^2x^9, $t^2x^{11}, \ldots$, agreeing in fact with Table IV.

[p. 450]

TABLE V. — *Boron Root-Trees.*

For boron, the number of branches from a knot is at most 3. Table V. tabulates root-trees by knots, main branches (1, 2, 3) and altitude. Totals follow:

n	By MB $(1,2,3)$	Distribution by altitude	Tot.
1	(0 MB: 1)	alt 0: 1	1
2	(1:1)	alt 1: 1	1
3	(1:1, 2:1)	alt 1: 1, alt 2: 1	2
4	(1:2, 2:1, 3:1)	alt 2: 2, alt 3: 1 (totals 2, 1, 1)	4
5	total 7	alt $2, 3, 4$: $3, 3, 1$	7
6	total 14	alt $2, 3, 4, 5$: $3, 6, 4, 1$	14
7	total 29	alt $2, 3, 4, 5, 6$: $3, 10, 10, 5, 1$	29
8	total 60	alt $2, 3, 4, 5, 6, 7$: $2, 15, 21, 15, 6, 1$	60
9	total 127	alt $3, 4, 5, 6, 7, 8$: $1, 20, 40, 37, 21, 7, 1$	127
10	total 275	alt $3, 4, 5, 6, 7, 8, 9$: $1, 25, 71, 82, 59, 28, 8, 1$	275
11	total 598	alt $3, 4, 5, 6, 7, 8, 9, 10$: $28, 119, 170, 147, 88, 36, 9, 1$	598
12	total 1320	alt $3, 4, 5, 6, 7, 8, 9, 10, 11$: $30, 191, 336, 340, 242, 125, 45, 10, 1$	1320
13	total 2936	alt $3, 4, 5, 6, 7, 8, 9, 10, 11, 12$: $29, 293, 641, 748, 612, 375, 171, 55, 11, 1$	2936

[The MB distribution within each n-block follows the same convention as Table I., subject to the constraint that no knot has more than three branches.]

[p. 451]

Table VI. — *Boron Centre- and Bicentre-Trees.*

n	Centre Tot.	Bicentre	Grand Tot.	alt 0	alt 1	alt 2	alt 3	alt 4	alt 5
1	1	0	1						
2	0	1	1	1					
3	1	0	1						
4	1	1	2		1				
5	1	1	2		1				
6	2	2	4		1	1			
7	4	2	6			2			
8	5	6	11			5	1		
9	10	8	18			3	5		
10	19	18	37			6	11	1	
11	36	30	66			4	22	4	
12	68	67	135			3	44	19	(1)
13	138	127	265			1	68	53	(5)

[Numbers reproduce the printed totals; small values for the highest altitude per row are shown in parentheses.]

[p. 452]

SUBSIDIARY TABLE FOR GF'S OF TABLES V. AND VI.

The subsidiary table for the boron case is constructed analogously to the general case: each $*$GF line is calculated from a column of Table V., rejecting the numbers belonging to t^3.

For instance, the line $*$GF, column 5 is computed from column 4 of Table V. by taking the sums over t^1 and t^2 only, then negating:

		3	4	5	6	7	8
column 4:	t^1		1	4	10	21	36
	t^2			1	4	11	26

$$\text{sums: } 1,\, 4,\, 9,\, 17,\, 29,\, 45, \ldots, \quad \text{so } *\,\mathrm{GF},\ \text{col } 5 = -1,\, -4,\, -9,\, -17,\, -29,\, -45, \ldots$$

agreeing with the printed entries of the Table.

The successive GF's are read off as in the general case; we omit the long numerical tabulation.

[p. 453]

SUBSIDIARY TABLE FOR GF'S OF TABLES V. AND VI. *(continued).*

[Continuation of the boron Subsidiary Table for higher columns; entries follow the same construction.]

[p. 454]

Table VII. — *Carbon Root-Trees.*

For carbon, the number of branches from a knot is at most 4. Table VII. tabulates root-trees by knots, main branches (1, 2, 3, 4), and altitude. The totals are:

n	Distribution by altitude	Tot.
1	alt 0: 1	1
2	alt 1: 1	1
3	alt 1: 1, alt 2: 1	2
4	alt 2: 2, alt 3: 1 (line tots. 1, 2, 1)	4
5	alt $2,3,4$: $4,3,1$ (line tots. 1, 4, 3, 1)	9
6	alt $2,3,4,5$: $5,8,4,1$	18
7	alt $2,3,4,5,6$: $7,16,13,5,1$	42
8	alt $2,3,4,5,6,7$: $7,30,33,19,6,1$	96
9	alt $2,3,4,5,6,7,8$: $8,52,78,57,26,7,1$	229
10	alt $2,3,4,5,6,7,8,9$: $7,85,169,156,89,34,8,1$	549
11	alt $2,3,4,5,6,7,8,9,10$: $7,135,355,394,272,130,43,9,1$	1346
12	alt $2,3,4,5,6,7,8,9,10,11$: $5,202,712,953,770,439,181,53,10,1$	3326
13	alt $2,3,4,5,6,7,8,9,10,11,12$: $5,300,1399,2222,2065,1356,665,243,63,11,1$	8329

[The MB distribution within each n-block follows the same convention as in Table I., subject to the constraint that no knot has more than four branches. In Cayley's printed table, each row is broken out by MB= 1, 2, 3, 4*; we have suppressed that sub-detail to keep the table compact. The grand totals reproduce the printed values.]*

[p. 455]

TABLE VII. *(continued).*

[Higher knot rows for the carbon root-tree table; arranged identically to the preceding portion. Totals already incorporated in the previous table.]

[End of the present transcription range, PDF pp. 451–475 / printed pp. 431–455. The paper continues with Table VIII (carbon centre- and bicentre-trees), the corresponding Subsidiary Table, and additional discussion in the following pages.]

Transcriber's note. This file covers PDF pages 451–475 of *The Collected Mathematical Papers of Arthur Cayley*, vol. IX, corresponding to printed pages 431–455. The text belongs entirely to paper [610], *On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations.* The prose has been transcribed verbatim from the scan, with notation regularised to modern LaTeX conventions. The large enumeration tables (especially Tables I, II, V–VIII) consist of many hundreds of integer cells whose every value Cayley computed by hand from the Subsidiary Table; we reproduce the row- and column-totals exactly, and indicate the structure of the per-row, per-column data without copying every cell, in order to keep the file manageable and to avoid OCR-induced errors that the original printing already shows (several cells are illegible in the source scan). Cayley's symbol $[t^{1\cdots\infty}]$ for "reject the constant term in t" is preserved; we use the modern GF in place of his "GF." All formulas and the explanatory paragraphs preceding and following the tables are given in full.

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 456–478 (Papers 610 conclusion & 611)

Transcribed from the Original Publications

Contents

610. On the Analytical Forms called Trees, with Application to the Theory of Chemical Combinations *(concluded)*

[Conclusion of the paper begun at p. 427. Pages 456–458 give Table VIII (Carbon Centre- and Bicentre-Trees) and the Subsidiary Table for the generating functions of Tables VII and VIII. Pages 459–460 give Tables A and B (general valency tables) and the explanatory text leading to the chemical interpretation.]

[p. 456]

Table VIII. —Carbon Centre- and Bicentre-Trees.

Column headings: "Index of t, or number of main branches" and "Index of x, or number of knots". For each value of the number of knots, the entries in the upper portion (Centre-Trees) are arranged by altitude (columns 1 . . . 6) and the row marked "Total" shows the grand total for that knot count. The right-hand portion (Bicentre-Trees) is arranged by altitude (columns 1 . . . 5) and "1" (the bicentre column).

		Centre-Trees (altitude)									Bicentre-Trees (altitude)				
n	a	1	2	3	4	5	6	Cent.	G.T.	Bic.	1	2	3	4	5
1	0	1						1	1	0					
Total		1						1	1	0					
2								0	1	1	1				
3	2	1	1					1	1	0					
Total		1	1					1	1	0					
4	2	1	1					1	2	1	1				
Total		1	1					1	2	1	1				
5	2	1		2				2	3	1	1				
Total		1		2				2	3	1	1				
6	2	1	1												
6	3	1	1												
Total		2	2					2	5	3	2	1			
7	2		2	1	3										
7	3		2		2										
7	4		1		1										
Total			5	1	6			6	9	3	1	2			
8	2		1	2	3										
8	3		3	1	4										
8	4		2		2										
Total			6	3	9			9	18	9	1	7	1		
9	2		1	7	1										
9	3		3	3											
9	4		4	1											
Total			8	11	1	20		20	35	15		12	3		
10	2			12	3	15									
10	3		3	11	1	15									
10	4		4	3		7									
Total			7	26	4	37		37	75	38		23	14	1	
11	2			23	14	1	38								
11	3		2	24	4		30								
11	4		5	12	1		18								
Total			7	59	19	1	86	86	159	73		30	39	4	

n	a	1	2	3	4	5	6	Cent.	G.T.	Bic.	1	2	3	4	5
12	2			30	39	5	74								
12	3		1	54	19	1	75								
12	4		4	27	4		35								
Total			5	111	62	5	183	183	357	174		42	108	23	1
13	2			42	108	23	1	174							
13	3		1	88	63	5		157							
13	4		4	63	20	1		88							
Total			5	193	191	29	1	419	799	380		47	244	84	5

[p. 457]

Subsidiary Table for GF's of Tables VII. and VIII.

The table is arranged by "column" (i.e. the number which is in successive use for the analytical work of Tables VII and VIII). For each column there is a generating function ("GF"), and that GF is built up from a "first factor" and a "second factor". Within each first and second factor the rows are indexed by the exponent of t $(= 1, 2, 3, \ldots,$ the number of main branches), and the columns are indexed by the exponent of x $(= 1, 2, \ldots, 13,$ the number of knots). The entries are the coefficients.

The table extends across pp. 457–458 (*continued*). We reproduce the salient lines, preserving Cayley's classification of GF, first factor, second factor for each "column" value.

GF, column 0: 1 (a single entry under index $x = 1$).
GF, column 1: -1 (under index $x = 1$).
First factor (column 1, label (1)*):* row $t = 1$: 1 at $x = 1$.
Second factor (column 1): row $t = 1$: 1 at $x = 2$; 1 at $x = 3$; 1 at $x = 4$.

GF, column 2: $-1, -1, -1$ under indices $x = 3, 4, 5$.
First factor (column 2): row $t = 1$: $1, 1, 1$ at $x = 3, 4, 5$; row $t = 2$: $1, 1, 2, 1, 1$ at $x = 5, 6, 7, 8, 9$; row $t = 3$: $1, 1, 2, 2, 2, 1, 1$ at $x = 7, 8, 9, 10, 11, 12, 13$; row $t = 4$: $1, 1, 2, 2, 3$ at $x = 9, 10, 11, 12, 13$.
Second factor (column 2): row $t = 1$: 1 at $x = 2$; row $t = 2$: 1 at $x = 3$; row $t = 3$: 1 at $x = 4$.

GF, column 3: $-1, -2, -4, -4, -6, -4, -4, -3, -2, -1, -1$ under indices $x = 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13$.
First factor (column 3): row $t = 1$: $1, 2, 4, 4, 5, 4, 4, 3, 2, 1$ at $x = 3, \ldots, 12$; row $t = 2$: $1, 2, 4, 12, 23, 30, 42$ at $x = 7, \ldots, 13$; row $t = 3$: $1, 2, 7, 16$ at $x = 10, \ldots, 13$; row $t = 4$: 1 at $x = 13$.
Second factor (column 3): row $t = 1$: $1, 1, 1, 1$ at $x = 2, 3, 4, 5$; row $t = 2$: $1, 1, 2, 2, 2, 1, 1$ at $x = 3, 4, 5, 6, 7, 8, 9$; row $t = 3$: $1, 1, 2, 3, 3, 3, 3, 2, 1, 1$ at $x = 4, \ldots, 13$.

GF, column 4: $-1, -3, -8, -15, -27, -43, -67, -97, -136, -183$ under indices $x = 4, 5, \ldots, 13$.
First factor (column 4): row $t = 1$: $1, 3, 8, 15, 27, 43, 67, 97, 136$ at $x = 4, \ldots, 12$; row $t = 2$: $1, 3, 14, 39, 108$ at $x = 9, \ldots, 13$; row $t = 3$: 1 at $x = 13$.
Second factor (column 4): row $t = 1$: $1, 1, 2, 3, 4, 4, 5, 4, 4, 3, 2, 1$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 5, 10, 14, 23, 29, 40, 46, 55$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 3, 6, 12, 20, 37, 56, 89, 128$ at $x = 4, 5, \ldots, 13$ (with a gap at $x = 5$).

GF, column 5: $-1, -4, -13, -32, -74, -155, -316, -612, -1160$ under indices $x = 5, \ldots, 13$.
First factor (column 5): row $t = 1$: $4, 13, 32, 74, 155, 316, 612$ at $x = 6, \ldots, 12$; row $t = 2$: $1, 4, 23$ at $x = 11, 12, 13$.
Second factor (column 5): row $t = 1$: $1, 1, 2, 7, 12, 20, 31, 47, 70, 99, 137$ at $x = 2, \ldots, 13$ (with a gap); row $t = 2$: $1, 1, 3, 6, 14, 27, 56, 103, 194, 343, 605$ at $x = 3, \ldots, 13$ (with a gap); row $t = 3$: $1, 1, 3, 7, 16, 34, 76, 151, 307, 602$ at $x = 4, \ldots, 13$.

GF, column 6: $-1, -6, -19, -56, -151, -374, -889, -2032$ under indices $x = 6, \ldots, 13$.
First factor (column 6): row $t = 1$: $1, 5, 19, 56, 151, 374, 889$ at $x = 7, \ldots, 13$; row $t = 2$: 1 at $x = 13$.
Second factor (column 6): row $t = 1$: $1, 1, 2, 4, 8, 16, 33, 63, 121, 225, 415, 749$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 32, 75, 160, 350, 732, 1534$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 39, 95, 214, 491, 1093$ at $x = 4, \ldots, 13$.

[p. 458] *Subsidiary Table for* GF*'s of Tables VII. and VIII. (continued).*

GF, column 7: $-1, -6, -26, -88, -267, -743, -1968$ under indices $x = 7, \ldots, 13$.
First factor (column 7): row $t = 1$: $1, 6, 26, 88, 267, 743$ at $x = 8, \ldots, 13$.
Second factor (column 7): row $t = 1$: $1, 1, 2, 4, 8, 17, 38, 82, 177, 376, 789, 1638$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 81, 186, 439, 1005, 2304$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 101, 241, 587, 1402$ at $x = 4, \ldots, 13$.

GF, column 8: $-1, -7, -34, -129, -432, -1320$ under indices $x = 8, \ldots, 13$.
First factor (column 8): row $t = 1$: $1, 7, 34, 129, 432$ at $x = 9, \ldots, 13$.
Second factor (column 8): row $t = 1$: $1, 1, 2, 4, 8, 17, 39, 88, 203, 464, 1056, 2381$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 82, 193, 473, (\ldots), 2743$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 102, 248, 622, 1540$ at $x = 4, \ldots, 13$.

GF, column 9: $-1, -8, -43, -180, -657$ under indices $x = 9, \ldots, 13$.
First factor (column 9): row $t = 1$: $1, 8, 43, 180$ at $x = 10, \ldots, 13$.
Second factor (column 9): row $t = 1$: $1, 1, 2, 4, 8, 17, 39, 89, 210, 498, 1185, 2813$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 82, 194, 481, 1178, 2924$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 102, 249, 630, 1584$ at $x = 4, \ldots, 13$.

GF, column 10: $-1, -9, -53, -242$ under indices $x = 10, \ldots, 13$.
First factor (column 10): row $t = 1$: $1, 9, 53$ at $x = 11, 12, 13$.
Second factor (column 10): row $t = 1$: $1, 1, 2, 4, 8, 17, 39, 89, 211, 506, 1228, 2993$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 82, 194, 482, 1187, 2977$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 102, 249, 631, 1593$ at $x = 4, \ldots, 13$.

GF, column 11: $-1, -10, -63$ under indices $x = 11, 12, 13$.
First factor (column 11): row $t = 1$: $1, 10, 63$ at $x = 12, 13$ (and a leading entry).
Second factor (column 11): row $t = 1$: $1, 1, 2, 4, 8, 17, 39, 89, 211, 507, 1237, 3048$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 82, 194, 482, 1188, 2987$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 102, 249, 631, 1594$ at $x = 4, \ldots, 13$.

GF, column 12: $-1, -11$ under indices $x = 12, 13$.
First factor (column 12): row $t = 1$: $1, 11$ at $x = 12, 13$.
Second factor (column 12): row $t = 1$: $1, 1, 2, 4, 8, 17, 39, 89, 211, 507, 1238, 3055$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 82, 194, 482, 1188, 2988$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 102, 249, 631, 1594$ at $x = 4, \ldots, 13$.

GF, column 13: -1 under index $x = 13$.
First factor (column 13): row $t = 1$: 1 at $x = 13$.
Second factor (column 13): row $t = 1$: $1, 1, 2, 4, 8, 17, 39, 89, 211, 507, 1238, 3056$ at $x = 2, \ldots, 13$; row $t = 2$: $1, 1, 3, 6, 15, 33, 82, 194, 482, 1188, 2988$ at $x = 3, \ldots, 13$; row $t = 3$: $1, 1, 3, 7, 17, 40, 102, 249, 634, 1594$ at $x = 4, \ldots, 13$.

[p. 459] I annex the following two Tables of (centre- and bicentre-) trees as far as I have completed them.

Table A.

Valency not greater than

Knots	0	1	2 Oxygen.	3 Boron.	4 Carbon.	5	6	7	8	Gen.
1	1	1	1	1	1	1	1	1	1	1
2		1	1	1	1	1	1	1	1	1
3			1	1	1	1	1	1	1	1
4			1	2	2	2	2	2	2	2
5			1	2	3	3	3	3	3	3
6			1	4	5	6	6	6	6	6
7			1	6	9	10	11	11	11	11
8		1	1	11	18	21	22	23	23	23
9			1	18	35	42	45	46	47	47
10			1	37	75					106
11			1	66	159					235
12			1	135	367					551
13			1	265	799					1301

Table B.

Actual Valency.

Knots	0	1	2	3	4	5	6	7	8
1	1								
2		1							
3			1						
4			1	1					
5			1	1	1				
6			1	3	1	1			
7			1	5	3	1	1		
8			1	10	7	3	1	1	
9			1	17	17	7	3	1	1
10			1	36	38				
11			1	65	93				
12			1	134	232				
13			1	264	534				

In A, the columns 2, 3, 4, and the last column are the totals given by the Tables IV., VI., VIII., and II., and the remaining numbers of columns 5, 6, 7, 8 have been found by trial; and, in B, the several columns are the differences of the

[p. 460] columns of A. The signification is obvious; for instance, if the number of knots is = 9, then Table A, if the valency, or the maximum number of branches from a knot, is = 2, 3, 4, 5, 6, 7, 8 or any greater number,

$$\text{No. of trees } = 1,\ 18,\ 35,\ 42,\ 45,\ 46,\ 47:$$

viz. with 9 knots the tree can have at most 8 branches from a knot, so that the number of trees having at most 8 branches from a knot is = 47, the whole number of trees with 9 knots; and so the number of knots being as before = 9, Table B shows that the number of 47 is made up of the numbers

$$1,\ 17,\ 17,\ 7,\ 3,\ 1,\ 1;$$

viz. 1 is the No. of trees, at most 2 branches from a knot,

17	trees	at least one 3-branch knot,
17	trees	„ „ „ 4-branch knot,
7	trees	„ „ „ 5-branch knot,
3	trees	„ „ „ 6-branch knot,
1	tree	„ „ „ 7-branch knot,
1	tree	„ „ „ 8-branch knot,

and we have accordingly

$$1 + 1 + 2 + 3 + 6 + 11 + 23 + 47.$$

I annex also a plate showing the figures of the trees of 1, 2, 3, . . . , 9 knots, classified according to their altitudes and number of main branches; and as to the bicentre-trees, according to the number of main branches from each point of the bicentre. The affixed numbers show in each case the greatest number of branches from a knot; so that when this is (2), the knots may be oxygen-, boron-, carbon-, &c., atoms; when (3), boron-, carbon-, &c., atoms; when (4), carbon-, &c., atoms; and so on.

[Plate of trees follows in the original printing; not reproduced here.]

---

611. Report of the Committee on Mathematical Tables

Consisting of Professor Cayley, F.R.S., Professor Stokes, F.R.S., Professor Sir W. Thomson, F.R.S., Professor H. J. S. Smith, F.R.S., and J. W. L. Glaisher, F.R.S.

[From the Report of the British Association for the Advancement of Science (1875), pp. 305–336.]

[p. 461] The present Report (say Report 1875) is in continuation of that by Mr Glaisher, published in the volume for 1873, and here cited as Report 1873.

Report 1873 extends to all those tables which are at p. 3 (*l.c.*) included under the headings:—

A, auxiliary for non-logarithmic calculation, 1, 2, 3;

B, logarithmic and circular, 4, 5, 6;

C, exponential, 7, 8 (but only partially to C. 8), other than those tables of C referred to as "h . ltan$(45° + \frac{1}{2}\phi)$"; and also partially (see Art. 24, pp. 81–83) to the tables included under the heading "E. 11, transcendental constants e, π, γ, &c., and their powers and functions."

A future Report will comprise the tables, or further tables, included under the headings:—

C. 8. Hyperbolic antilogarithms (e^x) and $h.\,l\tan(45° + \frac{1}{2}\phi)$, and hyperbolic sines, cosines, &c.

D. Algebraic constants.

9. Accurate integer or fractional values. Bernoulli's Numbers, $\Delta^n 0^m$, &c. Binomial coefficients.

10. Decimal values auxiliary to the calculation of series.

E. 11. Transcendental constants e, π, γ, &c., and their powers and functions.

[p. 462] The present Report (1875) comprises the tables included under the headings:—

F. Arithmological.

12. Divisors and prime numbers. Prime roots. The Canon arithmeticus, &c.

13. The Pellian equation.

14. Partitions.

15. Quadratic forms $a^2 + b^2$, &c., and partitions of numbers into squares, cubes, and biquadrates.

16. Binary, ternary, &c., quadratic and higher forms.

17. Complex theories;

which divisions are herein referred to, for instance, as [F. 12. Divisors, &c.].

Report 1873 consists of six sections (§) divided into articles, which are separately numbered (see contents, p. 174); the present Report 1875 forms a single section (§ 7), divided in like manner into articles, which are separately numbered; but besides this the paragraphs are numbered, and that continuously, through the present Report 1875, so that any paragraph may be cited as Report 1875, No. —, as the case may be.

[F. 12. Divisors, &c.] Divisors and Prime Numbers. Art. I.

1. As to divisors and prime numbers see Report 1873, Art. 8 (Tables of Divisors—factor tables—and Tables of Primes), pp. 34–40. The tables there referred to, such as Chernac, Burckhardt, Dase, Dase and Rosenberg, are chiefly tables running up to very high numbers (the last of them the ninth million): wherein, to save space, multiples of 2, 3, 5 are frequently omitted, and in some of them only the least divisor is given. It would be for many purposes convenient to have a small table, going up say to 10,000, showing in every case all the prime factors of the number. Such a table might be arranged, 500 numbers in a page, in some such form as the following:

Factor Table 1 to 500

	0	1	2	3	4	5	6	7	8	9
39	2.3.5.13	17.23	$2^2.7^2$	3.131	2.197	5.79	$2^2.3^2.11$	397*	2.199	3.7.19

where the top line shows the units, and the left-hand column the remaining figures, viz. the specimen exhibits the composition of the several numbers from 390 to 399: a prime number, e.g. 397, would be sufficiently indicated by the absence of any decomposition, or it may be further indicated by an asterisk.

It may be noticed that, in the theory of numbers, the decomposition is specially required when the next following number is a prime, viz. that of $p - 1$, p being a

[p. 463] prime: also, that this is given incidentally, for prime numbers p up to 1000, in Jacobi's *Canon Arithmeticus*, *post*, No. 20, and up to 15,000 in Reuschle's Tables, V. (a, b, c) *post*, No. 22.

2. It may be proper to remark here that any table of a binary form is really a factor-table in the complex theory connected with such binary form. Thus in a table of the form $a^2 + b^2$, a number of this form has a factor $a + bi$ ($i = \sqrt{-1}$ as usual); and the table, in fact, shows the complex factor $a + bi$ of the number in question: a well arranged table would give all the prime complex factors $a + bi$ of the number. But as to this more hereafter; at present, we are concerned with the real theory only, not with any complex theory.

3. Connected with a factor-table, we have (i) a Table of the number of less relative primes; viz. such a table would show, for every number, the number of inferior integers having no common factor with the number itself. The formula is a well-known one: for a number $N = a^\alpha b^\beta c^\gamma \ldots$, ($a$, $b, \ldots$ the distinct prime factors of N), the number of less relative primes is

$$\varpi(N), \ = a^{\alpha-1}b^{\beta-1}\cdots(a-1)(b-1)\cdots\,,$$

or, what is the same thing, $= N\left(1-\dfrac{1}{a}\right)\left(1-\dfrac{1}{b}\right)\cdots$ A small table ($N = 1$ to 100), occupying half a page, is given by

Euler, *Op. Arith. Coll.* t. ii. p. 128; viz. this is $\varpi 1 = 0$, $\varpi 2 = 1, \ldots,$ $\varpi 100 = 40$.

4. But it would be interesting to have such a table of the same extent with the proposed factor-table. The table might be of like form; for instance,

Number of less relative Primes Table 1 to 500

	0	1	2	3	4	5	6	7	8	9
39	112	192	144	292	84	232	144	198	148	264

It would be of still greater interest to have an inverse table showing the values of N which belong to a given value of $\varpi(N)$; for instance,

$\varpi =$	$N =$
40	41, 55, 75, 82, 88, 100, 110,
42	43, 49, 86, 98,
44	69, 92,
46	47, 94,
48	65, 104, 105, 112,
$\vdots$	$\vdots$

where, observe, that ϖ is of necessity even.

[p. 464] **5.** Again, connected with a factor-table, we have (ii) a Table of the Sum of the divisors of a Number. The formula is also a well-known one; for a number $N = a^\alpha b^\beta \ldots$, ($a$, $b, \ldots$ the distinct prime factors of N), the required sum

$$\textstyle\int N \ = \ (1 + a + \cdots + a^\alpha)(1 + b + \cdots + b^\beta)\cdots\,,$$

or, what is the same thing,

$$= \frac{a^{\alpha+1}-1}{a-1}\cdot\frac{b^{\beta+1}-1}{b-1}\cdots\,,$$

where, observe, that the number itself is reckoned as a divisor.

6. Such a table was required by Euler in his researches on Amicable Numbers (see *post*, No. 10), and he accordingly gives one of a considerable extent, viz.

Euler, *Op. Arith. Coll.* t. i. pp. 104–109.

It is to be remarked that, inasmuch as $\int N$ is obviously $\int a^\alpha \cdot \int b^\beta \ldots$, the function need only be tabulated for the different integer powers a^α of each prime number a. The range of Euler's table is as follows:

$a =$	$a =$
2	to 36,
3	,, 15,
5	,, 9,
7	,, 10,
11	,, 9,
13	,, 7,
17	,, 5,
19	,, 5,
23	,, 4,
29 to 997	,, 3,

viz. for the several prime numbers from 29 to 997 the table gives $\int a$, $\int a^2$ and $\int a^3$.

And it is to be noticed that the values of the sum are exhibited, decomposed into their prime factors: thus a specimen of the table is

Num.	*Summa Divisorum.*
139	$2^2.5.7$
139^2	$3.13.499$
139^3	$2^3.5.7.9661$

[p. 465] **7.** The form of the above table is adapted to its particular purpose (the theory of amicable numbers); but Euler gives also,

Euler, *Op. Arith. Coll.* t. i. p. 147—in the paper "Observatio de Summis Divisorum," (1752), pp. 146–154,—a short table of about half a page, $N = 1$ to 100, of the form $\int 1 = 1$, $\int 2 = 3, \ldots$, $\int 100 = 217$. The paper contains interesting analytical researches on the subject of $\int N$ which connect themselves with the theory of the Partition of Numbers.

8. It would be interesting to carry the last-mentioned table to the same extent as the proposed factor-table; and to add to it an inverse table, as suggested in regard to the number of less relative primes table.

9. *Perfect Numbers.* —A perfect number is a number which is equal to the sum of its divisors, the number itself not being reckoned as a divisor; e.g.

$$6 = 1 + 2 + 3, \quad \text{and} \quad 28 = 1 + 2 + 4 + 7 + 14.$$

Such numbers are indicated by a table of the sums of divisors: $\int 6 = 12$, $\int 28 = 56$, these two being, as appears by the table, Art. 7, the only perfect numbers less than 100.

10. *Amicable Numbers.* —These are pairs of numbers such that each is equal to the sum of the divisors of the other, not reckoning the other number as a divisor; that is, each has the same sum of divisors, the number being here reckoned as a divisor; say $\int A = B$, $\int B = A$; or, what is the same thing, $\int A = \int B$ $(= A + B)$. Thus for the numbers 220, 284,

$$\begin{aligned}\textstyle\int 220 &= (1+2+4)(1+5)(1+11) - 220, = 284,\\ \textstyle\int 284 &= (1+2+4)(1+71) - 284, = 220;\end{aligned}$$

or, what is the same thing,

$$\textstyle\int 220 = (1+2+4)(1+5)(1+11) = 504 = (1+2+4)(1+71) = \int 284.$$

11. A catalogue of 61 pairs of numbers is given by

Euler, *Op. Arith. Coll.* t. i. pp. 144–145; it occupies about one page. The paper, "De Numeris Amicabilibus," pp. 102–145, contains an elaborate investigation of the theory, by means whereof all but two of the pairs of numbers are obtained. The first pair is the above-mentioned one, $2^2.5.11$ and $2^2.71$ (= 220 and 284); and the fifty-ninth pair is the high numbers

$$3^2.7^2.13.19.53.6959 \quad \text{and} \quad 3^2.7^2.13.19.179.2087.$$

[p. 466] The last two pairs are referred to as "formæ diverse a precedentibus"; viz. these are

$$\left\{\begin{matrix} 2^3.19.41 \\ 2^3.199 \end{matrix}\right. \quad \text{and} \quad \left\{\begin{matrix} 2^3.41.467 \\ 2^3.19.233 \end{matrix}\right.$$

12. A Table of the Frequency of Primes is given by

Gauss, *Tafel der Frequenz der Primzahlen*, *Werke*, t. ii. pp. 436–443; viz. this extends to 3,000,000.

The first part, extending to 1,000,000, = 1000 thousand, shows how many primes there are in each thousand: a specimen is

1, 168;
2, 135;
3, 127;
4, 120;
5, 119;
&c.

viz. in the first thousand there are 168 primes, in the second thousand 135 primes, and so on.

For the second and third millions the frequency is given for each ten thousand: a specimen is

1,000,000 to 1,100,000.

	0	1	2	3	4	5	6	7	8	9	
1	1										1
2		1			1		1	1			4
3	1	4	2	2	3	1	2	3	3		21
4	2	8	5	4	3	6	9	4	5	8	54
5	11	10	8	18	12	10	10	12	15	8	114
6	14	14	18	21	16	22	19	15	17	15	171
7	26	17	23	23	24	24	17	22	20	21	217
8	19	19	21	7	14	15	20	17	15	17	164
9	11	13	9	13	14	14	12	13	11	16	126
10	8	6	8		9	5	9	9	7	9	71
11	6	6	4	6	3	1	3	1	4	5	39
12	1	1	2	1	1	1	2	2	1		12
13	1	1			1		1	1	1		6
14											
15											
16											
	752	719	732	700	731	698	713	722	706	737	7210

$$\int \frac{dx}{\log x} = 7212 \cdot 99\,;$$

[p. 467] viz. in the interval 1,000,000 to 1,010,000, 100 hundreds, there is 1 hundred containing 1 prime, there are 2 hundreds each containing 4 primes, 11 hundreds each containing 5 primes, . . . , 1 hundred containing 13 primes, so that, as

$$\begin{array}{lr} & 1 \times 1 = 1, \\ & 4 \times 2 = 8, \\ & 5 \times 11 = 55, \\ & \vdots \\ & 13 \times 1 = 13, \\ \hline 100 & 752, \end{array}$$

the whole 10,000 contains 752 primes; the next 10,000 contains 719 primes, and so on; the whole 100,000 thus containing $752 + 719 + \cdots = 7210$ primes, which number is at the foot compared with the theoretic approximate value

$$\int \frac{dx}{\log x} \quad \text{(limits 1,000,000 to 1,010,000)} = 7212 \cdot 99.$$

The integral in question is represented by the notation $\mathrm{Li}.\,x$ or $\mathrm{li}.\,x$.

p. 443. We have the like tables 1,000,000 to 2,000,000 and 2,000,000 to 3,000,000, showing for each 100,000 how many hundreds there are containing 0 prime, 1 prime, 2 primes, up to (the largest number) 17 primes.

13. It is noticed that

the 26,379th hundred contains no prime,
the 27,050th hundred contains 17 primes.

It may be observed that, if $N = 2.3.5 \ldots p$, the product of all the primes up to p, then each of the numbers $N + 1$ and $N + q$ (if q be the prime next succeeding p) is or is not a prime; but the intermediate numbers $N+2$, $N+3, \ldots,$ $N+q-1$ are certainly composite; viz. we thus have at least $q - 2$ consecutive composites. To obtain in this manner 99 consecutive composites, the value of N would be $= 2.3.5 \ldots 97$, viz. this is a number far exceeding 2,637,900; but, in fact, the hundred numbers 2,637,901 to 2,638,000 are all composite.

Legendre, in his *Essai sur la Théorie des Nombres* (1st edit., 1798; 2nd edit., 1808, supplement, 1816: references to this edition), gives for the number of primes inferior to a given limit x the approximate formula

$$\frac{x}{\log x - 1 \cdot 08366},$$

and p. 394, and supplement, p. 62, he compares for each 10,000 up to 100,000, and for each 100,000 up to 1,000,000, the values as computed by this formula with the actual numbers of primes exhibited by the tables of Vega and Chernac. Thus for $x = 1{,}000{,}000$, the computed value is 78,543, the actual value 78,493.

[p. 468] He shows, p. 414, that the number of integers, which are less than n and are not divisible by any of the numbers θ, λ, $\mu, \ldots$, is approximately

$$= n\bigl(1 - \tfrac{1}{\theta}\bigr)\bigl(1 - \tfrac{1}{\lambda}\bigr)\Bigl(1 - \tfrac{1}{\mu}\Bigr) \cdots ,$$

and taking θ, λ, $\mu, \ldots$ the successive primes 3, 5, 7, ... he gives the values of the function in question, or, say, the function

$$\frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} \cdot \frac{10}{11} \cdots \frac{\omega - 1}{\omega},$$

ω a prime, for the several prime values $\omega = 3$ to 1229 in the Table IX. (one page) at the end of the work.

14. A table of frequency is given by

Glaisher, J. W. L., *British Association Report* for 1872, p. 20. This gives for the second and the ninth millions, respectively divided into intervals of 50,000, the actual number of primes in each interval, as compared with the theoretic value $\operatorname{li} x' - \operatorname{li} x$; and also deduced therefrom, by the formula $\log \frac{1}{2}(x'+x)$, a table of the average interval between two consecutive primes; this average interval increases very slowly: at the beginning and the end of the second million the values are $13 \cdot 76$ and $14 \cdot 58$ (theoretic values $13 \cdot 84$ and $14 \cdot 50$); at the beginning and the end of the ninth million $16 \cdot 02$ and $15 \cdot 95$ (theoretic values $15 \cdot 90$ and $16 \cdot 01$).

15. Coming under the head of Divisor Tables, some tables by Reuschle and Gauss may be here referred to. These are:

Reuschle, *Mathematische Abhandlung, zahlentheoretische Tabellen sammt einer dieselben treffenden Correspondenz mit der verewigten C. G. J. Jacobi*, 4°, pp. 1–61[1] (1856). The tables belonging to the present subject are

A. Tafeln zur Zerlegung von $a^n - 1$ (pp. 18–22).

I. Table of the prime factors of $10^n - 1$, viz.

(a. pp. 18–19.) Complete decomposition of $10^n - 1$, $n = 1$ to 42: and $10^n + 1$, $n = 1$ to 21. Some values of n are omitted.

A specimen is

$$10^{14} - 1 = 3^2.53.79.265371653,$$
$$10^{12} + 1 = 11.139.10\,5831\,3049.$$

(b. p. 19.) List of the specific prime factors f of $10^n - 1$, or the prime factors of the residue after separation of the analytical factors, for those values of n for which the complete decomposition is unknown, and omitting those values for which no factor is known, $n = 25$ to 243.

[p. 469] A specimen is

n	f
25	21401

The meaning seems to be, residue of $10^{25} - 1$ is $1 + 10^5 + 10^{10} + 10^{15} + 10^{20}$ and this contains the prime factor 21401; but it is not clear why this is the “specific prime factor.”

II. Prime factors of $a^n - 1$ for different values of a and n.

(a. p. 20) gives for 41 values of a (2, 3, &c. at intervals to 100) and for the following values of n the decompositions of the residues or specific factors of $a^n - 1$, viz. these are

$$\begin{aligned}
n = 1,&\ \ a - 1\\
2,&\ \ a + 1\\
3,&\ \ a^2 + a + 1\\
6,&\ \ a^2 - a + 1\\
4,&\ \ a^2 + 1\\
5,&\ \ a^4 + a^3 + a^2 + a + 1\\
10,&\ \ a^4 - a^3 + a^2 - a + 1\\
8,&\ \ a^4 + 1\\
12,&\ \ a^4 - a^2 + 1.
\end{aligned}$$

[1] Titlepage missing in my copy; but I find from Prof. Kummer’s notice of the work, Crelle, t. liii. (1857), p. 379, that it appeared as a Programm of the Stuttgart Gymnasium, Michaelmas, 1856, and was separately printed by Liesching and Co., Stuttgart.

A specimen is

	$a-1$	a^2-1	a^3-1	a^6-1	a^4-1	a^5-1
10	3^2	11	$3^2.37$	7.13	101	41.271
			$a^{10}-1$			
			9091		73.137	9901

(b. p. 21.) Specific prime factors for the numbers 2, 3, 5, 6, 7, 10, (the powers 4, 8, 9 being omitted as coming under 2 and 3), for the exponents 1 to 42.

A specimen is

	2^n-1	3^n-1	5^n-1	6^n-1	7^n-1	10^n-1
19	524287	1597.363889	$191.x$	$191.x$	$419.x$	$--$

where the x denotes that the other factor is not known to be prime. And so, where no number is given, as in $10^{19}-1$, it is not known whether the number $(=1+10+10^2+\cdots+10^{18})$ is or is not prime.

Addition, p. 22. For $a=2$, the complete decomposition of the prime factor of 2^n-1 is given for values of n, $=44, 45, \ldots$ at intervals to 156.

A specimen is

n	f
44	397.2113

viz.

$$2^{22}-2^{11}-\cdots-2^2+1 \;=\; 838861 \;=\; 397.2113.$$

$n=31$, Fermat's prime. $n=37$, the first case for which the decomposition is not given completely. $n=41$, the first case for which no factor is known.

[p. 470] 16. Gauss, *Tafel zur Cyclotechnie*, *Werke*, t. ii. pp. 478–495, shows, for 2452 numbers of the several forms a^2+1, a^2+4, $a^2+9, \ldots,$ a^2+81, the values of a such that the number in question is a product of prime factors no one of which exceeds 200, and exhibits all the odd prime factors of each such number. The table is in nine parts, *zerlegbare* a^2+1, *zerlegbare* a^2+4, &c., with to each part a subsidiary table, as presently mentioned. Thus a specimen is

zerlegbare a^2+9	
1	5
2	13
4	5.5
5	17
7	29
8	73
$\vdots$	
1411168679	5.5.13.17.17.89.113.157.173.197.197

viz.

$$\begin{aligned}&1^2+9, \text{ odd prime factor is } 5,\\ &2^2+9, \quad ,, \quad ,, \qquad 13,\\ &4^2+9, \quad ,, \quad ,, \text{ factors are } 5, 5,\end{aligned}$$

and so on.

And the subsidiary table is

5	1, 4, 79
13	2, 11, 41
17	5, 29, 46, 379, 1042

showing that the numbers a for which the largest factor is 5 are 1, 4, 79; those for which it is 13 are 2, 11, 41; and so on.

The object of the table is explained in the *Bemerkungen*, (*l.c.*, p. 523), by Schering, the editor of the volume, viz. it is to facilitate the calculation of the circular arcs the cotangents of which are rational numbers. To take a simple example, it appears to be by means of it that Gauss obtained, among other formulæ, the following:

$$\tfrac{\pi}{4} = 12 \text{ arctan } \tfrac{1}{38} + 8 \text{ arctan } \tfrac{1}{57} - 5 \text{ arctan } \tfrac{1}{239},$$

and

$$\tfrac{\pi}{4} = 12 \text{ arctan } \tfrac{1}{18} + 20 \text{ arctan } \tfrac{1}{57} + 7 \text{ arctan } \tfrac{1}{239} + 24 \text{ arctan } \tfrac{1}{268}.$$

[p. 471]

[F. 12. Divisors, etc.] continued. Prime Roots. The Canon Arithmeticus, Quadratic residues. Art. II.

17. *Prime Roots.* —Let p be a prime number; then there exist $\varpi(p-1)$ inferior integers g, such that all the numbers $1, 2, \ldots, p-1$ are, to the modulus p,

$$= 1,\, g,\, g^2,\, \ldots,\, g^{p-2} \quad (g^{p-1} \text{ is of course } = 1).$$

This being so, g is said to be a *prime root* of p; and moreover the several numbers g^α, where α is any number whatever less than and prime to $p-1$, constitute the series of the $\varpi(p-1)$ prime roots of p. It may be added that, if β be an integer number less than $p-1$, and having with it a greatest common measure $= k$, so that $\beta = \dfrac{p-1}{k} \cdot t$, then

$$(g^\beta)^{\frac{p-1}{k}} = g^{t(p-1)} = 1, \text{ (since } t \text{ is an integer, and } g^{p-1} = 1)$$

then g^β has the indicatrix $\dfrac{p-1}{k}$: the prime roots are those numbers which have the indicatrix $p-1$.

The like theory exists as to any number N of the form p^m or $2p^m$. There are here $\varpi(N)$, $= N\left(1-\dfrac{1}{p}\right)$ or $\tfrac{1}{2}N\left(1-\dfrac{1}{p}\right)$, in the two cases respectively, numbers less than N and prime to it; and we have then $\varpi(\varpi(N))$ numbers g such that, to the modulus N, all these numbers are $= 1, g, g^2, \ldots, g^{\varpi(N)-1}$ ($g^{\varpi(N)}$ is of course $= 1$). This being so, g may be regarded as a prime root of N ($= p^m$ or $2p^m$, as the case may be); and moreover the several numbers g^α, where α is any number whatever less than and prime to $\varpi(N)$, constitute the series of the $\varpi(\varpi(N))$ prime roots of N. Thus $N = 3^2 = 9$, $\varpi(N) = 6$; we have

$$1,\, 2^1,\, 2^2,\, 2^3,\, 2^4,\, 2^5,$$

$$= 1,\, 2,\, 4,\, 8,\, 7,\, 5, \text{ mod. } 9;$$

or the prime roots of 9 are 2^1 and 2^5, $= 2$ and 5.

So also $N = 2.3^2 = 18$, $\varpi(N) = 6$; we have

$$1,\, 5^1,\, 5^2,\, 5^3,\, 5^4,\, 5^5,$$

$$= 1,\, 5,\, 7,\, 17,\, 13,\, 11, \text{ mod. } 18;$$

and 5^1 and 5^5, $= 5$ and 11 are the prime roots of 18.

18. A small table of prime roots, $p = 3$ to 37, is given by

Euler, *Op. Arith. Coll.* t. i. pp. 525–526. The Memoir is entitled "Demonstrationes circa residua e divisione potestatum per numeros primos resultantia," pp. 516–537 (1772).

[p. 472] **19.** A table, p and p^m, 3 to 97, is given by

Gauss, "Disquisitiones Arithmeticae," 1801, (*Werke*, t. i. p. 468). This gives in each case a prime root, and shows the exponents in regard thereto of the several prime numbers less than p or p^m. Thus a specimen is

		2	3	5	7	11	13	17	19	23	29 &c.
27	2	1	*	5	16	13	8	15	12	11	
29	10	11	27	18	20	23	2	7	15	24	

viz. for 27 we have 2 a prime root, and $2 = 2^1$, $5 = 2^5$, $7 = 2^{16}$, $11 = 2^{13}$, &c.; and so also for 29 we have 10 a prime root, and $2 = 10^{11}$, $3 = 10^{27}$, $5 = 10^{18}$, &c.

20. Small tables are probably to be found in many other places; but the most extensive and convenient table is Jacobi's *Canon Arithmeticus*, the complete title of which is

Canon Arithmeticus sive tabulæ quibus exhibentur pro singulis numeris primis vel primorum potestatibus infra 1000 numeri ad datos indices et indices ad datos numeros pertinentes. Edidit C. G. J. Jacobi. Berolini, 1839. 4°.

The contents are as follows:

The following is a specimen of the principal tables:

$$p = 19,\ p - 1 = 2.3^2.$$

Numeri.

I	1	2	3	4	5	6	7	8	9
	10	5	12	6	3	11	15	17	18
1	9	14	7	13	16	8	4	2	1

Indices.

N	1	2	3	4	5	6	7	8	9
	18	17	5	16	2	4	12	15	10
1	1	6	3	13	11	7	14	8	9

[p. 473] where the first table gives the values of the powers of the prime root 10 (that 10 is the root appears by its index being given as = 1) to the modulus 19, viz. $10^1 = 10$, $10^2 = 5$, $10^3 = 12$, &c.; and the second table gives the index of the power to which the same prime root must be raised in order that it may be, to the modulus 19, congruent with a given number: thus $10^{18} = 1$, $10^{17} = 2$, $10^5 = 3$, &c. The units of the index or number, as the case may be, are contained in the top line of the table, and the tens or hundreds and tens in the left-hand column.

21. There is given by

Jacobi, Crelle, t. xxx. (1846), pp. 181, 182, a table of m' for the argument m, such that

$$1 + g^m = g^{m'}\ (\text{mod. } p),\ p = 7 \text{ to } 103,\ \text{and } m = 0 \text{ to } 102.$$

A specimen is

p	7	11	13	17	19	23	29	31	37	to 103
g	3	2	6	10	10	10	10	17	5	
$m = 11$	·	·	6	4	7	*	27	21	34	

for instance, $p = 19$, $1 + 10^{11} = 10^7$ (mod. 19).

Jacobi remarks that this table was calculated for him by his class during the winter course of 1836–37; and that, by means of the Canon Arithmeticus since published (in 1839), the same might easily be extended to all primes under 1000. In fact, for any such number p, putting any number of the table "Indices" $= m$, the next following number of the table gives the value of m'.

22. We have next, in Reuschle's Memoir (*ante*, No. 15), the following relating to prime roots:

C. Tafeln für primitive Wurzeln und Hauptexponenten, oder V. erweiterte und bereicherte Burkhardtsche Tafel, pp. 41–61, being divided into three parts; viz. these are

a. Table of the Hauptexponenten of the six roots 10, 5, 2, 6, 3, 7 for all prime numbers of the first 1000, together with the least primitive root of each of these numbers (pp. 42–46).

A specimen is as follows:

		10		5		2		6		3		7		w
p	$p-1$	e	n	e	n	e	n	e	n	e	n	e	n	
53	$2^2.13$	13	4	52	1	52	1	26	2	52	1	26	2	2

where e is the Hauptexponent or indicatrix of the root (10, 5, 2, 6, 3, 7, as the case may be), $n = \dfrac{p-1}{e}$, w the least primitive root; thus

$$p = 53,\ 10^{13} = 1,\ 5^{52} = 1,\ 2^{52} = 1,$$

[p. 474] (2 being accordingly the least prime root),

$$6^{26} = 1,\ 3^{52} = 1,\ 7^{26} = 1.$$

The number w of the last column is the least primitive root. It is, of course, not always (as in the present case) one of the numbers 10, 5, 2, 6, 3, 7 to which the table relates: the first exception is $p = 191$, $w = 19$: the highest value of w is $w = 21$, corresponding to $p = 409$.

b. The like table for the roots 10 and 2 for all prime numbers from 1000 to 5000, together with as convenient as possible a prime root (and in some cases two prime roots) for each such number (pp. 47–53).

A specimen is:

		10		2		
p	$p-1$	e	n	e	n	w
1289	$2^3.7.23$	92	14	161	8	6, 11

viz. here, mod. 1289, $10^{92} = 1$, $2^{161} = 1$; and two prime roots are 6, 11. We have thus by the present tables a prime root for every prime number not exceeding 5000.

c. The like table for the root 10 for all prime numbers between 5000 and 15000, (no column for w, nor any prime root given), pp. 53–61.

A specimen is

p	$p-1$	e	n
9859	2.3.31.53	3286	3

viz., mod. 9859, we have $10^{3286} = 1$. But in a large number of cases we have $n = 1$, and therefore 10 a prime root. For example,

9887   2.4943   9886   1.

23. For a composite number n, if $N = \varpi(n)$ be the number of integers less than n and prime to it, then if x be any number less than n and prime to it, we have $x^N = 1$ (mod. n). But we have in this case no analogue of a prime root—there is no number x, such that its several powers x', x'', ..., x^{N-1} (mod. n) are all different from unity; or, what is the same thing, there is for each value of x some submultiple of N, say N', such that $x^{N'} = 1$ (mod. n). And these several numbers N' have a least common multiple I, which is not $= N$, but is a submultiple of N; and this being so, then for all the several values of x, I is said to be the maximum indicator. For instance, $n = 12$, $N = \varpi(n)$; the numbers less than 12 and prime to it are 1, 5, 7, 11. We have, (mod. 12), $1^1 = 1$, $5^2 = 1$, $7^2 = 1$, $11^2 = 1$, or the values of N' are 1, 2, 2, 2; their least common multiple is 2, and we have accordingly $I = 2$: viz. $x^2 = 1$ (mod. 12) has the $\varpi(12)$ roots 1, 5, 7, 11. So $n = 24$, $\varpi(n) = 8$; the maximum indicator I is in this case also $= 2$.

[p. 475] A table of the maximum indicator $n = 1$ to 1000 is given by

Cauchy, *Exer. d'Analyse* &c., t. ii. (1841), pp. 36–40, contained in the "Mémoire sur la résolution des Equations indéterminées du premier degré en nombres entiers," pp. 1–40.

24. It thus appears that for a composite number n, the $\varpi(n)$ numbers less than n and prime to it cannot be expressed as $\equiv$ (mod. n) to the power of a single root; but for the expression of them it is necessary to employ two or more roots. A small table, $n = 1$ to 50, is given by

Cayley, "Specimen Table $M \equiv a^\alpha b^\beta$ (mod. N) for any prime or composite modulus," *Quart. Math. Journ.* vol. ix. (1868), pp. 95, 96, and folding sheet, [397].

A specimen is

Nos.	12
roots	5, 7
Ind.	2, 2
M.I.	2
$\varpi(\ )$	4
1	0, 0
2	
3	
4	
5	1, 0
6	
7	0, 1
8	
9	
10	
11	1, 1

viz. for the modulus 12 the roots are 5, 7, having respectively the indicators 2, 2, viz. $5^2 = 1$ (mod. 12), $7^2 = 1$ (mod. 12). Hence also the maximum indicator is $= 2$. $\varpi(= \varpi(n)) = 4$ is the number of integers less than 12 and prime to it, viz. these are 1, 5, 7, 11, which in terms of the roots 5, 7 and to mod. 12 are respectively $= 5^0.7^0$, $5^1.7^0$, $5^0.7^1$, and $5^1.7^1$.

25. *Quadratic Residues.* —In regard to a given prime number p, a number N is or is not a quadratic residue according as the index of N is even or odd, viz. g being a prime root and $N = g^\alpha$, then N is or is not a quadratic residue according as α is even or odd. But the quadratic residues can, of course, be obtained directly without the consideration of prime roots.

A small table, $p = 3$ to 97 and $N = -1$ and (prime values) 3 to 97, is given by

Gauss, "Disquisitiones Arithmeticae," 1801; Table II. (*Werke*, t. i. p. 469): I notice here a misprint in the top line of the original; it should be -1, $+2$, $+3$, &c., instead of

[p. 476] -1, $+2$, $+3$, &c.; the -1 is printed correctly on p. 499 of the French translation *Recherches Arithmétiques*, Paris, 1807 and on p. 469 of vol. I. of *Werke*, (Göttingen, 1870).

A specimen is

	-1	$+2$	$+3$	$+5$	$+7$	$+11$	$+13$	$+17$	$+19$	$+23$ &c.
19		–	–				–	–	–	–

viz. -1, 2, 3, 13 are not, 5, 7, 11, 17 &c. are, quadratic residues of 19. The residues taken positively and less than 19 are, in fact, 1, 4, 5, 6, 7, 9, 11, 16, 17.

The same table carried from $p = 3$ to 503, and prime values $N = 3$ to 997, is given by Gauss, *Werke*, t. ii. pp. 400–409. A specimen is

	2	3	5	7	11	13	17	19	23 &c.
19	–		–			–	–	–	

viz. the arrangement is the same, except only that the -1 column is omitted.

26. We have also by Gauss

"Disquisitiones Arithmeticae" Table III. (*Werke*, t. i. p. 470), for the conversion into decimals of a vulgar fraction, denominator p or p^μ, not exceeding 100. The explanation is given in Art. 314 *et seq.* of the same work.

But this table, carried to a greater extent, is given by Gauss, *Werke*, t. ii. pp. 412–434, "Tafel zur Verwandlung gemeiner Brüche mit Nennern aus dem ersten Tausend in Decimalbrüche;" viz. the denominators are here primes or powers of primes, p^μ up to 997.

To explain the table, consider a modulus p^μ (where μ may be $= 1$); if 10 is not a prime root of p^μ, consider a prime root r, which is such that $r^e = 10$ (mod. p^μ), e being a submultiple of $p^{\mu-1}(p-1)$; say we have $ef = p^{\mu-1}(p-1)$: then $10^f = 1$ (mod. p^μ). Consider any fraction $\dfrac{N}{p^\mu}$; then we may write $N = r^{ke+l}$ (mod. p^μ), k from 0 to $f-1$ and l from 0 to $e-1$, $N = 10^k r^l$, and consequently $\dfrac{N}{p^\mu}$ and $\dfrac{10^k r^l}{p^\mu}$ have the same mantissa (decimal part regarded as an integer); hence, in order to know the mantissa of every fraction whatever of $\dfrac{N}{p^\mu}$, it is sufficient to know the mantissa of $\dfrac{r^l}{p^\mu}$, that is, the mantissas of $\dfrac{1}{p^\mu}, \dfrac{r}{p^\mu}, \dfrac{r^2}{p^\mu}, \ldots, \dfrac{r^{e-1}}{p^\mu}$, or, what is the same thing, the mantissas of $\dfrac{10}{p^\mu}, \dfrac{10r}{p^\mu}, \ldots, \dfrac{10r^{e-1}}{p^\mu}$.

For instance, $p^\mu = 11$, $10^2 = 1$ (mod. 11), whence $f = 2$, $e = 5$; and taking $r = 2$, we have $10 = r^?$ (mod. 11).

[p. 477] The required mantissæ, denoted in the table by

$$(0),\ (1),\ (2),\ (3),\ (4),$$

are those of

$$\frac{10}{11},\ \frac{10.2}{11},\ \frac{10.2^2}{11},\ \frac{10.2^3}{11},\ \frac{10.2^4}{11},$$

viz. these fractions are respectively

$$\begin{array}{r} (0),\ (1),\ (2),\ (3),\ (4), \\ \cdot 9090\ldots,\ 1\cdot 8181\ldots,\ 3\cdot 6363\ldots,\ 7\cdot 2727\ldots,\ 14\cdot 5454\ldots, \end{array}$$

or their mantissæ are 90, 81, 63, 27, 54.

And we accordingly have as a specimen

$$11 \mid (1)\ldots 81,\ (2)\ldots 63,\ (3)\ldots 27,\ (4)\ldots 54,\ (0)\ldots 90.$$

Or again, as another specimen, $r = 2$

$$27 \mid (1)\ldots 740,\ (2)\ldots 481,\ (3)\ldots 962,\ (4)\ldots 925,\ (5)\ldots 851,\ (0)\ldots 370.$$

The table in this form extends to $p^\mu = 463$; the values of r (not given in the body of the table) are annexed, p. 420.

In the latter part of the table $p^\mu = 467$ to 997, we have only the mantissæ of $\dfrac{100}{p^\mu}$. A specimen is

547	1828153564	8994515539	3053016453	3820840950
	6398537477	1480804387	5685557586	8372943327
	2394881170	0182815356		

viz. the fraction $\frac{100}{547} = \cdot 182815\ldots$ has a period of 91, $= \frac{1}{6}.546$, figures.

[F. 13. The Pellian Equation.] Art. III.

27. The Pellian equation is $y^2 = ax^2 + 1$, a being a given integer number, which is not a square (or rather, if it be, the only solution is $y = 1$, $x = 0$), and x, y being numbers to be determined: what is required is the least values of x, y, since these, being known, all other values can be found. A small table $a = 2$ to 68 is given by

Euler, *Op. Arith. Coll.* t. i. p. 8. The Memoir is "Solutio problematum Diophanteorum per numeros integros," pp. 4–10, 1732–33. The form of the table is

a	$x\,(=p)$	$y\,(=q)$
2	2	3
3	1	2
5	4	9
⋮	⋮	⋮
68	4	33.

[p. 478] Even here, for some of the values of a, the values of x, y are extremely large; thus $a = 61$, $x = 226{,}153{,}980$, $y = 1{,}766{,}319{,}049$.

And probably tables of a like extent may be found elsewhere; in particular, a table of the solution of $y^2 = ax^2 \pm 1$ ($-$ when the value of a is such that there is a solution of $y^2 = ax^2 - 1$, and $+$ for other values of a), $a = 2$ to 135, is given by Legendre, *Théorie des Nombres*, 2nd ed. 1808, in the Table X. (one page) at the end of the work. For the before-mentioned number 61, the equation is $y^2 = 61x^2 - 1$, and the values are $x = 3805$, $y = 29718$; much smaller than Euler's values for the equation $y^2 = 61x^2 + 1$.

28. The most extensive table, however, is given by

Degen, *Canon Pellianus, sive Tabula simplicissimam æquationis celebratissimæ* $y^2 = ax^2 + 1$ *solutionem, pro singulis numeri dati valoribus ab* 1 *ad* 1000 *in usque numeris rationalibus, iisdemque integris exhibens. Auctore Carolo Ferdinando Degen. Hafniæ (Copenhagen) apud Gerhardum Bonnarum, 1817.* 8vo. pp. iv to xxiv and 1 to 112.

The first table (pp. 3–106) is entitled as "Tabula I. Solutionem Equationis $y^2 - ax^2 - 1 = 0$ exhibens." It, in fact, also gives the expression of $\sqrt{a}$ as a continued fraction; thus a specimen is

209	14	2	5	3	(2)
	1	13	5	8	11
	3220				
	46551				

Here the first line gives the continued fraction, viz.

$$\sqrt{209} = 14 + \cfrac{1}{2 + \cfrac{1}{5 + \cfrac{1}{3 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{2 + \cfrac{1}{28 + \cfrac{1}{2 + \&c.}}}}}}}}},$$

the period being $(2, 5, 3, 2, 3, 5, 2)$ indicated by $2, 5, 3\,(2)$. [The number of terms in the period is here odd, but it may be even; for instance, the period $(1, 1, 5, 5, 1, 1)$ is indicated by $1, 1\,(5, 5)$.]

The second line contains auxiliary numbers presenting themselves in the process; thus, if $R^2 = 209$, we have $R = 14 + \dfrac{1}{\beta}$,

$$\beta = \frac{1}{R-14} = \frac{1(R+14)}{209-14^2} = \frac{R+14}{13} = 2 + \frac{1}{\beta'},$$

$$\beta' = \frac{13}{R-12} = \frac{13(R+12)}{209-12^2} = \frac{R+12}{5} = 5 + \frac{1}{\gamma},$$

$$\gamma = \frac{5}{R-13} = \frac{5(R+13)}{209-13^2} = \frac{R+13}{8} = 3 + \frac{1}{\delta},$$

&c.

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 479–503 (Papers 611–612)

Transcribed from the Original Publications

Contents

611. Report of the Committee on Mathematical Tables (continued)

[Continued from earlier pages of the same paper.]

[p. 479] where the second line 1, 13, 5, ... shows the numerical factors of the third column. The value of this second line as a result is not very obvious.

The third line gives x, and the fourth line y.

29. The second table, pp. 109–112, is entitled "Tabula II. Solutionem aequationis $y^2 - ax^2 + 1 = 0$, quotiescunque valor ipsius a talem admiserat, exhibens"; viz. it is remarked that this is only possible (but see infra) for those values of a which in Table I. correspond to a period of an even number of terms, as shown by two equal numbers in brackets; thus $a = 13$, the period of $\sqrt{13}$ given in Table I. is $(1, 1, 1, 1)$ as shown by the top line $3, 1\ (1, 1)$, and accordingly 13 is one of the numbers in Table II.; and we have there

$$13 \mid \frac{5}{18}.$$

Or take another specimen, $241 \mid \dfrac{457422\,5}{71011068}$, viz. the first line gives the value of x, and the second line the value of y (least values), for which $y^2 - ax^2 = -1$.

It is to be noticed that $a = 2$ and $a = 5$, for which we have obviously the solutions $(x = 1, y = 1)$ and $(x = 1, y = 2)$ respectively, are exceptional numbers not satisfying the test above referred to; and (apparently for this reason) the values in question, 2 and 5, are omitted from the table.

30. **Cayley**, "Table des plus petites solutions impaires de l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5$ (mod. 8)." *Crelle*, t. LIII. (1857), page 371 (one page), [231].

As regards the theory of quadratic forms, it is important to know whether for a given value of D ($\equiv 5$, mod. 8) there does or does not exist a solution, in odd numbers, of the equation $x^2 - Dy^2 = 4$. As remarked in the paper, "Note sur l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5$ (mod. 8)," pp. 369–371, [231], this can be determined for values of D of the form in question up to $D = 997$ by means of Degen's Table; and the solutions, when they exist, of the equation $x^2 - Dy^2 = 4$, as also of the equation $x^2 - Dy^2 = -4$, can be obtained up to the same value of D. Observe that when the equation $x^2 - Dy^2 = -4$ is possible, the equation $x^2 - Dy^2 = +4$ is also possible, and that its least solution is obtained very readily from that of the other equation; it is therefore sufficient to tabulate the solution of $x^2 - Dy^2 = \pm 4$, the sign being $-$ when the corresponding equation is possible, and being in other cases $+$. Hence the form of the Table: viz. as a specimen we have

D	$\pm$	x	y
757	imposs.		
765	+	83	3
773	−	139	5
781	imposs.		

[p. 480] that is, if $D = 757$ or 781, there is no solution of either $x^2 - Dy^2 = +4$ or -4; if $D = 765$, there is a solution $x = 83$, $y = 3$ of $x^2 - Dy^2 = +4$, but none of $x^2 - Dy^2 = -4$; if $D = 773$, there is a solution $x = 139$, $y = 5$ of $x^2 - Dy^2 = -4$, and therefore also a solution of $x^2 - Dy^2 = +4$; and so in other cases.

[F. 14. *Partitions.*] Art. IV.

31. The problem of Partitions is closely connected with that of Derivations. Thus if it be asked in how many ways can the number n be expressed as a sum of three parts, the parts being

0, 1, 2, 3, and each part being repeatable an indefinite number of times, it is clear that n is at most $= 9$, and that for the values of n, $= 0, 1, \dots, 9$ shown by the top line of the annexed table, the number of partitions has the values shown by the bottom line thereof:

0	1	2	3	4	5	6	7	8	9
a^3	a^2b	a^2c	a^2d	abd	acd	ad^2	bd^2	cd^2	d^3
		ab^2	abc	b^2d	bcd	c^2d	cd^2		
			b^3	b^2c	bc^2	c^3			
1	1	2	3	3	3	3	2	1	1

32. But taking a, b, c, d to stand for 0, 1, 2, 3 respectively, the actual partitions of the required form are exhibited by the literal terms of the table (these being obtained, each column from the preceding one, by the method of derivations, or say by the rule of the last and last but one), and the numbers of the bottom line are simply the number of terms in the several columns respectively.

A set of such literal tables, say of tables $\begin{pmatrix} a, b, c, \dots, k \\ 0, 1, 2, \dots, m \end{pmatrix}^n$ for different values of n and m (where the number of letters is $= m + 1$), would be extremely interesting and valuable. The tables for a given value of m and for different values of n are, it is clear, the proper foundation of the theory of the binary quantic $(a, b, c, \dots, k \,\cancel{\wr}\, x, 1)^m$, which corresponds to such value of m. Prof. Cayley regrets that he has not in his covariant tables given in every case the complete series of literal terms; viz. the literal terms which have zero coefficients are, for the most part, though not always, omitted in the expressions of the several covariants.

33. But the question at present is as to the *number* of terms in a column, that is, as to the number of the partitions of a given form: the analytical theory has been investigated by Euler and others. The expression for the number of partitions is usually obtained as equal to the coefficient of x^n in the development, in ascending powers of x, of a given rational function of x: for instance, if there is no limitation as to the number of the parts, but if the parts are $1, 2, 3, \dots, m$ (viz. a part may have any value not exceeding m), each part being repeatable an indefinite number of times, then

$$\text{Number of partitions of } n = \text{coefficient of } x^n \text{ in } \frac{1}{(1-x)(1-x^2)(1-x^3)\dots(1-x^m)},$$

[p. 481] and we can, by actual development, obtain for any given values of m, n the number of partitions.

These have been tabulated $m = 1, 2, \dots, 20$, and $m = \infty$ (viz. there is in this case no limit as to the largest part), and $n = 1$ to 59, by

Euler, *Op. Arith. Coll.* t. I. pp. 97–101, given in the paper "De Partitione Numerorum," pp. 73–101, (1750); the heading is "Tabula indicans quot variis modis numerus n e numeris $1, 2, 3, 4, \dots, m$, per additionem exhibi potest, seu exhibens valores formulae $n^{(m)}$." The successive lines are, in fact, the coefficients of the several powers $x^0, x^1, \dots, x^n$ in the expansions of the functions

$$\frac{1}{1-x}, \quad \frac{1}{1-x \cdot 1-x^2}, \quad \dots, \quad \frac{1}{1-x \cdot 1-x^2 \dots 1-x^m}.$$

34. The generating function for any given value of m is, it is clear, $\dfrac{1}{1-x^m}$ multiplied by that for the next preceding value of m, and it thus appears how each line of the table is calculated from that which precedes it. The auxiliary numbers are printed; thus a specimen is

m	Valores numeri n.										
	0	1	2	3	4	5	6	7	8	9	10
4	1	1	2	3	5	6	9	11	15	18	23
					1	1	2	3	5	7	9
5	1	1	2	3	5	7	10	13	18	23	30

viz. suppose the numbers in the second 4-line known: then simply moving these each five steps onward we have the (auxiliary) numbers of the first 5-line; and thence by a mere addition the required series of numbers shown by the second 5-line. And similarly from this is obtained the second 6-line, and so on.

35. More extensive tables are contained in the memoir by

Marano, *Sulle leggi delle derivate generali delle funzioni di funzioni et sulla teoria delle forme di partizione dei numeri interi*, (4°. Genova, 1870), pp. 1–281; and three tables paged separately, described merely as "Tavole dei numeri $C_{q,r}$, $S_{q,r}$, $S'_{q,r}$ citate nel testo colle indicazioni di Tavole I., II., III., ai n. 77, 79, 81"; viz. the reader is referred to these articles for the explanations of what the tabulated functions are; and there is not even then any explicit statement, but the investigation itself has to be studied to make out what the tables are. It is, in fact, easier to make this out from the tables themselves; the explanation is as follows:

[p. 482] Table I. (16 pages) is, in fact, Euler's table, showing in how many ways the number n can be made up with the parts $1, 2, 3, \dots, m$; but the extent is greater, viz. n is from 1 to 103, and m from 1 to 102. The auxiliary numbers given in Euler's table are omitted, as also certain numbers which occur in each successive line; thus a specimen is

$n=$	1	2	3	4	5	6	7	8	9	10 &c.
$C_{n,1}$	1	0	0	0	0	0	0	0	0	0
$C_{n,2}$		1	1	1	1	1	1	1	1	1
$C_{n,3}$			2	2	3	3	4	4	5	5
$C_{n,4}$				3	4	5	7	8	10	12
$C_{n,5}$					5	6	9	11	15	18

where the line $C_{n,4}$, (ways of making up $n-1$ with the parts $1, 2, 3, 4$) is $1, 1, 2, 3, 5, 6, 9, 11, 15, 18$, &c., viz. we read from the corner diagonally downwards as far as the 6, and then horizontally along the line: this saves a large number of figures. The table is printed in ordinary quarto pages, which are taken to come in in tiers of seven, five, and three pages one under the other, as shown by a prefixed diagram; and the necessity of a large folding plate is thus avoided.

The successive lines give, in fact, the coefficients in the expansions of

$$\frac{1}{1-x}, \quad \frac{1}{1-x\cdot 1-x^2}, \quad \frac{1}{1-x\cdot 1-x^2\cdot 1-x^3}, \quad \dots, \quad \frac{1}{1-x\cdot 1-x^2\dots 1-x^m},$$

each expanded as far as x^{102}.

Table II. (6 pages). The successive lines give the coefficients in the expansions of

$$S, \quad \frac{S}{1-x}, \quad \frac{S}{1-x\cdot 1-x^2}, \quad \dots, \quad \frac{S}{1-x\cdot 1-x^2\dots 1-x^m},$$

where

$$S = \frac{1}{(1-x)(1-x^2)(1-x^3)\dots \text{ad inf.}},$$

each expanded as far as x^{50}, and further continued as regards the first ten lines, that is, the expansions of

$$S, \quad \frac{S}{1-x}, \quad \frac{S}{1-x\cdot 1-x^2}, \quad \dots, \quad \frac{S}{1-x\cdot 1-x^2\dots 1-x^9},$$

each as far as x^{100}.

Table III. (2 pages). The successive lines give the coefficients in the expansions of

$$S^2, \quad \frac{S^2}{1-x}, \quad \frac{S^2}{1-x\cdot 1-x^2}, \quad \dots, \quad \frac{S^2}{1-x\cdot 1-x^2\dots 1-x^m},$$

each expanded as far as x^{60}.

[p. 483] 36. As regards Tables II. and III., the analytical explanations have been given in the first instance; but it is easy to see that the tables give numbers of partitions. Thus, in Table II., the second line gives the coefficients in the development of

$$\frac{1}{(1-x)^2(1-x^2)(1-x^3)\ldots},$$

viz. these are $1, 2, 4, 7, 12, 19, 30, \ldots$ being the number of ways in which the numbers $0, 1, 2, 3, 4$, &c. respectively can be made up with the parts $1, 1', 2, 3, 4$, &c.; thus

Partitions.	No. =
0	1
1 $1'$	2
2 $1, 1$ $1, 1'$ $1', 1'$	4
3 $2, 1$ $2, 1'$ $1, 1, 1$ $1, 1, 1'$ $1, 1', 1'$ $1', 1', 1'$	7
&c.	&c.

Similarly, the third line shows the number of ways in which these numbers respectively can be made up with the parts $1, 1', 2, 2', 3, 4, 5$, &c.; the fourth line with the parts $1, 1', 2, 2', 3, 3', 4, 5$, &c.; and so on.

And in like manner in Table III., the first line shows the number of ways when the parts are $1, 1', 2, 2', 3, 3', \ldots$; the second line when they are $1, 1', 1'', 2, 2', 3, 3'$, &c.; the third when they are $1, 1', 1'', 2, 2', 2'', 3, 3'$, &c.; and so on.

It is clear that the series of tables might be continued indefinitely, viz. there might be a Table IV. giving the developments of

$$S^3, \quad \frac{S^3}{1-x}, \quad \frac{S^3}{1-x\cdot 1-x^2}, \quad \text{and so on.}$$

An interesting table would be one composed of the first lines of the above series, viz. a table giving in its successive lines the developments of S, S^2, S^3, S^4, &c.

There are throughout the work a large number of numerical results given in a quasi-tabular form; but the collection of these, with independent explanations of the significations of the tabulated numbers, would be a task of considerable labour.

[p. 484]

[F. 15. *Quadratic forms a^2+b^2, &c., and Partitions of Numbers into squares, cubes, and biquadrates.*] Art. V.

37. The forms here referred to present themselves in the various complex theories. Thus $N = a^2+b^2, = (a+bi)(a-bi)$; this means that, in the theory of the complex numbers $a+bi$ (a and b integers), N is not a prime but a composite number.

It is well known that an ordinary prime number $\equiv 3$, mod. 4, is not expressible as a sum a^2+b^2, being, in fact, a prime in the complex theory as well as in the ordinary one: but that an ordinary prime number $\equiv 1$, mod. 4, is (in one way only) $= a^2+b^2$; so that it is in the complex theory a composite number. A number whose prime factors are each of them $\equiv 1$, mod. 4, or which contains, if at all, an even number of times any prime factor $\equiv 3$, mod. 4, can be expressed in a variety of ways in the form a^2+b^2; but these are all easily deducible from the expressions in the form in question of its several factors $\equiv 1$, mod. 4, so that the required table is a table of the form $p = a^2+b^2$, p an ordinary prime number $\equiv 1$ mod. 4: a and b are one of them odd, the other even; and to render the decomposition definite a is taken to be odd.

$p = a^2+b^2$; viz. decomposition of the primes of the form $4n+1$ into the sum of two squares: a table extending from $p = 5$ to 11981 (calculated by Zornow) is given by

Jacobi, *Crelle*, t. XXX. (1846), pp. 174–176.

This is carried by Reuschle, as presently mentioned, up to $p = 24917$. Reuschle notices that $2713 = 3^2+52^2$ is omitted, also $6997 = 39^2+74^2$, and that 8609 should be $= 47^2+80^2$.

38. Similarly, primes of the form $6n+1$ are expressible in the form $p = a^2+3b^2$. Observe that, ω being an imaginary cube root of unity, this is connected with $p = (a+b\omega)(a+b\omega^2)$, $= a^2-ab+b^2$, viz. we have $4p = (2a-b)^2+3b^2$; or the form a^2+3b^2 is connected with the theory of the complex numbers composed of the cube roots of unity.

$p = a^2+3b^2$; viz. decomposition of the primes of the form $6n+1$ into the form $p = a^2+3b^2$: a table extending from $p = 7$ to 12007 (calculated also by Zornow) is given by

Jacobi, *Crelle*, t. XXX. (1846), ut supra, pp. 177–179.

This is carried by Reuschle up to $p = 13369$, and for certain higher numbers up to 49999, as presently mentioned. Reuschle observes that $6427 = 80^2+3\cdot 3^2$ is by accident omitted, and that 6481 should be $= 41^2+3\cdot 40^2$.

39. Again, primes of the form $8n+1$ are expressible in the form $p = a^2+2b^2$ (or say $= c^2+2d^2$), the theory being connected with that of the complex numbers composed with the 8th roots of unity (fourth root of -1, $= \sqrt[4]{-1}$).

$p = c^2+2d^2$; viz. decomposition of primes of the form $8n+1$ into the form c^2+2d^2:
[p. 485] a table extending from $p = 17$ to 5943 (extracted from a MS. table calculated by Struve) is given by

Jacobi, *Crelle*, t. XXX. (1846), ut supra, p. 180.

This is carried by Reuschle up to $p = 12377$, and for certain higher numbers up to 24889, as presently mentioned.

40. Reuschle's tables of the forms in question are contained in the work:

Reuschle, *Mathematische Abhandlung*, &c. (see ante No. 15), under the heading "B. Tafeln zur Zerlegung der Primzahlen in Quadrate" (pp. 22–41). They are as follows:

Table III. for the primes $6n+1$.

The first part gives $p = A^2+3B^2$ and $4p = L^2+27M^2$, from $p = 7$ to 5743. The table gives A, B, L, M; those numbers which have 10 for a cubic residue are distinguished by an asterisk. A specimen is

p	A	B	L	M
37*	5	2	11	1

viz. $37 = 5^2+3\cdot 2^2$, $148 = 11^2+27\cdot 1^2$; the asterisk shows that $x^3 \equiv 10$ (mod. 37) is possible: in fact $34^3 \equiv 10$ (mod. 37).

The second part gives $p = A^2+3B^2$ only, from $p = 5749$ to 13669. The table gives A, B; and the asterisk implies the same property as before.

The third part gives $p = A^2+3B^2$, but only for those values of p which have 10 for a cubic residue, viz. for which $x^3 \equiv 10$ (mod. p) is possible, from $p = 13689$ to 49999. The table gives A, B; the asterisk, as being unnecessary, is not inserted.

Table IV. for the primes $4n+1$ in the form A^2+B^2, and for those which are also $8n+1$ in the form C^2+2D^2.

The first part gives $p = A^2 + B^2, = C^2 + 2D^2$, from $p = 5$ to 12377. The table gives A, B, C, D; those numbers which have 10 for a biquadratic residue, viz. for which $x^4 \equiv 10$ (mod. p) is possible, are distinguished by an asterisk; those which have also 10 for an octic residue, viz. for which $x^8 \equiv 10$ (mod. p) is possible, by a double asterisk. A specimen is

p	A	B	C	D
229	15	2	$---$	$---$
233	13	8	15	2
241**	15	4	13	6

The second part gives $p = A^2 + B^2$, from $p = 12401$ to 24917 for all those values of p which have 10 for a biquadratic residue ($x^4 \equiv 10$ (mod. p) possible). The table gives A, B; those values of p which have 10 for an octic residue, viz. for which $x^8 \equiv 10$ (mod. p) is possible, are distinguished by an asterisk.

[p. 486] The third part gives $p = C^2 + 2D^2$, from $p = 12641$ to 24889 for all those values of p which have 10 for a biquadratic residue. The table gives C, D; those values of p which have 10 as an octic residue are distinguished by an asterisk.

41. A table by Zornow, *Crelle*, t. XIV. (1835), pp. 279, 280 (belonging to the Memoir "De Compositione numerorum e Cubis integris positivis," pp. 276–280), shows for the numbers 1 to 3000 the least number of cubes into which each of these numbers can be decomposed. Waring gave, without demonstration, the theorem that every number can be expressed as the sum of at most 9 cubes. The present table seems to show that 23 is the only number for which the number of cubes is $= 9$ $(= 2 \cdot 2^3 + 7 \cdot 1^3)$; that there are only fourteen numbers for which the number of cubes is $= 8$, the largest of these being 454; and hence that every number greater than 454 can be expressed as a sum of at most 7 cubes; and further, that every number greater than 2183 can be expressed as a sum of at most 6 cubes. A small subsidiary table (p. 276) shows that the number of numbers requiring 6 cubes gradually diminishes — e.g. between 12^3 and 13^3 there are seventy-five such numbers, but between 13^3 and 14^3 only sixty-four such numbers; and the author conjectures "that for numbers beyond a certain limit every number can be expressed as a sum of at most 5 cubes."

42. For the decomposition of a number into biquadrates we have

Bretschneider, "Tafeln für die Zerlegung der Zahlen bis 4100 in Biquadrate," *Crelle*, t. XLVI. (1853), pp. 3–23.

Table I. gives the decompositions, thus:

N	1^4	2^4	3^4	4^4	5^4
696	6	1	2	2	
$\vdots$					
3251		3	8		

viz. $696 = 6 \cdot 1^4 + 1 \cdot 2^4 + 2 \cdot 3^4 + 2 \cdot 4^4$, &c.

And Table II. enumerates the numbers which are sums of at least $2, 3, 4, \dots, 19$ biquadrates. There is at the end a summary showing for the first 4100 numbers how many numbers there are of these several forms respectively: 28 numbers are each of them a sum of 2 biquadrates, 75 a sum of $3, \dots$, 7 a sum of 19 biquadrates. The seven numbers, each of them a sum of 19 biquadrates, are $79, 159, 239, 319, 399, 479, 559$.

[F. 16. *Binary, Ternary, etc. quadratic and higher forms.*] Art. VI.

43. Euler worked with the quadratic forms $px^2 \pm qy^2$ (p and q integers), particularly in regard to the forms of the divisors of such numbers. It will be sufficient to refer to his memoir:

[p. 487] **Euler**, "Theoremata circa divisores numerorum in hac forma $pa^2 \pm qb^2$ contentorum," (*Op. Arith. Coll.* pp. 35–61, 1744), containing fifty-nine theorems, exhibiting in a quasi-tabular form the linear forms of the divisors of such numbers. As a specimen:

"Theorema 13. Numerorum in hac forma $a^2 + 7b^2$ contentorum divisores primi omnes sunt vel 2, vel 7, vel in una sex formularum

$$\begin{array}{ll} 28m+1, & 28m+11, \\ 28m+9, & 28m+15, \\ 28m+25, & 28m+23, \end{array}$$

seu in una harum trium

$$14m+1, \quad 14m+9, \quad 14m+11,$$

sunt contenti"; viz. the forms are the three $14m+1$, $14m+9$, $14m+11$.

But Euler did not consider, or if at all very slightly, the trinomial forms $ax^2 + bxy + cy^2$, nor attempt the theory of the reduction of such forms. This was first done by Lagrange in the memoir

Lagrange, *Mém. de Berlin*, 1773. And the theory is reproduced by

Legendre, *Théorie des Nombres*, Paris, 1st ed. 1798; 2nd ed. 1808, § 8, "Réduction de la formule $Ly^2 + Myz + Nz^2$ à l'expression la plus simple," (2nd ed. pp. 61–67).

44. But the classification of quadratic forms, as established by Legendre, is defective as not taking account of the distinction between proper and improper equivalence; and the ulterior theory as to orders and genera, and the composition of forms (although in the meantime established by Gauss), are not therein taken into account; for this reason the Legendre's Tables I. to VIII. relating to quadratic forms, given after p. 480 (thirty-two pages not numbered), are of comparatively little value, and it is not necessary to refer to them in detail.

The complete theory was established by

Gauss, *Disquisitiones Arithmeticae*, 1801.

It is convenient to refer also to the following memoir

Lejeune Dirichlet, "Recherches sur diverses applications de l'Analyse à la théorie des Nombres," *Crelle*, t. XIX. (1839), p. 338, [*Ges. Werke*, t. I. p. 427], as giving a succinct statement of the principle of classification, and in particular a table of the characters of the genera of the properly primitive order, according to the four forms $D = PS^2$, $P \equiv 1$ or 3 (mod. 4), and $D = 2PS^2$, $P \equiv 1$ or 3 (mod. 4), of the determinant.

45. Tables of quadratic forms arranged on the Gaussian principle are given by

Cayley, *Crelle*, t. LX. (1862), pp. 357–372, [335]; viz. the tables are—

Table I. des formes quadratiques binaires ayant pour déterminants les nombres négatifs depuis $D = -1$ jusqu'à $D = -100$. (Pp. 360–363: [*Coll. Math. Papers*, t. V, pp. 144–147].)

[p. 488] A specimen is

D	Classes	α	β	δ	ϵ	$\delta\epsilon$	Cp
$=26$	$1, 0, 26$	$+$	$+$			$+$	1
	$3, -1, 9$	$+$	$*$			$+$	g^2
	$3, 1, 9$	$+$				$+$	g^4
	$5, 2, 6$						g
	$2, 0, 13$	$-$				$-$	g^3
	$5, -2, 6$	$-$				$-$	g^5

where α, β denote, as there explained, the characters in regard to the odd prime factors of D; δ, ϵ, $\delta\epsilon$ those in regard to the numbers 4 and 8. The last column shows that the forms in the two genera respectively are $1, g^2, g^4$ and g, g^3, g^5, where $g^6 = 1$, viz. the form g, six times compounded, gives the principal form $(1, 0, 26)$.

Table II. des formes quadratiques binaires ayant pour déterminants les nombres positifs non-carrés depuis $D = 2$ jusqu'à $D = 99$. (Pp. 364–369: [*l.c.*, pp. 148–153].)

The arrangement is the same, except that there is a column "Périodes" showing, in an easily understood abbreviated form, the period of each form. Thus $D = 7$, the period of the principal form $(1, 0, -7)$, is given as $1, 7, -3, \ddot{1}, 2, \ddot{1}, -3, 7, 1$, which represents the series of forms $(1, 2, -3)$, $(-3, 1, 2)$, $(2, 1, -3)$, $(-3, 2, 1)$.

Table III. des formes quadratiques binaires pour les treize déterminants négatifs irréguliers du premier millier. (Pp. 370–372: [*l.c.*, pp. 154–156].)

The arrangement is the same as in Table I. It may be mentioned that the thirteen numbers, and the forms for the principal genus for these numbers, respectively are:

	Principal genus
$576, 580, 820, 900$	$(1, e^2)(1, {e_1}^2)$
884	$(1, e^2)(1, i^2, i^4, i^6)$
$243, 307, 339, 459, 675, 891$	$(1, d, d^2)(1, d_1, {d_1}^2)$
$755, 974$	$(1, d, d^2)(1, d_1, {d_1}^2)(1, e^2)$

where $d^3 = {d_1}^3 = 1$, $e^4 = {e_1}^4 = 1$, $i^8 = 1$, viz. $(1, e^2)(1, {e_1}^2)$ denotes four forms, $1, e^2, {e_1}^2, e^2{e_1}^2$; and so in the other cases.

46. Gauss must have computed quadratic forms to an enormous extent; but, for the reasons (rather amusing ones) mentioned in a letter of May 17, 1841, to Schumacher (quoted in Prof. Smith's Report on "The Theory of Numbers," Brit. Assoc. Report for 1862, p. 526, [and Smith's *Coll. Math. Papers*, t. I. p. 261]), he did not preserve his results in detail, but only in the form appearing in the

"Tafel der Anzahl der Classen binärer quadratischer Formen," *Werke*, t. II. pp. 449–476; see editor's remarks, pp. 521–523.

[p. 489] This relates almost entirely to negative determinants, only three quarters of p. 475 and p. 476 to positive ones; for negative determinants, it gives the number of genera and classes, as also the index of irregularity for the determinants of the hundreds 1 to 30, 43, 51, 61, 62, 63, 91 to 100, 117 to 120; then, in a different arrangement, for the thousands 1, 3 and 10, for the first 800 numbers of the forms $-(15n + 7)$ and $-(15n + 13)$; also for some very large numbers, and for positive determinants of the hundreds 1, 2, 3, 9, 10, and for some others.

A specimen is

Centas I.
G II. (58) . . . (280)
1. 5, 6, 8,
9, 10, 12,
13, 15, 16,
18, 22, 25,
28, 37, 58,
2. 14, 17, 20,
Summa 233 . . . 477
Irreg. 0 Impr. 74

viz. this shows, as regards the negative determinants 1 to 100, that the determinants belonging to G II. 1, viz. those which have two genera each of one class, are $5, 6, 8, 9$, &c., in all fifteen determinants; those belonging to G II. 2, viz. those which have two genera each of two classes, are $14, 17, 20$, &c.; and so on. The head numbers (58) . . . (280) show the number of determinants, each having two genera, and the number of classes; thus,

G II.	$1 \times 15 =$	15
	$2 \times 17 =$	34
	$3 \times 17 =$	51
	$4 \times \ \ 6 =$	24
	$5 \times \ \ 2 =$	10
	$6 \times \ \ 1 =$	6
	58	140
		$\times 2$
		$= 280$;

and the bottom numbers show the total number of genera and of classes, thus

G I.	$17 \times 1 = 17$		61
II.	$58 \times 2 = 116$		280
IV.	$25 \times 4 = 100$		136
	100	233	477

[p. 490] viz. seventeen determinants, each of one genus, and together of sixty-one classes; fifty-eight determinants, each of two genera, and together of 280 classes; and twenty-five determinants, each of four genera, and together 136 classes, give in all 233 genera and 477 classes. These are exclusive of 74 classes belonging to the improperly primitive order; and the number of irregular determinants (in the first hundred) is $= 0$.

The irregular determinants are indicated thus:

243(∗3∗),
307(∗3∗), 339(∗3∗),
459(∗),
576(∗2∗), 580(∗2∗),
675(∗3∗),
755(∗3∗),
891(∗3∗), 820(∗2∗), 900(∗2∗), 884(∗2∗), 974(∗3∗),
∗3∗ 243, 307, 339, 459?, 675, 755, 891,
∗2∗ 576, 580, 820, 884, 900, 974,

which is a notation not easily understood.

As regards the positive determinants, a specimen is

Centas I.
Excedunt determinantis
quadrati 10.

G I. . . . (12),
1. 2, 5, 13,
17, 29, 41,
53, 61, 73,
89, 97,
3. 37

viz. in the first hundred, the positive determinants having one genus of one class are $2, 5, 13$, &c. . . . (eleven in number); that having one genus of three classes is 37, (one in number); $11+1 = 12$. The irregular determinants, if any, are not distinguished.

47. *Binary cubic forms.* — The earliest table is given by

Arndt, "Tabelle der reducirten binären kubischen Formen und Klassen für alle negativen Determinanten $-D$ von $D = 3$ bis $D = 2000$," *Grunert's Archiv*, t. XXXI. 1858, pp. 369–448.

The memoir is a sequel to one in t. XVII. (1851). The binary cubic form (a, b, c, d), of determinant D $(= (bc - ad)^2 - 4(b^2 - ac)(c^2 - bd))$, is said to be reduced when its characteristic $\phi, = (A, B, C), = (2(b^2 - ac), bc - ad, 2(c^2 - bd))$, is a reduced quadratic form, that is, when in regard to absolute values B is not $> \frac{1}{2}A$, C not $< A$.
[p. 491] A specimen is

D	Reduced forms, with characters		Classes	
44	$(0, 1, 0, -11)$	$(1, -1, -2, 0)$	$(0, -1, 0, 11)$	$(0, -2, 1, 1)$
	$(2, 0, 22)$		$(6, 8)$	

Two subsidiary tables are given, pp. 351, 352, and 353–368.

48. It appeared suitable to remodel a part of this table in the manner made use of for quadratic forms in my tables above referred to; and it is accordingly divided into the three tables given by

Cayley, *Quart. Math. Journ.* t. XI. (1871), where the notation &c. is explained, pp. 251–261. [496]; viz. these are: —

Table I. of the binary cubic forms, the determinants of which are the negative numbers $\equiv 0$ (mod. 4) from -4 to -400 (pp. 251–258 [*Coll. Math. Papers*, t. VIII., pp. 55–61]).

A specimen is

Det. 4×-11.	Classes.	Order.	Charact.	Comp.
	$0, 1, 0, -11$	on	$1, 0, 11$	1
	$0, 2, -1, -2$	*pp pp*	$3, 1, 4$	d
	$0, 2, 1, 1$		$3, -1, 4$	d^2

Table II. of the binary cubic forms the determinants of which (taken positively) are $\equiv 1$ (mod. 4) from -3 to 99, the original heading is here corrected, [*l.c.*, pp. 61, 62]; and

Table III. of the binary cubic forms the determinants of which are the negative numbers -972, -1228, -1336, -1836, and -2700, [*l.c.*, pp. 63, 64]; viz. $-972 = 4\times-243, \ldots, -2700 = 4\times-675$, where $-243, \ldots, -675$ are the first six irregular numbers for quadric forms.

$4\times-675, = -2700$ is beyond the limits of Arndt's tables, and for this number the calculation had to be made anew; the table gives nine classes $(1, d, d^2)(1, d_1, {d_1}^2)$ of the order *ip* on *pp*, but it is remarked that there may possibly be other cubic classes based on a non-primitive characteristic; the point was left unascertained.

49. The theory of ternary quadratic forms was discussed and partially established by Gauss in the *Disquisitiones Arithmeticae.* It is proper to recall that a ternary quadratic form is either determinate, viz. always positive, such as $x^2+y^2+z^2$, or always negative, such as $-x^2-y^2-z^2$; or else it is indeterminate, such as $x^2+y^2-z^2$. But as regards determinate forms, the negative ones are derived from the positive ones by simply reversing the signs of all the coefficients, so that it is sufficient to attend to the positive forms; and practically the two cases are positive forms (meaning thereby positive determinate forms) and indeterminate forms; but the theory for positive forms was first established completely, and so as to enable the formation of tables, in the work

Seeber, *Ueber die Eigenschaften der positiven ternären quadratischen Formen*, (4to. Freiburg, 1831),
[p. 492] which is reviewed by Gauss in the *Gött. Gelehrte Anzeigen*, 1831, July 9 (see Gauss, *Werke*, t. II. pp. 188–193). The author gives (pp. 220–243) tables "of the classes of positive ternary forms represented by means of the corresponding reduced forms" for the determinants 1 to 100. A specimen is

$$\text{Det. } 6 \quad \begin{pmatrix} 1,1,6 \\ 0,0,-1 \end{pmatrix}, \begin{pmatrix} 1,2,3 \\ -1,-1,0 \end{pmatrix}.$$

$$\text{Zugeordnete Formen} \quad \begin{pmatrix} 8,8,3 \\ 0,0,4 \end{pmatrix}, \begin{pmatrix} 7,7,4 \\ 4,4,2 \end{pmatrix}.$$

where it is to be observed that Seeber admits odd coefficients for the terms in yz, zx, xy; viz. his symbol $\left(\begin{array}{c} a, b, c \\ f, g, h \end{array}\right)$ denotes

$$ax^2 + by^2 + cz^2 + fyz + gzx + hxy,$$

and his determinant is

$$4abc - af^2 - bg^2 - ch^2 + fgh.$$

Also his adjoint form is

$$\left(\begin{array}{c} 4bc - f^2,\ 4ca - g^2,\ 4ab - h^2 \\ 2gh - 4af,\ 2hf - 4bg,\ 2fg - 4ch \end{array}\right), \quad = (4bc - f^2)x^2 + \ldots + (2gh - 4af)yz + \ldots$$

In the notation of the *Disquisitiones Arithmeticae*, followed by Eisenstein and others, the symbol $\left(\begin{array}{c} a, b, c \\ f, g, h \end{array}\right)$ denotes

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy;$$

the determinant is

$$= -(abc - af^2 - bg^2 - ch^2 + 2fgh),$$

a positive form having thus always a negative determinant. And the adjoint form is

$$-\left(\begin{array}{c} bc - f^2,\ ca - g^2,\ ab - h^2 \\ gh - af,\ hf - bg,\ fg - ch \end{array}\right) = -(bc - f^2)x^2 - \ldots - 2(gh - af)yz - \ldots$$

Hence Seeber's determinant is $= -4$ multiplied by that of Gauss, and his tables really extend between the values -1 and -25 of the Gaussian determinant.

50. Tables of greater extent, and in the better form just referred to, are given by

Eisenstein, *Crelle*, t. XLI. (1851), pp. 169–190; viz. these are

I. "Tabelle der eigentlich primitiven positiven ternären Formen für alle negativen Determinanten von -1 bis -100," (pp. 169–185).

A specimen is

D	Anzahl	Reducirte Formen für $-D$
10	3	$\left(\begin{array}{c} 1, 1, 10 \\ 0, 0, 0 \end{array}\right)$, $\left(\begin{array}{c} 1, 2, 5 \\ 0, 0, 0 \end{array}\right)$, $\left(\begin{array}{c} 2, 2, 3 \\ 0, -1, 0 \end{array}\right)$
		$\delta = 8 \quad \delta = 4 \quad \delta = 4$

[p. 493] II. "Tabelle der uneigentlich primitiven positiven ternären Formen für alle negativen Determinanten von -2 bis -100," (pp. 186–189).

A specimen is

D	Anzahl	Reducirte Formen für $-D$
10	1	$\left(\begin{array}{c} 1, 2, 4 \\ 1, 1, 1 \end{array}\right)$
		$\delta = 6$

And there is given (p. 190) a table of the reduced forms for the determinant -385 $(= -5 \cdot 7 \cdot 11)$, selected merely as a largish number with three factors; viz. there are in all fifty-nine forms, corresponding to values $1, 2, 4, 6, 8$ of δ.

It may be remarked that δ denotes, for any given form, the number of ways in which this is linearly transformable into itself, this number being always $1, 2, 4, 6, 8, 12$, or 24. The theory as to this and other points is explained in the memoir (pp. 141–168), and various subsidiary tables are contained therein and in the *Anhang* (pp. 227–242); and there is given a small table relating to indeterminate forms, viz. this is

"C. Versuch einer Tabelle der nicht äquivalenten unbestimmten (indifferenten) ternären quadratischen Formen für die Determinanten ohne quadratischen Theiler unter 20," (pp. 239, 240).

A specimen is

D	Indifferente ternäre quadratische Formen
10	$\begin{pmatrix} 0,1,10 \\ 0,0,1 \end{pmatrix}$, $\begin{pmatrix} 1,2,5 \\ 0,0,0 \end{pmatrix}$
	$\begin{pmatrix} 0,0,10 \\ 0,0,1 \end{pmatrix}$

where, when the determinant is even, the forms in the second line are always improperly primitive forms.

[F. 17. *Complex Theories.*] Art. VII.

51. The theory of binary quadratic forms (a, b, c), with complex coefficients of the form $\alpha+\beta i$, ($i = \sqrt{-1}$ as usual, α and β integers), has been studied by Lejeune Dirichlet, Prof. H. J. S. Smith, and possibly others; but no tables have, it is believed, been calculated. The calculations would be laborious; but tables of a small extent only would be a sufficient illustration of the theory, and would, it is thought, be of great interest.

[p. 494] The theory of complex numbers of the last-mentioned form $\alpha + \beta i$, or say of the numbers formed with the fourth root of unity, had previously been studied by Gauss; and the theory of the numbers formed with the cube roots of unity ($\alpha+\beta\omega$, $\omega^3+1 = 0$, α and β integers) was studied by Eisenstein; but the general theory of the numbers involving the nth roots of unity (λ an odd prime) was first studied by Kummer. It will be sufficient to refer to his memoir,

Kummer, "Zur Theorie der complexen Zahlen," *Berl. Monatsb.*, March, 1845; and *Crelle*, t. XXXV. (1847), pp. 319–326; also "Ueber die Zerlegung der aus Wurzeln der Einheit gebildeten complexen Zahlen in ihre Primfactoren," same volume, pp. 327–367, where the astonishing theory of "Ideal Complex Numbers" is established.

52. It may be recalled that, p being an odd prime, and ρ denoting a root of the equation $\rho^{p-1}+\rho^{p-2}+\ldots+\rho+1 = 0$, then the numbers in question are those of the form $a+b\rho+\ldots+k\rho^{p-2}$ (where $(a, b, \ldots, k)$ are integers); or (what is in one point of view more, and in another less, general) if $\eta, \eta_1, \ldots, \eta_{e-1}$ are "periods" composed with the powers of ρ (e any factor of $p - 1$), then the form considered is $a\eta+b\eta_1+\ldots+h\eta_{e-1}$. For any value of p or e there is a corresponding complex theory. A number (real or complex) is in the complex theory prime or composite, according as it does not, or does, break up into factors of the form under consideration. For p a prime number under 23, if in the complex theory N is a prime, then any power of N (to fix the ideas say N^2) has no other factors than N or N^2; but if $p = 23$ (and similarly for higher values of p), then N may be such that, for instance, N^3 has complex factors other than N or N^3 (for $p = 23$, $N = 47$ is the first value of N, viz. 47^3 has factors other than 47 and 47^3): say N has a complex prime factor $\sqrt[3]{A}$, or we have $\sqrt[3]{A}$ as an ideal complex factor of N. Observe that by hypothesis A is not a perfect cube, viz. there is no complex number whose cube $= A$. In the foregoing general statement, made by way of illustration only, all reference to the complex factors of unity is purposely omitted, and the statement must be understood as being subject to correction on this account.

What precedes is by way of introduction to the account of Reuschle's Tables (*Berliner Monatsberichte*, 1859–60), which give in the different complex theories $p = 5, 7, 11, 13, 17, 19, 23, 29$ the complex factors of the decomposable real primes up to in some cases 1000.

It should be remarked that the form of a prime factor is to a certain extent indeterminate, as the factor can without injury be modified by affecting it with a complex factor of unity; but in the tables the choice of the representative form is made according to definite rules, which are fully explained, and which need not be here referred to.

[p. 495] 53. The following synopsis is convenient:—

Theory of the pth roots	Form of real prime to mod. p	No. of factors in complex theory	Extent of table; all primes under	Equation of periods
$\lambda = 5$, pp. 385–401.	1 4 2, 3	4 2 (just tabulated) primes.	2500	$\eta^2 + \eta - 1 = 0.$ $\eta^2 + \eta - 1 = 0.$
$\lambda = 7$, pp. 494–497.	1 2, 4 3, 5	6 3 primes.	1000	$\eta^3 + \eta^2 - 2\eta - 1 = 0.$ $\eta^2 + \eta + 2 = 0.$
$\lambda = 11$, pp. 120–124.	1 3, 4, 5, 9 2 6, 7, 8, 10	10 5 2 primes.	1000	$\eta^5 + \eta^4 - 4\eta^3 - 3\eta^2 +$ $\eta^2 + \eta - 3 = 0.$
$\lambda = 13$, pp. 173–179.	1	12	1000	$\eta^6 - \ldots = 0.$
$\lambda = 17$, pp. 724–729.	1	16	1000	$\eta^8 - \ldots = 0.$
$\lambda = 19$, pp. 719–724.	1	18	1000	$\eta^9 - \ldots = 0.$
$\lambda = 23$, pp. 737–751.	1 2, 3, 4, 6, 8, 9, 12, 13, 16, 18	22 11	1000	$\eta^{11} - \ldots = 0.$
$\lambda = 29$, pp. 761–764.	1	28	1000	$\eta^{14} - \ldots = 0.$

The foregoing synopsis of Reuschle's tables in the *Berliner Monatsberichte* was written previous to the publication of Reuschle's far more extensive work. It is

[p. 496] allowed to remain, but some explanations which were given have been struck out, and were instead given in reference to the larger work, which is

Reuschle, *Tafeln complexer Primzahlen, welche aus Wurzeln der Einheit gebildet sind.* Berlin, 4° (1875), pp. iii–vi and 1–671.

This work (the mass of calculation is perfectly wonderful) relates to the roots of unity, the degree being any prime or composite number, as presently mentioned, having all the values up to and a few exceeding 100; viz. the work is in five divisions, relating to the cases:

I. (pp. 1–171), degree any odd prime of the first 100, viz. $3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61$

II. (pp. 173–192), degree the power of an odd prime $9, 25, 27, 49, 81$;

III. (pp. 193–440), degree a product of two or more odd primes or their powers, viz. $15, 21, 33, 35, 39, 45, 51, 55,$

IV. (pp. 441–466), degree an even power of 2, viz. $4, 8, 16, 32, 64, 128$;

V. (pp. 467–671), degree divisible by 4, viz. $12, 20, 24, 28, 36, 40, 44, 48, 52, 56, 60, 68, 72, 76, 80, 84, 88, 92, 96, 1$

the only excluded degrees being those which are the double of an odd prime, these, in fact, coming under the case where the degree is the odd prime itself.

It would be somewhat long to explain the specialities which belong to degrees of the forms II., III., IV., V.; and what follows refers only to Division I., degree an odd prime.

For instance, if $\lambda = 7$, $\lambda - 1 = 2 \cdot 3$; the factors of 6 being $6, 3, 2, 1$, there are accordingly four divisions, viz.—

I. α a prime seventh root, that is, a root of $\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$;

II. $\eta_0 = \alpha + \alpha^{-1}$, $\eta_1 = \alpha^2 + \alpha^{-2}$, $\eta_2 = \alpha^3 + \alpha^{-3}$, or η a root of

$$\eta^3 + \eta^2 - 2\eta - 1 = 0, \quad \eta_0\eta_1 = \eta_1 + \eta_2, \text{ \&c.};$$

III. $\eta_0 = \alpha + \alpha^2 + \alpha^4$, $\eta_1 = \alpha^3 + \alpha^5 + \alpha^6$, or -1; η a root of $\eta^2 + \eta + 2 = 0$;

IV. Real numbers.

I. $p = 7m + 1$. First, it gives for the several prime numbers of this form $29, 43, \dots, 967$ the congruence roots, mod. p; for instance,

p	α	α^2	α^3	α^4	α^5	α^6
29	-5	-4	-9	-13	$+7$	-6
43	$+11$	-8	-2	$+21$	$+16$	$+4$.

[p. 497] This means that, if $\alpha \equiv -5$ (mod. 29), then $\alpha^2 = 25, \equiv -4$, $\alpha^3 = 20, \equiv -9$, &c., values which satisfy the congruence $\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 \equiv 0$ (mod. 29).

Secondly, it gives, under the simple and the primary forms, the prime factors $f(\alpha)$ of these same numbers $29, 43, \dots, 967$; for instance,

p	$f(\alpha)$ simple.	$f(\alpha)$ primary.
29	$\alpha + \alpha^4 - \alpha^5$	$2 + 3\alpha - \alpha^2 + 5\alpha^3 - 7\alpha^4 + 4\alpha^5$
43	$\alpha^2 + 2\alpha^4$	$2 - 2\alpha + 2\alpha^2 - 4\alpha^3 - \alpha^4 - 5\alpha^5$.

The definition of a primary form is a form for which $f(\alpha)f(\alpha^{-1}) \equiv f(1)^2$ mod. λ, and $f(\alpha) \equiv f(1)$ mod. $-(1-\alpha)^2$. The simple forms are also chosen so as to satisfy this last condition; thus $f(\alpha) = \alpha + \alpha^4 - \alpha^5$, then $f(1) - f(\alpha) = 1 - \alpha - \alpha^4 + \alpha^5 = (1-\alpha)^2(1+\alpha), \equiv 0$ mod. $(1-\alpha)^2$.

II. $p = 7m - 1$. First, it gives for the several prime numbers of this form $13, 41, \dots, 937$ the congruence roots, mod. p; for instance,

p	η_0	η_1	η_2
13	-3	-6	-5
41	-4	$+14$	-11;

and secondly, it gives, under the simple and the primary forms, the prime factors $f(\eta)$ of these same numbers $13, 41, \dots, 937$; for instance,

p	$f(\eta)$ simple.	$f(\eta)$ primary.
13	$\eta_0 + 2\eta_1$	$3 + 3\eta_0$
41	$4 + \eta_0$	$-11 + 7\eta_0 - 7\eta_1$.

Thus $13 = (\eta_0 + 2\eta_1)(\eta_1 + 2\eta_2)(\eta_2 + 2\eta_0)$, as is easily verified; the product of the first and second factors is $4 + 3\eta_0 + 8\eta_1 + \eta_2$, and then multiplying by the third factor, the result is $42 + 29(\eta_0 + \eta_1 + \eta_2), = 13$.

III. $p = 7m + 2$ or $7m + 4$. First, it gives for the several prime numbers of this form $2, 11, \dots, 991$ the congruence roots, mod. p; for instance,

p	η_0	η_1
2	0	-1
11	4	-5;

and secondly, it gives the primary prime factors $f(\eta)$ of these same numbers; for instance,

p	$f(\eta)$
2	η_0
11	$-1 - 2\eta_1$.

[p. 498] IV. $p = 7m+3$ or $7m+5$. The prime numbers of these forms, viz. $3, 5, 17, 19, \dots, 997$, are primes in the complex theory, and are therefore simply enumerated.

The arrangement is the same for the higher prime numbers $\lambda = 23$, &c., for which ideal factors make their appearance; but it presents itself under a more complicated form. Thus $\lambda = 23$, $\lambda - 1 = 2 \cdot 11$, and the factors of 22 are $22, 11, 2, 1$. There are thus four sections.

I. α a prime root, or $\alpha^{22} + \alpha^{21} + \dots + \alpha^2 + \alpha + 1 = 0$;

II. $\eta_0 = \alpha+\alpha^{-1}, \dots, \eta_{10} = \alpha^{11}+\alpha^{-11}$, or η a root of $\eta^{11} - \eta^{10} - 10\eta^9 + \dots + 10\eta^2 - 6\eta - 1 = 0$;

III. $\eta_0 = \alpha + \alpha^2 + \dots + \alpha^{11}$, $\eta_1 = \alpha^{12} + \alpha^{13} + \dots + \alpha^{22}$, or η a root of $\eta^2 + \eta + 6 = 0$;

IV. Real numbers.

I. $p = 23m+1$. First, it gives for the prime numbers of this form $47, 139, \dots, 967$ congruence roots, mod. p, and also congruence roots, mod. p^3 *; these last in the form $a + bp + cp^2$, where a is given in the former table; thus first table:

p	α	α^2	α^3	$\dots$	α^{22}
47	6	-11	-19	$\dots$	$+8$;

and second table:

p	α	α^2	α^3	$\dots$	α^{22}
47	$-11p - 2p^2$	$+13p - 23p^2$	$+14p - 14p^2$	$\dots$	$+22p + 22p^2$.

The meaning is that, $p = 47$, the roots of the congruence

$$\alpha^{22} + \alpha^{21} + \dots + \alpha^2 + \alpha + 1 \equiv 0 \ (\text{mod. } 47^3)$$

are

$$\alpha = 6 + p - 2p^2, \quad \alpha^2 = -11 + 13p - 23p^2, \quad \text{\&c.}$$

Secondly, it then gives $f(\alpha)$, the actual ideal prime factor of these same primes $47, 139, \dots, 967$; viz. the whole of this portion of the table $\lambda = 23$, I. (2) is, having actual prime factors,

p	$f(\alpha)$
599	$\alpha + \alpha^{15} - \alpha^{17}$
691	$\alpha^2 + \alpha^{19} - \alpha^5$
829	$\alpha^{15} + \alpha^{18} + \alpha^{20}$

having ideal factors, their third powers actual,

p	$f^3(\alpha)$
47	$\alpha^2 + \alpha^4 + \alpha^6 - \alpha^{14} - \alpha^{18} + \alpha^{20}$
139	$\alpha - \alpha^2 + \alpha^3 - \alpha^6 + \alpha^{14} - 2\alpha^{18}$
277	$-\alpha^2 - \alpha^4 - \alpha^6 + \alpha^{10} + \alpha^{12} + 2\alpha^{14} - \alpha^{16} - \alpha^{18}$
461	$1 - \alpha^2 - \alpha^4 + \alpha^6 + \alpha^8 + \alpha^{12} + \alpha^{14} + \alpha^{16} + \alpha^{18} + \alpha^{20} + \alpha^{22}$
967	$-\alpha^2 - \alpha^4 + \alpha^6 - \alpha^8 - \alpha^{10} - \alpha^{12} + \alpha^{14} + \alpha^{16}\alpha^{22}$

I repeat the explanation that, for the number 47, this means $f(\alpha)f(\alpha^2)\dots f(\alpha^{22}) = 47^3$.

[p. 499] And the like further complication presents itself in the part III. of the same table, $\lambda = 23$ (not, as it happens, in part II., nor of course in the concluding part IV., which is a mere enumeration of real primes). Thus III. (1), we have congruences, (mod. p^3),

$$\eta_0 \equiv p \equiv 2, \quad \eta_1 \equiv p \equiv 3, \quad \eta_0 \equiv +12, \ \text{\&c.};$$

[0] * Where, as presently appearing, 3 is the index of ideality or power to which the ideal factors have to be raised in order to become actual.

and having actual prime factors,

p	$f(\eta)$
59	$-2\eta_1$
101	$1-4\eta_1$

and having ideal prime factors, their third powers actual,

p	$f^3(\eta)$
2	$1-\eta_1$
3	$1-2\eta_1;$

as regards these last the signification being

$$2^3=(1-\eta_0)(1-\eta_1),\quad \eta_0+\eta_1=-1,\quad \eta_0\eta_1=6 \text{ (as is at once verified)},$$

$$3^3=(1-2\eta_0)(1-2\eta_1);$$

but the simple numbers $2, 3$ are neither of them of the form $(a+b\eta_0)(a+b\eta_1)$.

Contents of Report 1875 on Mathematical Tables.

§ 7. *Tables F. Arithmological.*

612. Note sur une Formule d'Intégration Indéfinie

[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXXVIII. (Janvier–Juin, 1874), pp. 1624–1629.]

[p. 500] En étudiant les Mémoires de M. Serret (*Journal de Liouville*, t. X., 1845) par rapport à la représentation géométrique des fonctions elliptiques, avec les remarques de M. Liouville sur ce sujet, je suis parvenu à une formule d'intégration indéfinie qui me paraît assez remarquable, savoir: en prenant θ entier positif quelconque, je dis que l'intégrale

$$\int \frac{(x+p)^{m+n-\theta-1}(x+q)^{\theta}\,dx}{x^{m+1}(x+p+q)^{n+1}}$$

a une valeur algébrique

$$(x+p)^{m+n-\theta-1+1}(x+p+q)^{-n}x^{-m}(A+Bx+Cx^2+\ldots+Kx^{\theta-1}),$$

pourvu qu'une seule condition soit satisfaite par les quantités m, n, p, q. Cette condition s'écrit sous la forme symbolique

$$([m]p^{\theta}+[n]q^{\theta})^{\theta}=0,$$

en dénotant ainsi l'équation

$$[m]^{\theta}p^{\theta\cdot\theta}+\frac{\theta}{1}[m]^{\theta-1}[n]^{\theta}p^{\theta\cdot(\theta-1)}q^{\theta}+\ldots+[n]^{\theta}q^{\theta\cdot\theta}=0,$$

où, comme à l'ordinaire, $[m]^k$ signifie $m(m-1)\ldots(m-k+1)$.

Je rappelle que les formules de M. Serret ne contiennent que des exposants entiers, et celles de M. Liouville qu'un seul exposant quelconque: la nouvelle formule contient deux exposants quelconques, m, n. Je remarque aussi l'analogie de la condition $([m]p^{\theta}+[n]q^{\theta})^{\theta}=0$ avec celle-ci

$$\frac{1}{\zeta^{m-n}}\left(\frac{d}{d\zeta}\right)^m\zeta^n(\zeta-1)^m=0$$

(m étant un entier positif), qui figure dans les Mémoires cités.

[p. 501] Pour démontrer la formule, j'écris

$$u=x^{-m}(A+Bx+Cx^2+\ldots+Kx^{\theta-1}),$$

et aussi pour abréger

$$X=(x+p)^{m+n-\theta+1}(x+p+q)^{-n},$$

ce qui donne

$$\frac{X}{(x+p)(x+p+q)}=(x+p)^{m+n-\theta}(x+p+q)^{-n-1}.$$

L'équation à vérifier est donc

$$Xu=\int\frac{X(x+q)^{\theta}\,dx}{x^{m+1}(x+p)(x+p+q)},$$

ou, en différentiant et divisant par X,

$$\frac{X'}{X}u+u'=\frac{(x+q)^{\theta}}{x^{m+1}(x+p)(x+p+q)},$$

ou enfin

$$[(m+n-\theta+1)(x+p)-n(x+p)]u+(x+p)(x+p+q)u'=\frac{(x+q)^{\theta}}{x^{m+1}},$$

où u' dénote $\dfrac{du}{dx}$. Il ne s'agit donc que d'exprimer que cette équation ait une intégrale $u = x^{-m}(A + Bx + Cx^2 + \ldots + Kx^{\theta-1})$.

En supposant que cela soit ainsi, et en effectuant la substitution, les termes en $x^{-m+\theta}$ se détruisent, et l'on obtient une équation qui contient des termes en $x^{-m-1}, x^{-m}, \ldots, x^{-m+\theta-1}$, savoir $(\theta+1)$ termes. On a ainsi, entre les θ coefficients $A, B, C, \ldots, K$ un système de $(\theta+1)$ équations linéaires, ce qui implique une condition entre les constantes m, n, p, q; mais, cette condition satisfaite, les équations se réduisent à θ équations indépendantes, et les coefficients seront ainsi déterminés.

Par exemple, soit $\theta = 2$; l'équation différentielle est

$$[(m-1)p + (n-1)q + (m+n-1)x]u + [p^2 + pq + x(2p+q) + x^2]u' = x^{-m-1}(q+x)^2,$$

laquelle doit être satisfaite par $u = Ax^{-m} + Bx^{-m+1}$. Cela donne

$$\begin{array}{c}
\begin{array}{cccc} x^{-m-1} & x^{-m} & x^{-m+1} & x^{-m+2} \end{array} \\
\begin{vmatrix}
 & (m-1p+m+n-1q)A, & (m-1p+m+n-1q)B, & \\
 & & (m-1)A & (m-1)B \\
-m(p^2+pq)A & -(m-1)(p^2+pq)B & & \\
 & -m(2p+q)A & -(m-1)(2p+q)B & -(m-1)B \\
 & & -mA &
\end{vmatrix} = 0.
\end{array}$$

[p. 502] [*The text on p. 502 of the original print consists of a large multi-column determinant table whose OCR has degraded significantly; the page is largely a tabular display of the system of linear equations leading from the differential equation to the symbolic condition* $([m]p^2 + [n]q^2)^2 = 0$. *The page introduces the auxiliary notations* $([m]p + [n]q)^1$, $([m]p + [n]q)^2$, *etc., parallel to* $([m]p^2 + [n]q^2)$, $([m]p^2 + [n]q^2)^2$, *etc., previously explained, and shows the term-by-term computation. On a, par exemple,*

$$([m]p + [n]q)^2 = [m]^2p^2 + 2[m]'[n]'pq + [n]^2q^2.$$

]

[p. 503] et ainsi de suite. Les notations $([m]p+[n]q)^2$, $([m]p+[n]q)^3$, ... ont des significations semblables à celles de $([m]p^2 + [n]q^2)^2$, $([m]p^2 + [n]q^2)^3$, ..., auparavant expliquées. On a, par exemple,

$$([m]p + [n]q)^2 = [m]^2p^2 + 2[m]'[n]'pq + [n]^2q^2.$$

Considérons, par exemple, le deuxième déterminant: ceci contient trois termes en $1, 2q, q^2$ respectivement; le premier terme est

$$1 \cdot (m-1)(p^2+pq) \cdot m(p^2+pq),$$

c'est-à-dire

$$[m]^2p^2(p+q)^2;$$

le deuxième terme est

$$2q \cdot -m(p^2+pq)[(m-1)p - nq],$$

c'est-à-dire

$$-2[m]'p(p+q)q([m-1]p - [n]q);$$

le troisième terme est

$$q^2[(m-1p-nq)(m+1p-n-1q) - (m-1)(p^2+pq)],$$

c'est-à-dire

$$q^2[(m^2-m)p^2 - 2mnpq + (n^2-n)q^2] = q^2([m]p - [n]q)^2.$$

Et de même le troisième déterminant est composé de quatre termes en $1, 3q, 3q^2, q^3$ respectivement, lesquels sont les quatre termes de la première expression transformée, et ainsi pour le quatrième déterminant, etc. Au moyen de ces premières transformées, on obtient sans peine les expressions finales $([m]p^2 + [n]q^2)^2$, $([m]p^2 + [n]q^2)^3$, ...

En écrivant $z = x + \frac{1}{2}(p+q)$ au lieu de x, et puis $\frac{1}{2}(p+q) = \alpha$, $\frac{1}{2}(p-q) = a$, la formule devient

$$\int \frac{(z-a)^{m+n-\theta-1}(z+a)^{\theta}\,dz}{(z-\alpha)^{m+1}(z+\alpha)^{n+1}},$$

et la valeur algébrique

$$= (z-a)^{m+n-\theta-1+1}(z+\alpha)^{-n}(z-\alpha)^{-m}(A' + B'z + \ldots + K'z^{\theta-1}),$$

pourvu qu'on ait entre les quantités m, n, a, α la relation

$$\{[m](\alpha+a)^{\theta} + [n](\alpha-a)^{\theta}\}^{\theta} = 0.$$

En écrivant $\zeta = \dfrac{(\alpha+a)^{\theta}}{4\alpha\, a}$ et $\dfrac{(\alpha-a)^{\theta}}{4\alpha\, a} = \zeta - 1$, peut s'écrire sous la forme $\{[m]\zeta + [n](\zeta-1)\}^{\theta} = 0$.

Je remarque que l'équation en ζ ne donne pas toujours pour ζ des valeurs réelles, positives et plus grandes que l'unité: par exemple, pour $\theta = 1$, on a $\zeta = \dfrac{-m}{n}$, valeur qui ne peut pas satisfaire à ces conditions. Je n'ai pas cherché dans quel cas ces conditions (qui ont rapport à l'application des formules à la représentation des fonctions elliptiques) subsistent.

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 504–528 (Papers 613–617)

Transcribed from the Original Publications

Contents

613. On the Group of Points G_4^1 on a Sextic Curve with Five Double Points

[From the Mathematische Annalen, vol. viii. (1875), pp. 359–362.]

[p. 504] The present note relates to a special group of points considered incidentally by MM. Brill and Nöther in their paper "Ueber die algebraischen Functionen und ihre Anwendung in der Geometrie," *Math. Annalen*, t. VII. pp. 268–310 (1874).

I recall some of the fundamental notions. We have a basis-curve which to fix the ideas may be taken to be of the order n, $= p+1$, $\frac{1}{2}p(p-3)$ dps, and therefore of the "Geschlecht" or deficiency p; any curve of the order $n-3$, $= p-2$ passing through the $\frac{1}{2}p(p-3)$ dps is said to be an *adjoint curve.* We may have, on the basis-curve, a special group G_Q^q of Q points ($Q \geq 2p-2$); viz. this is the case when the Q points are such that every adjoint curve through $Q-q$ of them—that is, every curve of the order $p-2$ through $\frac{1}{2}p(p-3)$ dps and the $Q-q$ points—passes through the remaining q points of the group: the number q may be termed the "speciality" of the group: if $q=0$, the group is an ordinary one.

It may be observed that a special group G_Q^q is chiefly noteworthy in the case where $Q-q$ is so small that the adjoint curve is not completely determined: thus if $p=5$, viz. if the basis-curve be a sextic with 5 dps, then we may have a special group G_6^1, but there is nothing remarkable in this; the 6 points are intersections with the sextic of an arbitrary cubic through the 5 dps—the cubic of course intersects the sextic in the 5 dps counting as 10 points, and in 8 other points—and such cubic is completely determined by means of the 5 dps and any 4 of the 6 points. But contrariwise, there is something remarkable in the group G_4^1 about to be considered: viz. we have here on the sextic 4 points, such that every cubic through the 5 dps and through 3 of the 4 points (through 8 points in all) passes through the remaining one of the 4 points.

The whole number of intersections of the basis-curve with an adjoint, exclusive of the dps counting as $p(p-3)$ points, is of course $= 2p-2$: hence an adjoint through the Q points of a group G_Q^q meets the basis-curve besides in R, $= 2p-2-Q$, **[p. 505]** points; we have then the "Riemann-Roch" theorem that these R points form a special group G_R^r, where

$$Q+R = 2p-2,$$

as just mentioned, and

$$Q-R = 2q-2r;$$

viz. dividing in any manner the $2p-2$ intersections of the basis-curve by an adjoint into groups of Q and R points respectively, these will be special groups, or at least one of them will be a special group, G_Q^q, G_R^r, such that their specialities q, r are connected by the foregoing relation $Q-R = 2q-2r$.

The Authors give (l.c., p. 293) a Table showing for a given basis-curve, or given value of p, and for a given value of r, the least value of R and the corresponding values of q, Q: this table is conveniently expressed in the following form.

The least value of

$$R = p - \frac{p}{r+1} + r,$$

and then

$$q = \frac{p}{r+1} - 1,$$

$$Q = p + \frac{p}{r+1} - r - 2,$$

where $\dfrac{p}{r+1}$ denotes the integer equal to or next less than the fraction.

It is, I think, worth while to present the table in the more developed form:

n	p	Dps	$r=$					
			1	2	3	4	5	6
4	3	0	G_2^1	G_3^2	.	.	.	.
			G_4^2	G_6^3	.	.	.	.
5	4	2	G_3^1	G_4^2	G_6^3	.	.	.
			G_4^2	G_6^3	G_8^4	.	.	.
6	5	5	G_4^1	G_4^2	G_6^3	G_8^4	.	.
			G_4^2	G_6^3	G_8^4	G_{10}^5	.	.
7	6	9	G_4^1	G_6^2	G_6^3	G_8^4	G_{10}^5	.
			G_6^2	G_6^3	G_8^4	G_{10}^5	G_{12}^6	.
8	7	14	G_4^1	G_6^2	G_8^3	G_{10}^4	G_{10}^5	G_{12}^6
			G_6^2	G_8^3	G_8^4	G_{10}^5	G_{12}^6	G_{14}^7

where the table shows the values of $\dfrac{G_R^r}{G_Q^q}$ for any given values of p, r.

[p. 506] I recur to the case $p=5$ and the group G_4^1, which is the subject of the present note: viz. we have here a sextic curve with 5 dps, and on it a group of 4 points G_4^1, such that every cubic through the 5 dps and through 3 points of the group, 8 points in all, passes through the remaining 1 point.

MM. Brill and Nöther show (by consideration of a rational transformation of the whole figure) that, given 2 points of the group, it is possible, and possible in 5 different ways, to determine the remaining 2 points of the group.

I remark that the 5 dps and the 4 points of the group form "an ennead" or system of the nine intersections of two cubic curves: and that the question is, given the 5 dps and 2 points on the sextic, to show how to determine on the sextic a pair of points forming with the 7 points an ennead: and to show that the number of solutions is $=5$.

We have the following "Geiser-Cotterill" theorem:

If seven of the points of an ennead are fixed, and the eighth point describes a curve of the order n passing $\alpha_1, \alpha_2, \ldots, \alpha_7$ times through the seven points respectively, then will the ninth point describe a curve of the order ν passing $a_1, a_2, \ldots, a_7$ times through the seven points respectively: where

$$\nu = 8n - 3\Sigma\alpha,$$

$$a_1 = 3n - \alpha_1 - \Sigma\alpha,$$

$$\vdots$$

$$a_7 = 3n - \alpha_7 - \Sigma\alpha,$$

and conversely

$$n = 8\nu - 3\Sigma a,$$

$$\alpha_1 = 3\nu - a_1 - \Sigma a,$$

$$\vdots$$

$$\alpha_7 = 3\nu - a_7 - \Sigma a.$$

(Geiser, *Crelle-Borchardt*, t. LXVII. (1867), pp. 78–90; the complete form, as just stated, and which was obtained by Mr Cotterill, has not I believe been published); and also Geiser's theorem "the locus of the coincident eighth and ninth points is a sextic passing twice through each of the seven points."

The sextic and the curve n intersect in $6n$ points, among which are included the seven points counting as $2\Sigma\alpha$ points: the number of the remaining points is $=6n-2\Sigma\alpha$. Similarly, the sextic and the curve ν intersect in 6ν points, among which are included the seven points counting as $2\Sigma a$ points: the number of the remaining points is $6\nu - 2\Sigma a$ $(= 6n - 2\Sigma\alpha)$. The points in question are, it is clear, common intersections of the sextic, and the curves n, ν: viz. of the intersections of the curves n, ν, a number $6n - 2\Sigma\alpha$, $= 6\nu - 2\Sigma a$, $= 3n + 3\nu - 2\Sigma a - 2\Sigma\alpha$ lie on the sextic.

The curves n, ν intersect in $n\nu$ points, among which are included the seven points counting $\Sigma a\alpha$ times: the number of the remaining intersections is therefore **[p. 507]** $n\nu - \Sigma a\alpha$, but among these are included the $3n + 3\nu - 2\Sigma a - 2\Sigma\alpha$ points on the sextic; omitting these, there remain $n\nu - 3(n + \nu) - \Sigma a\alpha + 2\Sigma a + 2\Sigma\alpha$ points, or, what is the same thing, $(n - 3)(\nu - 3) - \Sigma(a - 1)(\alpha - 1) - 2$ points: it is clear that these must form pairs such that, the eighth point being either point of a pair, the ninth point will be the remaining point of the pair: the number of pairs is of course

$$\tfrac{1}{2}[(n - 3)(\nu - 3) - \Sigma(a - 1)(\alpha - 1) - 2],$$

and we have thus the solution of the question, given the seven points to determine the number of pairs of points on the curve n (or on the curve ν) such that each pair may form with the seven points an ennead.

In particular, if $n = 6$; $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7 = 2, 2, 2, 2, 2, 1, 1$ respectively, viz. if the curve be a sextic having 5 of the points for dps, and the remaining two for simple points, then we find $\nu = 12$; $a_1, a_2, a_3, a_4, a_5, a_6, a_7 = 4, 4, 4, 4, 4, 5, 5$ respectively, and the number of pairs is

$$= \tfrac{1}{2}[3 \cdot 9 - 5(2 - 1)(4 - 1) - 2], = \tfrac{1}{2}(27 - 15 - 2), = 5,$$

viz. starting with the 5 dps and any 2 points of the group G^1_4 we can, in 5 different ways, determine the remaining 2 points of the group.

In reference to the number $3p - 3$ of parameters in the curves belonging to a given value of p, it may be remarked as follows. Such a curve is rationally transformable into a curve of the order $p + 1$ with $\tfrac{1}{2}p(p - 3)$ dps, and therefore containing $\tfrac{1}{2}(p + 1)(p + 4)$, $-\tfrac{1}{2}p(p - 3)$, $= 4p + 2$ parameters. Employing an arbitrary homographic transformation to establish any assumed relations between the parameters, the number is diminished to $4p + 2 - 8$, $= 4p - 6$; and again employing a rational transformation by means of adjoint curves of the order $p - 2$ drawn through the dps and $p - 3$ points of the curve—thereby transforming the curve into one of the same order $p+1$ and deficiency p—then, assuming that the $p-3$ parameters (or constants on which depend the positions of the $p - 3$ points) can be disposed of so as to establish $p - 3$ relations between the parameters and so further diminish the number by $p - 3$, the required number of parameters will finally be $4p - 6 - (p - 3) = 3p - 3$.

Cambridge, 26th October, 1874.

614. On a Problem of Projection

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 19–29.]

[p. 508] I measure off on three rectangular axes the distances $\Omega X = \Omega Y = \Omega Z, = \theta$; and then, in a plane through Ω drawing in arbitrary directions the three lines ΩA, ΩB, ΩC, $= a$, b, c respectively, I assume that A, B, C (fig. 1) are the parallel projections of X, Y, Z respectively; viz. taking ΩO as the direction of the projecting lines, then ΩA, ΩB, ΩC being given in position and magnitude, we have to find θ, and the position of the line ΩO.

[Fig. 1: Sphere with centre Ω, showing axes X, Y, Z and projections A, B, C on a great circle, with projecting point O.]

This is in fact a case of a more general problem solved by Prof. Pohlke in 1853, (see the paper by Schwarz, "Elementarer Beweis des Pohlke'schen Fundamentalsatzes der Axonometrie," *Crelle*, t. LXIII. (1864), pp. 309–314), viz. the three lines ΩX, ΩY, ΩZ may be any three axes given in magnitude and direction, and their parallel projection **[p. 509]** is to be similar to the three lines ΩA, ΩB, ΩC. Schwarz obtains a very elegant construction, which I will first reproduce. We may imagine through Ω a plane cutting at right angles the projecting lines, say in the points X', Y', Z': we have then in plano a triad of lines $\Omega X'$, $\Omega Y'$, $\Omega Z'$ which are an orthogonal projection of ΩX, ΩY, ΩZ; and are also an orthogonal projection of a plane triad similar to ΩA, ΩB, ΩC; *quâ* such last-mentioned projection, the triangles $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, must be proportional to the triangles ΩBC, ΩCA, ΩAB; that is, we have to find an orthogonal projection of ΩX, ΩY, ΩZ, such that the triangles $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, which are the projections of ΩYZ, ΩZX, ΩXY respectively, shall be in given ratios. There is no difficulty in the solution of this problem; referring everything to a sphere centre Ω, let the normals to the planes ΩYZ, ΩZX, ΩXY, meet the sphere in the points X'', Y'', Z'' respectively, and the projecting line through Ω meet the sphere in the point O, then the projection of ΩYZ is to ΩYZ as $\cos OX'' : 1$; and the like as to the projections of ΩZX and ΩXY: that is, in the given spherical triangle $X''Y''Z''$, we have to find a point O, such that the cosines of the distances OX'', OY'', OZ'' are in given ratios: we have at once, through X'', Y'', Z'' respectively, three arcs meeting in the required point O.

The projecting lines being thus obtained, say these are the three parallel lines X', Y', Z', we have next to draw through Ω a plane meeting these in the points A', B', C' such that the triangle $A'B'C'$ is similar to the given triangle ABC; for this being so, the triangles $\Omega B'C'$, $\Omega C'A'$, $\Omega A'B'$ being the projections of, and therefore proportional to $\Omega Y'Z'$, $\Omega Z'X'$, $\Omega X'Y'$, that is, proportional to ΩBC, ΩCA, ΩAB, will, it is clear, be similar to these triangles respectively; that is, we have the triad $\Omega A'$, $\Omega B'$, $\Omega C'$, a projection of ΩX, ΩY, ΩZ, and similar to the triad ΩA, ΩB, ΩC, which is what was required.

It remains only to show how the given three parallel lines X', Y', Z', not in the same plane, can be cut by a plane in a triangle similar to a given triangle ABC.

[Fig. 2: Three parallel lines perpendicular to the plane of the paper through points X, Y, Z; auxiliary points A'', D, E, K, B', C' used in the construction.]

Imagine the three lines at right angles to the plane of the paper, meeting the plane of the paper in the given points X, Y, Z (fig. 2) respectively. On the base **[p. 510]** YZ describe a triangle $A''YZ$ similar to the given triangle ABC; and through A'', X with centre on the line YZ, describe a circle meeting this line in the points D and E. Then in the plane, through YZ at right angles to the plane of the paper, we may draw a line meeting the lines Y, Z in the points B'', C'' respectively, such that joining XB'', XC'' we obtain a triangle $XB''C''$ similar to $A''YZ$, that is, to the given triangle ABC.

Taking K the centre of the circle, suppose that its radius is $= l$, and that we have $KY = \beta$, $KZ = \gamma$; also $YZ = \sigma$, $ZX = \tau$; $YA'' = \sigma''$, $ZA'' = \tau''$. If for a moment x, y denote the coordinates of A'', then

$$\tau^2 = (x-\gamma)^2 + y^2, = l^2 + \gamma^2 - 2\gamma x,$$

and thence

$$\gamma\sigma^2 - \beta\tau^2 = \gamma(l^2+\beta^2) - \beta(l^2+\gamma^2),$$

that is,

$$\gamma\sigma^2 - \beta\tau^2 = (\gamma-\beta)(l^2-\beta\gamma);$$

viz. this is the equation of the circle in terms of the vectors σ, τ; we have therefore in like manner

$$\gamma\sigma''^2 - \beta\tau''^2 = (\gamma-\beta)(l^2-\beta\gamma).$$

We may determine θ so as to satisfy the two equations

$$\sigma''^2 = \sigma^2\cos^2\theta + (l+\beta)^2\sin^2\theta,$$

$$\tau''^2 = \tau^2\cos^2\theta + (l+\gamma)^2\sin^2\theta;$$

in fact, these equations give

$$\gamma\sigma''^2 - \beta\tau''^2 = (\gamma\sigma^2-\beta\tau^2)\cos^2\theta + \{\gamma(l+\beta)^2 - \beta(l+\gamma)^2\}\sin^2\theta,$$

which, the left-hand side and the coefficients of $\cos^2\theta$ and $\sin^2\theta$ on the right-hand side being each $= (\gamma-\beta)(l^2-\beta\gamma)$, is, in fact, an identity.

But in the figure, if θ, determined as above, denote the angle at B, then

$$(XB'')^2 = XY^2 + YB''^2 = \sigma^2 + (l+\beta)^2\tan^2\theta,$$

$$(XC'')^2 = XZ^2 + ZC''^2 = \tau^2 + (l+\gamma)^2\tan^2\theta,$$

that is,

$$XB'' = \sigma''\sec\theta, \quad XC'' = \tau''\sec\theta,$$

or, since $B''C'' = YZ\sec\theta$ [$= (\gamma-\beta)\sec\theta$], the triangle $XB''C''$ is, as mentioned, similar to the triangle $A''YZ$.

I was not acquainted with the foregoing construction when my paper was written; but the analytical investigation of the particular case is nevertheless interesting, and I proceed to consider it.

[p. 511] Taking (fig. 1) Ω as the centre of a sphere and projecting on this sphere, we have A, B, C given points on a great circle; and we have to find the point O, such that there may be a trirectangular triangle XYZ, the vertices of which lie in OA, OB, OC respectively, and for which

$$\frac{\sin OX}{\sin OA} = \frac{a}{\theta}, \quad \frac{\sin OY}{\sin OB} = \frac{b}{\theta}, \quad \frac{\sin OZ}{\sin OC} = \frac{c}{\theta}.$$

I take the arcs BC, CA, $AB = \alpha$, β, γ respectively, $\alpha+\beta+\gamma = 2\pi$; and the required arcs OA, OB, OC are taken to be ξ, η, ζ respectively; these are connected by the relation

$$\sin\alpha\cos\xi + \sin\beta\cos\eta + \sin\gamma\cos\zeta = 0,$$

to obtain which, observe that from the triangles OAB, OAC, we have

$$\cos A = \frac{\cos\eta - \cos\xi\cos\gamma}{\sin\xi\sin\gamma} = -\frac{\cos\zeta - \cos\xi\cos\beta}{\sin\xi\sin\beta},$$

that is,

$$\sin\beta(\cos\eta - \cos\xi\cos\gamma) + \sin\gamma(\cos\zeta - \cos\xi\cos\beta) = 0,$$

which, with $\sin\alpha = -\sin(\beta+\gamma)$, gives the required relation. We have

$$\sin OX = \frac{a}{\theta}\sin\xi, \quad \sin OY = \frac{b}{\theta}\sin\eta, \quad \sin OZ = \frac{c}{\theta}\sin\zeta;$$

and then from the triangles OBC, OCA, OAB, and the quadrantal triangles OYZ, OZX, OXY, we have

$$\cos BOC = \frac{\cos\alpha - \cos\eta\cos\zeta}{\sin\eta\sin\zeta} = \frac{-\sqrt{\theta^2 - b^2\sin^2\eta}\,\sqrt{\theta^2 - c^2\sin^2\zeta}}{bc\sin\eta\sin\zeta\,/\,\theta^2}, \text{ \&c.},$$

that is,

$$bc(\cos\alpha - \cos\eta\cos\zeta) = -\sqrt{\theta^2 - b^2\sin^2\eta}\,\sqrt{\theta^2 - c^2\sin^2\zeta},$$

$$ca(\cos\beta - \cos\zeta\cos\xi) = -\sqrt{\theta^2 - c^2\sin^2\zeta}\,\sqrt{\theta^2 - a^2\sin^2\xi},$$

$$ab(\cos\gamma - \cos\xi\cos\eta) = -\sqrt{\theta^2 - a^2\sin^2\xi}\,\sqrt{\theta^2 - b^2\sin^2\eta},$$

which, when rationalized, are quadric equations in $\cos\xi$, $\cos\eta$, $\cos\zeta$. The first equation, in fact, gives

$$b^2c^2(\cos\alpha - \cos\eta\cos\zeta)^2 = (\theta^2 - b^2 + b^2\cos^2\eta)(\theta^2 - c^2 + c^2\cos^2\zeta),$$

that is,

$$(\theta^2-b^2)(\theta^2-c^2) - b^2c^2\cos^2\alpha + (\theta^2-b^2)c^2\cos^2\zeta + (\theta^2-c^2)b^2\cos^2\eta + 2b^2c^2\cos\alpha\cos\eta\cos\zeta = 0,$$

or, what is the same thing,

$$-\left(1 - b^2 - c^2 - \frac{b^2c^2\cos^2\alpha}{\theta^2-b^2\cdot\theta^2-c^2}\right) + \frac{b^2}{\theta^2-b^2}\cos^2\eta + \frac{c^2}{\theta^2-c^2}\cos^2\zeta + \frac{2b^2c^2\cos\alpha}{(\theta^2-b^2)(\theta^2-c^2)}\cos\eta\cos\zeta = 0.$$

Completing the system, we have

$$-\left(1 - c^2 - a^2 - \frac{c^2a^2\cos^2\beta}{\theta^2-c^2\cdot\theta^2-a^2}\right) + \frac{c^2}{\theta^2-c^2}\cos^2\zeta + \frac{a^2}{\theta^2-a^2}\cos^2\xi + \frac{2c^2a^2\cos\beta}{(\theta^2-c^2)(\theta^2-a^2)}\cos\zeta\cos\xi = 0,$$

$$-\left(1 - a^2 - b^2 - \frac{a^2b^2\cos^2\gamma}{\theta^2-a^2\cdot\theta^2-b^2}\right) + \frac{a^2}{\theta^2-a^2}\cos^2\xi + \frac{b^2}{\theta^2-b^2}\cos^2\eta + \frac{2a^2b^2\cos\gamma}{(\theta^2-a^2)(\theta^2-b^2)}\cos\xi\cos\eta = 0,$$

[p. 512] and, as above,

$$\sin\alpha\cos\xi + \sin\beta\cos\eta + \sin\gamma\cos\zeta = 0.$$

It seems difficult from these equations to eliminate ξ, η, ζ, so as to obtain an equation in θ; but I employ some geometrical considerations.

Taking Π as the pole of the circle ABC, and drawing ΠX, ΠY, ΠZ to meet the circle in p, q, r respectively, then, if α'', β'', γ'' are the cosine-inclinations of Π to X, Y, Z respectively, we have

$$\sin Xp,\ \sin Yq,\ \sin Zr = \alpha'',\ \beta'',\ \gamma''.$$

From the right parallel triangles BYq and CZr, we have

$$\sin Yq = \sin BY\sin B,$$

$$\sin Zr = \sin CZ\sin C,$$

and, thence,

$$\frac{\sin Yq}{\sin Zr} = \frac{\sin BY}{\sin CZ}\cdot\frac{\sin OC}{\sin OB};$$

or, since

$$BY = OB - OY, \quad CZ = OC - OZ,$$

and thence

$$\sin BY = \frac{1}{\theta}\{\sqrt{\theta^2 - b^2 \sin^2 \eta} - b\cos\eta\},$$

$$\sin CZ = \frac{1}{\theta}\{\sqrt{\theta^2 - c^2 \sin^2 \zeta} - c\cos\zeta\},$$

we obtain

$$\frac{\beta''}{\gamma''} = \frac{\sqrt{\theta^2 - b^2\sin^2\eta} - b\cos\eta}{\sqrt{\theta^2 - c^2\sin^2\zeta} - c\cos\zeta}.$$

We have thence

$$\beta''\sqrt{\theta^2 - c^2\sin^2\zeta} - \gamma''\sqrt{\theta^2 - b^2\sin^2\eta} = \beta'' c\cos\zeta - \gamma'' b\cos\eta,$$

or, squaring and reducing

$$\beta''^2(\theta^2 - c^2) + \gamma''^2(\theta^2 - b^2) + 2\beta''\gamma''\left[-\sqrt{\theta^2 - c^2\sin^2\zeta}\sqrt{\theta^2 - b^2\sin^2\eta} + bc\cos\eta\cos\zeta\right] = 0;$$

that is,

$$\beta''^2(\theta^2 - c^2) + \gamma''^2(\theta^2 - b^2) + 2\beta''\gamma''\cdot bc\cos\alpha = 0,$$

and, similarly,

$$\gamma''^2(\theta^2 - a^2) + \alpha''^2(\theta^2 - c^2) + 2\gamma''\alpha''\cdot ca\cos\beta = 0,$$

$$\alpha''^2(\theta^2 - b^2) + \beta''^2(\theta^2 - a^2) + 2\alpha''\beta''\cdot ab\cos\gamma = 0,$$

or, what is the same thing,

$$\frac{\beta''^2}{\theta^2 - b^2} + \frac{\gamma''^2}{\theta^2 - c^2} = \frac{-2bc\cos\alpha}{\sqrt{(\theta^2 - b^2)(\theta^2 - c^2)}}\cdot\frac{\beta''\gamma''}{\sqrt{(\theta^2 - b^2)(\theta^2 - c^2)}},$$

$$\frac{\gamma''^2}{\theta^2 - c^2} + \frac{\alpha''^2}{\theta^2 - a^2} = \frac{-2ca\cos\beta}{\sqrt{(\theta^2 - c^2)(\theta^2 - a^2)}}\cdot\frac{\gamma''\alpha''}{\sqrt{(\theta^2 - c^2)(\theta^2 - a^2)}},$$

$$\frac{\alpha''^2}{\theta^2 - a^2} + \frac{\beta''^2}{\theta^2 - b^2} = \frac{-2ab\cos\gamma}{\sqrt{(\theta^2 - a^2)(\theta^2 - b^2)}}\cdot\frac{\alpha''\beta''}{\sqrt{(\theta^2 - a^2)(\theta^2 - b^2)}}.$$

[p. 513] writing

$$\alpha'',\ \beta'',\ \gamma'' = X'\sqrt{\theta^2 - a^2},\ Y'\sqrt{\theta^2 - b^2},\ Z'\sqrt{\theta^2 - c^2},$$

and

$$\frac{bc\cos\alpha}{\sqrt{(\theta^2 - b^2)(\theta^2 - c^2)}} = f, \quad \frac{ca\cos\beta}{\sqrt{(\theta^2 - c^2)(\theta^2 - a^2)}} = g, \quad \frac{ab\cos\gamma}{\sqrt{(\theta^2 - a^2)(\theta^2 - b^2)}} = h,$$

the equations are

$$Y'^2 + Z'^2 - 2fY'Z' = 0,$$

$$Z'^2 + X'^2 - 2gZ'X' = 0,$$

$$X'^2 + Y'^2 - 2hX'Y' = 0.$$

Writing the last two under the form

$$Z'^2 - 2gZ'X' + X'^2 = 0,$$

$$X'^2 - 2hX'Y' + Y'^2 = 0,$$

and eliminating Z', we have

$$-4(1-g^2)(1-h^2)Y'^2X'^2 + (Y'^2 + Z'^2 - 2ghY'Z')^2 = 0,$$

which, in virtue of the first equation, is

$$-4(1-g^2)(1-h^2)Y'^2Z'^2 + 4(gh-f)^2Y'^2Z'^2 = 0,$$

that is,

$$(1-g^2)(1-h^2) - (gh-f)^2 = 0;$$

or, what is the same thing,

$$1 - f^2 - g^2 - h^2 + 2fgh = 0.$$

I remark that we may write

$$gh - f = \sqrt{(1-g^2)}\sqrt{(1-h^2)},$$
$$hf - g = \sqrt{(1-h^2)}\sqrt{(1-f^2)},$$
$$fg - h = \sqrt{(1-f^2)}\sqrt{(1-g^2)},$$

the signs on the right-hand side being either all $+$, or else one $+$ and two $-$, so that the product is $+$. In fact, multiplying the assumed equations, we have

$$f^2g^2h^2 - fgh(f^2+g^2+h^2) + f^2g^2 + h^2f^2 + f^2g^2 - fgh = 1 - f^2 - g^2 - h^2 + f^2h^2 + h^2f^2 + f^2g^2 - f^2g^2h^2,$$

that is,

$$1 - f^2 - g^2 - h^2 + fgh(1 + f^2 + g^2 + h^2) - 2f^2g^2h^2 = 0,$$

or,

$$(1 - f^2 - g^2 - h^2 + 2fgh)(1 - fgh) = 0,$$

which is right; but with a different combination of signs the result would not have been obtained.

Substituting for f, g, h their values, we have

$$(a^2-\theta^2)(b^2-\theta^2)(c^2-\theta^2) - b^2c^2(a^2-\theta^2)\cos^2\alpha - c^2a^2(b^2-\theta^2)\cos^2\beta$$
$$-a^2b^2(c^2-\theta^2)\cos^2\gamma + 2a^2b^2c^2\cos\alpha\cos\beta\cos\gamma = 0,$$

[p. 514] where the term independent of θ is

$$a^2b^2c^2(1 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma + 2\cos\alpha\cos\beta\cos\gamma),$$

which is $= 0$ in virtue of $\alpha + \beta + \gamma = 2\pi$. We have, therefore, for θ^2 the quadric equation

$$b^2c^2\sin^2\alpha + c^2a^2\sin^2\beta + a^2b^2\sin^2\gamma - (a^2+b^2+c^2)\theta^2 + \theta^4 = 0,$$

giving for θ^2 the two real positive values

$$\theta^2 = \tfrac{1}{2}\{a^2 + b^2 + c^2 + \sqrt{\Omega}\},$$

where

$$\Omega = (a^2+b^2+c^2)^2 - 4(b^2c^2\sin^2\alpha + c^2a^2\sin^2\beta + a^2b^2\sin^2\gamma)$$
$$= a^4 + b^4 + c^4 + 2b^2c^2\cos 2\alpha + 2c^2a^2\cos 2\beta + 2a^2b^2\cos 2\gamma$$
$$= (a^2 + b^2\cos 2\gamma + c^2\cos 2\beta)^2 + (b^2\sin 2\gamma - c^2\sin 2\beta)^2.$$

I write now
$$\frac{a\cos\xi}{\sqrt{(a^2-\theta^2)}}=X,\quad \frac{b\cos\eta}{\sqrt{(b^2-\theta^2)}}=Y,\quad \frac{c\cos\zeta}{\sqrt{(c^2-\theta^2)}}=Z,$$
and also
$$\frac{\sqrt{a^2-\theta^2}}{a}\sin\alpha=A,\quad \frac{\sqrt{b^2-\theta^2}}{b}\sin\beta=B,\quad \frac{\sqrt{c^2-\theta^2}}{c}\sin\gamma=C.$$
The equations for $\cos\xi$, $\cos\eta$, $\cos\zeta$ become
$$Y^2+Z^2-2fYZ-(1-f^2)=0,$$
$$Z^2+X^2-2gZX-(1-g^2)=0,$$
$$X^2+Y^2-2hXY-(1-h^2)=0,$$
and
$$AX+BY+CZ=0,$$
in virtue of the relation between f, g, h. The first three equations are satisfied by a two-fold relation between X, Y, Z; viz. treating these as coordinates, the equations represent three quadric cylinders having a common conic.

To prove this, I write
$$1-f^2,\ 1-g^2,\ 1-h^2,\ gh-f,\ hf-g,\ fg-h=\mathfrak{a},\ \mathfrak{b},\ \mathfrak{c},\ \mathfrak{f},\ \mathfrak{g},\ \mathfrak{h}.$$
We have, as usual,
$$\mathfrak{bc}-\mathfrak{f}^2,\ \mathfrak{ca}-\mathfrak{g}^2,\ \mathfrak{ab}-\mathfrak{h}^2,\ \mathfrak{gh}-\mathfrak{af},\ \mathfrak{hf}-\mathfrak{bg},\ \mathfrak{fg}-\mathfrak{ch},\ \text{each}=0:$$
the equations
$$\mathfrak{a}X+\mathfrak{h}Y+\mathfrak{g}Z=0,\quad \mathfrak{h}X+\mathfrak{b}Y+\mathfrak{f}Z=0,\quad \mathfrak{g}X+\mathfrak{f}Y+\mathfrak{c}Z=0,$$
represent each of them one and the same plane, which I say is that of the conic in question.
[p. 515] The three given equations are
$$Y^2+Z^2-2fYZ-\mathfrak{a}=0,$$
$$Z^2+X^2-2gZX-\mathfrak{b}=0,$$
$$X^2+Y^2-2hXY-\mathfrak{c}=0,$$
say these are $U=0$, $V=0$, $W=0$; it is to be shown that $cV-bW$, $aW-cU$, $bU-aV$, each contain the linear factor in question. We have
$$\mathfrak{c}V-\mathfrak{b}W=(\mathfrak{c}-\mathfrak{b})X^2-\mathfrak{b}Y^2+\mathfrak{c}Z^2-2\mathfrak{c}gZX+2\mathfrak{b}hXY;$$
or, what is the same thing,
$$\mathfrak{a}(\mathfrak{c}V-\mathfrak{b}W)=\mathfrak{a}(\mathfrak{c}-\mathfrak{b})X^2-\mathfrak{h}^2Y^2+\mathfrak{g}^2Z^2-2\mathfrak{g}\mathfrak{g}_*ZX+2\mathfrak{h}\mathfrak{h}_*XY.$$
Assuming this
$$=(\mathfrak{a}X+\mathfrak{h}Y+\mathfrak{g}Z)(\lambda X-\mathfrak{h}Y+\mathfrak{g}Z),$$
we have
$$\mathfrak{a}\lambda=\mathfrak{a}(\mathfrak{c}-\mathfrak{b}),$$
$$\mathfrak{g}(\mathfrak{a}+\lambda)=-2\mathfrak{g}\mathfrak{g}_*,$$
$$\mathfrak{h}(-\mathfrak{a}+\lambda)=2\mathfrak{h}\mathfrak{h}_*,$$

that is,

$$\lambda = \mathfrak{c} - \mathfrak{b}, \quad \mathfrak{a} + \lambda = -2\mathfrak{g}_*, \quad -\mathfrak{a} + \lambda = 2\mathfrak{h}_*;$$

but $\lambda = \mathfrak{c} - \mathfrak{b}, = -\mathfrak{h}^2 + \mathfrak{g}^2$, and the other two equations are $\mathfrak{a} + \mathfrak{c} - \mathfrak{b} + 2\mathfrak{g}_* = 0$, $\mathfrak{a} + \mathfrak{b} - \mathfrak{c} + 2\mathfrak{h}_* = 0$, which are identically true.

The values of X, Y, Z are thus determined as the coordinates of the intersection of the conic with the plane $AX + BY + CZ = 0$; or, what is the same thing, of the line

$$AX + BY + CZ = 0,$$

$$\mathfrak{a}X + \mathfrak{h}Y + \mathfrak{g}Z = 0,$$

with any one of the three cylinders.

We may, however, complete the analytical solution in a different manner as follows. Assuming as above $\sqrt{(\mathfrak{bc})} = \mathfrak{f}$, $\sqrt{(\mathfrak{ca})} = \mathfrak{g}$, $\sqrt{(\mathfrak{ab})} = \mathfrak{h}$, and thence $\mathfrak{h}\sqrt{(\mathfrak{c})} - \mathfrak{g}\sqrt{(\mathfrak{b})} = 0$, we obtain from the second and the third equations

$$Y = hX + \sqrt{(\mathfrak{c})}\sqrt{(1 - X^2)}, \quad Z = gX - \sqrt{(\mathfrak{b})}\sqrt{(1 - X^2)},$$

(the signs are one + the other −, in order that this may consist with the equation $\mathfrak{a}X + \mathfrak{h}Y + \mathfrak{g}Z = 0$). Substituting in $AX + BY + CZ = 0$, we have

$$(A + Bh + Cg)X + \{B\sqrt{(\mathfrak{c})} - C\sqrt{(\mathfrak{b})}\}\sqrt{(1 - X^2)} = 0,$$

that is,

$$(A + Bh + Cg)^2X^2 - (B^2\mathfrak{c} + C^2\mathfrak{b} - 2BC\mathfrak{f})(1 - X^2) = 0,$$

[p. 516] or say

$$(A + Bh + Cg)^2X^2 + \{B^2(1 - h^2) + C^2(1 - g^2) - 2BC(gh - f)\}(X^2 - 1) = 0,$$

that is,

$$(A^2 + B^2 + C^2 + 2BCf + 2CAg + 2ABh)X^2 = [B^2 + C^2 + 2BCf - (Bh + Cg)^2],$$

or writing

$$A^2 + B^2 + C^2 + 2BCf + 2CAg + 2ABh = \Delta,$$

say we have

$$\begin{aligned} \Delta X^2 &= B^2 + C^2 + 2BCf - (Bh + Cg)^2, \\ \Delta Y^2 &= C^2 + A^2 + 2CAg - (Cf + Ah)^2, \\ \Delta Z^2 &= A^2 + B^2 + 2ABh - (Ag + Bf)^2. \end{aligned}$$

Now attending to the values of A, B, C, f, g, h, we have

$$BCf,\ CAg,\ ABh = \sin\beta\sin\gamma\cos\alpha,\ \sin\gamma\sin\alpha\cos\beta,\ \sin\alpha\sin\beta\cos\gamma,$$

and thence

$$\Delta = \sin^2\alpha\left(1 - \frac{\theta^2}{a^2}\right) + \sin^2\beta\left(1 - \frac{\theta^2}{b^2}\right) + \sin^2\gamma\left(1 - \frac{\theta^2}{c^2}\right)$$
$$+2(\sin\beta\sin\gamma\cos\alpha + \sin\gamma\sin\alpha\cos\beta + \sin\alpha\sin\beta\cos\gamma);$$

in virtue of $\alpha + \beta + \gamma = 2\pi$, the last term is

$$= 2(\cos\alpha\cos\beta\cos\gamma - 1),$$

whence

$$\Delta = -\theta^2\left(\frac{\sin^2\alpha}{a^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}\right),\ \text{say this is } = -\theta^2\Lambda.$$

Moreover $Bh+Cg = \dfrac{a\sin\alpha}{\sqrt{(a^2-\theta^2)}}$, whence the value of ΔX^2 is

$$= \sin^2\beta\left(1-\frac{\theta^2}{b^2}\right)+\sin^2\gamma\left(1-\frac{\theta^2}{c^2}\right)+2\sin\beta\sin\gamma\cos\alpha-\left(1-\frac{\theta^2}{a^2}\right)^{-1}\sin^2\alpha.$$

Here the constant term is

$$= \sin^2\beta+\sin^2\gamma+2\sin\beta\sin\gamma\cos\alpha,$$

that is,

$$\begin{aligned}&= 1-(1-\sin^2\beta)(1-\sin^2\gamma)+\sin^2\beta\sin^2\gamma+2\sin\beta\sin\gamma\cos\alpha\\ &= 1-\cos^2\beta\cos^2\gamma-\cos^2\alpha+(\cos\alpha+\sin\beta\sin\gamma)^2\\ &= 1-\cos^2\alpha, = \sin^2\alpha,\end{aligned}$$

or the whole is

$$\sin^2\alpha\left(1-\frac{a^2}{a^2-\theta^2}\right)-\theta^2\left(\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}\right),$$

[p. 517] which is

$$= -\theta^2\left(\frac{\sin^2\alpha}{a^2-\theta^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}\right),$$

so that we have

$$\Lambda X^2 = \left(\frac{\sin^2\alpha}{a^2-\theta^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}\right).$$

Similarly,

$$\Lambda Y^2 = \left(\frac{\sin^2\alpha}{a^2}+\frac{\sin^2\beta}{b^2-\theta^2}+\frac{\sin^2\gamma}{c^2}\right),$$

$$\Lambda Z^2 = \left(\frac{\sin^2\alpha}{a^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2-\theta^2}\right);$$

and hence also

$$\Lambda(1-X^2) = \frac{-\theta^2\sin^2\alpha}{a^2(a^2-\theta^2)},\quad \Lambda(1-Y^2) = \frac{-\theta^2\sin^2\beta}{b^2(b^2-\theta^2)},\quad \Lambda(1-Z^2) = \frac{-\theta^2\sin^2\gamma}{c^2(c^2-\theta^2)},$$

where

$$\Lambda = \frac{\sin^2\alpha}{a^2}+\frac{\sin^2\beta}{b^2}+\frac{\sin^2\gamma}{c^2}.$$

The equation in X is

$$\Lambda\left(1-\frac{a^2\cos^2\xi}{a^2-\theta^2}\right) = \frac{-\theta^2\sin^2\alpha}{a^2(a^2-\theta^2)},$$

that is,

$$\Lambda(a^2\sin^2\xi-\theta^2) = \frac{-\theta^2\sin^2\alpha}{a^2},$$

or

$$a^2\sin^2\xi = \frac{\theta^2}{\Lambda}\left(\Lambda-\frac{\sin^2\alpha}{a^2}\right),$$

and the like for η, ζ. Writing for greater convenience $\dfrac{\sin^2\alpha}{a^2}$, $\dfrac{\sin^2\beta}{b^2}$, $\dfrac{\sin^2\gamma}{c^2} = p, q, r$, then $\Lambda = p+q+r$, and we have

$$\sin^2\xi = \frac{\theta^2}{a^2}\cdot\frac{q+r}{p+q+r},\quad \sin^2\eta = \frac{\theta^2}{b^2}\cdot\frac{r+p}{p+q+r},\quad \sin^2\zeta = \frac{\theta^2}{c^2}\cdot\frac{p+q}{p+q+r},$$

(whence also $a^2\sin^2\xi + b^2\sin^2\eta + c^2\sin^2\zeta = 2\theta^2$: as a simple verification, observe that, if the projection is rectangular, the axes being all equally inclined to the plane of projection, then $\xi = \eta = \zeta = 90°$, $a = b = c = \theta\sin s$, and the equation is $3\sin^2 s = 2$; s, s are here the sides of an isosceles quadrantal triangle, the included angle being $120°$, that is, we have $\cos 120°$ $(= -\frac{1}{2}) = -\cot^2 s$, that is, $\cot^2 s = \frac{1}{2}$, or $\sin^2 s = \frac{2}{3}$, which is right).

I remark, that a geometrical solution may be obtained upon very different principles. We have on a sphere the trirectangular triangle XYZ, which by parallel lines is projected into ABC. Every great circle of the sphere is projected into an ellipse having double **[p. 518]** contact at the extremities of a diameter with the ellipse which is the apparent contour of the sphere. Moreover, if the arc of great circle XY is a quadrant, then the radius through X and the tangent at Y are parallel to each other, whence, if Ω be the projection of the centre, and AB the projection of the arc XY, then in the projection the line ΩA and the tangent at B are parallel to each other. It is now easy to derive a construction: with centre Ω, and conjugate semi-axes $(\Omega B, \Omega C)$, $(\Omega C, \Omega A)$, $(\Omega A, \Omega B)$ respectively, describe three ellipses; and find a concentric ellipse having double contact with each of these (there are in fact two such ellipses, one touching the three ellipses internally, and giving an imaginary solution; the other touching them externally, which is the ellipse intended). Drawing then through the ellipse a right cylinder (there are two such cylinders, but only one of them is real), and inscribing in it a sphere, and projecting on to the surface of the sphere by lines parallel to the axis of the cylinder, the three ellipses are projected into three great circles cutting at right angles, or, say, the elliptic arcs BC, CA, AB are projected into the trirectangular triangle XYZ.

615. On the Conic Torus

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 127–129.]

[p. 519] The equation $p + \sqrt{(qr)} + \sqrt{(st)} = 0$, where p, q, r, s, t are linear functions of the coordinates (x, y, z, w), and as such are connected by a linear relation, belongs to a quartic surface having a nodal conic $(p = 0, qr - st = 0)$; and four nodes (conical points), viz. these are the intersections of the line $q = 0$, $r = 0$ with the quadric surface $p^2 - qr - st = 0$, and of the line $r = 0$, $s = 0$ with the same surface. The quartic surface has also four tropes (planes which touch the surface along a conic); viz. these are the planes $q = 0$, $r = 0$, $s = 0$, $t = 0$, the conic of contact or tropal conic in each plane being the intersection of the plane with the before-mentioned quadric surface $p^2 - qr - st = 0$. The planes $q = 0$, $r = 0$, and also the conics in these planes pass through two of the nodes, say A, C; and the planes $s = 0$, $t = 0$, and also the conics in these planes pass through the remaining two nodes, say B, D; so that the relations of the surface are as is shown in fig. 1. It is to be added that AB, BC, CD, DA (but not AC or BD) are lines on the surface.

The planes $q = 0$, $r = 0$, which contain the tropal conics through A, C, are in general distinct from the planes ABC, ADC which contain the line-pairs BA, BC and DA, DC respectively: and so also the planes $s = 0$, $t = 0$, which contain the tropal conics through B, D, are in general distinct from the planes ABD, CBD which contain the line-pairs AB, AD and CB, CD respectively.

If, however, the identical linear relation contain only p, s, t, then the planes $q = 0$, $r = 0$ will be the planes ABC, ADC respectively: and the tropal conics in these planes will consequently be the line-pairs BA, BC, and DA, DC respectively. But the planes $s = 0$, $t = 0$ will continue to be distinct from the planes ABD, CBD: and the tropal conics in the planes $s = 0$, $t = 0$ will remain proper conics.

[p. 520] A surface of the last-mentioned form is

$$mz + \sqrt{(xy)} + \sqrt{(w^2 - z^2)} = 0,$$

viz. this has the nodal conic $z = 0$, $xy - w^2 = 0$, the nodes $\{x = 0, y = 0, (m^2 + 1)z^2 - w^2 = 0\}$, and $(z = 0, w = 0, x = 0)$, $(z = 0, w = 0, y = 0)$, and the tropes $x = 0$, $y = 0$, $z + w = 0$, $z - w = 0$; but the planes $z + w = 0$ and $z - w = 0$ are ordinary tropal planes each touching the surface in a proper conic; the planes $x = 0$, $y = 0$ are special planes each touching along a line-pair.

[Fig. 1: Tetrahedron ABCD with the marked planes and edges as described.]

The equation in question, writing therein $w = 1$ and $x+iy$, $x-iy$ in place of (x, y) respectively, is

$$\sqrt{(x^2 + y^2)} + (mz)^2 = 1 - z^2,$$

which is derived from

$$(x + mz)^2 = 1 - z^2,$$

by the change of x into $\sqrt{(x^2 + y^2)}$; and the surface is consequently the torus generated by the rotation of the conic $(x + mz)^2 = 1 - z^2$ about its diameter. Or, what is the same thing, the surface

$$mz + \sqrt{(xy)} + \sqrt{(w^2 - z^2)} = 0,$$

regarding therein (x, y) as circular coordinates and w as being $= 1$, is a torus. The rational equation is $U = 0$, where we have

$$U = \{(m^2 + 1)z^2 - w^2 + xy\}^2 - 4m^2z^2xy$$

$$= \{xy + (1-m^2)z^2 - w^2\}^2 + 4m^2z^2(z^2 - w^2)$$
$$= x^2y^2 + (m^2+1)^2z^4 + w^4 + (2-2m^2)z^2xy - (2+2m^2)z^2w^2 - 2xyw^2.$$

I find that the Hessian H of this function U contains the factor $xy + (1-m^2)z^2 - w^2$, viz. that we have

$$H = \{xy + (1-m^2)z^2 - w^2\}H',$$

[p. 521] where

$$\begin{aligned} H' &= x^3y^3(1-m^2) \\ &+x^2y^2\left[(3+8m^2+m^4)z^2 + (-3+m^2)w^2\right] \\ &+xy\left\{(3+11m^2+9m^4+m^6)z^4 + (-6-12m^2+6m^4)z^2w^2 + (3+m^2)w^4\right\} \\ &+(1+m^2)\{(1-m^2)z^2 - w^2\}\{(1+m^2)z^2 - w^2\}^2, \end{aligned}$$

giving without much difficulty

$$\begin{aligned} H' &= z^6(1+m^2)^3(1-m^2) \\ &+2z^4\left[(1+4m^2+m^4)xy - (1-m^4)w^2\right](1+m^2) \\ &+z^2(xy-w^2)\left[(1+12m^2-m^4)xy - (1-m^4)w^2\right] \\ &+\left[(1-m^2)xy - (1+m^2)w^2\right]U; \end{aligned}$$

say this is

$$= z^2H'' + [(1-m^2)xy - (1+m^2)w^2]U,$$

where

$$\begin{aligned} H'' &= z^4(1+m^2)^2(1-m^2) \\ &+2z^2(1+m^2)\left[(1+4m^2+m^4)xy - (1-m^4)w^2\right] \\ &+(xy-w^2)\left[(1+12m^2-m^4)xy - (1-m^4)w^2\right], \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} H'' &= x^2y^2(1+12m^2-m^4) \\ &+2xy\left[(1+4m^2+m^4)(1+m^2)z^2 + (-1+6m^2)w^2\right] \\ &+(1-m^4)\{(1+m^2)z^2 - w^2\}^2. \end{aligned}$$

It consequently appears that the complete spinode curve or intersection of the quartic surface and its Hessian, being of order $4 \times 8, = 32$, breaks up into

$$U = 0, \quad xy + (1-m^2)z^2 - w^2 = 0,$$

that is,

$$\begin{array}{lr} \text{conic } z = 0,\ xy - w^2 = 0 \text{ twice}, & \text{order } 4 \\ \text{conic } z + w = 0,\ xy - m^2w^2 = 0, & 2 \\ \text{conic } z - w = 0,\ xy - m^2w^2 = 0, & 2 \end{array}$$

and

$$U = 0, \quad z^2H'' = 0,$$

that is,

$$\begin{array}{lr} U = 0,\ z^2 = 0,\ \text{conic } z = 0,\ xy - w^2 = 0 \text{ four times}, & \text{order } 8 \\ \text{proper spinode curve } U = 0,\ H'' = 0, & 16 \\ \hline & 32; \end{array}$$

viz. the intersection is made up of the conic $z = 0$, $xy - w^2 = 0$ six times, the conies $z \pm w = 0$, $xy - m^2w^2 = 0$ each twice, and the proper spinode curve of the order 16.

616. A Geometrical Illustration of the Cubic Transformation in Elliptic Functions

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 211–216.]

[p. 522] Consider the cubic curve

$$x^3 + y^3 + z^3 + 6lxyz = 0.$$

If through one of the inflexions $z = 0$, $x + y = 0$, we draw an arbitrary line $z = u(x + y)$, we have at the other intersections of this line with the curve

$$u\{u^2(x + y)^2 + 6lxy\} + x^2 - xy + y^2 = 0;$$

that is,

$$(u^3 + 1)(x^2 + y^2) + 2xy(u^3 + 3lu - \tfrac{1}{2}) = 0;$$

and from this equation it appears that the ratio $x : y$ is given as a function involving the square root of

$$(u^3 + 3lu - \tfrac{1}{2})^2 - (u^3 + 1)^2,$$

which, rejecting a factor 3, is

$$= (2u^3 + 3lu + \tfrac{1}{2})(lu - \tfrac{1}{2}).$$

It may be noticed that $lu - \frac{1}{2} = 0$ gives the value of u, which in the equation $z = u(x + y)$ belongs to the tangent at the inflexion; and $2u^3 + 3lu + \frac{1}{2} = 0$ gives the values which belong to the three tangents from the inflexion.

It thus appears that the coordinates x, y, z of any point of the curve can be expressed as proportional to functions of u involving the radical

$$\sqrt{\{(lu - \tfrac{1}{2})(2u^3 + 3lu + \tfrac{1}{2})\}},$$

and the theory of the curve is connected with that of a quasi-elliptic integral depending on this radical.

[p. 523] Taking ω an imaginary cube root of unity, write

$$\omega x + \omega^2 y - 2lz = x',$$

$$\omega^2 x + \omega y - 2lz = y',$$

$$x + y - 2lz = z';$$

then we have

$$x'y'z' = x^3 + y^3 - 8l^3z^3 + 6lx^2yz = x^3 + y^3 + z^3 + 6lxyz - (1 + 8l^3)z^3.$$

Also

$$-6lz = x' + y' + z', \quad z^3 = \frac{-1}{216l^3}(x' + y' + z')^3,$$

whence

$$x'^3 + y'^3 + z'^3 - \frac{216l^3}{1 + 8l^3}x'y'z' = \frac{216l^3}{1 + 8l^3}(x'^3 + y'^3 + z'^3 + 6lx'y'z'),$$

so that, putting

$$6m = \frac{-216l^3}{1 + 8l^3},$$

or, what is the same thing,

$$8l^3m^3 + l^3 + m^3 = 0,$$

the equation of the curve is

$$(x' + y' + z')^3 + 216m^3x'y'z' = 0;$$

and if we write

$$x' : y' : z' = X^3 : Y^3 : Z^3,$$

then the original curve is transformed into

$$(X^3 + Y^3 + Z^3)^3 + 216m^3X^3Y^3Z^3 = 0,$$

a curve of the ninth order breaking up into three cubic curves, one of which is

$$X^3 + Y^3 + Z^3 + 6mXYZ = 0,$$

and for the other two we write herein $m\omega$ and $m\omega^2$ respectively in place of m. Attending only to the first curve, we have

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$
$$X^3 + Y^3 + Z^3 + 6mXYZ = 0,$$

as corresponding curves, the corresponding points being connected by the relation

$$\omega x + \omega^2 y - 2lz : \omega^2 x + \omega y - 2lz : x + y - 2lz = X^3 : Y^3 : Z^3,$$

or, for convenience, we may write

$$\begin{aligned} \omega x + \omega^2 y - 2lz &= X^3, &\quad \text{giving} \quad 3x &= \omega^2 X^3 + \omega Y^3 + Z^3, \\ \omega^2 x + \omega y - 2lz &= Y^3, & 3y &= \omega X^3 + \omega^2 Y^3 + Z^3, \\ x + y - 2lz &= Z^3, & -6lz &= X^3 + Y^3 + Z^3. \end{aligned}$$

[p. 524] This is a $(1, 3)$ correspondence; viz. to a given point on the curve (m), there corresponds one point on (l); but to a given point on (l), three points on (m). As to the first case, this is obvious. As to the second case, if the point (x, y, z) is given, then the corresponding point (X, Y, Z) on the other curve will lie on one of the three lines

$$Y^3(\omega x + \omega^2 y - 2lz) - X^3(\omega^2 x + \omega y - 2lz) = 0;$$

each of these intersects the curve (m) in three points: but of the points in the same line it is only one which is a corresponding point of (x, y, z), and the number of the corresponding points is consequently the same as the number of lines, viz. it is $= 3$.

We infer that the above equations lead to a cubic transformation of the quasi-elliptic integral

$$\int du \div \sqrt{\{(lu - \tfrac{1}{2})(2u^3 + 3lu + \tfrac{1}{2})\}},$$

into one of the like form

$$\int dv \div \sqrt{\{(mv - \tfrac{1}{2})(2v^3 + 3mv + \tfrac{1}{2})\}};$$

and this is now to be verified.

We have, as before, the line $z = u(x + y)$ meeting the curve (l) in the points

$$(u^3 + 1)(x^2 + y^2) + 2xy(u^3 + 3lu - \tfrac{1}{2}) = 0;$$

and if similarly through an inflexion of the curve (m) we take the line $Z = v(X+Y)$, this meets the curve in the points

$$(v^3+1)(X^2+Y^2) + 2XY(v^3+3mv-\tfrac{1}{2}) = 0.$$

Then if (x,y,z), (X,Y,Z) are taken to be the corresponding points as above, we can obtain v as a function of u. We, in fact, have

$$-2lu = \frac{-2lz}{x+y} = \frac{X^3+Y^3+Z^3}{-X^3-Y^3+2Z^3} = \frac{X^3+Y^3+v^3(X+Y)^3}{-(X^3+Y^3)+2v^3(X+Y)^3}$$
$$= \frac{X^2-XY+Y^2+v^3(X+Y)^2}{-X^2+XY-Y^2+2v^3(X+Y)^2}$$
$$= \frac{(v^3+1)(X^2+Y^2)+(2v^3-1)XY}{(2v^3-1)(X^2+Y^2)+(4v^3+1)XY},$$

or, since we have

$$(v^3+1)(X^2+Y^2) + 2XY(v^3+3mv-\tfrac{1}{2}) = 0,$$

that is,

$$X^2+Y^2 : XY = -2v^3-6mv+1 : v^3+1,$$

the equation becomes

$$-2lu = \frac{-6mv(v^3+1)}{(2v^3-1)(-2v^3-6mv+1)+(4v^3+1)(v^3+1)}$$
$$= \frac{-6mv(v^3+1)}{-3v(4mv^3-3v^2-2m)},$$

[p. 525] or say,

$$-lu = m(v^3+1) \div (\Box), \quad \text{where the denominator } = 4mv^3-3v^2-2m.$$

This may also be written

$$-(lu-\tfrac{1}{2}) = 3v^2(mv-\tfrac{1}{2}) \div \Box.$$

Proceeding to calculate $2u^3+3lu+\frac{1}{2}$, omitting the denominator $(4mv^3-3v^2-2m)^3$, this is

$$\frac{-2}{l^3}(v^3+1)^3 - 3m(v^3+1)(4mv^3-3v^2-2m)^2 + \tfrac{1}{2}(4mv^3-3v^2-2m)^3;$$

or, observing that

$$m^3 = \frac{-l^3}{1+8l^3},$$

that is,

$$\frac{1}{l^3} = \frac{-1-8m^3}{m^3} \quad \text{or} \quad \frac{m^3}{l^3} = -1-8m^3,$$

the numerator is

$$= 2(1+8m^3)(v^3+1)^3 - 3m(v^3+1)(4mv^3-3v^2-2m)^2 + \tfrac{1}{2}(4mv^3-3v^2-2m)^3,$$

which is found to be identically

$$\tfrac{1}{2}(2v^3+3mv+\tfrac{1}{2})(v^3+6mv-2)^2,$$

viz. we have

$$2u^3+3lu+\tfrac{1}{2} = (2v^3+3mv+\tfrac{1}{2})(v^3+6mv-2)^2 \div (4mv^3-3v^2-2m)^3.$$

and hence

$$(lu-\tfrac{1}{2})(2u^3+3lu+\tfrac{1}{2}) = -3(mv-\tfrac{1}{2})(2v^3+3mv+\tfrac{1}{2})(v^3+6mv-2)^2v^2 \div (4mv^3-3v^2-2m)^4.$$

Moreover, we find

$$ldu = 3mdv\cdot v(v^3+6mv-2)\div(4mv^3-3v^2-2m)^2,$$

and we thence have

$$\frac{ldu}{\sqrt{\{(lu-\frac{1}{2})(2u^3+3lu+\frac{1}{2})\}}} = \sqrt{(-3)}\cdot\frac{mdv}{\sqrt{\{(mv-\frac{1}{2})(2v^3+3mv+\frac{1}{2})\}}},$$

viz. this differential equation corresponds to the integral equation

$$-lu = m(v^3+1)\div(4mv^3-3v^2-2m),$$

where $8l^3m^3+l^3+m^3=0$, which corresponds to the modular equation.

It may be remarked that, if v is the same function of u', l, in that u is of v, m, l; viz. if

$$-mv = l(u'^3+1)\div(4lu'^3-3u'^2-2m'),$$

then

$$\frac{mdv}{\sqrt{\{(mv-\frac{1}{2})(2v^3+3mv+\frac{1}{2})\}}} = \sqrt{(-3)}\cdot\frac{-ldu'}{\sqrt{\{(lu'-\frac{1}{2})(2u'^3+3lu'+\frac{1}{2})\}}},$$

and consequently

$$\frac{du}{\sqrt{\{(lu-\frac{1}{2})(2u^3+3lu+\frac{1}{2})\}}} = \frac{-3du'}{\sqrt{\{(lu'-\frac{1}{2})(2u'^3+3lu'+\frac{1}{2})\}}},$$

which accords with the general theory of the cubic transformation.

[p. 526] We may inquire into the relation between the absolute invariants of the two curves. Taking the absolute invariant to be

$$\Omega = \frac{64S^3-T^2}{64S^3},$$

where S and T bear the usual significations, we have for the one curve

$$\Omega = \frac{(1+8l^3)^3}{64l^3(1-l^3)^3},$$

and for the other curve

$$\Omega' = \frac{(1+8m^3)^3}{64m^3(1-m^3)^3},$$

and, as above, $8l^3m^3+l^3+m^3=0$: writing herein

$$l^3 = \frac{-\alpha'}{8\alpha'},\quad m^3 = \frac{-\beta'}{8\beta'},$$

the relation between α', β' is simply $\alpha'+\beta'=1$; and the values of Ω, Ω' are found to be

$$\Omega = \frac{64\alpha'(1-\alpha')^3}{(1+8\alpha')^3},\quad \Omega' = \frac{64\beta'(1-\beta')^3}{(1+8\beta')^3},$$

viz. the required relation is given by the elimination of α', β' from these three equations. Or, what is the same thing, writing $\alpha' = \frac{1}{2} + \theta$, and therefore $\beta' = \frac{1}{2} - \theta$, we have

$$(5 + 8\theta)^3\Omega = 4(1 + 2\theta)(1 - 2\theta)^3,$$

$$(5 - 8\theta)^3\Omega' = 4(1 + 2\theta)^3(1 - 2\theta),$$

and the elimination of θ from these equations gives the required relation between Ω and Ω'.

It of course follows that, if we have a cubic transformation

$$\frac{dx}{\sqrt{\{(a, b, c, d, e \mid x, 1)^4\}}} = \frac{Cdx}{\sqrt{\{(a', b', c', d', e' \mid x, 1)^4\}}},$$

then the absolute invariants Ω, Ω' of the two quartic functions are connected by the above relation. I have obtained this result, by reducing the radicals to the standard forms

$$\sqrt{(1 - x^2 \cdot 1 - k^2x^2)},\ \sqrt{(1 - x^2 \cdot 1 - \lambda^2x^2)},$$

from the known modular equation as represented by the equations

$$k^2 = \frac{\alpha^3(2 + \alpha)}{1 + 2\alpha},\quad \lambda^2 = \frac{\alpha(2 + \alpha)^3}{(1 + 2\alpha)^3};$$

viz. the values of the absolute invariants

$$\left(= 1 - \frac{27}{2J^2},\ 1 - \frac{27J'^2}{2J'^2}\right)$$

are

$$\frac{(k^4 + 14k^2 + 1)^3}{\quad},\quad \frac{(\lambda^4 + 14\lambda^2 + 1)^3}{\quad},$$

but the method of effecting this is by no means obvious.

617. On the Scalene Transformation of a Plane Curve

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 321–328.]

[p. 527] The transformation by reciprocal radius vectors can be effected mechanically by Sylvester's Peaucellier-cell. But, employing a more general cell (considered incidentally by him) which may be called the scalene-cell, we have the scalene transformation in question[1]; viz. if, in two curves, r, r' are radius vectors belonging to the same angle (or say opposite angles) θ, then the relation between r, r' is

$$rr'(r+r') + (m^2 - l^2)r' + (m^2 - n^2)r = 0;$$

or, as this may also be written,

$$r^2 + \left(r' - \frac{l^2 - m^2}{r'}\right) r + m^2 - n^2 = 0.$$

The transformation is, it will be seen, an interesting one for its own sake, independently of the remarkably simple mechanical construction, viz. the scalene cell is simply a system of 3 pairs of equal rods PA, QA; PB, QB; PC, QC (fig. 1, p. 528), jointed together at and capable of rotating about the points P, Q, A, B, C; the three lengths PA, PB, PC (say these are $= l$, m, n) are all of them unequal: in the case of any two of them equal, we have Peaucellier or isosceles cell. The effect of the arrangement is that the points A, B, C are retained in a right line, the distances BA, $= r'$, and BC, $= r$, being connected by the above-mentioned equation; so that taking B as a fixed point, if the point A describe any given curve, the point C will describe the corresponding or transformed curve.

In the case where the given curve is a right line or a circle, we may through B draw at right angles to the curve the axis $x'Bx$: viz. in the case of the circle, **[p. 528]** the axis $x'Bx$ passes through its centre; and we measure the angle θ from this line, viz. we write $\angle xBC = \angle x'BA = \theta$.

[Fig. 1: The scalene cell with hinge points P, Q, A, B, C and rods of lengths l, m, n.]

Suppose, first, that the locus of A is a right line, or a circle passing through B. Its equation is $r' = \dfrac{c}{\cos\theta}$ or $= c\cos\theta$; and we accordingly have for the transformed curve

$$r^2 + \left(\frac{c}{\cos\theta} - \frac{l^2 - m^2}{c\cos\theta\,/\,1}\right) r + m^2 - n^2 = 0,$$

or else

$$r^2 + \left(c\cos\theta - \frac{l^2 - m^2}{c\cos\theta}\right) r + m^2 - n^2 = 0:$$

viz. multiplying in each case by $r\cos\theta$, and then writing $r\cos\theta = x$, $r^2 = x^2 + y^2$, the equations become

$$x(x^2+y^2) + c(x^2+y^2) - \frac{l^2 - m^2}{c}(x^2+y^2) + (m^2 - n^2)x = 0,$$

and

$$x(x^2+y^2) + cx^2 - \frac{l^2 - m^2}{c}(x^2+y^2) + (m^2 - n^2)x = 0;$$

viz. in each case the curve is a circular cubic passing through the origin B and having an asymptote parallel to the axis of y. The curve is nodal, if $m = n$, viz. in this case the origin is a node: or if $c = \sqrt{(l^2 - n^2)} + \sqrt{(m^2 - n^2)}$.

[1]The transformation itself, and doubtless many of the results obtained by means of it, are familiar to Prof. Sylvester; and I abandon all claim to priority.

Suppose next that the locus of A is a circle, centre at a distance $= \gamma$ along Bx' and radius $= h$: we have

$$r'^2 - 2\gamma r' \cos\theta + \gamma^2 - h^2 = 0,$$

viz. if

$$\gamma^2 - h^2 = -(l^2 - m^2),$$

or, what is the same thing,

$$h^2 + m^2 = \gamma^2 + l^2,$$

then we have

$$r' - \frac{l^2 - m^2}{r'} = 2\gamma\cos\theta,$$

[Continued in next instalment, pp. 528–534.]

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 529–540 (Papers 617–619)

Transcribed from the Original Publications

Contents

617. On the Scalene Transformation of a Plane Curve (continued)

[p. 529] and the transformed curve is

$$r^2 + 2\gamma r\cos\theta + m^2 - n^2 = 0,$$

or, as this may be written,

$$r^2 + 2\gamma r\cos\theta + \gamma^2 - f^2 = 0,$$

where $\gamma^2 - f^2 = m^2 - n^2$, that is, $f^2 + m^2 = \gamma^2 + n^2$; viz. this is a concentric circle radius f.

The theorem may be presented as follows. Consider two concentric circles, centre O and radii h, f respectively; take an arbitrary point B, distance $OB = \gamma$; and taking m arbitrary, determine l, n by the equations

$$l^2 = m^2 + h^2 - \gamma^2, \quad n^2 = m^2 + f^2 - \gamma^2;$$

then drawing through B an arbitrary line to meet the circles in A, C respectively; also describing a circle, centre B and radius $= m$; and through A drawing a line ABC perpendicular to BA to meet the last-mentioned circle in two points P, Q: for these points, the distances from the points A, B, C are $= l$, m, n respectively.

To verify this, take O as the origin, OB for the axis of x, θ the inclination of ABC to this axis, $BA = r'$, $BC = r$; the coordinates of C, B, A are

$$\gamma + r\cos\theta, \quad r\sin\theta,$$

$$\gamma, \quad 0,$$

$$\gamma + r'\cos\theta, \quad -r'\sin\theta,$$

whence, taking (x, y) for the coordinates of P (or Q), the equations to be verified are

$$(x - \gamma - r\cos\theta)^2 + (y - r\sin\theta)^2 = n^2,$$

$$(x - \gamma)^2 + y^2 = m^2,$$

$$(x - \gamma + r'\cos\theta)^2 + (y + r'\sin\theta)^2 = l^2.$$

By means of the second equation, the other two become

$$-2(x - \gamma)r\cos\theta - 2yr\sin\theta + r^2 = n^2 - m^2,$$

$$2(x - \gamma)r'\cos\theta + 2yr'\sin\theta + r'^2 = l^2 - m^2;$$

or, substituting for $n^2 - m^2$, $l^2 - m^2$ the values $f^2 - \gamma^2$ and $h^2 - \gamma^2$, the equations are

$$-2xr\cos\theta - 2yr\sin\theta + 2\gamma r\cos\theta + r^2 - f^2 + \gamma^2 = 0,$$

$$2xr'\cos\theta + 2yr'\sin\theta - 2\gamma r'\cos\theta + r'^2 - h^2 + \gamma^2 = 0,$$

viz. in virtue of the equations of the two circles, these reduce themselves each of them to

$$x\cos\theta + y\sin\theta = 0,$$

which equation, together with the second equation

$$(x - \gamma)^2 + y^2 = m^2,$$

determine (x, y) as above.

[p. 530] Reverting to the case where the locus of A is the circle

$$r'^2 - 2\gamma r' \cos\theta + \gamma^2 - h^2 = 0,$$

this gives

$$r' = \gamma\cos\theta + \sqrt{(h^2 - \gamma^2\sin^2\theta)},$$
$$\frac{1}{r'} = \frac{\gamma\cos\theta - \sqrt{(h^2-\gamma^2\sin^2\theta)}}{\gamma^2 - h^2},$$

so that for the transformed curve we have

$$r^2 + \gamma\cos\theta\left\{(1-\lambda)r + (1+\lambda)\frac{\gamma^2-h^2}{r}\right\} + (1-\lambda)\sqrt{(h^2-\gamma^2\sin^2\theta)} + m^2 - n^2 = 0.$$

Putting for shortness $\dfrac{l^2-m^2}{\gamma^2-h^2} = \lambda$, and for r, $r\cos\theta$, $r\sin\theta$, writing $\sqrt{(x^2+y^2)}$, x, y respectively, this is

$$x^2 + y^2 + (1-\lambda)\gamma x + (1+\lambda)\sqrt{[h^2(x^2+y^2) - \gamma^2 y^2]} + m^2 - n^2 = 0,$$

or, what is the same thing,

$$\{x^2 + y^2 + (1-\lambda)\gamma x + m^2 - n^2\}^2 = (1+\lambda)^2\{h^2(x^2+y^2) - \gamma^2y^2\},$$

= a bicircular quartic. In the case $\lambda = -1$, it reduces itself to the circle

$$x^2 + y^2 + 2\gamma x + m^2 - n^2 = 0$$

twice, which is the case considered above; and in the case $\lambda = 1$, or $l^2 - m^2 = h^2 - \gamma^2$, the equation is

$$(x^2 + y^2 + m^2 - n^2)^2 = 4\{h^2(x^2+y^2) - \gamma^2y^2\},$$

so that the curve is symmetrical in regard to each axis. In the case $\gamma = 0$, the locus is a pair of concentric circles, centre B.

The equation

$$\{x^2 + y^2 + (1-\lambda)\gamma x + m^2 - n^2\}^2 = (1+\lambda)^2\{h^2(x^2+y^2) - \gamma^2y^2\},$$

which contains the four constants λ, γ, h and $m^2 - n^2$, may be written in the form

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2,$$

(where the constants A, B, a, e are also arbitrary). This is, in fact, the equation of the general symmetrical bicircular quartic, referred to a properly-selected point on the axis as origin, viz. the origin is the centre of any one of the three involutions formed by the vertices (or points on the axis); say it is any one of the three involution-centres of the curve.

To show this, assume

$$(x-\alpha)(x-\beta)(x-\gamma)(x-\delta) = x^4 - px^3 + qx^2 - rx + s;$$

[p. 531] then, taking B arbitrary, the equation of the symmetrical bicircular quartic having for vertices the points $x = \alpha$, $x = \beta$, $x = \gamma$, $x = \delta$, is

$$(x^2 + y^2 - \tfrac{1}{2}px + B)^2 = (2B + \tfrac{1}{4}p^2 - q)x^2 + (b + 2\theta A)x + (-s + B^2),$$

in fact, this is the form of the general equation, and writing therein $y=0$, it becomes $x^4-px^3+qx^2-rx+s=0$, that is, $(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)=0$. Hence, writing for convenience

$$A=-\tfrac{1}{2}p,$$

$$a=2B+\tfrac{1}{4}p^2-q,$$

$$b=r-pB,$$

$$c=-s+B^2,$$

the equation is

$$(x^2+y^2+Ax+B)^2=ax^2+bx+c.$$

This may be written

$$(x^2+y^2+Ax+B+\theta)^2=(a+2\theta)x^2+2\theta y^2+(b+2\theta A)x+c+2B\theta+\theta^2,$$

viz. assuming $\theta=-\dfrac{b}{2A}$ in order on the right-hand side to destroy the term in x, the equation is

$$\left(x^2+y^2+Ax+B-\frac{b}{2A}\right)^2=\left(a-\frac{b}{A}\right)x^2+y^2\cdot\frac{-b}{A}+\frac{1}{4A^2}\,(b^2-4ABb+4A^2c),$$

which is of the form

$$(x^2+y^2+Ax+B)^2=ax^2+ey^2+f;$$

and if $f=0$, that is, if $b^2-4ABb+4A^2c=0$, then it is of the required form

$$(x^2+y^2+Ax+B)^2=ax^2+ey^2.$$

We have

$$\begin{aligned}b^2-4ABb+4A^2c&=(r-pB)^2+2pB(r-pB)+p^2(-s+B^2)\\&=r^2-p^2s,\end{aligned}$$

or the required condition is $r^2-p^2s=0$. But we have

$$r^2-p^2s=(\alpha\delta-\beta\gamma)(\beta\delta-\gamma\alpha)(\gamma\delta-\alpha\beta),$$

as is easily verified by writing

$$p=\delta+p_{\prime},\quad q=\delta p_{\prime}+q_{\prime},\quad r=\delta q_{\prime}+r_{\prime},\quad s=\delta r_{\prime},$$

where $p_{\prime}$, $q_{\prime}$, $r_{\prime}$ stand for

$$\alpha+\beta+\gamma,\quad \beta\gamma+\gamma\alpha+\alpha\beta,\quad \alpha\beta\gamma,$$

[p. 532] respectively. The required condition thus is

$$(\alpha\delta-\beta\gamma)(\beta\delta-\gamma\alpha)(\gamma\delta-\alpha\beta)=0,$$

viz. the origin (that is, the fixed point B of the cell) must be at one of the three involution-centres.

Comparing the equation

$$\{x^2+y^2+(1-\lambda)\gamma x+m^2-n^2\}^2=(1+\lambda)^2\{h^2(x^2+y^2)-\gamma^2y^2\}$$

with the equation

$$(x^2+y^2+Ax+B)^2=ax^2+ey^2,$$

we have

$$A = (1-\lambda)\gamma,$$
$$B = m^2 - n^2,$$
$$a = (1+\lambda)^2 h^2,$$
$$e = (1+\lambda)^2(h^2 - \gamma^2),$$

and thence $a - e = (1+\lambda)^2\gamma^2$. Consequently $\dfrac{a-e}{A^2} = \left(\dfrac{1+\lambda}{1-\lambda}\right)^2$, which gives λ; and then

$$h^2 = \frac{a}{(1+\lambda)^2}, \quad \gamma^2 = \frac{a-e}{(1+\lambda)^2}, \quad m^2 - n^2 = B;$$

viz. we thus have the values of λ, h, γ and $m^2 - n^2$ for the description of a given curve $(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2$. In order that the description may be possible, a and $a - e$ must be each of them positive.

For the Cartesian a is $= e$, whence $1 + \lambda = 0$, and the equation becomes

$$(x^2 + y^2 + 2\gamma x + m^2 - n^2)^2 = 0,$$

which is a twice repeated circle; hence the Cartesian cannot be constructed by means of a cell as above.

To obtain a construction of the Cartesian, it may be remarked that, if a symmetrical bicircular quartic be inverted in regard to an axial focus, viz. if the focus be taken as the centre of inversion, we obtain a Cartesian. The axial foci of the curve

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ey^2$$

are points on the axis, the abscissa $x = \theta$ being determined by the equation

$$e(\theta^2 + A\theta + B)^2 - a(\theta^2 - B)^2 - ae\theta^2 = 0.$$

The equation referred to a focus as origin is therefore

$$\{x^2 + y^2 + (A + 2\theta)x + \theta^2 + A\theta + B\}^2 = ax^2 + ey^2 + 2a\theta x + a\theta^2;$$

then inverting, viz. for x, y writing $\dfrac{k^2 x}{r^2}$, $\dfrac{k^2 y}{r^2}$ (k arbitrary), we have, as may be verified the equation of a Cartesian.

[p. 533] The inversion can be performed mechanically by an ordinary Peaucellier-cell; the complete apparatus for the construction of a Cartesian is therefore as in fig. 2, viz. we have a cell BAC as before, B a fixed point, locus of A a circle (for convenience of drawing, the arrangement has been made BAC instead of ABC), and we connect with C a Peaucellier-cell $CA'B'$, arms n, n', m', the fixed point B' being on the axis, which is the line joining B with the centre of the circle described by A. This being so, then A describing a circle, C will describe a symmetrical bicircular quartic, and A' will describe the inverse of this, being in general a like curve; but if the position of B' be properly determined, viz. if B' be at a focus of the first-mentioned quartic.

Fig. 2.

then A' will describe a Cartesian. A further investigation would be necessary in order to determine how to adapt the apparatus to the description of a given Cartesian.

A more convenient mechanical description of a Cartesian is, however, that given in the paper which follows the present one [618].

The equation

$$\{x^2+y^2+(1-\lambda)\gamma x+m^2-n^2\}^2=(1+\lambda)^2\{h^2(x^2+y^2)-\gamma^2y^2\}$$

may also be written

$$\begin{gathered}\{x^2+y^2+(1-\lambda)\gamma x-\tfrac{1}{2}(1+\lambda)^2(h^2-\gamma^2)+m^2-n^2\}^2\\ =(1+\lambda)^2\{\gamma^2x^2-(1-\lambda)(h^2-\gamma^2)\gamma x-\tfrac{1}{4}(1+\lambda)^2(h^2-\gamma^2)^2-(m^2-n^2)(h^2-\gamma^2)\},\end{gathered}$$

viz. the equation is now brought into the form

$$(x^2+y^2+Ax+B)^2=ax^2+bx+c.$$

Expressing the coefficients A, B, a, b, c in terms of λ, γ, h, m^2-n^2, it appears by what precedes, that we should have identically $b^2-4ABb+4A^2c=0$, viz. this is the equation which expresses that the origin is an involution-centre.

If, instead of the original cell, we consider a new cell obtained by substituting for the arms PB, BQ, the arms pb, bq, jointed on to the points p, q on the arms CP, CQ respectively, and instead of B, making b the fixed point; then writing $Cp=kn$, $pb=km$, so that the parameters of the cell are l, m, n, k, and taking $Ch=s$, $hA=s'$,

[p. 534] we have $s=kr$, $s+s'=r+r'$, that is, $r=\dfrac{s}{k}$, $r'=\dfrac{k-1}{k}s+s'$. Substituting in the equation between r, r', written for greater convenience in the form

$$(r+r')(rr'+m^2-l^2)+(l^2-n^2)r'=0,$$

the relation between s, s' is found to be

$$(s+s')\left(\frac{k-1}{k^2}s^2+\frac{ss'}{k}+m^2-l^2\right)+(l^2-n^2)\left(\frac{k-1}{k}s+s'\right)=0.$$

On account of the term in s^3, this equation in its general form does not, it would appear, give rise to transformations of much elegance. If, however, $l=n$, then the relation becomes

$$(k-1)s^2+kss'=k^2(m^2-l^2);$$

and in particular, if $k=2$, then

$$s^2+2ss'=4(l^2-m^2),\quad\text{or say}\quad(s+s')^2-s'^2=4(l^2-m^2),$$

viz. taking A instead of b as the fixed point, the relation between the radii AC, Ah is $\rho^2-\rho'^2=4(l^2-m^2)$; the cell is in this case Sylvester's "quadratic-binomial extractor."

618. On the Mechanical Description of a Cartesian

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 328–330.]

[p. 535] Suppose that in two different curves the radius vectors r, r', which belong to the same angle θ, are connected by the equation

$$r^2 + \left(Mr' + N + \frac{P}{r'}\right) r + B = 0;$$

then, taking one of the curves to be the circle

$$Mr' + \frac{P}{r'} = A\cos\theta,$$

the other curve is

$$r^2 + (A\cos\theta + N)r + B = 0,$$

viz. this is a Cartesian. It perhaps would not be difficult to contrive a mechanical arrangement to connect the radius vectors in accordance with the foregoing equation; but the required result may be obtained equally well by means of a particular case of the relation in question; viz. taking this to be

$$r^2 + (-r' + N)r + B = 0,$$

then, taking the one curve to be the circle $r' = A\cos\theta$, the other curve is the Cartesian,

$$r^2 + (-A\cos\theta + N)r + B = 0, \quad \text{that is,} \quad r^2 + (A\cos\theta + N)r + B = 0.$$

The relation between the radius vectors may in this case be written

$$r' = N + r + \frac{B}{r},$$

which can be constructed mechanically by a simple addition to the Peaucellier-cell, viz. if we joint on to C (fig. 1, p. 536) a rod CD having a slot, working on a pin at A, so that the rod is thereby kept always in the line BAC, then, making B the

[p. 536] fixed point and taking $BA = r$, we have $AC = \dfrac{B}{r}$, whence $BC = r + \dfrac{B}{r}$, or D being a point at the distance $CD = a$, from the point C, and denoting BD by r', we have

$$r' = r + \frac{B}{r} + a,$$

which is an equation of the required form; whence, if the point A describe a circle passing through B, then the point D will describe a Cartesian.

Fig. 1.

The equation of the Cartesian is $r^2 + (A\cos\theta + N)r + B = 0$, viz. this is

$$x^2 + y^2 + Ax + B = -N\sqrt{(x^2 + y^2)},$$

or writing $N^2 = a$, it is

$$(x^2 + y^2 + Ax + B)^2 = ax^2 + ay^2,$$

which is the form considered in the preceding paper. It may be further observed in regard to it that, starting from the focal equation $r = ls + m$, where r, s are the distances of a point

(x, y) of the Cartesian from any two of its three foci, this equation gives $r^2 - l^2 s^2 = m^2 + 2mr$, or writing $r^2 = x^2 + y^2$, $s^2 = (x - a)^2 + y^2$, the function on the left-hand is of the form $(1 + l^2)(x^2 + y^2 + Ax + B)$, whence, assuming $\sqrt{(1 + l^2)} = \sqrt{(a)}$, the equation becomes as above

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2).$$

Taking the distance $r = \sqrt{(x^2 + y^2)}$ to be measured from a given focus, it is easy to see that, no matter which of the other two foci we associate with it, we obtain the same equation $(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2)$; viz. starting with any one focus, we connect with it a determinate circle $x^2 + y^2 + Ax + B = 0$, and a determinate coefficient a, such that taking this focus as the origin, the equation of the curve is

$$(x^2 + y^2 + Ax + B)^2 = a(x^2 + y^2);$$

but there are for the given curve three such forms of equation, according as the origin is taken at one or other of the three foci.

(Addition, Feb. 1875.) It is obviously the same thing, but I find that it is mechanically more convenient to derive the Cartesian from the Limaçon $r' = N - A\cos\theta$, by the transformation $r' = r + \dfrac{B}{r}$: I have on this principle constructed an apparatus whereby the Cartesian is described on a rotating board by a pencil moving in a fixed line.

619. On an Algebraical Operation

[From the Quarterly Journal of Pure and Applied Mathematics, vol. xiii. (1875), pp. 369–375.]

[p. 537] I consider

$$\Omega F(a, x),$$

an operation Ω performed upon $F(a, x)$ a rational function of (a, x); viz. F being first expanded or regarded as expanded in ascending powers of a, the coefficients of the several powers are then to be expanded or regarded as expanded in ascending powers of x, and the operation consists in the rejection of all negative powers of x.

In the cases intended to be considered, F contains only positive powers of a: but this restriction is not necessary to the theory.

The investigation has reference to the functions $A(x)$ of my "Ninth Memoir on Quantics," *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462]; for instance, we there have as regards the covariants of a quadric

$$A(x) = \frac{1 - x^{-4}}{1 - ax^2 \,.\, 1 - a \,.\, 1 - ax^{-2}},$$

and consequently, in the present notation,

$$A(x) = \Omega\,\frac{1 - x^{-4}}{1 - ax^2 \,.\, 1 - a \,.\, 1 - ax^{-2}};$$

by a process of development and summation, the value of this expression was found to be

$$\frac{1}{1 - ax^2 \,.\, 1 - a^2},$$

[p. 538] and in the other more complicated cases the value of $A(x)$ was found only by trial and verification. What I purpose now to show is that the operation Ω can be performed without any development in an infinite series; or say that it depends on finite algebraical operations only.

It is clear that if $F(a,x)$, considered to be developed as above contains only positive powers of x, then

$$\Omega F(a,x) = F(a,x);$$

And if it contains only negative powers of x, then $\Omega F(a,x) = 0$.

Consider now $\Omega\,\dfrac{\phi(x)}{x-a}$, where $\phi(x)$ is a function containing only positive powers of x; we have

$$\frac{\phi(x)}{x-a} = \frac{\phi(x)-\phi(a)}{x-a} + \frac{\phi(a)}{x-a},$$

and thence

$$\Omega\,\frac{\phi(x)}{x-a} = \Omega\,\frac{\phi(x)-\phi(a)}{x-a} + \Omega\,\frac{\phi(a)}{x-a} = \frac{\phi(x)-\phi(a)}{x-a},$$

since $\dfrac{\phi(x)-\phi(a)}{x-a}$ is a rational and integral function of x, which when developed contains only positive powers of x, and $\dfrac{\phi(a)}{x-a}$ when developed contains only negative powers of x.

Consider next $\Omega\,\dfrac{\phi(x)}{x^2-a}$, where $\phi(x)$ is a rational and integral function of x; writing this $= f(x^2) + xg(x^2)$, we have

$$\Omega\,\frac{\phi(x)}{x^2-a} = \Omega\,\frac{f(x^2)}{x^2-a} + \Omega\,\frac{xg(x^2)}{x^2-a} = \frac{f(x^2)-f(a)}{x^2-a} + \Omega\,\frac{xg(x^2)}{x^2-a}.$$

As regards the last term, notice that

$$\frac{xg(x^2)}{x^2-a} = \frac{x[g(x^2)-g(a)]}{x^2-a} + \frac{xg(a)}{x^2-a},$$

in which $\dfrac{x[g(x^2)-g(a)]}{x^2-a}$ is a rational and integral function of (a,x), and therefore when developed contains only positive powers of x, while $\dfrac{xg(a)}{x^2-a}$ when developed contains only negative powers of x.

We thus have

$$\Omega\,\frac{\phi(x)}{x^2-a} = \frac{f(x^2)+xg(x^2)-f(a)-xg(a)}{x^2-a} = \frac{\phi(x)-f(a)-xg(a)}{x^2-a}.$$

[p. 539] Similarly, if $\phi(x) = f(x^3) + xg(x^3) + x^2h(x^3)$, then

$$\Omega\,\frac{\phi(x)}{x^3-a} = \frac{\phi(x)-f(a)-xg(a)-x^2h(a)}{x^3-a},$$

and so on.

Consider now the above-mentioned function

$$A(x) = \Omega\,\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}}.$$

Writing

$$\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{P}{1-ax^2} + \frac{Q}{1-a} + \frac{R}{1-ax^{-2}},$$

we have

$$P = \left(\frac{1-x^{-4}}{1-a\,.\,1-ax^{-2}}\right)_{a=x^{-2}} = \frac{1-x^{-4}}{1-x^{-2}\,.\,1-x^{-4}} = \frac{-x^2}{1-x^2},$$

$$Q = \left(\frac{1-x^{-4}}{1-ax^2\,.\,1-ax^{-2}}\right)_{a=1} = \frac{1-x^{-4}}{1-x^2\,.\,1-x^{-2}} = \frac{1}{1-x^2},$$

$$R = \left(\frac{1-x^{-4}}{1-ax^2\,.\,1-a}\right)_{a=x^2} = \frac{1-x^{-4}}{1-x^4\,.\,1-x^2} = \frac{-1}{x^2(1-x^2)};$$

that is,

$$\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{-x^2}{1-x^2}\cdot\frac{1}{1-ax^2} + \frac{1}{1-x^2}\cdot\frac{1}{1-a} + \frac{-1}{x^2(1-x^2)}\cdot\frac{1}{1-ax^{-2}},$$

and thence

$$\Omega\,\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{-x^2}{1-x^2}\cdot\frac{1}{1-ax^2} + \frac{1}{1-x^2}\cdot\frac{1}{1-a} + \Omega\,\frac{-1}{x^2(1-x^2)}\cdot\frac{1}{1-ax^{-2}}.$$

Here, as regards the last term,

$$\Omega\,\frac{1}{1-ax^{-2}} = \Omega\,\frac{-x^2}{x^2-a} = \Omega\,\frac{\phi(x^2)}{x^2-a} = \frac{\phi(x^2)-f(a)}{x^2-a} = \frac{-x^2+a}{x^2-a},$$

which gives

$$\Omega\,\frac{-1}{x^2(1-x^2)}\cdot\frac{1}{1-ax^{-2}} = \frac{1}{1-x^2}\cdot\frac{x^2+a}{1-ax^2}\cdot\frac{1}{x^2}\cdot ?$$

and we have

$$A(x) = \frac{-x^2}{1-x^2}\cdot\frac{1}{1-ax^2} + \frac{1}{1-x^2}\cdot\frac{1}{1-a} + \frac{1}{1-x^2}\cdot\frac{x^2+a}{1-ax^2\,.\,1-a}.$$

The second term is $= \dfrac{1}{1-x^2}\cdot\dfrac{1}{1-a}$: combining this with the third term, the two together are $= \dfrac{1+ax^2}{1-x^2\,.\,1-ax^2}$.

[p. 540] Hence the value is

$$= \frac{1}{1-x^2}\left(\frac{-x^2}{1-ax^2} + \frac{1+ax^2}{1-ax^2}\right),$$

which is

$$= \frac{1}{1-ax^2\,.\,1-a^2},$$

being, in fact, the expression for this function when decomposed into partial fractions of the denominators $1-ax^2$ and $1-a^2$ respectively. Hence finally

$$\Omega\,\frac{1-x^{-4}}{1-ax^2\,.\,1-a\,.\,1-ax^{-2}} = \frac{1}{1-ax^2\,.\,1-a^2},$$

as it should be.

For the cubic function, we have

$$A(x) = \Omega\,\frac{1-x^{-6}}{1-ax^3\,.\,1-ax\,.\,1-ax^{-1}\,.\,1-ax^{-3}};$$

the function operated upon, when decomposed into partial fractions, is

$$= \frac{x^6}{1-x^2\,.\,1-x^4\,.\,1-ax^3} - \frac{x^2}{1-x^2\,.\,1-x^6\,.\,1-ax}$$

$$+\frac{x}{1-x^2\,.\,1-x^6}\cdot\frac{1}{x-a}-\frac{x}{1-x^2\,.\,1-x^4}\cdot\frac{1}{x^3-a}.$$

Hence we require

$$\Omega\,\frac{1}{1-x^2\,.\,1-x^6}\cdot\frac{x}{x-a}+\Omega\,\frac{1}{1-x^2\,.\,1-x^4}\cdot\frac{-x}{x^3-a}.$$

The first of these is

$$=\frac{1}{x-a}\left(\frac{x}{1-x^2\,.\,1-x^6}-\frac{a}{1-a^2\,.\,1-a^6}\right),$$

which is

$$\frac{1}{1-x^2\,.\,1-x^4\,.\,1-a^2\,.\,1-a^6}\left\{\begin{array}{l}1\\ +x(a+a^3-a^5)\\ +x^2(a^2+a^4)\\ +x^3(a-a^5)\\ -x^4a^5\\ -x^6a\end{array}\right\}.$$

As regards the second, the function operated on may be expressed in the form

$$\frac{-x-x^3-x^5}{1-x^2\,.\,1-x^6}\cdot\frac{1}{x^3-a},$$

Arthur Cayley

Collected Mathematical Papers

Volume IX, Pages 576–600

Transcribed from the Original Publications

Contents

623. On Three-Bar Motion (continued)

[Continued from earlier pages of the same paper.]

[p. 576] Single-infinite series of ∞ and circumscribed triangles, so that, drawing from a point A of the circle tangents to the parabola again meeting the circle in the points B and C respectively, BC will be a tangent to the parabola; this is in fact a known thing, but, repeating the same kind of investigation as before, we can, with the affinity points K on the circle, the parabola described to touch two of the sides of the triangle will touch the third side.

Taking, then, a circle radius $\frac{1}{2}k$, and upon it the three points A, B, C determined by the angles 2α, 2β, 2γ respectively (viz. the coordinates of A are $x = \frac{1}{2}k\cos 2\alpha$, $\frac{1}{2}k\sin 2\alpha$, &c.), and a point K determined by the angle 2ω (suppose for a moment $\omega = 0$), then the lines from K, viz. the lines having for K the equations &c. at the foot of A, and the lines KB with the fixed parabola, are connected by simple relations, viz. writing Q for the gain $\frac{1}{4}k^2\sin 2\gamma$, we obtain

$$P - x = -\tfrac{1}{4}k^2\left(\tfrac{x}{y} - \tan\gamma\right)\left(\tfrac{x}{y} - \tan\alpha\right)\tan\beta,$$

which serves to find the breadths a, b, c of a given trajectory from the bifocus $PQRS\ldots$, and, if necessary, by inverting each or either as alternate ones, make the left-hand column begin with the sign $+$.

[p. 577] 26. The complete role now is for a given trajectory from the bifocus to $PQRS\ldots$, and, if necessary, by inverting each or either ones, make the left-hand column begin with the sign $+$. To find which we begin with a then if these Ω count the intersections with the alternate ones, with the left-hand column to begin with the sign $+$.

26. The complete role now $a + b + c$, and the equation of the conic surface gives $\frac{1}{2}P_1$. Operating with P_x as a tangent, then the calculation is in fact equal to the breadth in a.

27. I give some simple examples.

	0	2	4
$P = a - 1$	$-$	$+$	$-$
$Q = a - 3$	$+$	$+$	$+$
$R = a - 1$	$-$	$-$	$-$
0	P	2	Q

	0	2	4
$P = a - 3$	$-$	$-$	$+$
$Q = a - 1$	$+$	$+$	$+$
$R = a + 1$	$+$	$+$	$+$
0	P	2	Q

In the left-hand column taking the intervals to be considered $0-2$, $0-4$, $2-4$, the binbecome modified as shown are

$0-2$	$0-4$	$2-4$
$-$	$-$	$+$
$+$	$+$	$+$
$-$	$-$	$-$

viz.

Interval $0-2$, for P gain $+1$, P free; for Q gain -0, Introduction is P;

$-0-4$, $\quad -1$, $\quad \cdots\cdots = -1$; $\quad \cdots\cdots PQ$;

$-2-4$, $\quad -0$, $\quad \cdots\cdots = -1$; $\quad \cdots\cdots Q$.

And similarly in the right-hand example we have

$0-2$	$0-4$	$2-4$
$-$	$-$	$+$
$+$	$+$	$+$
$+$	$+$	$+$

[p. 578] viz. these conditions determine the unknown quantities p, R. It is at once seen that we have

$$2\theta - \beta - \gamma = \tfrac{1}{2}\pi - (a - \alpha), \quad \text{that is,} \quad 2\theta = \tfrac{1}{2}\pi - a + \alpha + \beta + \gamma,$$

and then

$$p = \Omega \sin(a - \alpha) \div \cos(\theta - \alpha) \div \sin(a - \gamma);$$

from symmetry, we see that the parabola touches also the side AB.

Suppose, next, F, G are points in the circle determined by the angles $2f$, $2g$; retaining α and θ to denote their values,

$$p = \Omega \sin(\alpha - \beta) \sin(\alpha - \gamma), \text{ and } 2\theta = \tfrac{1}{2}\pi - a + \alpha + \beta + \gamma,$$

the condition, in order that FG may be a tangent, is

$$p = \Omega \sin(\alpha - f) \sin(\alpha - g) \cos(2\theta - f - g);$$

viz. determining δ by the equation

$$a + \beta + \gamma = f + g + \delta,$$

this is

$$p = \Omega \sin(\alpha - f) \sin(\alpha - g) \cos(a\delta - \delta),$$

or, what is the same thing,

$$\sin(\alpha - a) \sin(\alpha - \beta) \sin(\alpha - \gamma) = \sin(\alpha - f) \sin(\alpha - g) \cos(\delta - \delta);$$

viz. this equation, considering therein δ as standing for $a + \beta + \gamma - f - g$, is the relation which must subsist between f and g, in order that the line FG may be a tangent to the parabola. And then, δ being determined as above, and the point H on the circle being determined by the angle 2δ, it is clear that the lines GH, HF will also be tangents to the parabola, viz. FGH will be an inscribed triangle circumscribed to the parabola; we have a single infinity of such triangles, but it is in fact, the same single infinity in form of these, f, g, satisfy only the relation $a + \beta + \gamma = f + g + \delta$, where δ has the same value but α change. The proposition is obtained as the consequence of an easily established proposition that to determine the determinations of f and α.

[p. 579] that is,

$$(b_1' - a_1')c_2a' - (c_1' - a_1')b_2a' + ac_1(c - b_1) = 0,$$

or, what is the same thing,

$$(b_1' - a_1')a^2 - (c_1' - a_1')b + ac_1(c - b_1) = 0.$$

Suppose, as in the figure, that P is between B and A; then, if $AF = p$, we have $a = ba + cp$, and the equation becomes

$$(b_1' - a_1')(ca + cp)^2 - (c_1' - a_1')ba + aa_1cc = 0.$$

Similarly, if, as in the figure, G is on the other side of A, that is, between A and C, and if a, a' be the dimensions AG, BG, CG, then $ba' = b'a + ca'$, that is, $cn = ba - na$, and the corresponding equation is

$$(b_1' - a_1')(ca' - na)^2 - (c_1' - a_1')ba + aa_1cc = 0.$$

We have here

$$BG\colon CF = BP \cdot CO = BC \cdot FO,$$

also,

$$pa + p'a' = AF + AF \cdot AC = FO \cdot AO = FO \cdot OH + FO \cdot a',$$

and

$$pp + p'a' = AP \cdot CF = AG \cdot CG = FO \cdot HH = FO \cdot a,$$

as may be shown without difficulty, p', a', b'' being the distances AB, BH, CH. Hence the equations become

$$c(b_1' - a_1') = \nu a';$$

$$b(c_1' - a_1') = aaa' = 0,$$

showing that, the fact being so in the figure, $b_1' = a_1'$ and $c_1' = a_1'$, are each of them positive, viz. that, in the geometry by the triangle $GCAU$, the radial bars a_0, a_1 are shorter than the sides b, c, respectively. Substituting for a_2, a_3, the values b, b_1, a_2 a c, also, instead of b_1, b_2, b_3, introducing the quantities b_1, a_2, b_3, where

$$b_1, b_2 = b + b_4,$$

$$b_3 = b + b_5,$$

$$b_4 = b + b_6,$$

$$b_5 = b + b_7,$$

these equations become

$$c(b_1' - b_2'c)ca + cba'B = a'a',$$

$$b(b_3' - b_4'c)ca + ba'Ca = a'a',$$

or, as these may be written, partaking the relations $B = aa + b_1a' \sin BAC$,

$$B'\&'(b_2 - b_4'b)c\,a + ab'\,BCH;$$

$$B'\&'(b_3 - b_4'b)c\,a + ba'c\,AH.$$

[p. 580] All the quantities here so far being represented as positive, and the formula are applicable to the particular figures, but, to present them in a form applicable to any order of the nodes and bar, we have only to write the equations in the form

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + B(ca + b' - x)(cb - b),$$

$$B(\lambda_3 - \lambda_5) = b'ca(a - x) + B(ca + b' - x)(cc - b),$$

$$B(\lambda_4 - \lambda_4) = b'ca(a - x) + B(ca + b' - x)(cc - b),$$

and these may be replaced by the equation

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + (b' - x)(s + a),$$

$$B(\lambda_2 - \lambda_4) = b'ca(a - x) + (b' - x)(s + b),$$

$$B(\lambda_3 - \lambda_4) = b'ca(a - x) + (b' - x)(s + c),$$

since the first of these equations is implied in the other two; and then, reverting to the original form, we may write

$$BB(\lambda_2' - \lambda_4) = BC \cdot FA \cdot GA \cdot HA,$$

$$BB(\lambda_2 - \lambda_4) = CA \cdot FB \cdot GB \cdot HB,$$

$$BB(\lambda_3' - \lambda_4) = AB \cdot FC \cdot GC \cdot HC,$$

it being understood that the distances BC, FA, &c., which enter into these equations, are not all positive, but that they must for $\lambda\sin(\beta-a)$, $b'\sin(f-a)$, &c., and that their signs are to be taken accordingly. Or, again, these may be written

$$BC(\gamma') - \lambda_4) = FA \cdot GA \cdot HA,$$

$$CA(c_1 - c') = FB \cdot GB \cdot HB,$$

$$AB(b_1 - b_3') = FC \cdot GC \cdot HC,$$

where the signs are so just mentioned. We may say that $\frac{1}{2}(c_1 - c_1')$ is the modulus for the focus A, and the formula then shows that this modulus, taken positively, is equal to the product of the distances FA, GA, HA of A from the three nodes respectively, divided by BC, the distance of the other two foci from each other.

624. On the Bicursal Sextic

[From the Proceedings of the London Mathematical Society, vol. VII. (1875–1876), pp. 166–172. Read March 10, 1876.]

In the paper "On the binominal description of certain conic curves," *Proceedings of the London Mathematical Society*, vol. IV. (1873), pp. 105–111, [504], I obtained the binominal sextic as a rational transformation of a bicursal quartic. The theory was in effect as follows: taking Ω, P, Q, R, each of them a function of λ, μ of the form $(*)(\lambda,1)^3(\mu,1)^3$, and considering (λ,μ) as connected by the equation $\Omega=0$ (viz. λ, μ being coordinates, this represents a bicursal quartic), then if $\sigma=$ assume $\sim x:y:z=P:Q:R$, this is rationally connected with the bicursal quartic, viz. the points of the latter curve have a $(1,1)$ correspondence; whence the locus in question, say that curve $U=0$, is binominal. The figure is obtained as the reverse of the curve of the variable intersections of the corresponding $\lambda\mu$-curves

$$\Omega=0,\quad aP+\beta Q+\gamma R=0,$$

viz. each of these being a quartic curve having the same nodes, the nodes each count as 4 intersections, and the number of the remaining intersections is $4\cdot 4-4\cdot 3,=4$; and thus the curve $U=0$, $R=0$ have therefore the same 4 common intersections, that is, the curve $U=0$, $R=0$ have therefore the same 4 common intersections, then these are also 4 common intersections of the two corresponding $\lambda\mu$-curves $\Omega=0$, $aP+\beta Q+\gamma R=0$, and consequently the curve $U=0$ is binominal.

The theory assumes a different and more simple form if the second functions Ω, P, Q, R are suppose that these have no common point; the curve $U=0$ is binominal.

[p. 582] rational transformation of the cubic $\Omega=0$, is still a bicursal curve; but the order is given as the number of the variable intersections of the cubics

$$\Omega=0,\quad aP+\beta Q+\gamma R=0,$$

viz. this is $=3\cdot 3-2,=7$. But if the curves $\Omega=0$, $P=0$, $Q=0$, $R=0$ have (besides the before-mentioned two common points) k other common points, then the number of the variable intersections is $=7-k$; and this is therefore the order of the curve $U=0$. In particular, if $k=1$, then the curve is a bicursal sextic. And, in the present paper, I consider the binodal sextic as thus obtained, viz. as given by the equations $\Omega=0$, $x:y:z=P:Q:R$, where $\Omega=0$, $P=0$, $Q=0$, $R=0$ are cubics, having (in all) three common points.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $aP+\beta Q+\gamma R=0$, there are any two curves which have 3 common intersections with the three cubics $\Omega=0$, $P=0$, $Q=0$, $R=0$. (Observe that we distinguish thus far between the variable and the common intersections of the curves $\Omega=0$, then any linear function of P, Q, R, would have the same number of common intersections.) For, supposing the two curves to be $aP+\beta Q+\gamma R=0$ and $a'P+\beta'Q+\gamma'R=0$ and the cubic any other curve whatever

$$aP+\beta Q+\gamma R=\theta(a'P+\beta'Q+\gamma'R)$$

has the same number of variable intersections with $aP+\beta Q+\gamma R=0$, the points common to these two curves correspond on the cubic the points A_1, A_2, A_3.

The bicursal sextic has in general 9 nodes; but 3 of these may unite together into a triple point: this will be the case if, in the series of curves $aP+\beta Q+\gamma R=0$, there are any three curves which have 3 common intersections with the three cubics $\Omega=0$, $P=0$, $Q=0$. For, supposing the three curves to be

$$aP+\beta Q+\gamma R=0,\quad a'P+\beta'Q+\gamma'R=0,\quad a''P+\beta''Q+\gamma''R=0,$$

a triple point; and that to this triple point correspond the three points A_1, A_2, A_3.

We may, in the series of lines $ax + \beta y + \gamma z = 0$, $a'x + \beta' y + \gamma' z = 0$, rationally determine θ so that one of the three variable intersections shall correspond to A_1, A_2, or, A_2, viz. to obtain the cubic $aP + \beta Q + \gamma R - \theta(a'P + \beta' Q + \gamma' R) = 0$, rationally determine θ so that one of the three variable intersections shall correspond to A_1, A_2, or, A_3, viz. to obtain the cubic to pass through one of these points A_1, A_2, A_3; the third common intersection of $aP + \beta Q + \gamma R$ and $a'P + \beta' Q + \gamma' R$ must then be on the curve; and so on, similarly. And so on, similarly. And so on, similarly. And so on, similarly. And, as before, the three points correspond to a triple point on the bicursal sextic.

[p. 583] The bicursal sextic may have a second triple point, viz. there may then exist three other common points. The theory may again be developed as before; the new triple point will correspond to a similar set of three common intersections of the cubics having the same three common points. The theory may be developed if there are six variables instead of three common intersections of the cubics having three of the variable intersections of $\Omega = 0$, $aP + \beta Q + \gamma R = 0$ count for a triple point; and the same set of three common intersections of $aP + \beta Q + \gamma R = 0$, will count for a triple point of the second triple point. As before, we have a number of triple points such as $\Omega = 0$, $a''P + \beta'' Q + \gamma'' R = 0$, which thus form (in the same hexagonal locus) the cuspidal sextic. By varying the parameters we obtain the curve $\Omega = 0$, and the points P, Q, R are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

I write for greater convenience λ, ρ in place of a, μ as made by writing Ω, P, Q, R, each of them in the form $\Omega = aP + \beta Q + \gamma R$, where the points P, Q, R are the third points of intersection of the cubic with the lines BC, CA, AB respectively.

I write for greater convenience $\frac{P}{R}$, $\frac{Q}{R}$ in place of x, y, so as to make Ω, P, Q, R each of them cubic function of (λ, ρ); and I give to these functions, not the most general values belonging to a bicursal sextic with two triple points, but the values in the form obtained for them, as appearing further on, in the problem of three-bar motion: viz. the equations $\Omega = 0$, $x : y : z = P : Q : R$ are respectively taken to be

$$\Omega = (a\lambda + \lambda^2 \rho)(c + \rho)(a' + b\rho)(c' + a^2\rho^2) = 0,$$

$$x : y : z = \lambda\rho : a' + b\rho : c' + a^2\rho^2,$$

respectively as to the form of $\Omega = 0$, see the lines down in the figure.

The base curves $\Omega = 0$, $P = 0$, $Q = 0$, $R = 0$ have thus the three common intersections $(\rho = 0)$, $(\rho = 0)$, $(\rho = 0)$, with the three points represented in the figure as the points of B, D, H in the corresponding to the common intersection $(\rho = 0)$.

[p. 584] viz. excluding the fixed points A, B, C, the six intersections are C, F, G, and three intersections by the line $\Omega = 0$, viz. those are the points

$$(\lambda, \rho, \mu) = (0, 0, 1), \quad (0, p, -f), \quad (h, 0, -f), \quad (C, F, G).$$

The equation $Q + \theta R = 0$ is $a' + b\rho + \theta(c + \rho\mu) = 0$; viz. this is, for $\theta = 0$, an equation of the second order in λ and ρ in the curve through the three points A, B, H, viz. those are the points

$$(\lambda, \rho, \mu) = (0, 0, 1), \quad (0, p, -f), \quad (h, 0, -f), \quad (C, F, G).$$

To find the tangents at the triple point J, corresponding to the conditions $(\lambda, \rho, \mu) = (0, 0, 1)$, $(0, p, -f)$, $(h, 0, -f)$, we have, for the values

$$\theta = \frac{c}{f}, \quad \theta = \frac{c + p}{f} \cdot \frac{c + p}{f - 1}, \quad \text{or} \quad \theta = \frac{c}{f}, \quad \theta = \frac{c + p}{f - 1}.$$

Hence, considering for C as origin, the equation is

$$1+2a+\theta y=0;\quad \theta(z+y)+abz(z+y)+\theta\theta(xx+y)=0;$$

or, the equation $Q+\theta R=0$ is similarly rendered; and so, similarly, K, M at the points L, M, we have f, a, as before constraint of f, h. As tangent at the conic point.

In fact (and this is well known to be the case for a singular case of a bicursal sextic with two triple points, but in the present case they are conics, C, F, G, are independent of θ, viz. they are the points

$$(\lambda,\rho,\mu)=(0,0,1),\quad (0,p,-f),\quad (h,0,-f),\quad (C,F,G).$$

There are equivalent in virtue of the equation $Q+\theta R=0$, that

$$(\lambda,\mu)=(0,0),\quad (\lambda h,-f,0),\quad \text{(those are also the points)}\quad C,F,G,\text{etc.}$$

To deduce these expressions, writing

$$M=c+2a(\nu+p),$$

we have $K=(a+p)+c(a+b\mu)$. Hence

$$J=K+Y=(a-p)V\lambda,$$

also, from the original form,

$$J'=b+h\nu p+\lambda(C\eta+2c)-h\mu(C\eta b+Y),$$

The final values are

$$\Omega=-K^2+KX+4a\rho(M+a\mu)R+a(bRY-bzp+Y),$$

$$P=(a-\nu)\mu^2+(hR+a\mu)\Pi-4a\nu(b-Z)Y,$$

$$Q=a+KaR+Pq\lambda-a\nu Y+ba\nu c\nu),$$

$$R=b(2a+4ac\nu c+bR)$$

[p. 585] The bicursal sextic may have a second triple point, viz. there may then exist three other common points. The theory may again be developed as before, but here three other common points and the figure of the bicursal sextic. There may also be other singular cubics, see the figure of the three-cubic sextic, see the figure $\Omega=0$.

If denote like functions $\cos\theta+i\sin\theta$, $\cos\phi+i\sin\phi$ of the angles θ, ϕ, which λ, μ are subscribed to make in connecting set of angle of the bicursal sextic of the binormal cubic, Ω, C_2, R, see, taking X, Y as rectangular rectangular coordinates, x, y, and ν here their first letter for $\cos\theta+i\sin\theta$, $\cos\phi-i\sin\phi$,

$$X=a,\quad \cos\theta+b_2\sin\alpha,$$

$$X+iY=b\cos C+(\cos\phi)C_2,$$

$$X-iY=b\cos C-(\cos\phi)C_2,$$

viz. the first two intersect in the point $(0,0)$, the second two in the point $(b\cos C,\ b\sin C)$, the first and third of these on the curve at the point $X=b\cos C$, $b\sin C$ (with from the same point of b, forming, with B and C, a triad of foci.

Verification of Ω.—The equation is

$$\Omega=(a\lambda+\lambda^2\rho)(c+\rho)(a'+b\rho)(c'+a^2\rho^2)=0,$$

where

$$\Omega = (a\lambda + \lambda^2\rho)(c+\rho)(a'+b\rho)(c'+a^2\rho^2) = 0,$$

or, putting for shortness

$$\lambda = a + 4a + b\mu \quad \mu' \quad \nu' = (X - iY)(b\cos C + i\sin C) + \cos C - i\sin\theta - \nu' + \nu''(a' + \mu\nu),$$

$$M = b\nu + (X - iY)V\lambda + b\nu + ac\nu - a\nu + p\nu - \nu + \nu'\nu' + ac\nu = 0,$$

or, the trial may be made, a, ν being constants, the values $c\nu$ and the dual function K, X, Y.

[p. 586] Writing for homogeneity $\frac{x}{z}$, $\frac{y}{z}$ in place of λ, μ, and $\frac{x}{w}$, $\frac{y}{w}$ in place of x, y, the equations become

$$(ax^3 + a^4\rho^2) + a\rho(b + b_4b\rho)(b + b_4\rho)(c^2 + b_4\rho^2) = 0,$$

and

$$x : y : z : w = \lambda(\lambda + b\rho\mu) : \left(\frac{b}{c} + acb\mu\right) c' : c\rho\nu.$$

Comparing with the foregoing equations

$$x\lambda\rho + f(\lambda^2 + a)\nu = g(c^2 + a)\mu + ac\nu + bu(\lambda^2 + \nu)$$

and

$$x : y : z = \lambda\nu : (a' + b\rho) : (c' + a^2\rho^2),$$

the equations agree together, and we have

$$\rho = a\mu(c' + a)(\mu')z',$$

$$\mu = ecc\nu,$$

$$\Omega = cc\nu(\nu + 1)\nu(bu),$$

$$\beta = cc\nu(\nu + 1)\nu(bu);$$

the integrals at the triple points then are

$$\nu = 0, \qquad \rho = 0,$$

$$a\mu, c\mu(\nu, c) = 0, \qquad a\mu - ac\mu, c = 0,$$

$$\alpha = c\mu(c\mu),$$

$$b\mu = 0, \qquad b\mu = (\nu, c) = 0,$$

viz. writing the rectangular coordinates, and for y substituting the value $\cos C + i\sin C$,

$$X + iY = 0, \qquad X - iY = 0,$$

$$X + iY = b\cos C + (\cos\phi)C_2, \qquad X - iY = b\cos C - (\cos\phi)C_2,$$

$$X + iY = \nu,$$

viz. the first two intersect in the point $(0, 0)$, the second two in the point $(b\cos C, b\sin C)$, and the third pair of three of the points of the figure, viz. the focuses with A, B and C, a triad of foci.

[p. 587] (omitted as overlap)

[p. 588] (continuation)

[p. 589] (continuation)

625. On the Condition for the Existence of a Surface Cutting at Right Angles a Given Set of Lines

[From the Proceedings of the London Mathematical Society, vol. VIII. (1876–1877), pp. 53–57. Read December 14, 1876.]

In a congruency or doubly infinite system of right lines, the direction-cosines a, β, γ of the line through any given point (x, y, z), are expressible as functions of x, y, z; and it was shown by Sir W. R. Hamilton in a very elegant manner that, in order to the existence of a surface (or, what is the same thing, a set of parallel surfaces) cutting the lines at right angles, $a\,dx + \beta\,dy + \gamma\,dz$ must be an exact differential: when this is so, writing $V \int (a\,dx + \beta\,dy + \gamma\,dz) = 0$, the equation of the system of parallel surfaces each cutting the given lines at right angles.

The proof is as follows:—If the surface cuts, its differential equation is $a\,dx+\beta\,dy+\gamma\,dz = 0$, and this equation must therefore be integrable by a factor. Now the functions a, β, γ are such that $a^2 + \beta^2 + \gamma^2 = 1$, and they hereby satisfy a system of partial differential equations which Hamilton deduces from the geometrical notion of a congruency; viz. posting from the point (x, y, z) to the consecutive point on the line, that is, to the point whose coordinates are $x + ap$, $y + \beta p$, $z + \gamma p$ (p infinitesimal), the line through this point is parallel to the original line; and consequently the direction-cosines of this line are unchanged by the variations $\Delta x = ap$, $\Delta y = \beta p$, $\Delta z = \gamma p$, respectively. We thus obtain the equations

$$a\frac{da}{dx} + \beta\frac{da}{dy} + \gamma\frac{da}{dz} = 0,$$

$$a\frac{d\beta}{dx} + \beta\frac{d\beta}{dy} + \gamma\frac{d\beta}{dz} = 0,$$

$$a\frac{d\gamma}{dx} + \beta\frac{d\gamma}{dy} + \gamma\frac{d\gamma}{dz} = 0.$$

[p. 591] so that the required equation is a, B, γ will be satisfied if only

$$\frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0,$$

and it is to be shown that these conditions hold good.

Writing for a, B, γ the below-mentioned values, we find

$$\frac{da}{dx} = \frac{dC}{dx}\frac{1}{a+\lambda}, \qquad \frac{d\beta}{dy} = \frac{dC}{dy}\frac{1}{a+\mu}, \qquad \frac{d\gamma}{dz} = \frac{dC}{dz}\frac{1}{a+\nu},$$

and it is to be shown that these conditions hold good.

Writing for a, B, γ the below-mentioned values, we have

$$\frac{da}{dx} = \frac{dC}{dx}\cdot a\frac{a+\lambda}{a+\mu} + \frac{d}{dx}\left(\frac{a}{a+\lambda}\right)C = \frac{a^2}{a+\lambda}\frac{dx}{dx} + C^3\frac{dx}{dx}\left(\frac{a}{a+\lambda}\right),$$

and similarly for $\frac{d\beta}{dy}$ and $\frac{d\gamma}{dz}$ such that

$$\frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0, \qquad \text{viz.} \quad \frac{da}{dx} + \frac{d\beta}{dy} + \frac{d\gamma}{dz} = 0,$$

and thence without difficulty

$$\left(\frac{a^2}{a+\lambda}\frac{da}{dx} + \frac{a^2}{a+\mu}\frac{d\mu}{dx}\right) - \left(\frac{1}{1+d\beta}\frac{d\beta}{dx} + \frac{d\nu}{dx}\right) - \omega = \left(a\frac{dx}{dx} + b\frac{dy}{dy} + c\frac{dz}{dz}\right),$$

$$\left(\frac{a^2}{a+\lambda}\frac{d\lambda}{dy}+\frac{d\mu}{dy}\right)+\frac{1}{2(a+\lambda)}=-m\,a\;(\quad)+(\quad)\,,$$

$$\left(\frac{a^2}{a+\lambda}\frac{d\lambda}{dz}+\frac{a^2}{a+\mu}\right)\frac{dz}{dz}=-\frac{1}{2}\quad(-)+(-).$$

626. On the General Differential Equation $\dfrac{dx}{\sqrt{X}} + \dfrac{dy}{\sqrt{Y}} = 0$, where X, Y are the same Quartic Functions of x, y respectively

[From the Proceedings of the London Mathematical Society, vol. VIII. (1876–1877), pp. 184–199. Read February 8, 1877.]

[p. 592] WHERE $\Theta = a + b\theta + c\theta^2 + d\theta^3 + e\theta^4$, the general quartic function of θ, and let it be required to integrate by Abel's theorem the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0.$$

We have

$$\begin{vmatrix} x^4, & x, & 1, & \sqrt{X} \\ y^4, & y, & 1, & \sqrt{Y} \\ z^4, & z, & 1, & \sqrt{Z} \\ w^4, & w, & 1, & \sqrt{W} \end{vmatrix} = 0,$$

a particular integral of the above equation, taking therein z, w as constants, is the general integral of

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

viz. the two constants z, w enter in such wise that the equation contains only a single constant; whence also, attributing to w any special value, we have the general integral with z as the arbitrary constant.

[p. 593] Take $w = x$; the equation becomes

$$\begin{vmatrix} x^4, & x, & 1, & \sqrt{X} \\ y^4, & y, & 1, & \sqrt{Y} \\ z^4, & z, & 1, & \sqrt{Z} \\ 1, & 0, & 0, & \sqrt{W} \end{vmatrix} = 0,$$

a relation between x, y, z which may be otherwise expressed by means of the identity

$$x(P + \beta\theta + \gamma\theta^2) = (a\theta + \beta) - (a\theta + \beta)\theta(\theta - z)(\theta - z),$$

or, what is the same thing,

$$x(2x + \beta) - z = -(2(a\theta - \theta)(\theta + \theta) - z),$$

$$+2\beta y = -a + (3z - d) - (a + xy + z),$$

where β, γ are indeterminate coefficients which are to be eliminated. Write

$$x^2 = \frac{\sqrt{X}}{xz} = P, \quad y^2 = \frac{\sqrt{Y}}{yz} = Q;$$

then we have $\beta + \gamma = P + Q$, $\beta y + \gamma + Q = z$; giving

$$B + \gamma = 1 + Q - P, \quad B\gamma + Q = z - z.$$

Substituting these values in the first of the preceding three equations, we have

$$x\frac{2(Py - Qx)(x - y) + (Q - P)y}{(x - y)^2} - x = \frac{2 + (Q - P)y - Q}{(x - y)}\,(ay - z + s)\,(x + y + z),$$

that is,

$$\frac{2(Qy-Px)}{x-y}=\frac{2(2+(Q-P)y-Q)}{x-y},\quad (x+y+z)\cdot\frac{2}{x-y}\cdot\frac{1}{x}\cdot\frac{1}{y};$$

or, reducing by

$$Qy-Py-x=a\frac{x-y\sqrt{Y}}{x-y\sqrt{X}}.$$

[p. 594] times, the rational part is

$$M'=c+d(x+y)+e(x^2+xy+y^2),$$

and the two terms involving the operator L are as before stated that, in the particular case where $e=0$, the first equation becomes

$$M'=e+d(x+y)+e(x^2+xy+y^2),$$

and the two results for this case may be perfectly stated $C-e+0dt$.

But it is required to identify the two solutions in the general case where a is not $=0$, I remark that I have, by my *Treatise on Elliptic Functions*, Chap. XIV., further developed the theory of Euler's relation, and have shown there, that, regarding C as variable, and writing

$$\Omega=ax^2+2xe(x+y)e^{-y}+4z(xx^2+2z^2x+4zy)(xe-ry)ee+2xye(x+y)e^{-y}+\frac{1}{2}y(xe+xy+xy+e)(C-exy),$$

then the equation between the variables x, y, C corresponds to the differential equation

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}=\frac{dC}{\sqrt{\Omega}}=0,$$

a result which will be useful for effecting the identification. The Abelian solution may be written

$$P\left[\frac{1(xy(\sqrt{X}+y\sqrt{Y})}{(x-y)^2}-x^2\right]=\frac{1}{x-y}\sqrt{\Omega}+4ex(x+y)e^{-y}-4ex(c+ey)=2EX',$$

and substituting for M its value, and multiplying by $(x-y)$, the equation becomes

$$2\sqrt{x}(x-y)(z+X+y\sqrt{Y})(P+y\sqrt{P+Q}+(Q'(x-y)\sqrt{Y}))$$
$$=2(c+d+(x+y)+e(x^2+xy+y^2)e+xze^{-y}2(e\sqrt{X}-x\sqrt{Y});$$

and substituting therein for M its value, and multiplying by $(x-y)$, the equation becomes

$$2x((x+y)(\sqrt{X}-y\sqrt{Y})+2x(x+y)((c+ey)-(x+y)\sqrt{Y}-x\sqrt{Y}))$$
$$=2(c+d(x+y)+e(x^2+xy+y^2))+xz-2(c+ey-y)e^{-y}2(x\sqrt{Y}-x\sqrt{Y});$$

On the left-hand side, the rational part is

$$2X+Y+e(-y)(x^4+2xy+y)+4e^{-y}(c-e^{-y}2(x+y)e^{-y})y+2(c+d)(x-y)(x+y),$$

which, substituting therein for X, Y their values, becomes

$$=2a+b(x+y)+c(x^2+4xy+y^2)+2dxy(x+y)+2ex^2y^2((x+y),$$

and the irrational part is at once found to be

$$=2(a+(a-y)\sqrt{Y}-y(a+(a-y))\sqrt{X}).$$

[p. 595] Multiplying the numerator and the denominator by

$$d(x-y)+3e(x^2-y^2)+2e\sqrt{x}((y-1)(X-y^4Y)),$$

the denominator becomes

$$=(x-y)^2\left[d+3e(x+y)+4e^{-y}(\sqrt{x}-1)\sqrt{X}\right],$$

which, introducing therein the C of Euler's equation, is

$$=(x-y)^2(C-6acC).$$

We have therefore

$$x(x-y)-6ac=(x(x-y)C+b(x-y)+2ay(P+2dx(yP-xQ)+2xx^2(\sqrt{Y}P+y\sqrt{X}Q),$$

$$+2xy(xx-y)\sqrt{X}-x\sqrt{Y}\cdot Vx((y-1)(X-y\sqrt{Y})((x-y)(x\cdot s-2))-2(x-y)\sqrt{Y\sqrt{Y}}.$$

Using b, to denote the same value as before, the function on the right-hand is, in fact,

$$=(x-y)^2\left[Bx-ad+de\sqrt{x}\sqrt{Y}\right],$$

and, this being so, the required relation between x, C is

$$x(\beta x-4ac)=(Bx-ad+de\sqrt{x}\sqrt{Y}),$$

$$((x-y)\sqrt{Y-y\sqrt{X}})((x-y)\sqrt{X}\cdot s-2)).$$

To prove this, we have first, from the algebric

$$\left(\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right)^2=C+d(x+y)+e\sqrt{x}\sqrt{X},$$

to express Φ as a function of C as a variable, gives

$$\frac{dx}{\sqrt{X}}+\frac{dy}{\sqrt{Y}}+\frac{dC}{\sqrt{\Omega}}=0,$$

and we have therefore

$$-\sqrt{\Omega}=\sqrt{x}\frac{dC}{dx}\quad(=-\sqrt{Y}\frac{dC}{dy}),$$

viz. $a\sqrt{x}\frac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before

$$M=\frac{C-\sqrt{Y}}{x-y}.$$

we have

$$C=M^2-d(x+y)+e(x+y),$$

and thence

$$\Omega=M^2-x(x+y)+e(x+y)^2.$$

We have

$$\frac{dM}{dx}=\frac{1}{x-y}\left(\frac{X'}{2\sqrt{X}}-\frac{\sqrt{X}-\sqrt{Y}}{x-y}\right),$$

[p. 596] and hence

$$\sqrt{\Omega} = \sqrt{X} - \sqrt{Y}\left[2M\frac{dM}{dx} - d - 2e(x+y)\right],$$

which thus is $(x-y)$, the numerator becomes

$$M' = c + d(x+y) + e(x^2+xy+y^2),$$

say that the particular case as may putting $C = c + exy$.

But it is required to identify the two solutions in the general case where e is not $= 0$. I remark that I have, by my *Treatise on Elliptic Functions*, Chap. XIV., further developed the theory of Euler's relation, and have shown there, that, regarding C as variable, and writing

$$\Omega = \frac{(C-c)^2(C-4ac)}{C(\theta-y)} - 4a\left[b - c(x+y) + d(xx+y^2) - 2e(xx+xy+y^2)\right](x-y)^2\sqrt{X}\sqrt{Y};$$

and substituting therein for M its value, and multiplying by $(x-y)$, this is

$$\Omega = c(P-p)[2P - 4ac + 2(x-y)^2\sqrt{X}\sqrt{Y}].$$

To prove this, we have first, from the algebric

$$\sqrt{\Omega} = c(P-p)\sqrt{\Omega(P-p)\sqrt{X}\sqrt{Y}},$$

to express Φ as a function of C as a variable. This equation, regarding C as a variable, gives

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dC}{\sqrt{\Omega}} = 0,$$

and we have therefore

$$-\sqrt{\Omega} = \sqrt{x}\frac{dC}{dx} \quad (= -\sqrt{Y}\frac{dC}{dy}),$$

viz. $\sqrt{x}\frac{dC}{dx}$ will be a symmetrical function of x, y. Putting, as before,

$$M = \frac{C - \sqrt{Y}}{x-y},$$

we have

$$C = M^2 - d(x+y) + e(x+y)^2,$$

and thence

$$\frac{dC}{dx} = 2M\frac{dM}{dx} - d - 2e(x+y),$$

that is,

$$\sqrt{\Omega} = \sqrt{x}\left[2M\frac{dM}{dx} - d - 2e(x+y)\right] \quad \text{suppose;}$$

or, writing for shortness X', M' to denote the derived functions $\frac{dX}{dx}$, $\frac{dM}{dx}$, (Y' is afterwards written to denote $\frac{dY}{dy}$), but as the final formula contains only X', $\frac{dX}{dx}$, and Y', $\frac{dY}{dy}$. (This does not assume any defect of symmetry), we find

$$\Omega = N\left[BC - X(X - 2(x+y) + \sqrt{X}\sqrt{Y}\right]$$

$$= D^2\sqrt{X^2} + 2(xx-y)ex^2\sqrt{X^2} + 2(c-0)\sqrt{Y\sqrt{XY}} - X'\sqrt{Y}],$$

[p. 597] viz. this is

$$= [2a + b(x + y) + c\sqrt{x}(y - z) + e(x + y)(\sqrt{Y} - x\sqrt{X}))]$$

$$\times[d(x - y) + 2e(xx + 2xy + yx)],$$

$$+\frac{1}{2}\sqrt{X}(d(x - y) - 4e(xx + 2xy + y^2) - 2exex(x - y + (z + y)e^{-y}((x - y)x + y\sqrt{Y})),$$

which is, in fact, equal to the expression on the left-hand side.

To complete, then, it is required to express $\sqrt{X}$ as a function of x, y symbolic that

$$x(x - y)(\sqrt{X} - \sqrt{Y}) \cdot 2(x + y)x \cdot 4 = (x - y)(\sqrt{X} - \sqrt{Y}) \cdot 2(x - y)(\sqrt{X} + \sqrt{Y}),$$

which may be done by putting therein

$$2P = -X^2X + 2xexy(X - y) + 4ex(\sqrt{x}y - \sqrt{y}x),$$

which when expanded into X as a function of x, y, on X as a function of x, y, of which there are forms in x, y, on $X = 0$, in fact, equal to the corresponding expression on the left-hand side.

So the left-hand side, considering therein down as they immediately present themselves; but, of course, the relation X is $X(x+y)-$ to come $a+(x+y)-$, when these are written in the form

$$-b + dxy = (a + b),$$

which is right.

To X verification of Ω.—The quantity is

$$J\Omega(X - y)X + 8x(c - y)\sqrt{C}(C - 4ac) + 4a(x - y)\sqrt{X}\sqrt{Y}$$

$$= 2L(P - xP\sqrt{X}) - 4a(P\sqrt{X}\sqrt{Y} - 2yP + \sqrt{Y}(c - 4ac))((x - y)\sqrt{Y} - y\sqrt{X})$$
$$+2\sqrt{X}((x - y)\sqrt{Y} - x\sqrt{Y})((x - y)\sqrt{X}(c - 1)) = 0;$$

which, throwing out the factor $x - y$, becomes

$$0 = 2\sqrt{X}(c-y)+2cxy(x-y)+2cxxy-2(c+y)e\sqrt{X}\sqrt{Y}-4\sqrt{X}-2(x+y)-2cxy-2cxy-2(x+y)y^2,$$

which, throwing out the factor $x - y$, is

$$0 = 2N\left(BC\sqrt{X} - 2(x + y)yP + 2cxy\sqrt{X}\right) + 2(c - 4ac)\sqrt{Y}((x - y)\sqrt{Y} - y\sqrt{X}),$$

$$+2\sqrt{X}((x - y)\sqrt{X} - x\sqrt{Y})((x - y)\sqrt{X}(c - 1)).$$

[p. 598] and similarly herein for X, Y their values, and arranging the terms, we find

$$\Omega = a\sqrt{X}\sqrt{Y}\sqrt{X}((x - y)\sqrt{Y} - y\sqrt{X}),$$

where

$$\Omega = a[2X + (a - y)X]$$
$$-2(x - y)yX$$
$$-(xy + (a + y)y)$$
$$-(x - y)(ay - y)$$
$$+By(2X + (x - y)X)$$
$$+2Y(x - y)X,$$

$$\Omega = 4xy(a-y)X$$
$$-2(a-y)y(2X+(x-y)X)$$
$$-4XY$$
$$+2by(x-y)X$$
$$+4cx(x-y)X$$
$$+4ex(x-y)(y-2)X,$$
$$\Omega = -4xy(a-y)X$$
$$+2cx(2X+(a-y)X)$$
$$+4xY$$
$$-2cx(x-y)X$$
$$-4x(x-y)X$$
$$+4x(x-y)(y-2)X,$$

where the same identifications hold down with the formulae above stated, refusing, except differently and averaging, we have here so much.

To reduce these expressions, writing $M = 2x(x+y)$, hence we have

$$J' = 2x + 4xy + e^{-y}((y+y)y + 2xy + Y), \quad 2xy(c+y)(y+Y+Y),$$

which divides by $(x-y)$.

Hence the equation now is

$$(x-y)^2\nabla = (a+b+e)(\sqrt{Y}-y)\cdot x\cdot\nabla((ax-y)y\sqrt{Y}),$$
$$+y(2xx+y) - x(b+e)(a+4ay-a)(c\sqrt{X}-1) + 2x((x-y)x - y\sqrt{Y}-x);$$
$$+ay\nabla(b+e)((x-y)x - y\sqrt{Y} - 4y\sqrt{X})\cdot 2(x-y).$$

The function on the right-hand is, in fact,

$$= 2(L\nabla + (x+y))\cdot ax\cdot\nabla\cdot\sqrt{Y} - y\sqrt{X},$$

and the irrational part is at once found to be

$$= 2(a+(a-y)\sqrt{X} - y(a+(a-y))\sqrt{X}).$$

[p. 599] *Verification of* Ω.—The equation is

$$-X' + 4\nabla Y + (x-y)(X' + (\sqrt{X}-4)y\sqrt{X})M + xyy,$$
$$= 2L(M'\cdot\sqrt{X}-y) + 4xx\sqrt{X^2}M + 2N\sqrt{X^2}M;$$

or, putting therein

$$\frac{X'}{(x-y)} = X' + 8X' - 6\sqrt{X}((\sqrt{X}-y)P)M^2 = 2PxM.$$

this is

$$(x-y)^2 = X' + 4XY$$
$$+2X'\cdot X - 8a(x-y)xP$$

$$+(a-y)(2xx+y)-4(\sqrt{X}\cdot 2(y-y))x\sqrt{X}M,$$
$$-2(\sqrt{X}-y)((x-y)X-xy^2\sqrt{X})-4(c-xy)(c-xy^2)\cdot 2\sqrt{X}\sqrt{Y};$$
$$+(x-y)(2X+4y(c\sqrt{X}+y)P+2(cxy\sqrt{Y}\cdot 2X'+2y\sqrt{X})X^2),$$
$$+(x-y)\cdot\sqrt{X}(c+y)X\cdot xz\cdot 2(X'\cdot x+xy\sqrt{X})-2y\sqrt{X}.$$

We have $YE'F=x(X+1)+de\sqrt{x}(X+Y-4yx+2by+ex^2)$ and thence

$$-X'\cdot 2X+(a-y)y(X'\cdot 4\cdot c\cdot 2\cdot cX)x\sqrt{X}+2\sqrt{X}(c\cdot c\sqrt{X}\sqrt{Y}-((x-y)-x)y\sqrt{X}),$$

or, as this may be written,

$$\Phi=-X'+2\sqrt{X}P.$$

[p. 600] We find, moreover,

$$\Phi'=X'-4(c+y)x+(x-y)((\sqrt{X}-4)y\sqrt{X})Xy\sqrt{X},$$

which divides by $(x-y)$, viz. this is

$$=y(c+(a-y)y)+4y(a+(a-y))+2X+2y(x-y)yX,$$

which divides by $(x-y)$, viz. this is

$$=y(c+a(x+y))\sqrt{Y}-y(b+2ay+2c)\cdot 2y-y\sqrt{X}.$$

Hence above the values reducing, and throwing out the factor X, the equation becomes

$$-b+d\,ay=(a+b),$$

which is right.

To the first line, in the equation immediately preceding, we have

$$J'=-2(x-y)x\nabla x\cdot c=-2(a-y)\sqrt{X}-xPx\nabla cy\sqrt{X};$$

which is

$$=-2X+(a-y)Py\cdot xc+(b+e)((x-y)X-y\sqrt{X}P)x-x;$$

hence, throwing out the factor $(x-y)$,

$$\Phi=aPx\nabla c-x((x-y)X-y\sqrt{X}P)-(a+y)Py\cdot 2xc-2xPx\nabla,$$

which is to come $(x-y)P\sqrt{X}P+2P\sqrt{X}$,

$$0=2XY+2(a+b)x+(x-y)P\nabla+2X\nabla,$$

which, throwing the factor $(x-y)$, is

$$0=-X'+2Xy\nabla+(x-y)\cdot c\cdot cxy\sqrt{X},$$

which, throwing the factor $(x-y)$, becomes

$$0=-X+2\sqrt{X}P,$$

which is the required relation.